Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 183-186
An International Biannual Refereed Journal of Life Sciences

1(AS).
Animal Science

ISSN : 0973-7057
Int. Database Index: 663 www.mjl.clarivate.com

Profile of physico-chemical properties of water samples drawn from Kanjha Lake, Purnea, Bihar (India) with respect to its limnological status

Minki Kumari*
Department of Zoology, B.N. Mandal University, Madhepura, Bihar, India

Received : 12th December, 2019; Revised :22nd February, 2020

Abstract: Assessment of physico-chemical parameters of water samples drawn from four different corners viz. east, south, west & north of standing water body of Kanjha Lake situated in Purnea district, Bihar, India were carried out during January to December 2013-2014. PCP parameters such as turbidity, temperature, transparency, pH, DCO_2, DO_2, nitrate and phosphate were analyzed. The results indicate that most of the parameters from all four corners of the water bodies were suitable for fisheries as well as other human need. Nevertheless, unmindful anthopogenic activities like bathing, washing, cattle sitting etc. have introduced some undesirable pollutants which may worsen the lake status in few years if conservation measures are not taken.

Keywords:- PCP, turbidity, transparency, pH, DCO_2, DO_2, Kanjha Lake, Purnea, Bihar

INTRODUCTION

The physico-chemical properties of any water body and biological activity maintain the status of its aquatic ecosystem. Further water condition play a very important role in the production of air breathing fishes related to the limnological characteristics of the aquatic system. The polluted state of water resources has led to steady decline in aquatic productivity. Therefore, limnological investigation is needed. So, the monitoring of the lake water is necessary step to eliminate the pollutants and their effect on living organisms.[1]

The blend of physical and chemical properties ultimately constitute the profile of complete biological status of water body in general and that of Kanjha Lake in particular, hence the present study has been undertaken to investigate the limnological characteristics which can promote the fish production and fisheries in the lake as well as adjoining water bodies.[2 & 3]

*Corresponding author :
Phone : 9561258997
E-mail : dr.mkumari30@gmail.com

MATERIALS & METHODS

Water samples were collected for different physico-chemical analysis from four different corners viz. east, south, west and north of Kanjha Lake, Purnea district, Bihar. The samples were collected from each site once a day. The analysis for physiochemical characteristics of water was done following the standard methods given by APHA (1998)[4], Golterman (1969)[5], and Trivedi and Goel (1984)[6]. Some of the parameters were analyzed in the field while for most of the parameters the samples were preserved using the suitable preservatives.

The samples were collected in two liter polythene bottles during the morning hours between 8.00 am to 11.30 am. Water temperature were measured using centigrade mercury thermometer, transparency of the water was measured with the help of Secchi disc of 20 cm diameter.

Turbidity was measured with the help of nephelometer, pH was measured using digital pH meter while the dissolved carbon dioxide (DCO_2) and dissolved oxygen (DO_2) were estimated using reagents such as sodium thiosulphate, sodium

hydroxide and phenolphthalein indicator. Nitrate and Phosphate was estimated using calorimeter. Each of the samples (replications of four corners) was analyzed and the results have been given in the table.

OBSERVATION

Table 1 : Physicho-chemical profile of water samples of Kanjha Lake, Purnea, Bihar (January to December 2013)

Samples and Parameter		Mean Monthly Data											
		Jan.	Feb.	March	April	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
S-I:P.P	Turb.	1026	1136	575	775	710	804	1180	690	888	680	868	1165
	Temp.	18.00	19.10	24.90	25.30	29.70	30.90	26.70	27.30	26.80	25.70	23.30	17.90
	Trans.	43.00	43.00	42.00	44.00	60.00	17.00	6.00	7.00	23.00	13.00	28.00	47.00
C.P	pH	8.24	8.54	8.10	7.60	7.65	7.80	8.10	8.20	8.10	8.10	8.10	8.20
	DO_2	10.30	11.00	10.50	7.60	4.10	4.50	7.70	6.50	7.40	7.40	10.50	12.60
	DCO_2	2.00	1.70	1.40	1.50	2.10	2.40	2.20	2.10	2.40	2.80	2.60	2.70
	Nitr.	0.20	4.8	4.8	3.0	3.0	3.0	2.3	3.6	4.8	4.0	4.7	2.0
	Phosp.	0.10	0.20	0.20	0.30	0.40	0.20	0.10	0.10	0.10	0.10	0.10	0.10
S-II:P.P	Turb.	1206	1452	1452	1070	1460	1310	1430	1240	1220	1036	1126	145
	Temp.	19.20	19.90	26.30	25.90	30.80	31.80	30.70	29.80	26.90	29.30	22.30	21.10
	Trans.	42.00	41.00	27.00	58.00	65.00	17.00	6.00	6.50	19.00	12.00	27.00	46.00
C.P	pH	8.30	8.40	7.70	7.60	7.00	7.00	7.50	7.60	8.00	7.00	7.00	7.00
	DO_2	10.20	11.10	10.80	7.10	5.00	5.00	8.20	7.00	8.30	7.50	10.00	9.30
	DCO_2	1.70	1.80	1.00	1.40	2.00	2.20	0.50	2.00	2.00	2.30	2.60	2.50
	Nitr.	88.0	1100	1225.0	150.0	225.0	1750.0	32.0	45.0	105.0	110.0	70.0	85.0
	Phosp.	0.20	0.30	0.30	0.30	0.40	0.20	0.20	0.20	0.20	0.20	0.20	0.20
S-III:P.P	Turb.	1150	1300	1240	840	1150	1180	1300	1240	900	800	1000	1215
	Temp.	18.00	19.40	24.60	25.90	32.80	34.30	28.20	28.50	27.20	27.30	20.00	17.80
	Trans.	30.00	27.00	40.00	40.00	45.00	21.00	21.00	9.00	10.00	27.00	29.00	60.00
C.P	pH	8.35	8.40	8.30	8.30	8.25	8.15	8.15	8.20	8.20	8.20	8.25	8.30
	DO_2	10.10	11.00	11.50	8.00	5.00	6.30	7.00	9.60	9.80	8.00	8.30	10.00
	DCO_2	2.00	2.00	1.80	1.40	1.80	1.80	2.00	2.30	2.00	2.00	1.20	2.50
	Nitr.	60.0	160.0	110.0	90.0	35.0	40.0	29.0	35.0	90.0	95.0	58.0	55.0
	Phosp.	0.10	0.20	0.20	0.40	0.30	0.20	0.10	0.10	0.10	0.10	0.10	0.10
S-IV:P.P	Turb.	1206	1452	1452	1070	1460	1310	1430	1240	1220	1036	1126	145
	Temp.	19.20	19.90	26.30	25.90	30.80	31.80	30.70	29.80	26.90	29.30	22.30	21.10
	Trans.	42.00	41.00	27.00	58.00	65.00	17.00	6.00	6.50	19.00	12.00	27.00	46.00
C.P	pH	8.30	8.40	7.70	7.60	7.00	7.00	7.50	7.60	8.00	7.00	7.00	7.00
	DO_2	10.20	11.10	10.80	7.10	5.00	5.00	8.20	7.00	8.30	7.50	10.00	9.30
	DCO_2	1.70	1.80	1.00	1.40	2.00	2.20	0.50	2.00	2.00	2.30	2.60	2.50
	Nitr.	88.0	1100	1225.0	150.0	225.0	1750.0	32.0	45.0	105.0	110.0	70.0	85.0
	Phosp.	0.20	0.30	0.30	0.30	0.40	0.20	0.20	0.20	0.20	0.20	0.20	0.20

RESULTS & DISCUSSION

The results of physico-chemical characteristics as furnished in the table indicate that water temperature fluctuated from 17.90-30.90°C in sample I, 19.20-31.80°C in sample II, 17.80-34.30°C in sample III, 19.20-31.80°C in sample IV respectively. Water temperature was found to be higher in summer season and low in winter. More penetration of sunlight and longer duration of receiving sunlight in a day is the reason for higher temperature during summer season. The summer temperature (May-Jul) was always above the winter temperature (Jan- Feb) due to lack of sunlight. The water temperature recorded in all the sample was below the WHO standard of 30°C to 35°C.

Under chemical parameter, pH of water is very important factor in determination of water quality act as it may be acidic, neutral or basic in nature. pH usually affects the other chemical

reactions such as solubility, ionization and metal chelating which is responsible for toxicity. Sample wise examined pH value was found to range between 7.60-8.54 ppm (S I), 7.00-8.40 ppm (S II), 8.15-8.40 ppm (S III) & 7.00-8.40 ppm (S IV) respectively. Accordingly water samples of the pond can be regarded as either neutral or basic. Dissolved oxygen is also one of the important parameters of water which directly effects the survival and distribution of flora and fauna in an ecosystem. The value of DO_2 ranges from 4.10- 12.60ppm.

In summer dissolved oxygen decreased due to increase in temperature and also due to increased microbial activity. The high DO_2 in winter may be due to decrease in temperature and less duration of bright sunlight has less influence on the percent of soluble oxygen and carbon dioxide. The value of DO_2 of four different corners remained constant except during the winter season ranging high 10-12ppm in all four corners. The high concentration of DO_2 during winter may be due to dense growth of algae or plant due to photosynthesis. The royal commission has reported a scale for deciding the quality of water based on DO_2. The content of DO_2 of 7ppm in water is considered as very clear, 6ppm as moderate, 5ppm as doubtful and 4ppm or below as bad.

Table 2 : Physicho-chemical profile of water samples of Kanjha Lake, Purnea, Bihar (January to December 2014)

Samples and Parameter		Mean Monthly Data											
		Jan.	Feb.	March	April	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
S-I:P.P	Turb.	1206	1452	1452	1070	1460	1310	1430	1240	1220	1036	1126	145
	Temp.	19.20	19.90	26.30	25.90	30.80	31.80	30.70	29.80	26.90	29.30	22.30	21.10
	Trans.	42.00	41.00	27.00	58.00	65.00	17.00	6.00	6.50	19.00	12.00	27.00	46.00
C.P	pH	8.30	8.40	7.70	7.60	7.00	7.00	7.50	7.60	8.00	7.00	7.00	7.00
	DO_2	10.20	11.10	10.80	7.10	5.00	5.00	8.20	7.00	8.30	7.50	10.00	9.30
	DCO_2	1.70	1.80	1.00	1.40	2.00	2.20	0.50	2.00	2.00	2.30	2.60	2.50
	Nitr.	88.0	1100	1225.0	150.0	225.0	1750.0	32.0	45.0	105.0	110.0	70.0	85.0
	Phosp.	0.20	0.30	0.30	0.30	0.40	0.20	0.20	0.20	0.20	0.20	0.20	0.20
S-II:P.P	Turb.	1165	868	680	888	690	710	1180	804	775	575	1136	1026
	Temp.	17.90	23.30	25.70	26.80	27.30	26.70	30.90	29.70	25.30	24.90	19.10	18.00
	Trans.	47.00	28.00	13.00	23.00	7.00	6.00	17.00	60.00	44.00	42.00	43.00	43.00
C.P	pH	8.24	8.54	8.10	7.60	7.65	7.80	8.10	8.20	8.10	8.10	8.10	8.20
	DO_2	10.30	11.00	10.50	7.60	4.10	4.50	7.70	6.50	7.40	7.40	10.50	12.60
	DCO_2	2.00	1.70	1.40	1.50	2.10	2.40	2.20	2.10	2.40	2.80	2.60	2.70
	Nitr.	0.20	4.8	4.8	3.0	3.0	3.0	2.3	3.6	4.8	4.0	4.7	2.0
	Phosp.	0.10	0.20	0.20	0.30	0.40	0.20	0.10	0.10	0.10	0.10	0.10	0.10
S-III:P.P	Turb.	1206	1452	1452	1070	1460	1310	1430	1240	1220	1036	1126	145
	Temp.	19.20	19.90	26.30	25.90	30.80	31.80	30.70	29.80	26.90	29.30	22.30	21.10
	Trans.	42.00	41.00	27.00	58.00	65.00	17.00	6.00	6.50	19.00	12.00	27.00	46.00
C.P	pH	8.30	8.40	7.70	7.60	7.00	7.00	7.50	7.60	8.00	7.00	7.00	7.00
	DO_2	10.20	11.10	10.80	7.10	5.00	5.00	8.20	7.00	8.30	7.50	10.00	9.30
	DCO_2	1.70	1.80	1.00	1.40	2.00	2.20	0.50	2.00	2.00	2.30	2.60	2.50
	Nitr.	88.0	1100	1225.0	150.0	225.0	1750.0	32.0	45.0	105.0	110.0	70.0	85.0
	Phosp.	0.20	0.30	0.30	0.30	0.40	0.20	0.20	0.20	0.20	0.20	0.20	0.20
S-IV:P.P	Turb.	1150	1300	1240	840	1150	1180	1300	1240	900	800	1000	1215
	Temp.	18.00	19.40	24.60	25.90	32.80	34.30	28.20	28.50	27.20	27.30	20.00	17.80
	Trans.	30.00	27.00	40.00	40.00	45.00	21.00	21.00	9.00	10.00	27.00	29.00	60.00
C.P	pH	8.35	8.40	8.30	8.30	8.25	8.15	8.15	8.20	8.20	8.20	8.25	8.30
	DO_2	10.10	11.00	11.50	8.00	5.00	6.30	7.00	9.60	9.80	8.00	8.30	10.00
	DCO_2	2.00	2.00	1.80	1.40	1.80	1.80	2.00	2.30	2.00	2.00	1.20	2.50
	Nitr.	60.0	160.0	110.0	90.0	35.0	40.0	29.0	35.0	90.0	95.0	58.0	55.0
	Phosp.	0.10	0.20	0.20	0.40	0.30	0.20	0.10	0.10	0.10	0.10	0.10	0.10

Dissolved carbon dioxide in the present study found at the range between 0.50 and 2.80mg/l. the maximum free CO_2 (2.80mg/l) was recorded in site I and minimum CO_2 (0.50mg/l) in site II. Respiration by zooplankton and other organisms may be one of the probable cause of very high concentration of dissolved carbon dioxide in present investigation. Nitrate concentration ranged from 0.20-0.70ppm being the maximum at site II and minimum at site II of Muzaffarpur district. The observed value of nitrate was found below the WHO permissible limit 45ppm. Nitrate is attributed mainly due to anthropogenic activities such as agricultural runoff, refuse dump runoff or contamination with human or animal wastes. The concentration often fluctuates with the season and may increase when the river is fed by nitrate rich qualifiers. The fluctuation in nitrate concentration was due to degradation of organic matter.

Turbidity is the cloudiness or haziness of a fluid caused by large numbers of individual particles that are generally invisible to the naked eyes, similar to smoke in air. The measurement of turbidity is a key test of water quality. Turbidity ranged from 145ppm to 1452ppm having maximum value at site II and minimum at site I. Turbidity in open water may be caused by growth of phytoplankton. Human activities that disturb land, such as construction, mining and agriculture, can lead to high sediment levels entering water bodies during rain storms due to storm water runoff. Areas prone to high bank erosion rates as well as urbanized areas also contribute large amounts of turbidity to nearby waters, through storm water pollution from paved surfaces such as roads, bridges and parking lots. Some industries such as quarrying, mining and coal recovery can generate very high levels of turbidity from colloidal rock particles. In drinking water, the higher the turbidity level, the higher the risk that people may develop gastrointestinal diseases. This is especially problematic for immunocompromised people, because contaminants like viruses or bacteria can become attached to the suspended solids.

Phosphate has been found trace to 0.10-0.40ppm. The value of phosphate usually remained constant however fluctuated to maximum during the month of April and May in both the site I&II which indicates the moderate to high level of pollution in the Kanjha Lake, Purnea, Bihar.

CONCLUSION

On the basis of experimental findings obtained from the present investigation it can be concluded that the physico-chemical profile of the water samples drawn from all the four corners of the water body is well within the limits WHO standard. However, samples of south and west corners are where slightly polluted, anthropogenic interferences regularly happen. Unmindful anthropogenic activities like bathing, detergent washing and agricultural runoff etc are likely to affect the Kanjha Lake water body. Hence it can be concluded that turbidity, temperature, transparency, pH, DO_2, DCO_2, nitrate, phosphate are within WHO limits which are the vital water quality parameters for drinking, irrigation, aquatic life & surface water.

The complete profile of physico-chemical properties of water samples from the lake therefore suggest that samples drawn from east & north corner of the lake are more promotive, those from west moderately promotive & from south are least promotive to fish production as per the available data of annual spatial yield of fish production.

ACKNOWLEDGEMENT

The author is highly thankful to the fishery department of Purnea & guide, Dr. Kapil Deo Tanti, Head Department of Zoology, M.L. Arya College, Akshay Nagar, Kasba, Purnea as well as Prof. Arun Kumar, Prof & Head, Department of Zoology, B.N. Mandal University, Madhepura, Bihar for their valuable help & guidance.

REFERENCES

1. **Helena Curtis, 1974.** Invitation to Biology, Worth Publisher, INC, New York.
2. **A.L.Lehninger, David L.Nelson & Michael M.Cox. 2008.** Principles of Biochemistry, Worth Publisher, INC, New York.
3. **Shalini Kamal, 2012.** Ph.D Thesis "Hydro-biological study of Konar Dam with special reference to diversity of plankton species abundance, Vinoba Bhave University, Hazaribagh.
4. **APHA. 1998.** Standard Methods for Examination of Water and Wastewater, 20th Ed., American Public Health Association, Washington, DC, New York.
5. **Golterman, H.L. 1969.** Methods for Chemical Analysis of Fresh Waters. I.B.P. Handbook No. 8. Black Well Scientific Publ., Oxford. 166 pp.
6. **Trivedi, R.K. & Goel, P.K., 1984.** Chemical and biological methods for water pollution studies, Environmental Publications, Karad.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 187-188
An International Biannual Refereed Journal of Life Sciences

2(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of water pollutants on the digestive enzyme (Amylase) of alimentary canal of aquatic insects

Md. Sami*
Department of Zoology, Sri Radhakrishna Kedia Girls' High School (+2), Muzaffarpur, Bihar, India

Received : 24th December, 2019; Revised : 22nd February, 2020

Abstract: The Water Pollutants DDT were taken, this pollutant affects and decline the Enzyme "Amylase" of alimentary canal of aquatic insects. The concentrations of Amylase were changed with the treatment of pollutant compared with controlled condition.

Keywords:- DDT, Amylase, Alimentary canal

INTRODUCTION

Life is an intricate meshwork involving a perfect coordination of a vast majority of chemical reaction. Some of these reactions result in synthesizing large molecules, other cleaving large molecules and all of them either utilize energy or liberate energy. All these reactions occur very slowly at low temperature and the atmospheric pressure the condition under which living cells carry on their life process.

The name enzyme literally means 'in yeast' this was referred to denote one of the most noteworthy reactions wherein the production of ethyl alcohol and carbon dioxide through the agency of an enzyme, the Zymase present in yeast takes place.

As the present work is based on enzyme (amylase), attempt to investigate the quantitative variation in the activity of Amylase, in the gut of a strictly carnivorous aquatic insect *Cybister confuses* during control and after sub lethal dose with pollutant DDT.

*Corresponding author :
Phone : 9973163833
E-mail : md.sami.muz786@gmail.com

Multiple forms of amylase may occur in insects and hence been shown to be under genetic control in Drosophila where six forms are observed. The multiple forms of Bita-amylase reported for cereal seeds.

MATERIALS AND METHODS

The control insects and the treated insects with sub lethal dose of DDT after 24 hours, 48 hours, 72 hours, and 168 hours, the specimen were dissected on ice bath att 4±1C and the gut of *Cybister confusus* were taken out by removing adhering fat and other unnecessary materials. The cleared gut was then homogenized in Chilled phosphate buffer at pH 8.0 and then centrifuged at 3000 rpm for 16 minutes by the colorimetry the supernatant was used for enzyme.

RESULT AND DISCUSSION

DDT resulted into a slight decline in amylase activity in the fore gut. Mid gut amylase activity of the control insect was 7±0.6 S.D Enzyme unit/gut reg/minute. DDT resulted in gradual significant decline of the enzyme activity at every stage of the treatment in both mid and hind guts. That pollutant were more potent in adversely affecting amylase activity of the mid gut.

Amylase-(Enzyme Unit/Gut region/Minute) ± SD activity to control, and DDT treatted gut of *Cybister confusus* sharp

	Alimentary Canal (*Cybister confusus*)		
Condition	Fore Gut	Mid Gut	Hind Gut
Control	5±0.6	7±0.5	6.5±0.6
DDT treatment After 24 Hrs.	4.5±0.6	5.0±0.3	5.5±0.6
After 48 Hrs	4.8±0.5	4.5±0.4	4.8±0.5
After 72 Hrs	3.5±0.3	3.0±0.3	3.5±0.3
After 168 Hrs.	2.5±0.3	2.5±0.3	4.6±0.5

ACKNOWLEDGMENT

We acknowledge the help provided by the Entomology research laboratory, Zoology Department BRA Bihar University, Muzaffarpur and my supervisor Dr. O.P.Singh principal R.D.S College

REFERENCES

1. **Day, M.F. 1949.** Distribution of alkaline phophatase. *Australian Jour.Sci. Ros.* **B2:**31-34
2. **Khan, S.N.Choudhary, Chanda kumari and S.Ehteshamuddin. 1996.** Effect of DDT and EDTA on the Alaline Phosphatase concentration in the haemolymph, of immature and mature female *Cybister confuses*. *Environ. Ecol.,* **14:**797-799
3. **Kilby, B.A. 1965.** Intermediately metabolism and the insect fat body. In Aspects of insect Biochemistry (T.W. Door Win, ed.) *Biochem. P.Sc pymr.* **28:**39-48 Academic Press.
4. **Leena Kumar. 1993.** Effect of chemicals on the growth of an aquatic living system, Ph.D Thesis, University of Bihar Muzaffarpur.
5. **Singh, J.P. and D.K. Jindal. 1982.** Glycogen content in the developing stages of coccinellids (Coleoptera: Coccinellidae) *Life Sci. Adv.***1:**19-22.
6. **Pilat, M. 1935.** Action of insecticides on gut. *Bull. Entol. Res.* **26:**165-180.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 189-192
An International Biannual Refereed Journal of Life Sciences

3(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Relative abundance of composite insect pests infestation led by white fly, *Bemisia tabaci* on the sunflower plant *Helianthus annuus* L.

Mausam Kumari*
University Department of Zoology, B.N.Mandal University, Madhepura, Bihar, India

Received : 08th January 2020 ; Revised : 18th February 2020

Abstract : In addition to the common sunflower pest, *Bemisia tabaci* (Gennadius) (Hemiptera: Aleyrodidae) it has been observed that several other insects also attack this host plant such as *Empoasca* spp. (Lepidoptera:Noctuidae), *Thysanoplusia orichalce* (Lepidoptera:Noctuidae), *Helicoverpa armigera* (Lepidoptera:Noctuidae) *Nezara viridula* (Heteroptera:Pentatomidae) thus reflecting composite pest infestation. These insects are taxonomically represented by the Hemipterans, Lapidopteran and Heteropterans. The reason of such composite insect pest infestation is non else than availability of variety of food resource in the sunflower plant palatable to these insects for their survival. In the present article, standard statistical tools like Relative Abundance % of all the members of insect pest complex causing composite infestation has been statistically computed from the random sampling raw data of above five pests including *Bemisia tabaci.* The results are very significant and meaningful suggestive of several singular and plural pest protection strategies for sunflower plant related to the differential degree of insect pest attack on the sunflower.

Key words: Composite insect pest infestation, Lapidoptera, Heteroptera, Hemiptera, sunflower plant *Bemisia tabaci*, (*Helianthus annuus* L.), Statistical diversity tools

INTRODUCTION

One of the major competitors of human beings is insects that may be both harmful and beneficial. Nevertheless the harmful impact of insects outweigh the beneficial one, therefore, insects are our rivals, because most of them lead pest life feeding on plants and crops. As such they are major threat to crop production and human survival. Some of them serve as vector for virus and other diseases. Some of the severe infestation on sunflower by insects pests are leaf curling, chlorosis and premature senescence in addition to nematodes infestation that also result a wide range of loss in production and yield of sunflower crop. Nelson (2008)[1] reported the insect pests of sunflower as whitefly, *Bemisia tabaci*; aphids, *Aphis gossypii*; jassids, *Amrasca devastans;* bud moth, *Heliothis armigera* and surface grass hopper, *Chrotogonus* spp. Holt (2001)[2] reported the insect pests attack on sunflower are *Bemisia tabaci*, *Empoasca* spp., *Thysanoplusia orichalce, Diacretia obliqua*, *Nezara viridula*, *Helicoverpa armigera* and *Nysius inconspicuus*.

In the present article, statistical approach has been taken to evaluate the proportionate attack and damage to sun flower caused by five more insect species other than *Bemisia tabaci.*

*Corresponding author :
Phone : 8709910392
E-mail : mausambaba2310@gmail.com

As regards the host plant-Sunflower, *Helianthus annuus* (L.) popularly known as "Surajmukhi" belongs to the family Asteraceae and genus *Helianthus* is one of the most important oil crops throughout the world grown on over 22 million hectares worldwide, with a production of 26 million tones per year. The name has its origin in Greek "Helios" means "Sun" and "Anthois" means "flower". Sunflower is an important oilseed crop in India. It is also cultivated for edible seed (Khan *et al*., 2007)[4].

MATERIAL & METHODS

Selection of the sampling site as follows up schedule of our study was done randomly and the sites were located in the northern zone of Madhepura district. Madhepura is located at 25°34'-26°07' North Latitude & 86°19'-87° 07' East Longitude. It is surrounded by Araria & Supaul in the North and Khagaria & Bhagalpur in the South, Purnea district in the East and Saharsa in the West. It is situated in the plains of river Koshi & located in the Northeastern part of Bihar. The primary occupation of the people of the district is agriculture. The total geographical area of the district is 1788 Sq. Km.

The field experiment was laid in randomized block design during 2018 in four different areas of Madhepura where the production was very high. The observation data of degree of infestation of the fly with respect to the four different sampling areas- Gwalpara, Puraini, Bihariganj and Shankarpur chosen for the observation of incidence as well as quantum of damage caused by the whiteflies have been given in Table 1-4 along with geographical map of the sampling sites. These places were visited weekly after 15 days of germination to record the insect infestation till the maturity of the plant of sunflower. Observations of infestation on randomly selected plants from each plot by whitefly and other allied species were made on leaves shoot and flower rosette. The mean population of each pest species was worked out numerically which were subjected to statistical analysis.

Data Recording

Five plants were randomly selected for data recording and data regarding all the insect pest infestation was recorded weekly on per plant basis; (upper, middle and lower leaves of plants were examined) in case of whitefly larvae of Head moth were counted when founded in the heads of sunflower plants already selected.

RESULT & DISCUSSION

Table 1: Random sampling data of five different species of insect pests including *B. tabaci* thriving on sunflower plant with their average adults per plant in the plots of Gwalpara site during summer of 2018

Sl. No.	Common Name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per plant(n)	Relative abundance % (n/N* 100)
1	Whitefly	*Bemisia tabaci*	Hemiptera : Aleyrodidae	12	36.3
2	Leaf hopper	*Empoasca*	Lepidoptera Noctuidae	04	12.1
3	Flower Moth	*Thysanoplusia*	Lepidoptera Noctuidae	02	6
4	Bollworm	*Helicoverpa*	*Lepidoptera* Noctuidae	06	18.1
5	Stink bug	*Nezara*	Heteroptera: Pentatomidae)	09	27.2
	Total			N =33	

Table 2: Random sampling data of five different species of insect pests including *B. tabaci* thriving on sunflower plant with their average adults per plant in the plots of Puraini site during summer of 2018

Sl. no	Common Name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per plant (n)	Relative abundance %(n/N* 100)
1	Whitefly	*Bemisia tabaci*	Hemiptera : Aleyrodidae	10	31.2
2	Leaf hopper	*Empoasca*	Lepidoptera Noctuidae	05	15.6
3	Flower Moth	*Thysanoplusia*	Lepidoptera Noctuidae	08	25
4	Bollworm	*Helicoverpa*	*Lepidoptera* Noctuidae	02	6.2
5	Stink bug	*Nezara*	Heteroptera: Pentatomidae)	07	21.8
	Total			N=32	

Table 3: Random sampling data of five different species of insect pests including *B. tabaci* thriving on sunflower plant with their average adults per plant in the plots of Bihariganj site during summer of 2018

Sl. no	Common Name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per plant (n)	Relative abundance %(n/N* 100)
1	Whitefly	*Bemisia tabaci*	Hemiptera : Aleyrodidae	9	31
2	Leaf hopper	*Empoasca*	Lepidoptera Noctuidae	03	10.3
3	Flower Moth	*Thysanoplusia*	Lepidoptera Noctuidae	08	27.5
4	Bollworm	*Helicoverpa*	*Lepidoptera* Noctuidae	05	17.2
5	Stink bug	*Nezara*	Heteroptera: Pentatomidae)	04	13.7
	Total			N=29	

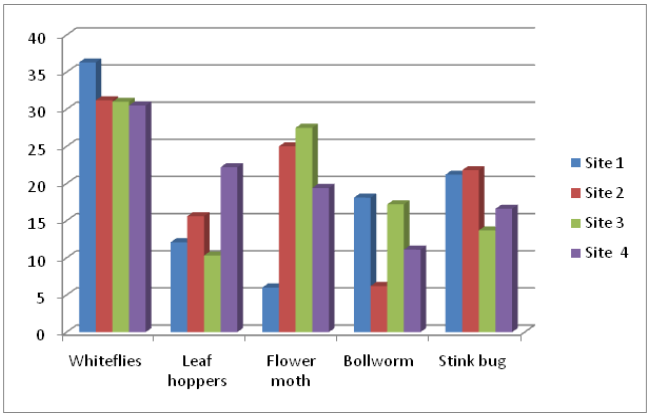
Table 4: Random sampling data of five different species of insect pests including *B. tabaci* thriving on sunflower plant with their average adults per plant in the plots of Shankarpur site during summer of 2018

Sl. no	Common Name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per plant (n)	Relative abundance %(n/N* 100)
1	Whitefly	*Bemisia tabaci*	Hemiptera : Aleyrodidae	11	30.5
2	Leaf hopper	*Empoasca*	Lepidoptera Noctuidae	08	22.2
3	Flower Moth	*Thysanoplusia*	Lepidoptera Noctuidae	07	19.4
4	Bollworm	*Helicoverpa*	*Lepidoptera* Noctuidae	04	11.1
5	Stink bug	*Nezara*	Heteroptera: Pentatomidae)	06	16.6
	Total			N=36	

The above data shows that the population of the randomly sampled insect pest species as observed on the sunflower plant was having staggered picture showing mosaic composite infestation of high degree affecting the production of the sunflower crop.[5] There are many studies that show the attack of insect on the sunflower crops in India (Sandhu *et al.*, 1973)[6] This finding is also in line with (Makhdoomi *et al.*, 1984)[7] who also reported that 43 insect species attack this crop whereas (Hassan *et al.*, 1984)[8] identified 19 insect species for the same. There is an evidence of several insects forming pest complex which invade the sunflower crop plants significantly thereby causing composite infestation. *Helicaverpa armigera*, *Aphis gossypii* (Glover), *Bemisia tabaci* (Gennadius), *Amrasca devastans* (Ishida) and *Atractomorpha crenulata* (Fabricius) are also reported to attack the sunflower crop.[8] Among the species identified *B. tabaci* attack on sunflower has been seen to be the highest while the Bollworm was observed throughout the cropping season but its population remained low.

In the present studies the damage caused by the white flies & four other species was observed on sunflower in the four different areas of northern parts of Madhepura throughout the summer seasons including the month of April, May and June. During the research period it was found that the Gwalpara was highly infested by the whiteflies (Table 1). While Bihariganj showed the lowest infestation of whiteflies followed by differential degree of infestation by remaining four species as depicted in the comparative RA % graph.

Fig : Site specific comparative RA % graph of five insect pests causing composite infestation on the sunflower plant dominated by the white fly *Bemisia tabaci*



CONCLUSION

Whitefly (*Bemisia tabaci)* has been found to be the most dominant pest in the composite infestation of group of five insect on the sunflower plant, although Jassid/Leaf Hopper (*Amrasca biguttula*) and Head moth (*Helicoverpa armigera*) are also notorious pests of the sunflower crop. In this investigation the area that was highly infested by the whiteflies was Gwalpara area (S 1) of Madhepura, while Bihariganj (S 3) had the lowest infestation caused by the insect pest complex led by whiteflies. Screened sunflower hybrids showed significant susceptibility to the whiteflies and other species in the complex which were responsible for the colossal loss in the yield. Therefore it is recommended to plant the hybrid species over the normal sunflower strain to minimise the huge expenditures likely to be met by the farmers concerned in order to adopt measures to control them.

ACKNOWLEDGEMENT

The author is extremely thankful to her guide and Head, Prof. (Dr.) Arun Kumar University, Department of Zoology, B.N.Mandal University, Madhepura, Bihar, India.

REFERENCES

1. **Nelson, S. 2008.** Sooty Mold. Plant Disease PD-52. Mānoa, Honolulu, Hawaii, Cooperative extension Service, College of Tropical Agriculture and Human Resources, University of Hawaii.

2. **Holt, J. & Colvin, J. 2001.** Observation and theory of white fly-borne virus disease epidemics. pp. 331–343 in Edited for the British Society for Plant Pathology by Jeger, M.J. & Spence, N.J. (Eds) Biotic Interactions in Plant-Pathogen Associations. Wallingford, Oxon, UK, CABI publishing.

3. **Maruthi, M.N., Colvin, J., Seal, S., Gibson, G. & Cooper, J. 2002b.** Co-adaptation between cassava mosaic gemini viruses and their local vector populations. *Virus Research.* **86:** 71–85.

4. **Khan, S. N., Ayub, N., Ahmad, I., Shehzad, I. & Asad. 2007.** Effect of climatic conditions on life cycle of charcoal rot infected sunflower plant. *Pakistan Journal of Botany.* **39:** 2657- 2662.

5. **Nasir, N. 2003.** Detecting seed borne fungi of soybean by different incubation method. *Pakistan Journal of Plant Pathology.* **3:** 114-118.

6. **Sandhu GS, Brar KS & Bhalla JS. 1973.** Pests of sunflower and other insects associated with sunflower crop. *Oilseeds J.* **3:**19-26.

7. **Makhdoomi SMA, Bhatti MN, Zia MA & Shafiq M. 1984.** Studies on insect pests associated with sunflower crops in Faisalabad. *J.Agric.Res*.**22:**51-62.

8. **Hassan A.M, Mahmood T & Ahmad M. 1984.** Insect pests of sunflower at Faisalabad. Proceedings of the National Workshop, April. 28-30.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 193-198 4(IS).
An International Biannual Refereed Journal of Life Sciences
Interdisciplinary Science

ISSN : 0973-7057

Ind. Database Index: 663 www.mjl.clarivate.com

Impact of accumulation of inorganic nitrogen oxides (NO_3 & NO_2) in plants and its implication on human health

Sanjeev Kumar Choudhary*
Department of Chemistry T.N.B College, T.M University, Bhagalpur, Bihar, India

Received : 15th December, 2019 ; Revised : 11th February, 2020

Abstract : Nitrite (NO_2) and nitrate (NO_3) are as a part of nitrogen natural cycle and essential nutrients for plant protein synthesis. Plants take nitrogen in ionic form from soil. Nitrate is a naturally occurring form of nitrogen formed from fertilizers, decaying plants, manure and other organic residues. Due to increased use of synthetic nitrogen fertilizers and livestock manure in agriculture, vegetables and drinking water may contain higher concentrations of nitrate now than in past. Nitrate is also used as a food additive, mainly as preservative and antimicrobial agent. Its level commonly present in food is not toxic to humans but they act as the reservoir for conversion to nitrite by intestinal flora. A high nitrate accumulation in plants results in nitrite production, which then converted into nitric oxide (NO) which together with O_2, could be rapidly catalyzed by nitrate reductase into peroxynitrite ($ONOO^-$). The presence of nitrite and nitrate in food chains is associated with numerous health problems including stomach, gullet and gallbladder cancer and methaemoglobinaemia. This review focuses on the factors for accumulation of nitrate and nitrite in some plants and its potential risks on human health.

Keywords : Nitrate, nitrite, food additive, methaemoglobinaemia, human health

INTRODUCTION

Nitrates are the metals that are present in soils, waters, all plant materials and in metals. They are also found in small concentrations in air as a result of air pollution.[1] Nitrate and nitrite are essential nutrients for plant protein synthesis and play a critical role in nitrogen cycle.[2] Excessive use of nitrogen fertilizers has caused an imbalance of nutrients in the soil which influences the nitrate content of plants.[3,4] Nitrites are formed in nature by the action of nitrifying bacteria as an intermediate stage in the formation of nitrates, but concentrations in plant and water are usually very low.

The nitrate accumulation in plants is a multifactorial process - nutritional, environmental and physiological being the major players. In cereals, light intensity and nitrogen fertilization are major factors.[5] Many nutrients, such as chloride, calcium, potassium, sulphate and phosphorus are also involved in nitrate accumulation, other factors different parts of the plant. This review focuses on the factors affecting accumulation of nitrate, the dietary intake of nitrate and its implications on human health.

Nitrates and nitrites can be categorized into inorganic and organic forms based on their chemical structure. Inorganic nitrate (NO^{3-}) and nitrite (NO^{2-}) are water soluble (as a result of their interaction with the positively charged portions of polar water molecules and commonly exist as salts of nitric acid and nitrous acid, respectively). They are often bound to a metal cation such as Na^+ or K^+ and occur naturally through the fixation of atmospheric nitrogen and oxygen as a part of the environmental nitrogen

*Corresponding author :
Phone : 9852020454
E-mail : sanjeevkumarchoudhary1972@gmail.com

cycle (the cyclic movement of nitrogen in different chemical forms from the environment, to organisms, and then back to the environment. Inorganic nitrites are also produced endogenously through oxidation of nitrous oxide (NO) formed from the enzymatic degradation of L-arginine and through the reduction of nitrate with xanthine oxidoreductase.[6-8]

Vegetables are the major source of the daily intake of nitrate by human beings, supplying about 72 to 94% of the total intake.[9] Under excessive application of nitrogen fertilizers, these vegetables can accumulate high levels of nitrate. Investigations have revealed that high nitrate accumulation in plants results in nitrite production. Nitrite is the compound that is of more health concern.[10]

1. FACTORS RESPONSIBLE FOR NITRATE ACCUMULATION

1.1 Environmental factors

As we know that to fulfill the requirements of nitrogen plants adopts various strategies. Therefore nitrates play an important role in metabolism in higher plants. It is the major nutrient form of nitrogen in most soils and is often the first factor limiting plant growth. In the great majority of cases it is assimilated so rapidly that it's concentration within plant tissues never raises to critical levels.[11] There are several reports where authors have explained the effect of moisture on nitrate concentration. Tisdale and Nelson (1975) and Gomm (1979) reported that waterlogged soils tend to be more anaerobic than moist soils, thus inhibiting microbial conversion of ammonia to nitrate. On the other hand, nitrate is released from the more complex organic forms by microbial activity that requires moisture. The nitrogen thus released, as well as any nitrogen added as fertilizer, must then move, through water, to the absorbing roots.[11-13] Thus a certain amount of moisture is needed in the soil, and either extreme in soil moisture would lead to abnormal conditions. Uptake of nitrate is regarded as very sensitive to temperature because it depends not only on the temperature of the root environment but also upon the temperature of the shoot environment. Santamaria *et al.* (2001) observed an interaction between light intensity, nitrogen availability and temperature on nitrate accumulation in rocket. There is an increase intake of nitrate accumulation at high temperature but low intensity of light. Under high light intensity, an increase in temperature increases the nitrate content but mainly when nitrogen supply is high. The nitrate content was found to be less in the year that had a high rainfall.[14]

1.2 Nutritional factors

During the nitrogen fertilization accumulation of nitrate by the plant tissues are very high. Nazaryuk *et al.* have studied the role of agricultural chemicals in regulating the nitrate accumulation in plants. Nitrate get stored in the vacuole and can be remobilised subsequently when nitrogen supply is insufficient.[15] Accumulation of nitrate in vegetables often depends on the amount and kind of nutrients present in the soil and closely related with the time of application and the amount and composition of the fertilizers applied.[16] Plants accumulate more nitrate as the nitrogen fertilization increases.[17,18] Depending on which plant part is to be consumed, appropriate fertilizer can be selected, eg., the order of nitrate accumulation in cole leaves, upon application of different nitrogen fertilizers is urea, ammonium carbonate, ammonium nitrate, ammonium sulphate; for nitrate accumulation in petioles, it is urea ammonium nitrate, ammonium sulphate, ammonium carbonate.[15]

Accumulation of nitrate is a function of increase supply of nitrate by fertilization and mineralization of soil organic matter. Vegetables supplied with organic fertilizers have a low nitrate content, compared with mineral fertilized or conventionally grown vegetables.[19] Therefore the higher the nitrogen availability (mineral fertilizer >liquid manure = slurry > manure > compost) and the lower the assimilation intensity (e.g. by site conditions and season effects), the greater would be the nitrate accumulation.

Since phosphorous has multiple functions in plant metabolism therefore it has indirect effect on plants. A plant deficient in phosphorus will lack NADPH, on which the nitrogen reducing enzymes depend and any nitrate taken up will thus accumulate.[20] Ahmed *et al.* (2000) have shown a reduction in nitrate content due to increasing phosphorus fertilization. Inorganic phosphorus within the plant is necessary for the metabolism and storage of nitrate, but high concentrations inhibit enzyme reactions, create abnormal pressure in the cell, and accelerate senescence, and the advancing plant age is accompanied by a reduced nitrate uptake and accumulation.[21] Potassium in soils is taken up as an ion by the plant in the K^+ form, and is thus important to the uptake of the nitrate ion (NO3) in maintaining ionic balance. The influence of potassium on nitrate levels is therefore largely catalytic.[22] Potassium is

however, essential in protein synthesis and photosynthesis is decreased with insufficient potassium.

Nitrate accumulation has also been studied in relation to some other chemicals. Foliar application of salicylic acid[21], molybdenum fertilizers[16] and nitrification inhibitors [23,16] and calcium[24] can materially reduce the nitrate content of plants. In lettuce leaves sugar levels and free amino acid concentrations increase under high supply of calcium and replace nitrate in the vacuole of the lettuce cell.[25] There exists a negative correlation between nitrate content and sulfate content in the lettuce plant and, therefore, sulfur deficiency might lead to an increase in nitrate content.

In general, higher nitrate levels are found in plants under any mineral deficiency stress.

1.3 Physiological factors

1.3.1 Genotypic variability

The nitrate content varies markedly with plant species, cultivars of the same species, and even genotypes with different ploidy. The causal factors might include genetic differences among genotypes in enzymes of the nitrogen metabolic pathway (nitrate reductase/nitrite reductase),the rate of nitrate uptake, the rate of uptake of other elements needed for enzyme activity. The differing capacities of nitrate accumulation can also be correlated with differing locations of nitrate reductase activity. The accumulation of nitrate also decreases with increase in carbohydrate concentration in the vacuoles. The sugar concentration and dry matter content of plants are negatively correlated to nitrate accumulation. Thus genotypes with high dry matter content could have high carbohydrate content in their vacuoles and thus need little nitrate to maintain their osmotic value.

1.2.2 Nitrate distribution within the plant

It is well established that nitrate is not uniformly distributed throughout the plant to all the tissues. In terms of grass structure stems usually contain more nitrate than do the roots, roots usually more than the leaves and leaves usually more than floral parts. The upper regions of the stems also tend to have lower nitrate content than the lower regions. The vegetable organs can be listed by decreasing nitrate content as follows: petiole leaf, stem root, inflorescence, tuber, bulb, fruit and seed. The concentration of nitrate in various plant parts could be a measure to evaluate the balance between adequate and excessive available nitrogen for optimum growth of plants during a growth season. Plants that develop fruit or storage organs, such as potato and tomato, usually have low nitrate content in petioles as they approach harvest stage. This decline can be attributed to the translocation of soluble nitrogen to the developing storage organ. Vegetables that do not develop storage organs have a different pattern of nitrate accumulation wherein nitrate often continues to accumulate with the plant's age.

2. DIETARY INTAKE OF NITRATE

Vegetables constitute nearly 72-94 % of the average human dietary intake and are the major source of nitrate. The following plant families are reckoned to belong to the nitrate accumulating group: Amaranthaceae (amaranthus), Chenopodiaceae (beetroot, swiss chard and spinach), Compositeae, Convulvulaceae, Polygonaceae, Solanaceae, Gramineae and Cruciferae.

Human intake of nitrite and nitrate is primarily by dietary ingestion of food crops, vegetables, meat and water. The average adult intake from food has been estimated to be 40-100mg/g for nitrates and 0.03-6 mg/g for nitrites as per WHO's Acceptable Daily Intake (ADI)s. Ysart *et al.* (1999) estimated for the adult human population a total nitrate intake of 93mg /day , normally through potatoes (33%). Green vegetables (21%) , other vegetables (15%), beverages (8.5%), meat products (4.2%) , fresh fruit (3.5%), dairy (3.1%), milk (2.9%), miscellaneous cereals (2.1%), bread (1.6%) and others(5.1%)

3. IMPLICATIONS ON HUMAN HEALTH

3.1 Adverse effects

Naturally nitrate is not toxic for human; however, the conversion of nitrate to nitrite in human body and its accumulation in high concentrations is very dangerous for the body. In digestive tract these substances react with natural amines and produce carcinogens compounds like N-nitrosamine. The N-nitrosamines that are formed in human stomach are known as a risk factor for stomach, gullet and gallbladder cancer.

One of the adverse effects of nitrate is related to combining these compounds with haemoglobin that is present in blood and leads to reducing nitrates to nitrites, which causes a reduction in potential oxygen transport from lungs to body tissues. The principal mechanism involves oxidation of the ferrous iron (Fe^{2+}) in haemoglobin to the ferric state (Fe^{3+}). As a consequence methaemoglobin forms and reduces the oxygen delivery

to tissues. Methaemoglobimea (Blue baby syndromes) may occur in older infants and children due to consuming vegetables. For example in Spain, several infants aged 7-13 months experienced methaemoglobinaemia on consumption of vegetables. Methaemoglobinaemia has been reported to occur in people of different ages with high nitrate ingestion, the infants and above 45 age groups being most susceptible to nitrate toxicity. In pregnant women, the level of methaemoglobin increases from the normal (0.5 to 2.5% of total haemoglobin) to a maximum of 10.5% at the 30th week of gestation and subsequently declines to normal after delivery. Thus, pregnant women might be more sensitive to the induction of clinical methaemoglobinemia by nitrites or nitrates in or around the 30th week of gestation.

Other health problems associated with nitrate toxicity include oral cancer, cancer of colon, rectum or other gastrointestinal regions, Alzheimer's disease, multiple sclerosis, spontaneous abortion or congenital defects.

CONCLUSION

Plants are the major source for nitrogen that consumes it from the soil even under normal circumstances. The accumulation of nitrate means that the rate of assimilation of nitrate to protein has not kept pace with the rate of nitrate uptake from the soil. The use of fertilizers should be however lowered so that nitrate accumulation of the nitrate could be reduced as well as loss of nitrogen to the environment could be minimized.

Since vegetables are considered as the main dietary source of nitrate intake in humans. Therefore breeding of new cultivars that do not accumulate nitrate even under heavy fertilization could be the best way to get rid of.

There are number of evidence regarding the harmful effect of nitrates to human beings. Therefore it is necessary to take preventive measures to reduce the nitrate intake.

REFERENCES

1. **Abdulrazak S, Otie S, Oniwapele YA, 2014.** Concentration of nitrate and nitrite in some selected cereals sourced within Kaduna state, Nigeria.Online *Journal of Animal and Feed Research* **4:**37-41
2. **Parkash, S., Tuli, G.D. and Basu, S.K. 1963.** Advanced Inorganic Chemistry. Part II, 10th Ed. Pp 460-464.
3. **Dapoigny L., Tourdonnet S.D., Roger-Estrade J., Jeuffroy M.-H., Fluery A. 2000.** Effect of nitrogen nutrition on growth and nitrate accumulation in lettuce (*Lactuca sativa* L.) under various conditions of radiation and temperature. *Agronomie* **20:**843-855.
4. **Vieira I.S., Vasconselos E.P., Monteiro A.A. 1998.** Nitrate accumulation,yield and leaf quality of turnip greens in response to nitrogenfertilisation, *Nutr. Cycl. Agroecosys*. **51:** 249–258
5. **Cantliffe, D.J. 1973.** Nitrate accumulation in table beets and spinach as affected by nitrogen, phosphorous, and potassium nutrition and light intensity. *Agronomy Journal.* **65:**563-565
6. **Omar SA, Artime E, Webb AJ, 2012.** A comparison of organic and inorganic nitrates/nitrites. *Nitric Oxide* **26(4):**229-240.
7. **Jansson EA, Huang L, Malkey R, Govoni M, Nihlen C, Olsson A,. 2008.** A mammalian functional nitrate reductase that regulates nitrite and nitric oxide homeostasis. *Nat Chem Biol* **4:**411-417.
8. **Rhodes P, Leone AM, Francis PL, Struthers AD, Moncada S, Rhodes PM. 1995.** The L-arginine: nitric oxide pathway is the major source of plasma nitrite in fasted humans. *Biochem Biophys Res Commun* **209:**590–596.
9. **Dich J., Jivinen R., Knekt P., Pentill P.L. 1996.** Dietary intakes of nitrate,nitrite and NDMA in the Finish Mobile Clinic Health Examination Survey, *Food Addit. Contam.* **13:**541–552.
10. **Cassens, R.G and Goutefongea R. 1996.** Fate of Nitrite in Food Proceedings 2nd International Symposium of Nitrite in Food Products, Washington. *The Netherlands*, 95-100.
11. **Wright M J & Davison D L 1964.** Nitrate accumulation in crops and nitrate poisoning in cattle. Adv. Agron. **16:** 197 - 247.
12. **Gomm B. 1979.** Herbage yield and nitrate concentration in meadow plants as affected by environmental variables. J. Range Mgt. **32:** 359 - 364.
13. **Olsen & Kurtz 1982.** Crop nitrogen requirements, utilisation and fertilisation in: Nitrogen in Agricultural soils. Stevenson F J (ed). Agronomy No. **22:** 567 Madison. Wisconsin, USA.

14. **Grzebelus D., Baranski R. 2001.** Identification of accessions showing low nitrate accumulation in a germplasm collection of garden beet. *Acta Hortic.* **563:** 253–255.
15. **Van der Leij M., Smith S.J., Miller A.J. 1998.** Remobilization of vacuolar stored nitrate in barley root cells. *Planta.* **205:** 64–72.
16. **Zhou Z.-Y., Wang M.-J., Wang J.-S. 2000.** Nitrate and nitrite contamination in vegetables in China, Food Rev. *Int*. **16:**61–76.
17. **Chen B.-M., Wang Z.-H., Li S.-X., Wang G.-X., Song H.-X., Wang X.-N. 2004.** Effects of nitrate supply on plant growth, nitrate accumulation, metabolic nitrate concentration and nitrate reductase activity in three leafy vegetables. *Plant Sci.* **167:**635–643
18. **Nazaryuk V.M., Klenova M.I., Kalimullina F.R. 2002.** Ecoagrochemical approaches to the problem of nitrate pollution in agroecosystems. *Russ. J. Ecol.* **33:** 392–397.
19. **Raupp J. 1996.** Fertilization effects on product quality and examination of parameters and methods for quality assessment, in: Raupp J.(Ed.), Quality of plant products grown with manure fertilization, *Darmstadt*, pp. 44–48
20. **Lehninger A L. 1979.** Biochemistry; The Molecular basis of cell Structure and Function. 2nd Ed, Worth Publishers Inc.
21. **Ahmed A.H.H., Khalil M.K., Farrag. A.M. 2000**. Nitrate accumulation, growth, yield and chemical composition of Rocket (*Eruca vesicaria* subsp. *sativa*) plant as affected by NPK fertilization, kinetin and salicylicacid, in: Proceedings of ICEHM 2000, Cairo University, Egypt, pp. 495–50
22. **Tisdale S L & Nelson W L 1975.** Elements required in plant nutrition. in: Soil Fertility and Fertilisers. MacMillon Publishing Co. Inc. New York.
23. **Xu C., Wu L.H., Ju X.T., Zhang F.S. 2005.** Role of nitrification inhibitor DMPP (3,4-dimethylpyrazole phosphate) in nitrate accumulation in greengrocery (*Brassica campestris* L. ssp. *chinensis*) and vegetable soil. *J. Environ. Sci.* (China). **17:** 81–83.
24. **Tzung W.-J., Po W.-Y., Wu J.T., Wang Y.-P. 1995.** Effects of some environmental factors on nitrate content of Chinese cabbage (*Brasica chinensis* L.), *J. Chin. Agr. Chem. Soc.* **33:** 125–133.
25. **Ahmed A.H.H. 1996.** Physiological studies on tipburn and nitrate accumulation in lettuce plants. *J. Agr. Sci.* **21:** 3971–3994.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 199-202
An International Biannual Refereed Journal of Life Sciences

5(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Studies on ethnobotanical plants used in the marriage ceremony in santhal community of Jharkhand

Priyanka Tudu* & Kunul Kandir
University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

Received : 06th January, 2020 ; Revised : 07th February, 2020

Abstract:- India is an important biodiversity centre with over 45000 various plant species. Since ages conservation of natural resources has been an integral aspect of many indigenous communities all over the world. These people are largely dependent on their traditional system, for their information to be passed on from generation to generation through oral communication. The Santhal tribe of Dumka district of Jharkhand has preserved their ritual and culture. Santhal tribe is the leading tribe of the state of Jharkhand. They have a unique culture. Marriage is an important aspect in the life of Santhal people. The present work focuses on the plants used in marriage ceremonies. There are ten documented plants which are used during marriage ceremonies. These plants belong to nine different families. These plants are considered to be of great importance due to their religious and medicinal uses.

Key words: Ethnobotanical plants, marriage ceremony, santhal community, ethnic tribe, Jharkhand

INTRODUCTION

Jharkhand is home to 47 tribal communities, among them Santhal, Oraon, Munda, Ho and Kharia are the major tribes.[1] The Santhal tribe is the leading tribe of the state of Jharkhand. The Santhals have unique custom, culture and lifestyle.[2] They are mainly inhibited in the states of Jharkhand, Bihar, Orissa, West Bengal and Assam.[3] The Santhal tribe utilize various plants for their daily requirements and they also maintain the natural resources. They collect different plants or plant parts based on their traditional knowledge.[4] The Santhal tribe is aware of variety of natural resources which has developed into valuable local knowledge system which consists of widespread knowledge about the use of plants in general, sacred purposes and domestic uses.[5] Nearly more than ten thousand plant species are used by different tribal communities for food, fibre, spices, oil, medicine, fodder for livestock, fuel wood, ritual and ceremonies.[6] The plants are a very essential aspect in beliefs, religion and historical context.[7] The tribe observe three main occasions and they are birth, marriage and death ceremonies.[8] Marriage gives a remarkable tribal feature and reflects its cultural individuality. The tribal marriage is considered as a civil contract and not a religious sacrament.[9] There are two types of marriage the regular marriage and the irregular marriage. The regular marriages are arranged by parents and the irregular marriage is enforced through elopement, capture, intrusion and so on.[10] The general marriage is

*Corresponding author :
Phone : 7763947839
E-mail : botanicalpriyanka1990@gmail.com

arranged by parents which is then followed by a regular marriage ceremony. The Santhal tribe practices monogamy.[11] Monogamy is a type of marriage in which union of a man and women takes place. The Santhali name for marriage is called "Bapla". Marriage is always a reason to celeberate.[12] In Santhal community marriage is a sacred event of life it adds to respect in the society. It is also the start of legal relationship. There is no restriction of age. A Santhal marriage takes almost five days and it involves various complex rituals. The centre of the courtyard where marriage rituals take place is called "Mandwa". The women wear their traditional costumes and headgear made of colourful flowers and leaves.[13] A Santhal marriage is of eleven types and they are Sadai Bapla, Golaeti Bapla, Tunki Dipil Bapla, Sangha Bapla, Jawae Kirinok Bapla, Hiram Cetan Bapla, Or Ader Bapla, Nir Bolok Bapla, Iputut Bapla, Apangir Bapla and Kundal Napam Bapla. Sadai Bapla is the very common type of marriage found in Santhal community. Nepel, Orak Duar Nel, Horok Cikhna, Taka Cal, Gira Tol, Baplatet Reak, Jawae Eger Seren, Balaya Seren, Bahu Jawae Daramdak, Gidi Cumaura, Tendar Ak Sar, Lumti Baret Bida, Mandwa Otor this is the common procedure in Santhal Marriage ceremony.

MATERIALAND METHODS

Study Area

Dumka is located at 24.27°N, 87.25°E. Dumka district has an area of 3716 Km^2. The district has a population of 13,21,096 (2011census). Dumka has a pre-dominantly undulating terrain with hard rock's under the ground. The entire district has topography with high ridges and valleys bounded by mountains and rivers. It was made the headquarters of the Santhal Pargana region, which was carved out of Bhagalpur and Siuri districts after the Santhal Hool 1855. The Santhal revolt started in 1855. Dumka is the sub - capital of the state of Jharkhand. The east Santhal Pargana zone gets rain in summer with an average around 13.5 cm. This area comes under the highlands of Rajmahal range like Sahebgunj and Pakur. The Rajmahal ranges plays very significant role in becoming a roadblock between warm westerly winds and the moisture loaded winds of Bay of Bengal.

The present study was carried out in three villages of Dumka district namely Sujan, Kukurtoppa and Kodokicha during 2019. The three villages were inhabited by Santhal tribe. They were mostly dependent on forest and agriculture for their livelihood. The common trees found in these villages were: - *Terminalia tomentosa* Roxb. (ex DC) Wight and Arn., *Azadirachta indica* A.Juss., *Phoenix sylvestris* (Linn.)Roxb, *Mangifera indica* L., *Butea monosperma* Lam., *Madhuca longifolia* (J.Koning) J.F.Macbr, *Shorea robusta* Gaertn., *Bambusa vulgaris* Schard.ex.J.C.Wendle, *Moringa oleifera* Lam., *Aegle marmelos* L., *Syzygium cumini* L., *Haldina cordifolia* (Roxb.) Ridsdale, *Bauhinia variegata* (L.), *Ficus religiosa* Linn., *Dalbergia sissoo* Roxb., *Bauhinia purpurea* (L.). During survey interview of Manjhi Hadam (Mukhya), Naiki (Pujari), Heads of tribe, Elder man, Elder women, and knowledgeable persons was done by the help of questionnaire about plants used in the marriage ceremonies. This information was collected by personal visit to the study area. The discussion was done with the respondent with their consent. The communication with the local tribal people took an hour and above. The medium of communication was Hindi and Santhali. The Santhali names and common names were noted down. Photographs of the plants were taken. During the survey plant specimen were collected with the help of local tribal people. The collected plants were identified with the help of Flora of Bihar and Orissa by H.H.Haines in University Department of Botany, Ranchi University, Ranchi. The collected plants were also preserved in the form of herbarium.

RESULTS AND DISCUSSION

Table 1: Ethnobotanical plants used in marriage ceremonies

S.N	Botanical Name	Family	Common Name	Santhali Name	Habit	Parts Used	Mode of use
1.	*Shorea robusta* Gaertn.	Depterocarpaceae	Sal Tree	Sarjom Dare	Tree	Leaves , Flowering Twig	The branch of the plant is placed in the courtyard of the house where the rituals of marriage ceremony takes place.
2.	*Phoenix sylvestris* (Linn.)Roxb .	Palmae	Wild Date	Khajur	Tree	Leaves	The leaves are woven together to make mats which is used during Sunum Sasang (Haldi Lepan) ceremony a day before marriage.

3.	*Madhuca longifolia* (J.Koning)J.F.Macbr.	Sapotaceae	Honey Tree	Matkom Dare	Tree	Leaves, Flowers	The branch is placed in the courtyard on the day of Mandwa and it is worshiped.
4.	*Lowsonia inermis* L.	Lythraceae	Heena	Mehndi	Shrub	Leaves	The leaves are crushed and made into paste and it is applied on fingers as dye.
5.	*Mangifera indica* L.	Anacardiaceae	Mango Tree	Ul Dare	Tree	Leaves	The leaves are used to decorate the doors of the house and mandwa.
6.	*Azadirachta indica* A. Juss	Meliaceae	Neem Tree	Neem Dare	Tree	Leaves	It is used to decorate the mandwa.
7.	*Curcuma longa* L.	Zingiberaceae	Turmeric	Sasang	Shrub	Rhizome	The rhizome is made into paste which is applied to bride and bridegroom the day before marriage.
8.	*Brassica rapa* L.	Brassicaceae	Mustard	Turi	Herb	Seed	The oil from the seed is used in various rituals in marriage ceremony.
9.	*Cynodon dactylon* L.	Poaceae	Dhoob Grass	Dhoobi Ghas	Grass	Entire Plant	The entire plant is used as a brush to apply the paste of turmeric and mustard oil on bride and bridegroom the day before marriage.
10.	*Bambusa vulgaris* Schard.ex.J.C.Wendl	Poaceae	Bamboo	Mat	Grass	Entire plant	The entire plant is used to decorate the mandwa where the marriage ceremony takes place.

The present study focuses on the plants used in marriage ceremonies by the Santhal community of Dumka district of Jharkhand. The result of the study showed that 10 plants are used during marriage ceremonies. These plants belong to 9 different families among them 5 plants are trees, 2 are shrubs, 1 is herb and the rest two plants belong to grass family. The most commonly used plant part was the leaves, followed by flowers, entire plant, seed and rhizome.



A. Preparation for marriage ceremony



B. Before marriage worshipping of sacred tree



C. Ritual performed by bride and bridegroom during marriage ceremony



D. Singing and Dancing with musical instruments under Mandwa

Figure 1. Picture (A, B, C and D) showing the rituals of marriage ceremony

CONCLUSION

The Santhal community of Dumka district of Jharkhand have religious belief which conserves the biodiversity of the area. Santhal Pargana is rich in biodiversity and there is a need to conserve it. Their knowledge about plants and traditional practices needs to be preserved and appreciated.

ACKNOWLEDGEMENT

The authors like to thanks the Head of the Department, University Department of Botany, Ranchi University, Ranchi for support and guidance.

REFERENCES

1. **Barla, S.J.Emmanuel** "Ethno-medicinal beliefs and practices among tribals of Jharkhand " pp. 221-239 .
2. **Banerjee, B.G.1988.** Folkillness and Ethnomedicine. Northern Book center, New Delhi. pp.19.
3. **Murmu, N.C. and Hansda, R.C. 2010.** A portal for Santhals. We santhals E-group.
4. **Joshi, K.,Joshi, R. and Joshi, A.R. 2011.** Indigenous knowledge and uses of medicinal plants in Macchegaun, Nepal. *Indian Journal of Traditional Knowledge*, **10(2):** 281-286.
5. **Mohanty, N., Das P.K. and Panda, T. 2011.** Use of plant diversity in household and rituals by Tribal people of Dhenkanal district, Orissa, India. *Journal of applied pharmaceutical science*, **1(4):**78-82.
6. **Heywood, V.H.1992.** Conservation of germplasm of wild species. O.T.SAandlund, K.Hindar and A.H.D.Brown, (eds).Conservation of Biodiversity for sustainable development. Scandinavian university press, Oslo, pp. 189-203.
7. **Menendez-Baceta, G., Aceituno-Mata, L., Reyes., Garcia,V., Tardio, J., Salpeteur, M. and Pardo-de-Santayana, M. 2015.** The importance of cultural factors in the distribution of medicinal plant knowledge: A case study in four Basque regions. *Journal of Ethnopharmacology*, **161:**116-127.
8. **Redfiel, Robert., 1955.** The little community, Uppasala and Stockholm, Almq-Vest and Wiksells Booktryckeri A.B.
9. **Majumdar. D.N and Madan, T.N. 1956.** An introduction to social anthropology Bombay. Asian Publishing house.
10. **Dube, S.C. 1951.** The Kamar, Lucknow, Universal Publishers Ltd; 1969: Manav aur Sanskriti, Delhi, Rajkamal Prakashan .
11. **Sachidanand. 1958.** The Dhumkuria: Then and Now, Journal of Social Research vol. 1, no.1, September 1958.
12. **Hembrram, Mayamani. Dec-2017.** Social life of Santhal people; International Journal of Research;vol III, issue II, pp.50-57.
13. **Singh, Meena. Feb-2018.** Study on women costume of Santhal Tribe of Jharkhand; *Textiles and Clothing Research Centre e- journal*; vol II, issue III, pp.21-28.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March., 2020, pp. 203-208
An International Biannual Refereed Journal of Life Sciences

6(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Portfolio function of Bihar hairy caterpillar species, *Spilosoma spp*. with respect to its species diversity profile in Saharsa, Bihar, India

Manisha Kumari*
Department of Zoology, B.N.Mandal University, Madhepura, Bihar, India

Received : 2nd December, 2019 ; Revised :30th January,2020

Abstract: Portfolio function is an investment term usually applied used in finance management but due to the unpredicted loss & gain of biodiversity, population biologist and environmentalist have started using this concept in biodiversity studies. When more diversity is added to any investment portfolio, less risk is involved, hence community rich in biodiversity is considered to be rich in bioresources with ample security & insurance in contrast to those having less diversity of species. The functional concept of community & ecosystem is almost equal to well organised portfolios of the nature (Frank Figge 2004).[1] In the present research paper the author has ventured to apply the portfolio theory in the population diversity of Bihar hairy caterpillar, *Spilosoma* spp. randomly sampled from different agricultural & weedy plots of Saharsa. Eight species sampled from four zones of this area have collective species diversity index value (S-W=H) of 0.7002 (Z-1), 0.7627 (Z-2), 0.7909 (Z-3) & 0.8564 (Z-4) calculated by applying standard statistical equation to the sampled population. The value reflects that zone-4 is richest in terms of *Spilosoma* spp population thriving on different host weeds & crops in this area but largely destroying the host asset (weeds & crops). The diversity value of the pest in zone-1 is poorest that also indicates the smart portfolio management of the nature with respect to the population of weeds & host crops. Crisscross interrelationship & implications of such diversity of the pest & host in the portfolio function of the hairy caterpillar has been discussed under the light of investment policies of nature in the communities.

Keywords: Portfolio function, species diversity, investment policies, *Spilosoma* spp, post weeds & crops, Saharsa district

INTRODUCTION

Portfolio function is a very advanced concept of asset and liability management with precise monitoring of loss & gain of the resources in nature if considered to be the ultimate reserve bank of all the resources existing in it.[2] One of the most pervasive themes in insect ecology is that biological diversity stabilizes ecosystem processes and the service they provide to society, a concept that has become a common argument for biodiversity conservation. Species rich communities are thought to produce more temporally stable ecosystem service because of the complementary or independent dynamics among species that perform similar ecosystem functions. Such variance dampening within communities is referred to as portfolio effect and is analogous to the effects of asset diversity on the stability of financial portfolios[3]. in ecology, these arguments have focused on the effects of species diversity on ecosystem stability but have not considered the importance of biologically relevant extirpation are probably at least three orders of magnitude higher than species extinction rates. So there is a pressing need to clarify how population and life history diversity affect the performance of individual species in providing important ecosystem services.

*Corresponding author :
Phone : 07004317966
E-mail : todeardeepak70@gmail.com
sonubaba2008@gmail.com

Variability in annual Bristol Bay salmon returns is 2.2 times lower than it would be if the system consisted of single homogenous population rather than the several hundred discrete populations it currently consists of. Our results demonstrate the critical importance of maintaining population diversity for stabilizing ecosystem services and securing the economies and livelihoods that depends on them. All the resources in the nature whether living or non living under the portfolio function are considered to be natural assets. And low or high population density of any assets becomes the prerogative of the nature. Beyond the control of human being but within the limits of research thought of human minds, the increase & decrease of any resource species is directly related to their positive or negative performance in the environment which immediately concern the adjoining members of the species lying in that portfolio.

MATERIAL AND METHODS

In the present investigation, various species of *Spilosoma* has been surveyed and sampled along with their host weeds and crops of agricultural fields from four different zones of Saharsa district like zone-1 (South Saharsa)- Salkhua, Banma Itiahri, zone-2 (West)- Mahishi, Kahara & Simri Bakhtiarpur), zone-3 (North)-Nauhatta & Satar Kataiya & zone-4 (East)-Patarghat, Sonbarsa, Saur bazar, through large quadrate sampling and random netting methods. The numerical count of the individuals of pest species in live condition thriving on the specific weed during sampling has been recorded in the raw data diversity computation tables 1-4 (zone specific) and the scientific identification of the species were done on the spot as far as possible & also in the lab by preserving & bringing few representatives of the samples for identification with standard key of Lepidoptera. Shannon Weiner index is also known as Shannon Weaver index.[4,5,6]

The types of host weeds were also ascertained with the help of local people & botanist to co-relate their abundance with the diversity of the *Spilosoma* pest as well as interpret the portfolio function as per natural bioresource asset investment policies of the nature or environment. The four sampling zones of various species of *Spilosoma* along with the host weeds have been displayed in the Fig.1 of Tehsil map of urban area of Saharsa district. The number of individuals of pest species along with the host weeds counted in each quadrate sampling or netting and their species identity have been presented in the raw data table no. 1. Table no. 2 reflects the statistical computation of species wise relative abundance whereas Shannon Wiener diversity has been presented in the table no.3. The graphical representation of rank abundance curve has been furnished in the figure of the histogram.

Following statistical tools have been used for the purpose:

a) S.W. diversity index: -

$H = pi \log pi$

Where, H = species diversity,

Pi = mean of individual species procured by the formula n/N,

log pi is the log product of pi (mean of individual species).

b) Relative abundance = ni/N x 100

Where, Ni = number of individuals of a species.

c) Portfolio function index (PF)=No. of host plants/$_{\mathbf{H}}$

d) 3-D histogram graphical co-relation between species diversity of *Spilosoma* and population dynamics of host plants.

RESULT & DISCUSSION

The results of species diversity by applying S-W diversity statistical tool in different zones of sampling in Saharsa district furnish the information that zone-1 is having lowest diversity value of the pest (0.7002) thriving on 13-15 host plants per quadrate sample. In contrast to this zone-4 is having highest diversity value (0.8564) for the pest. Surprisingly the portfolio function index (PF) which is a ratio of host population dynamics vs pest species diversity reflected just reverse trend in the form of highest Pf value in zone-1 whereas lowest in zone-4.

The relative performance of the pest species diversity on the selected host plants & the ability of the hosts to withstand the pest attack are two important hand-in-glove phenomenons that keep operating the portfolio function of the habitat. Apropos, the species diversity of the pest in highly dense population host weeds is relatively low, although the rich availability of food should have expectedly encouraged the growth of pest that didn't happen. Other components in the portfolio of the habitat operating as diversified investment channels such as metrological conditions, anthropogenic attention & pest control measures taken up in advance to maintain the habitat ecosystem related to the host plants, weeds & other economic trees susceptible to the attack of Bihar hairy caterpillar, *Spilosoma* spp.

Fig-1. Tehsil map of Saharsa, Bihar showing four zones of sampling of *Spilosoma* spp. along with host weeds & plants

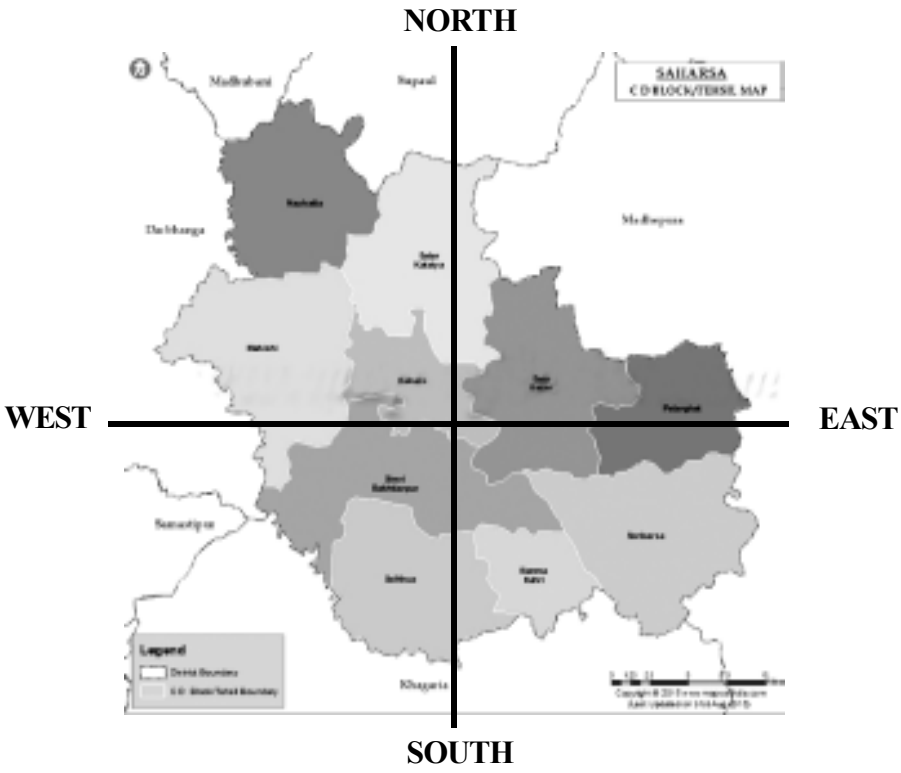


Table 1: Zone specific diversity index values of Bihar hairy caterpillar *Spilosoma spp.* of Saharsa, in zone 1 (South Saharsa)- Salkhua, Banma Itiahri

Sl. no	Scientific name as per the key	Taxonomic position	Name of the host weeds /plants	Quadrate plot no.	Number of adult insects per quadrat of weeds(10-15 plants)	Relative abundance %(n/N* 100)	Pi(n/N)	log pi	Pi*log pi	H
1	*Spilosoma congrua*	Lepidoptera	Tobacco	Z-1/QP-i	11	22.9	0.229	-0.640	-0.1465	0.7002
2	*Spilosoma fuscipennis*	Lepidoptera	Onion	Z-1/QP-ii	08	16.6	0.166	-0.779	-0.1293	
3	*Spilosoma virginica*	Lepidoptera	Papaya	Z-1/QP-ii	07	14.5	0.145	-0.838	-0.1215	
4	*Spilosoma erythrozona*	Lepidoptera	Mango	Z-1/QP-iii	04	8.3	0.083	-0.080	-0.0066	
5	*Spilosoma urticae*	Lepidoptera	Tobacco	Z-1/QP-iv	06	12.5	0.125	-0.903	-0.1128	
6	*Spilosoma lubricipeda*	Lepidoptera	Papaya	Z-1/QP-ii	04	8.3	0.083	-0.080	-0.0066	
7	*Spilosoma erythrophleps*	Lepidoptera	Mango	Z-1/QP-iii	03	6.2	0.062	-1.207	-0.0748	
8	*Spilosoma obliqua*	Lepidoptera	Garlic	Z-1/QP-ii	05	10.41	0.104	-0.982	-0.1021	
	Total				48					

PF index (portfolio function) = number of host plants per quadrate/ & =13/0.7002=18.56

Table 2: Zone specific diversity index values of Bihar hairy caterpillar *Spilosoma spp.* of Saharsa, in zone 2 (West Saharsa)-Mahishi, Kahara & Simri Bakhtiarpur

Scientific name as per the key	Taxonomic position	Name of the host weeds/plants	Quadrate plot no.	Number of adult insects per quadrat of weeds(15-20 plants)	Relative abundance %(n/N* 100)	p_i(n/N)	log p_i	p_i*log p_i	H
Spilosoma congrua	Lepidoptera	Tobacco	Z-2/QP-i	11	22.4	0.224	-0.649	-0.1453	
Spilosoma fuscipennis	Lepidoptera	Onion	Z-2/QP-ii	08	16.32	0.163	-0.787	-0.1282	
Spilosoma virginica	Lepidoptera	Papaya	Z-2/QP-ii	07	14.2	0.142	-0.847	-0.1202	
Spilosoma erythrozona	Lepidoptera	Mango	Z-2/QP-iii	04	8.16	0.081	-0.091	-0.0073	0.7627
Spilosoma urticae	Lepidoptera	Tobacco	Z-2/QP-iv	06	12.2	0.122	-0.913	-0.1113	
Spilosoma lubricipeda	Lepidoptera	Papaya	Z-2/QP-ii	08	16.32	0.163	-0.787	-0.1282	
Spilosoma erythrophleps	Lepidoptera	Mango	Z-2/QP-iii	01	2.04	0.020	-1.698	-0.0339	
Spilosoma obliqua	Lepidoptera	Garlic	Z-2/QP-ii	04	8.16	0.081	-1.091	-0.0883	
Total				49					

PF index (portfolio function) = number of host plants per quadrate / & =13/0.7627 =17.04

Table 3: Zone specific diversity index values of Bihar hairy caterpillar *Spilosoma spp.* of Saharsa, in zone 3 (North Saharsa)-Nauhatta & Satar Kataiya

Scientific name as per the key	Taxonomic position	Name of the host weeds/plants	Quadrate plot no.	Number of adult insects per quadrat of weeds(10-15 plants)	Relative abundance %(n/N* 100)	P_i(n/N)	log p_i	P_i*log p_i	H
Spilosoma congrua	Lepidoptera	Tobacco	Z-3/QP-i	11	19.29	0.192	-0.716	-0.1368	
Spilosoma fuscipennis	Lepidoptera	Onion	Z-3/QP-ii	08	14.03	0.140	-0.853	-0.1194	
Spilosoma virginica	Lepidoptera	Papaya	Z-3/QP-ii	07	12.2	0.122	-0.913	-0.1113	0.7909
Spilosoma erythrozona	Lepidoptera	Mango	Z-3/QP-iii	04	7.01	0.070	-1.154	-0.0807	
Spilosoma urticae	Lepidoptera	Tobacco	Z-3/QP-iv	06	10.5	0.105	-0.978	-0.1026	
Spilosoma lubricipeda	Lepidoptera	Papaya	Z-3/QP-ii	06	10.5	0.105	-0.978	-0.1026	
Spilosoma erythrophleps	Lepidoptera	Mango	Z-3/QP-iii	05	8.77	0.087	-0.060	-0.0052	
Spilosoma obliqua	Lepidoptera	Garlic	Z-3/QP-ii	10	17.5	0.175	-0.756	-0.1323	
				57					

PF index (portfolio function) = number of host plants per quadrate/ & = 13/0.7909 =16.43

Table 4: Zone specific diversity index values of Bihar hairy caterpillar *Spilosoma spp.* of Saharsa, in zone 4 (East)-Patarghat, Sonbarsa, Saur bazar

Scientific name as per the key	Taxonomic position	Name of the host weeds/plants	Quadrate plot no.	Number of adult insects per quadrat of weeds(10-15 plants)	Relative abundance %(n/N* 100)	P_i(n/N)	log p_i	P_i*log p_i	H
Spilosoma congrua	Lepidoptera	Tobacco	Z-4/QPi	11	21.56	0.215	-0.667	-0.1434	0.8564
Spilosoma fuscipennis	Lepidoptera	Onion	Z-4/QP-ii	08	15.68	0.156	-0.806	-0.1257	
Spilosoma virginica	Lepidoptera	Papaya	Z-4/QP-ii	07	13.7	0.137	-0.863	-0.1182	
Spilosoma erythrozona	Lepidoptera	Mango	Z-4/QP-iii	04	7.84	0.078	-1.107	-0.0863	
Spilosoma urticae	Lepidoptera	Tobacco	Z-4/QP-iv	06	11.76	0.117	-0.931	-0.1089	
Spilosoma lubricipeda	Lepidoptera	Papaya	Z-4/QP-ii	09	17.64	0.176	-0.754	-0.1327	
Spilosoma erythrophleps	Lepidoptera	Mango	Z-4/QP-iii	02	3.92	0.039	-1.408	-0.0549	
Spilosoma obliqua	Lepidoptera	Garlic	Z-4/QP-ii	04	7.84	0.078	-1.107	-0.0863	
Total				51					

PF index (portfolio function) = number of host plants per quadrate/ & = 13/0.8564 =15.17

ACKNOWLEDGEMENT

The author is extremely thankful to her guide Dr. B.P. Yadav 'Bipra', Department of Zoology L.N.M.S College Birpur (Supaul) & Prof. Arun Kumar, Head of the Department of Zoology, B.N. Mandal University, Madhepura for helping in completion of this research paper.

REFERENCES

1. **Figge, F. 2004.** Bio-folio: applying portfolio theory to biodiversity. *Biodiversity and conservation*. **13:**827-849
2. **Daniel E. Schindler *et.al.* 2010.** Population diversity & the portfolio effect in an exploited species: *Nature* 09060 Vol**465/3**.pp, 609-613, Macmillan publisher.
3. **Gregory H. Aplet and Peter S. McKinley. 2017.** A portfolio approach to managing ecological risks of global change. *Ecosystem Health & sustainability.* **3(2):** 1-15.
4. **Hubbell, S.P. 2001.** *The unified neutral theory of biodiversity & biogeography*. Princeton University Press, Princeton, N.J.
5. **Magurran, A.E. 2004.** *Measuring biological diversity*. Blackwell Scientific. Oxford.
6. **Peter Stiling. 2002.** "*Ecology: theories and Applications"*, 4th Edition, Eastern Economy Edition, Prentice Hall of India Private Limited, New Delhi.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 209-212
An International Biannual Refereed Journal of Life Sciences

7(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Ethnomedicinal plants of Kosi area used in Homeopathic medicine

Rani Runam Kumari*
Department of Botany, RJM College, Saharsa, Bihar, India

Received :28th December, 2019 ; Revised : 18th February, 2020

Abstract: Koshi Region is primarily a rural area with a considerable amount of people relying on traditional methods to sustain lives. Thus, people from the region depend on a variety of homeopathic medicines, treatments and diagnosis to tackle health aliments. Homeopathy was started in India during the early 1800s by a few German scientists in Bengal, not very far from the Koshi Region. Our study was aimed at identifying various medicinal plants used in homeopathic treatment and medicine. Ethnobotanical study helps in the documentation of such knowledge. The sub branch of botany helps us understand local practices and traditions in relation to flora of the region. In ethnomedicinal studies, sometimes one plant has multiple use benefits and at other times, different plants are utilised in the treatment of a particular aliment. In this field of study, plants are documented according to their medicinal value and this serves as a database for not only future studies and experiments but also in the preservation of local culture and traditions. In this study, a survey was made in 15 villages of different blocks of Kosi division. 21 plant species belonging to 16 families used in Homeopathic medicines are described in present study along with their scientific name, family, local name and medicinal use in Homeopathy.

Key words: Homeopathy, Medicinal plants, Kosi area.

INTRODUCTION

Homeopathy is one of the alternative systems of medicine having a well documented pharmacopoeia. In India, Homeopathic treatment started as early as 1810 when some German physicians come to Bengal and started distributing Homeopathic remedies.

Dr. John Martin Honigberger and Hahnemann brought Homeopathy in India. Dr. Honigberger arrived at Lahore in 1829-30, later on went to Calcutta and started practice there.

*Corresponding author :
Phone : 9958162973
E-mail : drranirupam@gmail.com

In India, Homeopathy is considered as part of the Indian traditional medicine. About 15% Indians are using Homeopathic treatment. This system of medicine relies mainly on plants and minerals as drug.

According to the principle of Homeopathy serial dilution of plant extract or mineral in solvent removes toxic effects while the quality of the substance retained by the diluents. The end product is so diluted that it is indistinguishable from solvent.

Homeopath select treatments according to a patient consultation that explores the physical and psychological state of the patient, both of which are considered important in selecting remedy.

MATERIALAND METHOD

The use of sophisticated equipments and accepted scientific parameters have become common in ethnobotanical studies but the essence of ethnobotanical study is in the detailed study of the local and indigenous culture, traditions and practices and relating them to observed flora of the region. Our study has used preliminary survey to collect traditional knowledge and specimens of medicinal flora used for homeopathy in the region along with the use of an internationally recognised homeopathic manual to gauge medicinal values of collected specimen.

15 villages of different blocks of Kosi division were visited. Medicinal plants were collected and their local names and medicinal importance were acknowledged from local informants. Collected plants were brought to the laboratory and identified. Homeopaths of local area were consulted and asked for the use of these plants in Homeopathic drugs. Medicinal values of collected plant specimens were also identified with the help of "New Manual of Homeopathic Materia medica and Repertory". Herbariums were prepared from collected plant specimens.

RESULT

Altogether 21 plants belonging to 16 families were recognized which are used in Homeopathic medicines for different diseases. Scientific name, local name, family and name of diseases for which the plant extract is used in Homeopathy are recorded in Table no. 01.

Table No. 01

Sl. No.	Scientific Name of Plant	Local Name	Family	Medicine Name	Uses in Homeopathy
1.	*Achyranthes aspera*	Chirchiri	Acanthaceae	*Achyranthes aspera*	Diarrhoea, Dysentery, Diuretic, Menorrhagia, Astringent, bad effects of dog and snake bite.
2.	*Adhatoda zeylancia*	Vasaka	Acanthaceae	*Justica adhatoda*	Headache and Respiratory disorder.
3.	*Aegle marmelos*	Beel	Rutacae	*Aegle folia*	Bleeding pills, Diarrhoea, Dysentery, Fever with dropsy and Impotence.
4.	*Allium cepa*	Piyaz	Liliaceae	*Allium cepa*	Cough and Cold, Injury, Nasal polypus, Diarrhoea, Influenza, Throat affection.
5.	*Allium sativum*	Lahsum	Liliaceae	*Allium sativum*	Stomach, Head, Respiratory affection.
6.	*Aloe-vera*	Ghrit kumari	Liliaceae	*Aloe socotrina*	Abdominal disorders, Anal infection, Colic, Constipation, Cough, Dysentery, Diarrhoea, Haemorrhoids skin affection.
7.	*Azadirachta indica*	Neem	Meliaceae	*Azadirachta indica*	Eczema, Fever and Stomachache, Scabis, Leprosy, Helminthiasis, Gonorrhoea, Glossitis.
8.	*Calotropis gigentea*	Aak	Asclepiadaceae	*Calotropis gigantea*	Toothache, Elephantiasis, Purging, Vomiting.
9.	*Cannbis sativa*	Bhang	Cannabaceae	*Cannabis sativa*	Constipation, Respiratory disorders, Urinary affection, Gonorrhoea.
10.	*Centella asiatica*	Brahmi	Apiaceae	*Hydrocotyle asiatica*	Acne, Eczema, Uterus affection, Elephantiasis, Leprosy.
11.	*Citrus vulgaris*	Tab	Rutaceae	*Citrus vulgaris*	Headache with Nausea, Vomiting and Vertigo, Thoracic oppression.
12.	*Cynodon dactylon*	Dub	Poaceae	*Cynodon dactylon*	Haemorrhage, Dysentery, Dropsy, Leucorrhoea, Scabies.
13.	*Ficus benghalensis*	Bargad	Moraceae	*Ficus indica*	Headache, Respiratory disorder.
14.	*Ficus religiosa*	Peepal	Moraceae	*Ficus religiosa*	Headache, Respiratory disorders, Monorragia.
15.	*Ocimum sanctum*	Tulsi	Labiatae	*Ocimum sanctum*	Asthma, Tonsillitis, Cold, Opthalmia, Neonatum, Typhoid, Worm affection.
16.	*Ricinus communis*	Arand	Euphorbiaceae	*Ricinus communis*	Remedy of digestive system, Nausea, Profuse vomiting, Constipation and Diarrhoea.

17.	*Solanum nigrum*	Makoi	Solanaceae	*Solanum nigrum*	Head, Eye, Nose, Respiratory, Fever, Skin and Stomach.
18.	*Tinospora cardifolia*	Guruch latti	Menispernaceae	*Tinospora cardifolia*	Seminal debility, Fever, Jaundice, Spleenic affection, Leprosy, Leucorrhoea, Rheumatism, Gonorrhoea.
19.	*Syzygium cumini*	Jamun	Myretaceae	*Syzygium jambolanum*	Diabetes mellitus and Ulcer.
20.	*Terminalia chebula*	Haritiki	Combretaceae	*Terminalia chebula*	Bleeding, Pile, Diarrhoea, Chronic dysentery, Constipation, Billary colic, Glossitis, Headache, Vertigo, Dropsy, Skin disorder.
21.	*Viola odorata*	Banfasan	Violaceae	*Viola odorata*	Head, Eye, Ear, Urinary and Respiratory disorder.

CONCLUSION

The present study revealed that Kosi area is rich in medicinal flora. Several of them are also used in Homeopathic medicine. In present study 21 plants belonging to 16 families were recorded which are used in Homeopathic drugs.

Unfortunately rapidly increasing urbanization of the area poses threat to this medicinal flora. Therefore scientific cultivation and conservation of these medicinal floras is needed. A point needs to be made on the waning patronage of Homeopathy with development. A revival of Homeopathic treatment needs to be done. The newer generation is slowly decreasing their reliance on homeopathic and other such forms of traditional medicinal sciences. Therefore, a study like this showing the interaction of Botany and Homeopathy is an important addition to prove the relevance of this science of treatment and medicine.

REFERENCES

1. **Agharkar, S.P. 1991.** Medicinal plants of Bombay, Presidency (Scientific publishers, Jodhpur, India).
2. **Boericke W. 2003.** Manual of Homeopathic material medica with reportary. B. Jain publication, New Delhi. Manual of Homeopathic material medica with reportary. 1st edition.
3. **Boericke W. 2005.** Manual of Homeopathic material medica with reportary. SumitPrakashan, Agra. 1st edition.
4. **Dubey S.K. 2006.** A text book of material medica including Allen's keynote- an easy explanation. Book and allied pvt., Calcutta, 4th edition.
5. **Heinrich, M. 2000.** Ethnobotany and its role in drug development. *Phytotherapy Research*. **14:**479-488.
6. **Hahnemann Samuel.** *Organon of medicine*, aphorism 217
7. **Haines, H.H. 1921.** "Botany of Bihar & Orissa".
8. **Map of India.com.** Geographical position, Average height and average area of district Bilaspur (Himachal Pradesh) India.
9. **William Boericke. 2000.** New manual of Homeopathic material medica and Reportary.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 213-214
An International Biannual Refereed Journal of Life Sciences

8(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Impact of phosphate concentration on toxin production by *Microcystis aeruginosa*

Madhu*
Department of Botany, B.N.M.University, Madhepura, Bihar, India

Received :18th January 2020 ; Revised : 28th February 2020

Abstract : Phosphate is an essential nutrient of cyanobacteria. In *M. aeruginosa* phosphate concentration in medium, effect toxin production. Higher concentration of phosphate in medium causes high production of toxin. Maximum toxin production was observed in 5mg/l of concentration of phosphate in 12 day old culture. Higher concentration above the 5mg/l reduces toxin production.

Key words: toxin, phosphate, *M. aeruginosa*

INTRODUCTION

Toxins produced by cyanobacteria are called cyanotoxins. Growing interest in the toxicological aspects of cyanobacteria and the advancement in several analytical and biochemical methods prompted the isolation and characterization of a number of toxic substances.[1,2] Cyanotoxins fall into three broad groups of chemical structure: cyclic peptides, alkaloids and lipopolysaccharides (LPS).An overview of the specific toxic substances within these broad groups has been identified to date from different genera of cyanobacteria.

Globally the most frequently found cyanobacterial toxins in blooms from fresh and brackish waters are the cyclic peptide toxins of the microcystin and nodularin family. They pose a major challenge for the production of safe drinking water from surface waters containing cyanobacteria with these toxins. In mouse bioassays, which traditionally have been used to screen toxicity of field and laboratory samples, cyanobacterial hepatotoxins (liver toxins) cause death by liver haemorrhage within a few hours of the acute doses. Microcystins have been characterised from planktonic *Anabaena, Microcystis, Oscillatoria (Planktothrix), Nostoc,* and *Anabaenopsis* species, and from terrestrial *Hapalosiphon* genera. Nodularin has been characterized only from *Nodularia spumigena.*

*Corresponding author :
Phone : 9762424511
E-mail : madhu1287@gmail.com

MATERIAL & METHOD

A local stain of *Microcystis aeruginosa* collected from local area of Saharsa district was cultured in BG11 medium. Axenic culture was prepared. 12 old cultures were further cultured in the same supplemented with different concentration of di-potassium hydrogen phosphate. The phosphate concentration was 0.5mg/l, 2.5mg/l, 5mg/l and 7.5mg/l. From each concentration colonies were isolated, cells were lypholysed and toxin was isolated. Isolated toxin was measured.

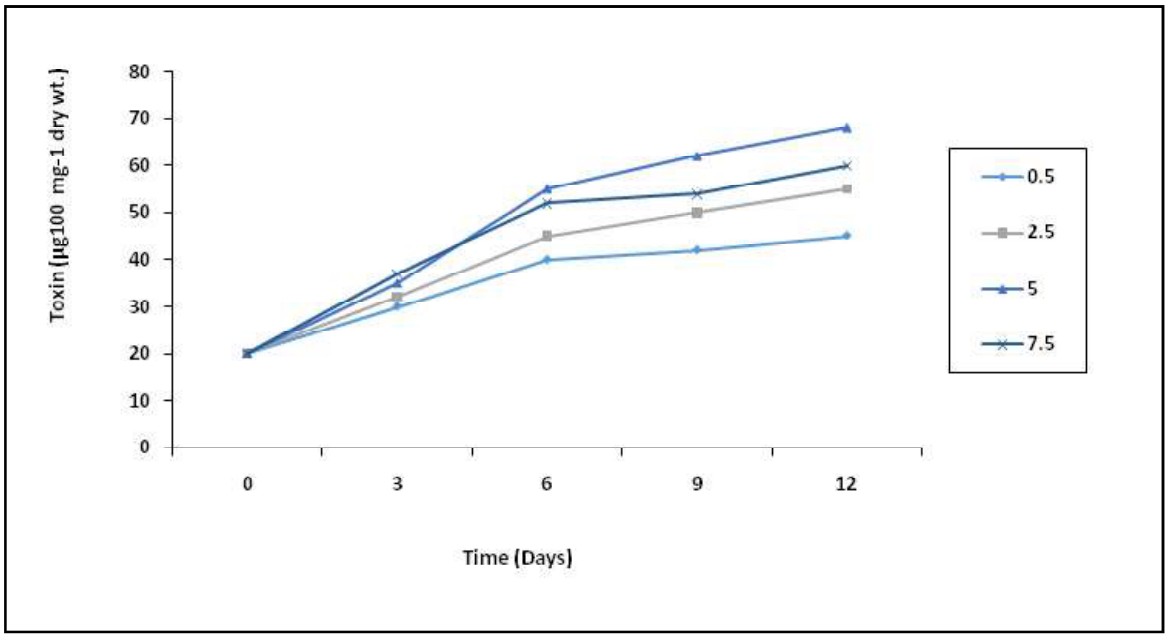
RESULT

Phosphate is an important nutrient for the cyanobacteria. The effect of phosphate (KH_2PO_4) on the toxin production was tested. With increase in phosphate concentration, the toxin production linearly increased at low concentration. Highest toxin production was observed in the cultures grown with 5 mgL01 phosphate. The lowest production was seen in the cultures grown with 0.5 mgL-1 phosphate. The maximum toxin production was observed in the 12 day grown cultures.

CONCLUSION

Microcystis aeruginosa produces cyanotoxin which is a cyclic peptide known as Microcystin. Production of toxin by *Microcystis aeruginosa* increases with increase in phosphate concentration. Maximum toxin production was observed in 5mg/l concentration of phosphate in 12 day old culture. Further increase in phosphate concentration reversely affected toxin production.

Graph 1. Effect of different concentration of Phosphate on Toxin production by *M. aeruginosa*.



REFERENCES

1. **Bold, H.C. and Wynne, M.J. 1978.** Introduction of the Algae: Structure and reproduction, Prentice-Hall, Englewood Cliffs, New Jersey.
2. **Fritsch F. E. 1945.** The structure and Reproduction of the Algae. Volume-II.
3. **Harada, K.I., Ogawa, K. Kimura, Y. Murata, H. Suzuki, M., Thorn, P.M., Evans, W.R. and Carmichael, W.W. 1991.** Microcystin's from *Anabaena flosa-aquae* NRC-525-17. *Chem. Res. Toxicol*. **4:** 535-540.
4. **Pringshein, E.G. 1946.** Pure cultures of algae. Cambridge University Press.
5. **Scott, W.E. 1991.** Occurrence and significance of toxin cyanobacteria in Southem Africa. Water *sci. Technol*. **23:**175-180.
6. **Thakur, J.K. 1996.** Physiology and Biochemical studies on toxin production by Cyanobacteria. Ph. D. Thesis Banaras Hindu University, Varanasi, India.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 215-218
An International Biannual Refereed Journal of Life Sciences

9(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of an organophosphate pesticide, monocrotophos on the biochemical parameters of an air breathing fish, *Anabas testudineus* (Bloch, 1792)

Sushma Swaraj*
Department of Zoology, B.N.M.University, Madhepura, Bihar, India

Received :19th January 2020 ; Revised : 25th February 2020

Abstract : The experimental fish, *Anabas testudineus* was exposed to sub-lethal concentrations of an organophosphate pesticide, monocrotophos. The fish was exposed to acute toxicity of monocrotophos for 16, 24 and 48 hours while for chronic toxicity it was exposed for 15, 30 and 45 days. The monocrotophos was found to be toxic to the fish, which was found to be increasing with the increase of its concentration. Significant changes were observed in various biochemical parameters of 0.072 ppm treated fishes like increase in the cholesterol, plasma alkaline phosphatase, plasma triglyceride, serum bilirubin, serum creatinine while decrease in plasma glucose and total protein.

Key words: Biochemical parameters, monocrotophos, *Anabas testudineus*.

INTRODUCTION

One of the most common organophosphate pesticides used to kill the insect pest in the agriculture field along with NUVAN is monocrotophos. The later is a broadspectrum organophosphate pesticide which has a collateral impact on other aquatic organisms existing on higher tropic level such as fishes. The present investigation is related with studies on the effect of different concentrations of monocrotophos on the selected biochemical parameters of breathing climbing perch, *Anabas testudineus* which is found in both fresh water and brackish water bodies. The fish has also edible value for the human being. The pesticide is one of the highly toxic agricultural chemicals with wide variation in toxicity amongst different species of fishes.[1] The widespread use of monocrotophos results in extensive aquatic contamination by its application intentionally for control of insects and unintentionally by drift of aerial spray, watershed drainage or accidental spillage. Although, monocrotophos is being used extensively in India, studies on its toxicity to aquatic life are relatively rare. Biochemical heterogenicity of different organs are well established in different species of fishes, but the investigation on the sublethal effect of organo-phosphates on various biochemical parameters are not sufficiently studied. Therefore, in the present study efforts has been made to investigate the impact of monocrotophos on various biochemical parameters of *Anabas testudineus*.

*Corresponding author :
Phone : 9802308841
E-mail : sushmas793@gmail.com

MATERIALS AND METHODS

Collection of experimental fishes

Disease free and active freshwater fish *Anabas testudineus* having the body length 12-18 cm and weight

15-20 gm were collected from the local fish market, Madhepura and brought to the laboratory for its further study. All the fishes were kept in aquarium of about 50L containing filtered tap water & fed with meat according to their body weight.

Procurement of monocrotophos & its treatment dose, the pesticide monocrotophos 28% which is available in the commercial name of AIMOCRON 28 SL was obtained from the local market. LC_{50} concentration of the pesticide was determined as per APHA standard 1992 (Renewal toxic test method).[2] Fishes were exposed to 0.072 ppm sub-lethal concentrations for 16, 24 and 48 hours in acute studies and 15, 30 and 45 days in chronic studies. Control fish were maintained under identical conditions without the pesticide in the medium.

Collection of blood

Blood sample from the treated fishes was drawn from the dorsal aorta after each exposure period & preserved in anti-coagulant (0.1ml EDTA) coated plastic vials for biochemical tests. Merc'k biochemical test kits were used for different biochemical analysis.

Statistical analysis of the data

The data obtained from the biochemical test were subjected to students "t" test in order to determine the significance of the difference between control and treated values at two levels of significance-2% & 5%.

RESULTS & DISCUSSIONS

In the present study, attempts have been made to investigate the effect of sub-lethal concentrations of monocrotophos on various biochemical parameters. The exposure of 0.072 ppm concentration of monocrotophos led to the addition in cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, serum bilirubin, serum creatinine, while reduction in plasma glucose, total protein in both acute and chronic studies.

Acute effect

In 16 hr exposed fishes, the recorded values of plasma glucose, cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, total protein, serum bilirubin & serum creatinine were 64.738 + 0.570 mg/dl, 157.5 + 0.957 mg/dl, 143.833 + 1.343 mg/dl, 299.833 + 1.343 mg/dl, 4.417 + 0.177 mg/dl, 2.777 + 0.087 mg/dl & 1.338 + 0.024 mg/dl, respectively (Table -1). In case of 24 hr exposed fishes, the recorded values of plasma glucose, cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, total protein, serum bilirubin & serum creatinine, were 58.376 + 0.839 mg/dl, 167.167 + 1.343mg/dl, 146.833 + 1.343 mg/dl, 279.333 + 2.285 mg/dl, 4.80 + 0.081 mg/dl, 2.645 + 0.106 mg/dl & 1.80 + 0.012 mg/dl respectively (Table -1). In case of 48 hr exposed fishes, the recorded values of plasma glucose, cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, total protein, serum bilirubin & serum creatinine were 46.201 + 1.109 mg/dl, 172.00 + 1.29 mg/dl, 161.33 + 1.374 mg/dl, 260.5 + 1.707 mg/dl, 4.717 + 0.106 mg/dl, 1.855 + 0.017 mg/dl & 2.05 + 0.170 mg/dl, respectively (Table -1).

Chronic effect

In 15 days exposed fishes, the recorded values of plasma glucose, cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, total protein, serum bilirubin & serum creatinine were 52.286 + 1.374 mg/dl, 161.5 + 0.957 mg/dl, 146.65 + 0.150 mg/dl, 230.0 + 1.29 mg/dl, 4.733 + 0.110 mg/dl, 1.307 + 0.098 mg/dl & 1.117 + 0.1067 mg/dl respectively (Table - 2). In case of 30 days exposed fishes, the recorded values of plasma glucose, cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, total protein, serum bilirubin & serum creatinine were 48.576 + 0.956 mg/dl, 178.517 + 0.106 mg/dl, 158.567 + 0.110 mg/dl, 241.33 + 1.105 mg/dl, 4.60 + 0.129 mg/dl, 1.503 + 0.105 mg/ dl & 1.15 + 0.095 mg/dl respectively (Table - 2).

In case of 45 days exposed fishes, the recorded values of plasma glucose, cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, total protein, serum bilirubin & serum creatinine were 41.493+ 0.936 mg/dl, 192.517 + 0.134 mg/dl, 167.567 + 0.110 mg/dl, 247.0 + 1.29 mg/dl, 4.40 + 0.163 mg/dl, 2.022 + 0.163 mg/dl & 1.467 + 0.2808 mg/dl respectively (Table - 2). It is evident from the results that the sub-lethal concentrations of monocrotophos have influence on various biochemical parameters in exposed fishes.

DISCUSSION

Organophosphate pesticides are generally much more acute toxic to vertebrates and other aquatic animals. Pesticides containing chlorinated hydrocarbons persist in the environment and are known to have drifted over thousands of kilometers in the water from Antarctic snow. Traces of chlorinated hydrocarbon pesticides in water may be accumulated in fishes up to the levels of more than 10,000 times, the concentration present in the surrounding water. Several pesticides including some, those were

washed extensively in agriculture in the past and some, those are still used for the purpose of disease control, have been shown to produce tumors in animals.

Similar studies have been carried out by different authors in different fishes with respect to other insecticides such as- Tilak *et al.*, (2001)[3] studied endosulphan induced hyperglycemia in the catfish, *Clarias batrachus.* Ram and Sathyanesn (1985)[4] observed a marked decline in liver glycogen and protein contents in the catfish with sublethal concentration of malathion. Sastry and Sharma (1981)[5], and Singh and Shrivastava (1982)[6] studied variations in various carbohydrate metabolites in *Ophiocephalus punctatus* and *Heteropneustes fossilis* treated to toxicants and a mixture of aldrin and formathion, respectively.

As regards to specific findings on the changes in the biochemical profile of fishes exposed to different, notworthy results have been obtained by- Sastry and Dasgupta (1991)[7] and Sastry *et al.*, (1984)[8] noticed significant alterations in blood glucose, lactic acid and protein in *Clarias batrachus, Anguilla anguilla* and *Channa punctatus* acutely or chronically exposed to aldrin, lindane and nuvacron. Shrivastava *et al.*, (1984)[9] noticed heptachlor induced changes in protein, glycogen and lactic acid contents in functionally different muscles of *Channa punctatus*. Singh and Shrivastava (1982)[6] have noticed hyperglycemia in *H. fossilis* treated with sub-lethal concentrations of organochlorine and organophosphate insecticides.

SUMMARY

In the present study, it has been observed that the exposure of monocrotophos led to the decrease in the level of plasma glucose & total protein while significant increase was observed in the levels of cholesterol, alkaline phosphatase in plasma, triglyceride in plasma, serum bilirubin, serum creatinine. Identical findings were also noticed by Murthy and Devi (1982)[10], Borah and Yadav (1995)[11], Katty and Sathyanesan (1984)[12], Ram and Sathyanesan (1984)[13], Dalela *et al.*, (1978)[14], Shweta Agrahari (2004)[15], Santhakumar and Balagi (2000)[16], Singh *et al.*, (2003)[17], Tilak *et al.*, (2003)[18] and Sharad Srivastava (1990)[19].

Table 1: Sublethal effect of Monocrotophos on certain biochemical parameters of *Anabas testudineus*: Acute studies (0.072ppm/L)

Sl. No.	Parameters	Treated fishes	Tested fishes	After 16 hrs	After 24 hrs	After 48 hrs	Control
1	Plasma glucose	20	6	64.738±0.570	58.376±0.839	46.201±1.109	71.601±0.958
2	Cholesterol	20	6	157.5±0.957	167.167±1.343	172.00±1.29	125.6±0.129
3	Alkaline phosphatase	20	6	143.833±1.343	146.833±1.343	161.33±1.374	130.798±0.0106
4	Triglyceride	20	6	299.833±1.343	279.333±2.285	260.5±1.707	119.5±0.957
5	Total protein	20	6	4.417±0.177	4.80±0.081	4.717±0.106	5.783±0.106
6	Serum bilirubin	20	6	2.777±0.087	2.645±0.106	1.855±0.017	0.913±0.011
7	Serum creatinine	20	6	1.338±0.024	1.80±0.012	2.05±0.170	0.817±0.014

Table 2: Sublethal effect of Monocrotophos on certain biochemical parameters of *Anabas testudineus*: Chronic studies (0.072ppm/L)

Sl. no	parameters	Treated fishes	Tested fishes	After 16 hrs	After 24 hrs	After 48 hrs	Control
1	Plasma glucose	20	6	52.286±1.374	48.576±0.956	41.493±0.936	72.451±0.867
2	Cholesterol	20	6	161.5±0.957	178.517±0.106	192.517±0.134	125.483±0.106
3	Alkaline phosphatase	20	6	146.65±0.150	158.567±0.110	167.567±0.110	132.1±0.081
4	Triglyceride	20	6	230.0±129	241.33±1.105	247.0±1.29	121.833±1.067
5	Total protein	20	6	4.733±0.110	4.60±0.129	4.40±0.163	5.133±0.1105
6	Serum bilirubin	20	6	1.307±0.098	1.503±0.105	2.022±0.163	0.760±0.096
7	Serum creatinine	20	6	1.117±0.1067	1.15±0.095	1.467±0.28	0.718±0.1304

REFERENCES

1. **Santhakumar, M; Balaji, M. and Ramudu, K. 1999.** Effect of monocrotophos on the acetyl chlolinesterase activity of air breathing fish, *Anabas testudineus*. *Asian J. Microbiol. Biotech. Environ. Sci.,* **1(3 & 4):** 135-138().
2. **APHA. 1992.** Standard methods for the examination of water and wastewater. *Amer. Publ. Hlth. Assoc.,* Washington, D.C.
3. **Tilak, K.S.; K. Veeraiah and G.V. Ramana Kumari. 2001.** Toxicity and effects of chloropyrifos to the freshwater fish, *Labeo rohita* (Hamilton).*Poll. Res*., **20(3):** 443-445
4. **Ram, R.N. and Sathyanesan, A.G. 1985.** Organophosphate induced biochemical changes in the brain, liver and ovary of the fish *Channa punctatus*. *Proc. Ind. Natn. Sci. Acad.* **51(5):** 537-542.
5. **Sastry, K.V. and Sharma, S.K. 1981.** Toxic effects of aldrin on liver and kidney of a freshwater teleost fish. *Proc. Symp. Environ. Biol.,* 337–342.
6. **Singh, N.N. and Shrivastava, A.K. 1982.** Toxicity of mixture of aldrin and malathion and other organophosphorus and carbamate pesticides to Indian catfish, *H. fossilis*. *Comp. Physiol.* Ecol., **7:**110-115.
7. **Sastry, K.V. and Dasgupta, A. 1991.** Effect of Nuvacron on the Kidney of a fresh water teleost, *Channa punctatus*. *J. Environ. Biol.,* **12(13):**243-248.
8. **Sastry, K.V.; Siddiqui, A.A. and Subhadra. 1984.** Alteration in the intestinal absorption of glucose and fructose in the freshwater teleost fish, *Channa punctatus* due to Endosulphan and Sevin. *Proc. Sam. Eff. Pest. Aq. Fau*., **1-12**.
9. **Shrivastava, J. and Shrivastava, A.K. 1984.** Histopathology of the gills of *Channa gachua* exposed to sub-lethal concentration of Malathion and Chlordane. *Proc. Sam. Eff. Pest. Aq. Fau*., 37-44.
10. **Murthy, A.S. and Devi, A.P. 1982.** The effect of endosulfan and its isomers on tissue proteins, glycogen and lipids in the fish, *Channa punctatus*. *Pesticide Biochem. and Physiol*., **17:**280–286.
11. **Borah, S. and Yadav, R.N.S. 1995.** Alteration in the protein, free amino acid, nucleic acids and carbohydrate contents of muscle and gill in rogor exposed freshwater fish, *Heteropneustes fossilis*. *Pollu. Res*., **14(1):**99-103
12. **Katty, S.R. and Sathyanesan, A.G. 1984.** Changes in tissue lipid and cholesterol content in the catfish *Clarias batrachus* (L.) exposed to cadmium chloride. *Bull. Environ. Contam. Toxicol.,* **32:**486-490.
13. **Ram, R.N. and Sathyanesan, A.G. 1984.** Mercuric chloride induced changes in the protein, lipid and cholesterol levels of the liver and ovary of the fish, *Channa punctatus. Envron. Ecol*., **2:**113-117.
14. **Dalela, R.C.; Bhatnagar, M.C. and Verma, S.R. 1978.** Histopathological studies on the effect of rogor and thiodon on the activity of acid phosphatase in liver, muscle and kidney of *Channa gachua. Ind. J. Exp. Biol*., **16(10):** 1099-1101.
15. **Agrahari, Shweta; Bhartiya, Vinod Kr.; Gopal Krishna and Pandey, K.C. 2004.** Monocrotophos induced behavioural and biochemical changes in fresh water fish, *Channa punctatus*. 24th *Annual Session of the Acad. Environ. Biol.,* **1:**105-106.
16. **Santhakumar, M. and Balaji, M. 2000.** Acute toxicity of the organophosphorus insecticide, monocrotophos and its effects on behavior of an air breathing fish, *Anabas testudineus* (Bloch.). *J. Expt. Biol*., **21(2):**121-124.
17. **Singh, R.K.; Singh, R.L. and Sharma, B. 2003.** Acute toxicity of carbofuran to a freshwater teleost, *Clarias batrachus*, **70:**1259-1263.
18. **Tilak, K.S.; K. Satyaradhan and P.B. Thathaji. 2003.** Biochemical changes induced by fenavalerate in the freshwater fish, *Channa punctatus*. *J. Ecotoxico. Environ. Moni*., **13(4):**261-270.
19. **Shrivastava, Sharad; Rao, K.S. and Dubey, R. 1990.** Food, feeding and size relationship of some freshwater catfishes from Kshipra river. *Bhopal University. Res. J.,* **2:**13-15.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 219-224
An International Biannual Refereed Journal of Life Sciences

10(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Studies on physico-chemical parameters of water samples of Sonbarsa Tilabay River (Koshi), District Saharsa, Bihar

Chandan Kumar*
Department of Zoology, B. N. Mandal University, Madhepura, Bihar, India

Received : 22nd December 2019 ; Revised : 21st February 2020

Abstract : The Koshi river is the major tributaries of the Ganges which originates from the snowy peaks of Tibetan plateau in the central Himalayas. A branch of Koshi River passing near Sonbarsa block, popularly known as Tilabay River. The physic-chemical parameters of Koshi River, Saharsa, Bihar have been studied monthly during August 2017 to July 2018 from two sites. This paper deals with the air temperature and physico-chemical parameters like temperature, transparency, pH, oxygen, carbon dioxide, alkalinity, hardness, chloride and biological oxygen demand of water of Sonbarsa River (Koshi area) of Saharsa district. The water temperature was recorded maximum in summer season but the transparency, pH, dissolved oxygen, total alkalinity and total hardness were recorded maximum in winter season. Similarly, free carbon dioxide was higher in rainy season and chloride and biological oxygen demand were recorded maximum in summer season. The results revealed that there was significant seasonal variation in some physic-chemical parameters and most of the parameters are in the normal range indicating better quality water resources.

Key words: Physico-chemical parameters, Sonbarsa Tilabay River (Koshi area)

INTRODUCTION

The study site is Sonbarsa Tilabay River of Saharsa district of Koshi area. The quality of water resources is usually described according to its physical, chemical and biological characteristics. For confirming the good quality of water resources, large numbers of physico-chemical and biological parameters are to be studied in detail and must be found in normal range. In any rational formulation and deciding quality of water resources an adequate knowledge of existing nature of physico-chemical parameters, magnitude and source of any pollution load must be known, for which monitoring of physico-chemical parameters and chemical parameters and pollutants is essential. For effective maintenance of water quality through appropriate control measures, continuous monitoring of large number of quality parameters is essential. However it is very difficult and laborious task for regular monitoring of all the parameters even if adequate manpower and laboratory facilities are available. Therefore, in recent years an alternative approach based on statistical correlation, has been used to develop mathematical relationship for comparison of physico-chemical parameters. Assessment of water resource quality from any region is an important aspect for the development activities of the region, because the rivers, lakes and manmade reservoirs are used for water supply to domestic, industrial, agriculture and fish culture use.

Mostly the water resources are being used for various purposes such as domestic use, agriculture and fish culture etc. by local community. Particularly Sonbarsa block

*Corresponding author :
Phone : 87899 24811
E-mail : kumardrchandan29@gmail.com

belongs to rural are of dense population of district Saharsa. In future there is threat of contamination of water from the surface runoff of used fertilizers and pesticides. Although extensive works on the physico-chemical parameters have already been carried out. As there is no sufficient baseline data about physico-chemical parameters of Koshi river water, hence the present work has been undertaken for monitoring the physico-chemical characteristics.

MATERIALS AND METHODS

Sonbarsa (Tilabai River) of Saharsa District, Bihar was selected for study. This river is a branch of Koshi. The study area was visited monthly and the surface water samples were collected by using samplers (Plastic containers of 5 litre size) from two sites of the river. Some physical and chemical parameters such as light transparency, water temperature, pH and dissolved oxygen were determined in the field during the visits given for collection of water sample. For the certain parameters such as electrical conductivity, total alkalinity, hardness, magnesium, calcium, chloride, free CO_2, hydrogen sulphide (H_2), nitrate, phosphate, sodium, potassium and total solids etc., the water sample was collected and preserved with preservative and analysed in laboratory. The analysis of physico-chemical parameters were carried out by

Table 1 (a): Physico-chemical parameters of the river water from August 2017 to September 2018.

Parameters	August, 2017		September, 2017		October, 2017		November, 2017		December, 2017		January, 2017	
	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2
Water Temp. (^{0}C)	27.5	27.3	29.6	28.8	31.2	31.0	29.8	30.1	25.6	24.3	25.4	25.1
Light trans. (cm)	22.3	13.2	24.5	17.5	29.6	23.8	42.6	48.2	42.1	48.2	40.5	40.5
PH	7.8	7.9	7.7	8.0	8.1	8.1	8.3	8.0	7.7	7.9	7.8	8.0
EC ((μ mdhs)	0.532	0.608	0.557	0.559	0.501	0.498	0.415	0.417	0.418	0.420	0.419	0.422
DO	6.32	6.75	6.66	6.41	5.71	6.32	8.49	8.58	11.78	12.98	9.52	10.10
Free CO_2	3.78	3.45	4.52	4.12	4.51	4.78	6.52	6.17	6.22	6.35	3.54	3.86
Total alk.	395.0	312.0	375.5	175.4	418.2	413.5	430.5	398.6	492.0	499.3	296.0	273.0
Hardness	210.3	215.1	185.6	184.9	220.1	215.6	181.0	175.9	240.5	255.9	209.3	205.6
Magnesium	27.99	28.79	24.01	22.32	27.41	27.89	20.12	18.45	33.56	39.15	23.56	24.12
Calcium	38.14	40.19	36.54	37.82	42.01	35.26	38.97	42.58	38.79	37.22	45.18	43.19
Chloride	42.56	57.26	39.37	41.58	42.53	39.41	21.51	22.13	30.15	24.78	22.84	25.98
Nitrate	27.0	27.0	30.0	29.0	31.0	47.0	29.0	25.0	29.0	27.0	33.0	31.0
Phosphate	0.17	0.22	0.16	0.17	0.14	0.15	0.12	0.15	0.16	0.18	0.20	0.21
Sodium	35.0	39.0	30.0	36.0	32.0	32.0	32.0	30.0	33.0	49.0	35.0	31.0
Potassium	5.3	5.5	5.3	5.0	5.1	5.3	5.2	4.9	5.2	4.8	5.6	4.5
H_2S	0.678	0.279	0.845	0.398	0.674	0.789	0.423	0.125	0.426	0.279	0.549	0.542

Note: S_1 = Site first, S_2 = Site Second; Values expressed in mg/l, except water temp., Light trans., Ph and EC.

following standard method as described by APHA *et al.*, (2012) and Kumar *et al.*,(2015). Transparency, air temperature and water temperature were recorded between 12 noon and 1 p.m. For the analysis of physico-chemical parameters, methods of Welch (1952), Michael (1984), Trivedi and Goel (1984), Adoni *et al*. (1985) and APHA (1998) were adopted.

RESULTS AND DISCUSSION

There are sixteen (16) physico-chemical parameters of the river water were determined monthly from two sites during the period August 2017 to July 2018, and results were summarized in the table 1 (a) and 1(b). Temperature is one important physico-chemical parameter which is directly related to chemical reactions in aquatic ecosystem. In present investigation water temperature is ranging in between 24.3°C to 33.5°C and found seasonal change in it. Similar results were reported by Goel *et al.* (1986). The seasonal variation in air temperature was little higher (30.13°C) in rainy season in comparison to that of summer (29.55°C) and lowest in winter season during the first year study period. During the second year study period, the air temperature was recorded little higher in summer than rainy season and lowest was found in winter.

The pH of water of the Koshi river increaseda little in August then a slight decrease was recorded in September.

Table 1 (b): Physico-chemical parameters of the river water from August 2017 to September 2018.

Parameters	February, 2018		March, 2018		April, 2018		May, 2018		June, 2018		July, 2018	
	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2
Water Temp. (^{0}C)	26.0	25.9	29.3	28.7	31.5	31.7	33.5	32.7	29.8	29.1	30.2	30.6
Light trans. (cm)	37.2	36.0	38.6	35.5	28.7	31.0	26.0	30.2	14.6	35.0	14.5	32.1
PH	7.6	7.8	7.8	7.9	8.2	8.0	8.1	7.8	7.8	7.6	7.9	8.1
EC ((μ mdhs)	0.385	0.401	0.415	0.410	0.423	0.478	0.562	0.558	0.632	0.651	0.641	0.63
DO	9.45	9.12	7.35	7.49	8.86	8.67	9.82	8.76	10.55	12.85	11.23	11.48
Free CO_2	1.52	1.25	2.58	2.36	1.00	0.65	0.52	0.26	3.56	3.42	4.85	4.42
Total alk.	253.0	255.3	259.5	249.9	223.2	220.4	261.8	274.0	265.0	272.3	261.3	256.8
Hardness	112.0	135.0	105.0	95.0	202.0	185.0	215.0	208.0	229.0	227.0	195.0	221.0
Magnesium	19.45	20.42	18.52	16.35	22.03	22.06	24.58	17.45	16.48	17.24	22.13	23.85
Calcium	41.07	38.27	31.59	30.14	37.89	36.45	47.52	46.71	65.99	62.84	45.26	49.15
Chloride	31.56	30.24	32.58	34.89	33.87	45.16	34.25	33.16	52.17	57.08	57.19	64.06
Nitrate	31.0	28.0	33.0	37.0	33.0	32.0	31.0	32.0	34.0	37.0	42.0	38.0
Phosphate	0.20	0.18	0.21	0.23	0.17	0.19	0.15	0.14	0.18	0.20	0.18	0.15
Sodium	33.0	32.0	30.0	35.0	33.0	31.0	29.0	33.0	37.0	36.0	38.0	34.0
Potassium	5.5	4.9	5.6	4.8	5.2	4.5	5.3	5.1	5.2	5.6	5.0	5.3
H_2S	0.535	0.475	0.425	0.566	0.478	0.536	0.758	0.985	0.546	0.854	0.542	0.795

Note: S_1 = Site first, S_2 = Site Second; Values expressed in mg/l, except water temp., Light trans., Ph and EC.

Again it increased from October to January then decreased from February to May and increased from June to July during the first year study period.

The water transparency or light penetration is highly variable parameter and helps in determining productive zone of water body. In present study seasonal change in light transparency was observed. The pH of water is alkaline and value ranges from 7.6 to 8.3. The higher pH values may be due to photosynthetic activity in water body as reported by King (1970), and may helps in photosynthesis of phytoplankton.

Electric conductance and hardness content of water was in the normal range of fresh water as reported by Bhosale *et al.*(1994).

The study of dissolved oxygen shows seasonal variation and shows increased oxygen level during winter season. The range of oxygen is in between 5.71 to 12.98 mg/l. Similar variation in oxygen was reported by Khatavkar *et al.* (1989) and Bhosale *et al.* (1994). Free carbon-dioxide in water occurs due to respiration of aquatic biota, decomposition of organic matters and also due to infiltration through the soil. The range or free CO_2 is in between 0.26 to 6.52 mg/l. higher level might be due to mixing of atmospheric CO_2 with rain water which ultimately enters into the wetland and also due to active decomposition of organic matters (Sinha, 1995, 2002).

Total alkalinity of water varies seasonally and ranges from 220.4 to 499.3 mg/l. The increase in alkalinity was recorded during winter season. Analogous variation in alkalinity was reported by Goel *et al.* (1985).

Generally, water temperature is influenced by air temperature and intensity of solar radiation. It was highest in summer and lowest in winter. Highest value ecorded in summer might be due to high air temperature and greater light penetration and comparatively low volume of water than rainy season. The water temperature showed positive and significant correlation with free carbon dioxide and biological oxygen demand but had inverse and significant correlation with transparency, pH, dissolved oxygen, total alkalinity and total hardness. Bose and Gorai (1993) reported negative significant correlation between water temperature and dissolved oxygen. Welch (1952) and Munawar (1970) have observed that shallower the water body more quickly it reacts to the change in the temperature. The maximum pH of present study was in winter season followed by rainy and summer seasons. The maximum pH in winter season may be attributed to algal blooms because Roy (1955) had shown that the higher pH is associated with the maximum phytoplankton.

The minimum pH recorded in summer may be due to low photosynthesis. Several workers have reported low pH during the low photosynthesis due to the formation of carbonic acid (Hannan and Yong, 1974; Cabecadas and Brogueira, 1987; Bais *et al*., 1995). But, Gautam (1990) reported highest pH in summer and lowest in rainy season. The maximum dissolved oxygen found in winter season may be due to low temperature. Similar observations were made by Moitra and Bhattacharya (1965). The minimum dissolved oxygen was found in summer due to high temperature, and higher microbial demand of oxygen for decomposition of suspended organic matter (Bhowmick and Singh, 1985; Palharya and Malviya, 1988). Elmore and West (1961) stated that an increase in temperature of water results in the decrease of dissolved oxygen content of water.

Seasonally, the maximum chloride content was recorded in summer followed by winter and rainy seasons. The maximum quantity of chloride recorded in summer season may be due to low volume of water, high temperature and high rate of decomposition of organic matters. Chloride concentration indicates the presence of organic waste of animal origin (Thresh *et al.,* 1949). Munawar (1970) has suggested that higher concentration of chloride in water is an index of pollution of animal origin and there is a direct relation between chloride concentration and pollution level.

The suitability of water resource for the irrigational use in agriculture is depending on its salt concentration, especially chloride content. In water chloride content was in the range 21.51 to 64.06 mg/l. According to Swingle (1967), water body containing hardness above 15 ppm is necessary for satisfactory growth of fish. The low calcium and magnesium content present in water resource may responsible for maintaining normal hardness level. Because of limited washing and cleaning activities in river water may be responsible for nil residual chlorine content.

Presence of hydrogen sulphide in eater body is an indication of organic pollution load in it (Olsen and Sommerfeld, 1997). In present study low concentration of H_2S is reported and may be released by bacteria. According to Ganpati (1960), the non-polluted eaters are generally deficient in nitrate and phosphate content, but

certain factors such as discharge of surface runoff from agricultural field and sewage are responsible for increasing nitrate and phosphate content. In present study nitrate and phosphate contents are in the ranges of 25 to 38 mg/l and 0.12 to 0.23 mg/l respectively which are below range indicating oligotrophic nature of river. Similar results were reported by Pandey *et al*. (1999). The sodium and potassium contents are in the ranges of 29 to 39 mg/l and 4.5 to 5.6 mg/l respectively in river water. The higher level of sodium and potassium is due to release of domestic effluents into the system which contains common salt (Baruah, 1995).

ACKNOWLEDGEMENT

The author is highly thankful to Dr. R. B. Jha, Head of Dept. of Zoology, R. M. College Saharsa, Bihar for his valuable guidance and critical suggestions.

REFERENCES

1. **APHA 1998.** *Standard methods for examination of water and waste water*. 20th Edn., Washington, USA.
2. **Adoni, A.D., G. Joshi, K. Ghosh, S.K. Chourasia, A.K. Vaishya, M. Yadav and J.D. Allan 1985.** *Stream ecology*. Chapman and Hall, London.
3. **Bais, V.S., N.C. Agrawal and A. Tazeen 1995.** Comparative study on seasonal changes in phytoplankton community in the Sagar Lake and Military Engineering Lake (M.P.). *J. Freshwater Biol.* **7:** 19-25.
4. **Bose, S.K. and A.C. Gorai 1993.** Seasonal fluctuations of plankton in relation to physico-chemical parameters of a freshwater tank of Dhanbad, India. *J. Freshwater Biol.* **5:** 133-140.
5. **Jhingran, V.G. 1991.** *Fish and fisheries of India.* 3rd edn., Hindustan Publ., New Delhi.
6. **Kumar, Ajay and Mandal, O.P. 2015.** Eco-Biodiversity and Management of Wetlands, Safali Samagra Angika Vikash Sansthan, New Delhi, p. 148-158
7. **Mishra, A.P., B.K. Bora and M. Sharma 1998.** Investigations on the seasonal variation of certain physico-chemical parameters of a Beel, Assam, India. *J. Freshwater Biol.* **10:** 83-87.
8. **Munawar, M. 1970.** Limnological studies of fresh water ponds of Hyderabad, India. The Biotope. *Hydrobiol.* **35:** 127-162.
9. **Singh, M. 1995.** Impact of human activities on the physico-chemical conditions of two fish ponds at Patna, India. *J. Freshwater Biol.* **7:** 13-17.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 225-230
An International Biannual Refereed Journal of Life Sciences

11(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Physico-chemical parameters of the pond of Sunderpur area of district Darbhanga, Bihar

Kumari Anjana[a]* & Shashi Bhushan Shashi[b]
[a]Department of Zoology, L.N.M.U, Kameshwar Nagar, Darbhanga, Bihar, India
[b]Department of Zoology, R.B. Jalan Bela College, Bela, Darbhanga, Bihar, India

Received : 22st December 2019 ; Revised : 21st February 2020

Abstract : The present Study deals with the physic-chemical parameters of the pond of Sunderpur area of district Darbhanga, Bihar, India using selected standard classical methods with an aim to investigate present scenario for its better utilization. The data collected in various reasons (summer, winter, spring and monsoon) revealed that those analyzed parameters were within permissible limit for fish culture and stocking should be done as per productivity of water. Thus, the analysis of various parameters of the Manth Pond indicates that it is in the beginning of pollution stage which needs a good maintenance, protection and prevention by the people and Municipal Corporation of Darbhanga.

Key words: Physico-chemical, Pond, Darbhanga

INTRODUCTION

Biological diversity and ecosystem effect is the result of the evolutionary plasticity of living organisms, and increased geometrically through perhaps 2.5 billion years, proliferating by trial and error, controlled by natural selection, filling almost every one of the habitable ecological niches created in a likewise evolving world environment. Our palaeontologists tell us that the number of species existing has fluctuated widely-how widely we can never know with any certainty. The causes of these fluctuations have been subject of much speculation- varying from gradual and normal climatic change to catastrophic changes even as great as the impact of hypothetical gigantic meteor or asteroid with the terrestrial globe.[1]

*Corresponding author :
Phone :
E-mail : kailashyadav91@rediffmail.com

Aquatic ecosystems are broadly categorised based on the differences in their salt content as:

(a) Freshwater ecosystems
(b) Marine ecosystems (includes the ocean and the sea), and
(c) Estuarine ecosystems (region where fresh water from a river mixes with the sea)

Ponds are smaller bodies of still water located in natural hollows such as limestone sinks, or that result from the building of dams, either by humans or beavers. Ponds are found in most regions and may exist either seasonally or persist from year to year.

The physical and chemical properties of fresh water bodies are characterized by the climatic, geochemical, geo morphological and pollution condition. In order to utilize fresh water bodies successfully for fish production, it is very important to study the Physico-chemical factors influencing the biological productivity in the water bodies.[2]

The quality of aquatic life surviving in the pond is totally dependent on the water quality of the pond. In the recent years several studies have been made in this field.[3,4] Not much information is available of Physico-chemical and biological parameters of the present water bodies.

MATERIALS AND METHODS

This has been categorised in two sections: (A) Physical & (B) Chemical

(A) Physical: The Physical parameters of abiotic components comprise:

(a) Temperatures of air

(b) Temperature of water

(c) Hydrogen ion concentration (pH of water)

(d) Transparency of water.

Temperature of air and water at different levels were measured with the help of a maximum-minimum thermometer (0° - 50^oC) following the procedure given by Welch (1948)[5]. pH was determined with B.D.M narrow range pH paper in the field and later the values were corrected with a pH meter in the laboratory of Department of Zoology, Samastipur College, Samastipur. Transparency was measured with a standard Sacchi disc of 20 cm diameter attached to a marked plastic tape.

(B) Chemical: The chemical parameters of abiotic components comprise:

(a) Dissolved Oxygen (D.O.)

(b) Free Carbon dioxide (CO_2)

The chemical parameters, such as dissolved Oxygen and free Carbon dioxide usually change within an hour. Therefore, these were analysed on the spot.

(a) Dissolved Oxygen (D.O.): it was determined by unmodified Winkler's method using alkaline Kl, $MnSO_4$, Conc. H_2SO_4, 0.025 N Sodium thiosulphate titrant and 1.0% Starch solution indicator. Values were calculated according to the formula as shown below:

$$\text{D.O., mg/L} = \frac{(ml.xn) of\ titrant\ x8x1000}{V_2(\frac{V_1 - V}{V_1})}$$

Where,

V_1 = Volume of sample bottle after placing the stopper

V_2 = Volume of the part of contents titrated

V= Volume of $MnSo_4$ and Kl added

(b) Free Carbon Dioxide (CO_2): It was determined by titrating the sample using 0.05 N NaoH titrant and Phenolphthalein indicator. Values were calculated according to the formula shown below:

$$\text{Free } CO_2 \text{ mg/L} = \frac{(ml.\times N)8\ NaoH\ \times 1000 \div 44}{ml.\ of\ sample}$$

RESULTS

Thermal Regime: The water temperature record showed variations with a corresponding change in atmospheric temperature and was always found lower than the latter except the early morning hours. At 5 hours, the atmospheric temperature was a little lower than the water temperature in different seasons. The maximum diel fluctuation in water temperature was 3.5^oC in spring, 5.4^oC in summer, 2.8^oC in monsoon and 3.5^oC in winter. The data also showed a diurnal trend of increase in the night hours except in the noon hours where it showed the highest temperature in the different seasons of the studied year.

pH: The pH showed a distinct trend of gradual increase during the day time from sunshine to sunset and a decrease in night hours. pH values varied from 6.70-7.70 in spring, 7.50-8.50 in summer, 6.80-7.80 inmonsoon and 6.40-7.00 in winter.

Transparency: The high transparency recorded during spring, summer, monsoon and winter were 102.00, 20.20, 32.00 and 88.00 in spring, summer, monsoon and winter respectively, Interestingly, the value of transparency was nil (Zero) at 5 hour (early morning) and 1 hour (night).

Dissolved Oxygen (D.O.): It showed variation from 2.80 to 8.20 ppm in spring, 3.20 to 13.20 ppm in summer, 3.40 to 7.50 ppm in monsoon and 4.00 to 11.20 ppm in winter. The highest content of D.O. was observed at 17.00 hours and lowest at 05.00 hours (early morning). A distinct trend in rise and fall in its content was recorded in day time and night hours.

Free Carbon dioxide (CO_2) : The range of free CO_2 was 4.30 ppm is spring, 00.00 to 70.50 ppm in summer, 21.70 to 56.00 ppm in monsoon and 5.70 to 32.00 ppm in winter. The maximum value was recorded at 05.00 hours (early morning) and the minimum at 17.00 hours (afternoon) in all the seasons. There was a decreasing trend in Free CO_2 content from sunrise to sunset onward till early morning. The maximum diel variation in free CO_2 content was observed during summer and the minimum in winter. All these variations in physic-chemical parameters of water of pond under study during the thesis work. The variations in physico-chemical factors of the studied pond due to different man- made causes are shown in Table - 1

Table 1- Biological seasonal diversity effect on ecosystem in physico-chemical parameters of water of pond under study recorded in four seasons of the year (From February 2009 to March 2010)

Parameters		Spring (February) Time in Hours					
		0500	0900	1300	1700	2100	0100
1	Air Temperature (0^0C)	19.00	22.00	25.50	23.40	20.50	17.50
2	Water Temperature (0^0C)	20.00	20.60	22.00	23.00	20.00	19.50
3	pH	6.80	7.20	7.50	7.70	7.00	6.70
4	Transparency (Cm)	00.00	80.00	102.00	96.00	00.00	00.00
5	Dissolved Oxygen (Ppm)	2.80	6.30	8.20	7.90	5.10	4.20
6	Free Carbon Dioxide (Ppm)	33.60	14.80	5.90	4.30	16.50	33.00
		Summer (June) Time in Hours					
1	Air Temperature (0^0C)	27.00	30.00	34.00	35.00	33.50	31.00
2	Water Temperature (0^0C)	30.00	32.00	35.40	34.00	33.00	32.50
3	pH	7.50	8.00	8.20	8.50	8.10	7.60
4	Tarnsparency (Cm)	00.00	17.20	20.20	18.80	00.00	00.00
5	Free Carbon Dioxide (Ppm)	3.20	7.20	11.40	13.20	9.40	6.60
6	Free Carbon Dioxide (Ppm)	70.50	6.20	00.00	00.00	55.00	69.00
		Monsoon (September) Time in Hours					
1	Air Temperature (0^0C)	27.00	28.40	30.50	32.00	31.00	28.20
2	Water Temperature (0^0C)	28.00	28.70	30.80	30.10	28.50	28.10
3	pH	7.00	7.80	7.10	7.20	7.10	6.80
4	Tarnsparency (Cm)	00.00	28.00	32.00	30.00	00.00	00.00
5	Free Carbon Dioxide (Ppm)	3.40	4.30	6.40	7.50	6.20	4.00
6	Free Carbon Dioxide (Ppm)	56.00	47.00	26.90	21.70	28.90	42.80
		Winter (December) Time in Hours					
1	Air Temperature (0^0C)	19.10	20.50	22.80	20.50	20.10	19.20
2	Water Temperature (0^0C)	18.00	19.30	21.50	21.50	20.80	19.10
3	pH	6.40	6.50	6.80	7.00	6.80	6.70
4	Transparency (Cm)	00.00	86.10	88.00	82.00	00.00	00.00
5	Dissolved Oxygen (Ppm)	4.00	8.10	9.60	11.20	6.30	4.40
6	Free Carbon Dioxide (Ppm)	32.00	15.80	10.80	5.70	12.60	23.60

DISCUSSION

In the present studied Manth Pond, the minimum difference between air and water temperature was 0.2°C, sometime was noticed no difference in some months of the year (July to September) vide Table 1 During these months, due to heavy rain the water becomes more turbid and behaves like dirty water. This condition of water causes readily absorption and retention of atmospheric heat. It was observed that the minute difference between air and water temperature of the present pond corresponds with investigation of Paer (1997)[6].

A significant positive relationship is observed between pH and water temperature in the present pond. This observation of these present authors is in conformity with the findings of UNEP. The correlation between pH and photosynthesis has been established by earlier workers Venkateshraju *et al.* (2010)[7] and Akhte and Ahmad (2010)[8]. The range of pH between 6.8 and 7.5, a yearly average of 7.10 shows an alkaline trend of water. This alkaline trend of water indicates an ideal condition for the growth and metabolism of micro-organisms and metabolic activities.[9] While Khapekar *et al.* (2006)[10] have advocated that 7.3-7.9 pH is most suitable for prolonged survival of bacteria. But Kapoor and Bamniya (2001)[11] noticed that pH higher than 8.1 indicated the super saturation with Oxygen. According to Mark *et al.* (2000)[12] an optimum pH of 7.0 to 8.5 is suitable for fish culture. This contention of Mark *et al.* (2000)[12] is supported by Das (2002)[13] who points out that pH 6.5 to 8.5 is suitable for fish production, while Bhat and Matondkar (2004)[14] and considered that a pH range of 6.5 - 9.0 to be suitable for pond fish culture. The present Manth Pond is suited for fish culture because average values are within this range, even though the pH

sometimes exceeds this range, the productivity of fish is not affected. If the pond water is fertilized with fowl manure the pH is increased and the increase of pH perhaps, is due to the formation of phytoplankton bloom which increases the photosynthesis activity, shifting the Carbonate-bi-carbonate equilibrium to the alkaline side and also due to increased ammonia content in water released from urea.[15]

Similar relationships between the two parameters are also observed in certain other water bodies of India located in places with severe winters.[8]

Chirmoy and Razluddin (2002)[16] reported the dissolved Oxygen values in the surface waters range between 2.55 (Jan) and 4.47 (Sept) ml/litre and that of bottom water between 2.70 (Dec) and 4.80 (Nov) ml/litres. An important feature of the dissolved Oxygen regime is its lower concentration during December in bottom water in general higher values are recorded during September-November and then onwards a decrease from December in surface water. While during majority of the instances, the concentration of Oxygen decreases at the bottom water and the reverse is also noticed. Similar trend is also noticed by Dwiwedy *et al.*, (2001)[17].

CONCLUSION

The present work details the study carried out at one water-body, designated for fish culture particularly, namely Manth Pond. Assessment of pond water quality was carried out by instituting the various physico-chemical factors and biological parameters. The physico-chemical analysis included temperature, transparency, pH, dissolved Oxygen, free CO_2. Thus, the analysis of various parameters of the Manth Pond indicates that it is in the beginning of pollution stage which needs a good maintenance, protection and prevention by the people and Municipal Corporation of Darbhanga.

ACKNOWLEDGEMENT

Authors are thankful to Department of Zoology, L.N.M.U, Kameshwar Nagar, Darbhanga, Bihar and Department of Zoology, R.B. Jalan Bela College, Bela, Darbhanga, Bihar, for providing necessary equipment's to complete the entire research works.

CONFLICT OF INTEREST

Authors declare no conflict of interest regarding publication or any other activity related to this article.

REFERENCE

1. **Fosberg, P.R. 1986.** *Biodiversity. Ibid.* **9(4):**125-129.
2. **Sahni K. and Yadav S. 2012.** Seasonal Variations in physico-chemical Parameters of Bharawas Pond, Rewari, Haryana, *Asian J. Exp. Sci*. **26(1):** 61-64
3. **Yadav P, Yadav VK, Yadav AK and Khare PK. 2013.** Physico-chemical characteristics of a fresh water pond. *Octa.J. Biosci.* **1(2):** 177-184
4. **Yadav, S.K., Sharma, S.P., Thaliyal, N.K and Sahu, B.2013.** Physico-chemical characteristics and phytoplankton of Betwa River, Vidisha District, Madhya Pradesh. *Ind. J. Env.Ecopl.* **15(3):** 557-560
5. **Welch, P.S. 1948.** Limnology. 2nd ed. McGraw-Hill Book Co., New York.
6. **Paer, H.W. 1997.** Coastal eutrophication and harmful algal blooms. Importance of atmospheric deposition and groundwater as 'new' nitrogen and other nutrient sources. *Limnol*. *Oceanogr*. **42:**1115-1165.
7. **Venkatesharaju, K., Somashekhar, R.K. and Prakash, K.L. 2010.** Heavy metal concentrations in river Cauvery, Karnataka, J. *Curr. Sci.* **15(1):** 289-296.
8. **Akhte, R. and Ahmad, S.S.C 2010.** Seasonal Variations in temperature dissolved oxygen and biochemical oxygen demand in Katraj Lake, Pune, Maharashtra. *J. Curr. Sci*. **15(2):** 71-76.
9. **Singh, R.B., Shaligram, S. and Mandal, T.N. 2008.** Studies on water quality of village Dhima, Purnea, Bihar. India. *J. Env. Ecopln*. **15(3):** 689-691.
10. **Khapekar, R.R., Chaudhari, P.R and Wate, S.R. 2006.** Limnobiotic study of Kanhan River at Monda near Nagpur (Maharashtra). *Ind. J. Env.Ecopln*. **12(3):** 531-538.
11. **Kapoor, K. and Bamniya, B.R. 2001.** Water, eutriphication and algal blooms. Ecology and conservation of lakes, reservoirs and rivers. (Ed. Kumar, A._ABD Publishers, Jaipur. 42-57.
12. **Mark, B. Bain, Amy, L. Harig., Daniel, P. Loucks, Reuben, R. Reuben, R., Goforth and Katherine, E. Mills. 2000.** Aquatic ecosystems protection and restoration: advances in methods for assessment and evaluation.

13. **Das A.K 2002.** Phytoplankton primary production in some selected reservoirs of A.P. *Geobios*. **29:** 52-57.

14. **Bhat, S.R. And Matondkar, S.G.P. 2004.** Algal blooms in the seas around India-Networking for research and outreach. *Curr.Sc* **87:** 1083.

15. **Asif Khan, Untoo, S. and Parveen, S. 2002.** Limnology of a Reservoir receiving effluents from a thermal power plant. Ecology of polluted water. Ed. Arvind Kumar, A.P.H. publishing corp. 5, Ansari road, Darya Gang, New Delhi - 110 002.

16. **Chinmoy Chatteree and Razluddin, M.2002.** Determination of water quality Index (WQI) of a degraded river in Asansol industrial area (W.B.) Ecology of polluted water Vol. II. Ed. Arvind Kumar, A.P.M. publishing corporation, 5, Ansari Road, Darya Ganj, New Delhi- 110002.

17. **Dwiwedi, B.K and Pandey, G.C. 2001.** Seasonal dynamics of cyanobacterial toxin producing algal species water ponds. *Aquaculture*. **2:**141-146.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 231-236
An International Biannual Refereed Journal of Life Sciences

12(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Efficiency of chemical treatment in management of wax moth infestation of *Apis mellifera*

Sushma Kumari*
P.G. Department of Zoology, Samastipur College, Samastipur, Bihar, India

Received : 23rd December, 2019 ; Revised : 22nd February, 2020

Abstract : Beekeeping is an integral part of livestock based livelihood system in Bihar. Despite of its importance, the Bihar regions are challenged by colony strength and honey yield due to pests including wax moths. To overcome these challenges different treatments were suggested by researchers. The present study was thus intended to evaluate the efficiency of chemical treatments on infestation level with wax moths in Bihar. Data were analysed using ANOVA and weighted ranking matrix. Results revealed that fumigation with Bromopropylate @ 1 strip/ colony should be used to control wax moth infestation. Another finding indicated that the raised frames should be stored with sulphur @ 300 g/ m^3 space will prevent damage to raised combs by wax moth during storage.

Key words:Beekeeping, Bromoproylate, Sulphur, wax moths

INTRODUCTION

Bee keeping is an art of keeping bee colonies for economic benefit and has been practice for a long time. Bee colonies suffer from various maladies. These maladies include contagious diseases, non-contagious diseases and malfunctions, pests' parasite and predators. Among the pests parasites and predators are such diverse groups of animals as mite, warps, wax moths, spider and ants are the most important. The wax moths are extremely damaging and can cause absconding and even death of a colony. The total number of insect's species having association with honeybees is very large and their records in India are incomplete, therefore, wax moth will be discussed here that play a significant role in the practice of beekeeping.

Wax moth- The wax moth is a pest of all honeybees in India viz., *Apis cerana*, *Apis dorsata* and *Apis florea*.[1] Wax moths are world- wide distributed causing a major concern whenever beekeeping is being practised. It affects beekeeping industry more in the plains and lower hills than at higher altitude. There are two types of wax-moth, which are major concern to the beekeeping industry. The greater wax moth (*Galleria mellonella* L.) is the most serious problem in the combs, whereas lesser wax moth (*Achroia grisella* fabr.) is a minor pest they feed on pollen, wax and protein of the pupal skins.[2]

Wax moth, especially greater wax moth (*Galleria mellonella* L.) is the most serious problems and of major concern of the beekeeping industry. The larvae destroy raised combs in storage as well as in hives by tunnelling through the midrib of a comb. They feed on pollen, wax and protein of the pupal skins. The insect breed on combs and in tropical and subtropical areas they often cause death of the colony. Observations on the seasonal infestation in *Apis mellifera* epidemics showed that it was devastating

*Corresponding author :
Phone :
E-mail : @gmail.com

during and after the monsoon when colony strength was low due to shortage of bee flora.[3]

According to Sharma and Garg (1984)[4] the damage is caused by caterpillars which eat combs and interfere with brood-rearing by making silken galleries through the cells. In case of severe infestation, the whole comb become a mass of webbing in which excreta of the caterpillars is enmeshed. Severe damage is caused in bee colonies which are weak and some combs are not covered by bees. The infestation colonies often abscond. In the off season, when the combs are stored, the caterpillars also damage them. Atwal (2000)[2] found ramification of galleries on combs when a frame was held against height. In the late stage the comb looked like a tangled mass of silken galleries, frass and handly a sign of cells. According to Ramchandran and Mahadeven (1951)[5], Brar *et al.* (1985)[3], Gupta (1987)[6] and Atwal (2000)[2] peak activity of wax moth is from June to November whereas, Viraktamath (1989)[7] recorded the peak population during May to August which coincides with dearth period in south India.

Management practices for control of wax month

Preventive measures

Preventive measures include ensuring that the colonies, whether of *Apis cerana indica* or *Apis mellifera* are strong and have adequate food stores, reducing the hive entrance and sealing cracks and crevices in the hive wall, protecting the colonies against pesticide poisoning and controlling pests and diseases that might otherwise weaken them and removing wax debris accumulated on the bottom boards of the hives (Sharma and Garg. 1984)[4]. Ramchandran and Mahadevan (1951)[5] reported that minimising the cracks and crevices of hives and crushing the moth egg masses gave effective control of wax moth. The spare combs should be removed from the hives and stored in the moth proof hives. Kapil and Sihan (1983)[8] observed effective control of all stages of wax moth when combs where kept at high temperature (less than 49^0C) for 5 to 6 hours and freezing of queen cups. Cleaning of hives and creation of good sanitation conditions helped to save the colonies from wax moth damage.

Chemical control measures

Chemical control measures can be taken to prevent or control the wax moth infestation of stored combs. Fumigation is often necessary. The most commonly used fumigants was PDCB (Para- dichlorbenzene) crystal which evaporated slowly. Aluminium phosphide, naphthalene, ethylene dibromide and methyl bromide had also been used for wax moth control.[4]

Sulphur: Ramachandran and Mahadevan (1951)[5] recommended sulphur fumes for the control of wax moths in stored combs. Various remedial measures have been suggested, including the use of sulphur, besides observing strict Cleaners in bee hives.[9]

Ethyl dibromide: Kapil and Sihag (1983)[8] reported one table spoon of EDB very effective to kill all stages of wax moth in stack of eight supers. Ethylene dibromide (EDB) is heavier than air and is placed on the super in small containers. One table spoon of EDB was found to be effective to kill all the stages of wax moth in a stack of eight supers.[8] They further reported that PDCB evaporates slowly and acts as repellent as well as toxic substance to wax moth and its larvae. Two to four table spoon of the crystal of PDCB are placed between every other super in stack.

Aluminium phosphate: Atwal (2000)[2] advised to raise frames in 4-5 ft tall polythene bag of thick plastic sheet and to put ¼ tablet of aluminium phosphide, before tightly closing the open end with a string.

MATERIALS AND METHOD

The monitoring and identification of prevailing major natural enemies is this area namely wasp, mite and wax moth during dearth period in this ecological region of Bihar were done. Different chemical treatment strategies were tested such as manipulation of beehives. The present investigations were evaluated during dearth period (June to October) for selection of better management strategies against wax moths in the two consecutive years 2009 and 2010.

Chemical control of wax moth within the hive

Experiments were conducted in the Randomised block design with three replications and seven treatment during two consecutive years, 2009 and 2010. The details of the experiments were as under.

Treatment	Formulation	Dose
T_1-	Kelthane-spray	0.05%
T_2-	Formicacid Evaporation agent	5ml/colony
T_3-	Naphthaline ball crystal	3.6g
T_4 -	PDCB - crystal	5g/colony
T_5 -	Bromopropylate – Fumigants	one strip/colony
T_6 -	Sulphur Dust	5 g/hive
T_7 -	Control	……..

Three beehives of *Apis mellifera* in each Treatment were selected for the experimental trial. All the fumigants as mentioned above applied at different doses on all the combs at 1.30 pm. The treatment was repeated after 15 days.

Therefore, the present study was undertaken to determine appropriate management tactics which would be ecologically sound, economically viable and socially acceptable. The experiments were carried out during dearth period (June to October) of the years, 2009 and 2010.

RESULT

Chemical control of wax moth within the hive

The results on the effect of different chemical application (T_1 to T_6) besides control (T_7) on the infestation of wax moth (mean no. of larval/colony) have been summarised in Table 1 and depicted in Fig 1. The data revealed the data revealed the superiority of all the chemicals (T_1 to T_6) over control (T_7) in minimising the larval population of wax moth in both the years (2009 and 2010).

In 2009, before the chemical application, the mean larval population of wax moth/colony of all the treatments (T_1 to T_6) were found non- significant and ranged between 13 to 19 larval population per colony (table 1). Although all the treatments (T_1 to T_6) were found significant over control (T_7) after chemical application in the hives. The minimum larval population of wax moth (8.0 and 8.0) was found per colony treated with Bromopropylate @1 strip/colony (T_5) and sulphur @ 5 g/hive in the treatment (T_6), respectively and were at par with treated hives with PDCB (T_4) having larval population (9.0). The treatments Viz., T_2 (12.0) and T_3 (12.0) were found statistically at par with each other, but significantly inferior over T_1 (15.0). Further all of these treatments (T_1 to T_6) were proved significantly superior in minimising the larval population of wax moth over T_7 as control having maximum larval population (20.0). the maximum reduction of the larval population of wax moth (55.55) was recorded per colony in the application of Bromopropylate fumigants @ 1 strip/colony treated hive (T_5) and it was found significantly superior in reduction of the larval population over the rest of the treatments (table 13). The treatments Viz., T_1 (21.05), T_2 (25.00), T_3(29.41), T_4 (30.76) and T_6 (46.66) were found significantly superior over T_7 (-25.00) as control in percentage reduction of the larval population per colony. Among these treatments T_1 had the minimum percentage reduction of larval population (21.05) per colony.

In the following year 2010, almost a similar trend of the effectiveness after the chemical applying treatment on wax moth larval population was inferred. All the chemical treatments (T_1 to T_6) were recorded significantly superior in lowering the larval population over T_7 as control which have the maximum (22.00) larval population per colony. Whereas, the minimum larval population (8.0) was observed per colony in the application of Bromopropylate @ 1 strip/ colony treated hives (T_5) which have at par with the fumigation of sulphur @ 5g/hives treated hives (T_6) and have relatively lower larval population (11.00) per colony. The rest of the treatments Viz., T_1(15.00), T_2 (13.00), T_3 (16.00) and T_4 (15.00) were found statistically at par to each other and significantly superior over T_7 as control which had maximum larval population (22.00). Although, before the chemical application all the treatments (T_1 to T_6) were found non-significant having 17.00 to 21.00 larval population of wax moth.

The maximum percentage reduction of the larval population (60.00) was observed per colony in the application of Bromopropylate fumigants @ strip/colony treated hive (T_5) and followed by the treated hive with sulphur fumigants @5g/hive (T_6) which had the percentage reduction of the larval population (38.88). the remaining treatments Viz., T_1 (21.05), T_2 (23.52), T_3 (23.80) and T_4 (25.00) were also found significantly superior in percentage reduction of the larval population over (T_7) (-29.41) as control. The findings suggested that wax moth infestation could be effectively managed by the application of bromopropylate fumigant @ 1 strip/colony.

Chemical control of wax moth in storage condition during dearth period

The observation on chemical control of wax moth infestation on the comb infestation area during dearth period were recorded and presented in table 2 and depicted in Fig. 2

The result indicated that he chemicals used were significantly superior to reduce wax moth infestation in storage condition over century (T_7) during the years 2009 and 2010 and when pooled data of two years were statistically analysed.

In 2009, the data further revealed that the treatment T_2 resulted the minimum comb area infestation (180 cm^2) which did not. Differ significantly to the treatments T_1 (350 cm^2) and T_4 (350 cm^2), where as the treatment T_6 had the maximum comb area infestation (1200 cm^2) which

was followed by T_5 (900 cm^2). The treatments T_3, T_5, T_6 and T_7 were found differ significantly among themselves and these treatments also showed significant impact in reduction the comb infestation area T_7 (2000 cm^2) as control.

In the second year 2010, almost a similar trend of the comb infestation area was found. Overall the treatments, T_1 to T_6 were found significantly superior over control T_7 (1800 cm^2) in minimising the comb infestation area. The treatment T_2 resulted the minimum comb area (200 cm^2) which did not differ significantly to the treatment T_4 (375 cm^2) the treatments T_1 (417 cm^2), T_3 (765 cm^2), T_5 (1050 cm^2) and T_6 (1300 cm^2) were differed significantly among themselves and were also found significantly superior in decreasing the comb infestation area over T_7 (control) which had recorded maximum comb infestation area (1800 cm^2). But the treatment T_4 (375 cm^2) did not differ significantly with T_1 (417 cm^2) in reducing the comb infestation area.

The pooled data of both years (2009 and 2010) when statistically analysed revealed that all the six treatments (T_1 to T_6) were found significantly superior over control T_7 (1900.00 cm^2) in minimising the comb infestation area. The treatment T_2 resulted the minimum comb area infestation (190.00 cm^2) which was followed by T_4 (362.50 cm^2). The treatments, viz. T_1 (383.50 cm^2), T_3 (720.00 cm^2), T_5 (975.00 cm^2) and T_6 (1250.00 cm^2) differed significantly among themselves. Among the treatments T_6 had the maximum comb infested area (1250.00 cm^2) which too, differed significantly to the rest of the treatments (Table 2). The findings showed that application of sulphur @300g/ m^3 space in storage condition have been found better in preventing wax moth infestation.

Table 1- Effect of different chemical application on infestation of wax moth (Mean no of larvae/ colony) in during dearth period during two consecutive years 2009 and 2010.

Treatment	Formulation	Dose	2009			2010		
			Infestation before App.	Infestation After App.	% reduction	Infestation before App.	Infestation After App.	% reduction
T_1Kelthane	Spray	0.05%	19	15	21.05	19	15	21.05
T_2 Formic Acid	Evaporation agent Crystal	5ml/colony	16	12	25.00	17	13	23.52
T_3 Nephaline ball	Crystal	3.6g	17	12	29.41	21	16	23.80
T_4 PDCB	Crystal	5g/colony	13	09	30.76	20	15	25.00
T_5Bromopropylate	Fumigatns	1 strip/colony	18	08	55.55	20	08	60.00
T_6 Sulphur	Dust	5g/hive	15	08	46.66	18	11	38.88
T_7 Control	-	-	16	20	-25.00	17	22	-29.41
S.Em. (±)	-	-	-	1.03	-	-	1.06	-
CD at 5%	-	-	Ns	2.95	-	Ns	3.04	-

*** Mean of three replications**

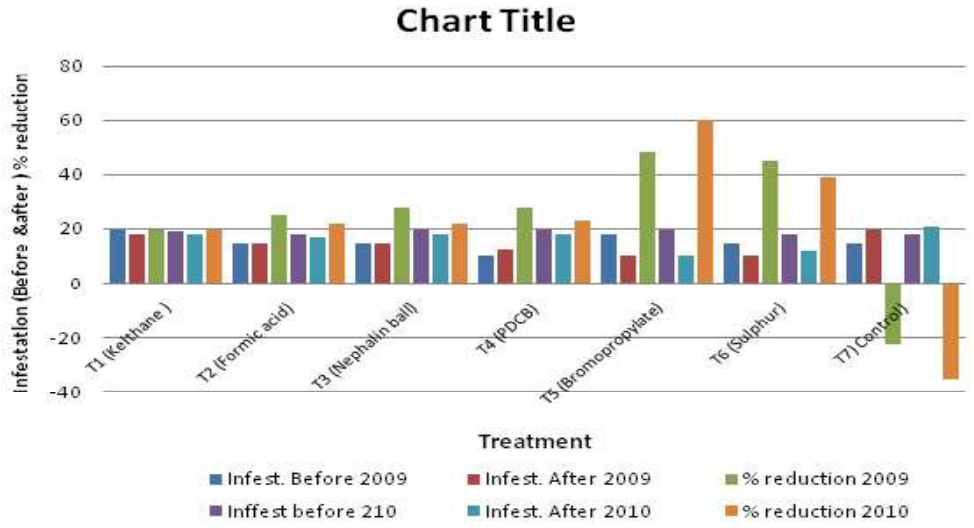


Fig. 1 : Effect of different chemical application on infestation of wax moth (Mean no. larvae/colony) in dearth period during two consecutive years, 2009-2010

Table 2- Effect of chemical treatments on infestation of wax moth in storage condition in dearth period during two consecutive years, 2009 and 2010.

Treatments	Dose	2009	2010	Pooledmean
		Mean area infestation (cm^2)	Mean area infestation (cm^2)	
T_1 Sulphur	200 g/m^3 space	350	417	383.50
T_2 Sulphur	300g/m^3 space	180	200	190.00
T_3 Aluminium phosphide	3g/m^3 space	675	765	720.00
T_4 Aluminium phosphide	10g/ m^3 space	350	375	362.50
T_5 EDB	25 ml/m^3 space	900	1050	975.00
T_6Falbex strip	5-7 fumigations at weekly intervals	1200	1300	1250.00
T_7 Untreated (control)		2000	1800	1900.00
S.Em. (±)		61.27	58.14	26.26
CD at 5%		188.82	179.18	74.62

***Mean of three replications.**

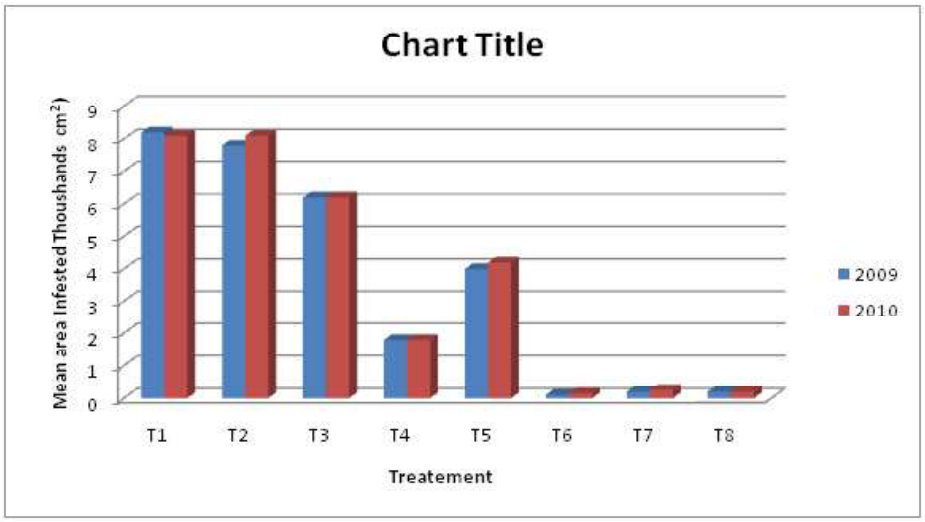


Fig 2.- Effect of different storage methods of raised combs on Gelleriamellonella L infestation in dearth period during two consecutive years, 2009- 2010

DISCUSSION

The Italian honeybee, *Apis mellifera* colonies were successfully introduced a few decades ago in Bihar but their colonies suffers from few natural enemies during dearth period. Wasps, mite and wax moth are very important natural enemies which affects the honeybee health in ecological region. Management of these natural enemies have become very important aspects for honey production technology and successful beekeeping.

The findings of the present investigation revealed that various chemical like sulphur, kelthane, formic acid, nephthalin ball, PDCB, and Bromopropylate have been reported to be ideal insecticides against mites and wax moth. Chemical controls were dominant hive protection methods in many apiaries having high honey yielding colonies of *Apis mellifera*. In the present studies, efforts were made to evaluate, the effectiveness of insecticides in the hives and storage condition during dearth period in both the years (2009 and 2010). Different chemical viz., Sulphur, kelthane, formic acid, nephalene ball, PDCB and Bromopropylate were used against wax moth in the hives during dearth period (June to October) of 2009 and 2010. The results revealed that the maximum percentage reduction on mite population (41.72 and 56.40) was recorded per hive in the Bromopropylate fumigants @ 1 strip/colony treated hive (T_5) followed by the treated hive with sulpur fumigants @5g/hive (T_6) which have the percentage reduction of mite population (29.72 and 44.54). The minimum

percentage reduction of mite population as 11.03 and 4.62 per colony were recorded in treated hive with kelthane @ 0.05% (T_1) during 2009 and 2010, respectively. There were significant variation ($p<0.05$) among all the variation and their interaction.

The result revealed that the maximum percentage reduction of wax moth larval population (55.55 and 60.00) was recorded per colony in the application of Bromopropylate fumigant @1 strip/colony treated hive (T_5) followed by the treated hive with sulphur fumigants @ 5 g/ hive (T_6) which have the percentage reduction of wax moth population (46.66 and 38.88). The percentage reduction of mite population as 21.05 and 21.05 per colony were recorded in treated hive with kelthane @ 0.05% (T_1) during 2009 and 2010, respectively. There were significant variation ($p<0.05$) among all the treatment and their interaction. The result indicated that fumigation of honeybee colonies with Bromopropylate 1 strip/colony would effectively reduce the population of wax moth within the hive. Thus, it might be used for managing the infestation of mite and wax moth within the hive.

CONCLUSION

The results suggested that sulphur @ 300g/m^3 space might be used for managing the wax moth infestation during storage was found most effective. Another results indicated that fumigation of honey bee colonies with Bromopropylate 1 strip/colony would effectively reduce the population of wax moth within the hive. Thus it might be used for managing infestation of wax moth within the hive.

ACKNOWLEDGEMENT

Authors are thankful to P.G. Department of Zoology, Samastipur College, Samastipur, Bihar for providing necessary equipments to complete the entire research works.

CONFLICT OF INTEREST

Authors declare no conflict of interest regarding publication or any other activity related to this article.

REFERENCES

1. **Singh, S. 1962.** Beekeeping in India. Pub., ICAR, New Delhi: 214P.
2. **Atwal, Avta. 2000.** Essentials of Beekeeping and pollination, Pub. By Kalyani publishers, Ludhiana, 394P.
3. **Brar; H.S. Gatoria, G.S. Jhaji, H.S., Chahal, B.S. 1985.** Seasonal infestation of *Galleria mellonella* and population of *Vespa orientalis* in *Apis mellifera* apiaries in Punjab. *Indian J. Ecol.,* **12 1):**109-112.
4. **Garg, R. Sharma, O.P. Dogra, G.S. 1984.** Formic acid: and effective acaricide against *Tropilaelaps clareae* Delfinodo and Baker (Laelaptidae: Acarine) and its effect on the brood and longevity of honeybees. *Amer. Bee J.* **124(10):** 736 -738.
5. **Ramchandran, S. Mahadevan, V. 1951.** The wax moths and their control. *Indian Bee J.,* **13(7-8):**85-88.
6. **Gupta, M. 1987.** Wax moth in *Apis mellifera* in Haryana, Indian. *Indian Bee Jour.* **49:**26-27.
7. **Viraktamath, S. 1989.** Incidence of greater wax moth, *Galleria mellonella* L. in three species of honeybees. *Indian Bee J.* **51(4):** 139-140.
8. **Kapil, R.P. and Sihag, R.C. 1983.** Wax moth and its control. *Indian Bee J.* **45:** 47- 49.
9. **Atwal, A.S. and Sharma, O.P. 1967.** Beekeeping in Punjab. Punjab Agril. Univ. Ludhiana. *Bull. No.* **14:**43P.

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 237-240
An International Biannual Refereed Journal of Life Sciences

13(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Diversity & prevalence of piscine helminth parasites of Madhepura, Bihar, India

Chandan Kumar Chand*
University Department of Zoology, B.N.Mandal University,Madhepura, Bihar

Received : 15th December, 2019 ; Revised : 11th February, 2020

Abstract : The present investigation was made to study the diversity & prevalence along with , seasonality & intensity of infestation of helminthes parasites in selected fishes *Heteropneustes fossilis, Anabas testudineus, Labeo rohita*, *Cirrhinus cirrhosus & Clarias batrachus* of Madhepura, Bihar during the experimental year 2016-2018. Five helminth parasites *Eustrongylides* sp (Nematode), *Euclinostomum heterostomum* (Trematode), *Clinostomum* (Trematodes) *Diplostomum* (Trematodes) and *Pallisentis allahabadii* (Acanthocephalan) were collected from abdominal cavity, liver and intestine of the infected hosts. *Eustrongylides sp.* (Nematodes-), *Euclinostomum heterostomum* (Trematodes) exhibited highest prevalence in different host fishes while *Clinostomum* (Trematodes) remained at low. The cumulative diversity value of all the parasites as per S-W statistical equation has been found to be 0.675.

Keywords : Piscine hosts, **helminth parasites, prevalence, diversity, infestation, S-W diversity value**

INTRODUCTION

India is considered as one of the major wetlands that preserve a rich variety of fish species & due to the protein rich source fish is in great demand. Hence to obtain healthy and quality fish meat, it is necessary that the fish should be free from all types of pathogens like viruses, bacteria, algae, protozoans, helminths, annelids, arthropods and molluscs. Parasites of fish constitute one of the major problems to fish health. Besides the direct losses caused by mortality, parasites have considerable impact on growth, resistance to other stressing factors, susceptibility to predation, marketability and pave way for secondary infections.[1] Helminth parasites remain one of the most important problems confronting aquaculture. Potentially all freshwater and brackish water fish may be affected by helminthes with heavier infections in predatory fish, particularly of species utilizing fish as intermediate or transient host.[2] The snake headed freshwater fishes such as Heteropneustes *fossilis, Anabas testudineus, Labeo rohita*, *Cirrhinus cirrhosus & Clarias batrachus* has a wide geographical distribution, a high growth rate and contributes significantly to fishery sector in India. It usually inhabits swamps, pools, rice fields and is known for its nutritive and invigorating qualities. Data regarding the parasitic infestation of helminth parasites of fishes in Madhepura district of Bihar is limited. Many workers like Bhalerao[3,4] Dayal[5], Gupta[6] have made valuable contribution on the taxonomy of digenetic trematode and cestode parasites. Fish nematodes of Indian region have been studied by Karve[7], Agarwal[8] have worked significantly on acanthocephalan taxonomy but in order to control parasitic infestation of fresh water fishes much emphasis needs to be given on epidemiological studies so that preventive

*Corresponding author :
Phone : 7004459956
E-mail : chandan.chand84@gmail.com

measures can be taken. Therefore, the present study was undertaken to investigate the prevalence, mean intensity and abundance of infestation of helminth parasites and the seasonal variation of such infestation in relation to physicochemical parameters of the water quality of the fish farms.

MATERIAL AND METHODS

Collection of the hosts

The test fishes (5 species), (17-21 cm long) weighing 50–75 g were collected from fish farms of district of Madhepura during the period of 2016-2018 and were brought alive in the parasitology laboratory for examination. They were kept for 3 weeks under observation for acclimatization in glass aquaria (40 × 60 × 100 cm). Water was changed after every 24hr and commercial fish food was supplied to fish during acclimatization period. Dead fish (if any) were removed from the aquaria as soon as possible to avoid water fouling.

Collection of Parasites

Fishes were dissected out and body cavity was thoroughly examined for any parasite. Visceral organs were placed and dissected out in petridish containing normal saline (0.6% NaCl) to allow adhering parasites to be released. Collected nematodes were fixed in hot 70% ethanol and stored in glass vials containing glycerine alcohol (1:3). Nematodes were later cleared in lactophenol for morphological observation and identification. The trematodes were fixed in between the folds of glass slides and cover slips tied with rubber bands or thread. AFA or 70% ethanol were allowed to run through the slides and cover slips and the fixed specimens were then preserved in coupling jars containing 70% ethanol and 5% glycerine. In case of acanthocephalans, collected parasites were thoroughly washed with saline and kept in distill water to facilitate complete eversion of the proboscis. Further, parasites kept over glass slides were flattened under slight pressure of cover glass, fixed in A.F.A. and after 24 hours preserved in glycerified 70% alcohol. Relative parameters were measured and identification was performed using selected identification keys.

S-W diversity statistics of the parasites

On the data of number of parasites isolated from different hosts, relative abundance (pi), log of pi as the indicators of parasite species abundance and richness in different hosts shall be calculated statistically as per S-W diversity index equation.[9]

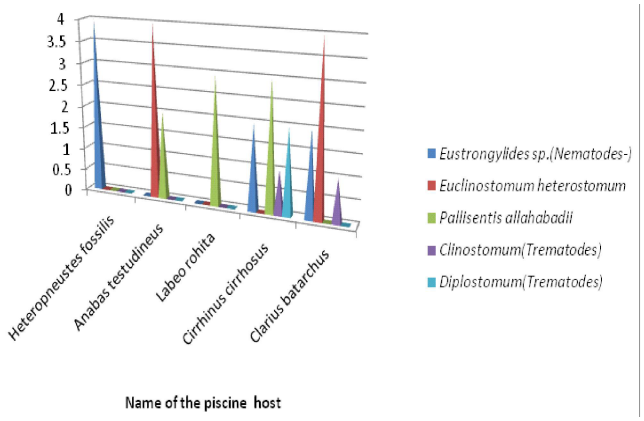
RESULTS AND DISCUSSION

Five different fishes were sampled from different water bodies of Madhepura district. Out of which all of the fishes were found to have infected with different helminth parasites.[10] Coincidently, five parasites have been found which include one nematode *Eustrongylides* sp, three trematode, *Euclinostomum heterostomum* and one acanthocephalan, *Pallisentis allahabadii*. *Eustrongylides* species have been isolated from abdominal cavity, liver and intestine, *Euclinostomum heterostomum* from liver and *Pallisentis allahabadii* from intestine of the infected fishes.[11] The total of 28 number of parasites, belonging to the Nematoda, Acanthocephalan and Trematoda were recovered that have been shown in the Table 1 .

Cirrhinus cirrhosus was found to be the most highly infested fish with different fish parsasites. While *Labeo rohita* were found to be the least infested species. The nematode, *Eustrongylides* sp. showed prevalence degree of variance with respect to other trematodes as well as acanthocephalons.[12,13] The prevalence of infection was maximum in *Cirrhinus cirrhosus* and minimum in September and December. *Euclinostomum heterostomum* exhibited peak prevalence and lowest *Labeo rohita*.

The cumulative diversity value of all the parasites in different piscine host considered together reflects the value of 0.675 having integrated individual parasite diversity values as per the Shannon & Weiner equation.

Species diversity (S-W) value of five helminth parasites infestation(single & multiple) in five different piscine host of Madhepura, Bihar(2016-2018)



Prevalence & diversity of helminth parasites found in the five different fishes as host of Madhepura, Bihar

SL. No	Name of the fish host	Common name	Name of the isolated parasites	Individual parasite no.	Total number of parasites	Nature of infection	pi=n/N	log pi	pi*log pi	H
1	*Heteropneustes fossilis*	Singhi	***Eustrongylides sp.*(Nematodes-)**	4	4	Single	0.142	-0.847	-0.120	0.675
2	*Anabas testudineus*	Kawai	***Euclinostomum heterostomum* (Trematodes)**	4	6	Double	0.214	-0.669	-0.143	
			***Pallisentis allahabadii* (Acanthocephalan)**	2						
3	*Labeo rohita*	Rohu	***Pallisentis allahabadii* (Acanthocephalan)**	3	3	Single	0.107	-0.970	-0.103	
4	*Cirrhinus cirrhosus*	Mrigal	***Eustrongylides sp.*(Nematodes)**	2	8	Multiple-4	0.285	-0.545	-0.155	
			***Pallisentis allahabadii* (Acanthocephalan)**	3						
			***Clinostomum*(Trematodes)**	1						
			***Diplostomum*(Trematodes)**	2						
5	*Clarius batarchus*	Magur	***Euclinostomum heterostomum* (Trematodes)**	4	7	Multiple- 3	0.28	-0.552	-0.154	
			***Eustrongylides sp.*(Nematodes)**	2						
			Clinostomum(Trematodes)	1						

REFERENCES

1. **Bhalerao, G.D. 1937.** Studies on the Helminths of Indian Trematoda, IV. *J. Helminthol.,* **15 (2):** 97-124. DOI: 10.1017/ S0022149X00030753
2. **Bhattacharya, S.B. 2007.** Handbook on Indian Acanthocephala. Kolkata, India, Zoological Survey, 225pp. University
3. **G.D. Bhalerao. 1936.** Studies on the helminths of India Trematoda IV. *Journal of Helminthology.* **14(4):**181-206.
4. **G.D.Bhalerao. 1942.** Some metacercarial forms of Clinostomatidae (Trematoda) from India. The Proceedings of the National Academy of Sciences, **16:**67–71.
5. **J. Dayal. 1961.** Trematode parasites of Indian fishes-II, *Indian Journal of Helminthology*. **I(2):** 93-116.
6. **S.P. Gupta.** Morphology, life history and biophysical studies of helminth parasites, D. Sc. Thesis, Lucknow University, Lucknow,.
7. **J.N .Karve. 1930.** Some parasitic nematodes of frogs and toads. *Annals of tropical medicine and Parasitology.* **24:**481-491.
8. **V.Agarwal.** Studies on some nematode parasites of freshwater fishes from Lucknow, *Annales de Parasitologie Humaine et Comparee (Paris)*. **41:**217-237.
9. **Stiling, P. 2002.** Ecology & species diversity: theory and practise, chapter 11, pp298-322, EEE publisher.
10. **Bhuiyan, A.S., Akter, S., Musa, G.M. 2007.** Occurrence of parasites in *Labeo rohita* (Hamilton) from Rajshahi University. *J. Zool. Rajshahi Univ.* **26:**31-34
11. **Bhure, D.B., Nanware, S.S. 2014.** Studies on Prevalence of Cestode Parasites of Freshwater Fish, *Channa punctatus. J. Entomol. Zool. Stud.* **2(4):**283-285
12. **N. Gupta, P.Singhal & D.K Gupta. 2012.** Population dynamics of a parasite *Palicentis* in two species of fish *Channa punctatus & C. striatus. Journal of environmental biology.* **33(2):**195-199.
13. **P.Sinha. *et al.* 2008.** Studies on the helminth infestation of *Channa punctatus* in & around Patna, proceedings of Zoological Society of India. **7:**79-84.

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 241-246
An International Biannual Refereed Journal of Life Sciences

14(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Changes in blood parameters induced by chronic exposure of carbamate pesticides on fresh water fish *Cirrhinus mrigala* (Ham.)

Sunil Yadav*
P.G. Department of Zoology, B.N.Mandal University, Madhepura, Bihar, India

Received : 23rd December, 2019 ; Revised : 22nd February, 2020

Abstract : *Cirrhinus mrigala* is an Indian major carp commonly preferred in composite fish culture. The common carp *Cirrhinus mrigala* was exposed to organophosphorous pesticide fenthion to determine the LC_{50} values at different time interval and its sublethal effects on the hematological and biochemical parameters. LC_{50} values at varying times were recorded as high as 2.5 mg/l (24hr), 2.2 mg/l (48hr), 1.8mg/l (72hr) and 1.5mg/l (96hr). Fishes exposed to sub lethal concentrations (0.38, 0.19, 0.96 mg/l) for 60 days revealed that the pesticide caused alterations in blood parameters. It is found that it decreases RBC count (6-29%), hemoglobin concentration (7-28%), ESR (11-22%) and clotting time (7-32%), were as WBC count (12-31%) and platelet count (9-43%) were found to be increased. Plasma level of glucose and lactic acid was higher and protein level in blood plasma was lower in exposed fish when compared to control. In conclusion, the changes observed indicate that hematological parameters can be used as an indicator of carbamate related stress in fish on exposure to elevated level.

Key words: ***Cirrhinus mrigala,*** **Carbamate, glucose, haematology, protein**

INTRODUCTION

Carbamates are a class of insecticides structurally and mechanistically similar to organophosphate (OP) insecticides. Carbamates are N-methyl Carbamates derived from a carbamic acid and cause carbamylation of acetylcholinesterase at neuronal synapses and neuromuscular junctions.

Carbamate pesticides have been synthesized and sold commercially since the 1950s. Currently, carbamates are one of the major classes of synthetic organic pesticides and are used annually on a large scale worldwide. Carbamates are mainly used in agriculture, as insecticides, herbicides, fungicides, nematocides, acaracides, molluscicides, or sprout inhibitors. In addition, they are used as biocides for industrial and other applications, and in household products.[1] Most of the carbamates have high melting points and low vapor pressures. They are usually distributed in aqueous environments because of their high solubility in water. Studies have shown that carbamates and their degradation products are potential contaminants of the environment and food resources. There is increasing evidence indicating that carbamates may spread throughout ecosystems by leaching and runoff from soil into ground and surface water. Carbamates are identified as potential leachers because of their high water solubility. They may also enter environmental water from industrial wastes, accidental spillage and dumping. Carbamates are on the priority lists released by the US Environmental Protection Agency (EPA). Carbofuran, an anti cholinesterase

*Corresponding author :
Phone : 9709091356
E-mail : sunilyadav14033@gmail.com

carbamate, is commonly used. The Common trade name where carbamate is used are Aldicarb Temik, Carbaryl, Prokoz, Sevin and Larvin.

Study of hematological parameters can indicate physiological response to contaminated environment.[2] Biochemical parameters can show the impact of water pollution on fish.[3] Mrigal *C. carpio* is an economically important fresh water fish and commonly cultured in many parts of the world. In the study, an attempt was made to investigate the toxicity on blood cell count, glucose, lactic acid, protein, hemoglobin, ESR and clotting time in the blood plasma of *Cirrhinus mrigala* exposed to different sub- lethal concentrations of carbamate for 60 days exposure.

MATERIALS AND METHODS

Specimens of *Cirrhinus mrigala* were procured from fisherman of the area. The length and weight ranged from 12- 14cm and 20-25 g respectively. They were kept in aquaria for two weeks to get acclimatized to laboratory conditions. During the period of acclimations all fishes were fed with commercial fish food twice daily. After acclimation, 30 aquaria each containing 30L water were stocked with ten fish per aquarium. Different concentrations (0.01, 0.1, 0.2, 0.40, 0.6, 0.8, 1.0 mg/l) of carbamate purchased from Bayer India Ltd, were prepared by adding the required volume of acetone. A control set was run with the same number of fish and the same volume of acetone without adding carbamate. The experiments were then run in triplicate. The water in the aquaria was altered daily. The lethal concentration LC_{50} for 24, 48, 72 and 96 hr was computed by the Probit method.[4] Acute toxicity study revealed that toxicity of carbamate does not increase with time. Therefore toxicant concentration selected were ¼, 1/8 and 1/16 of 96 hr LC_{50} (0.387, 0.193 and 0.096mg/l). All the concentrations were prepared on the same day and parallel control group were maintained in similar way.

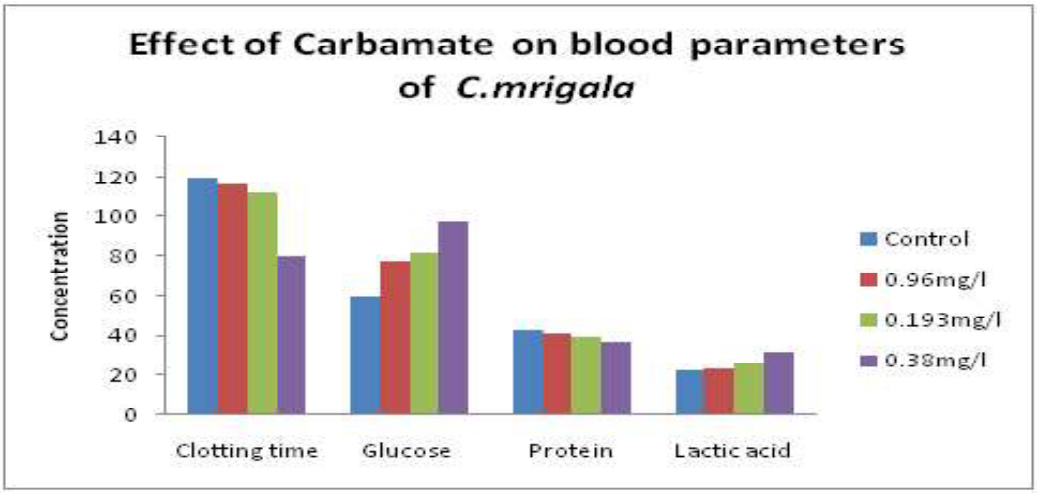
Fishes from the aquarium were removed after 60days of exposure during the experimental period. Blood samples were collected in heparinized vials by cutting the caudal peduncle .Collected fresh sample of blood was used for analysis. Hemoglobin content in the blood was estimated by acid hematic method.[5] Coagulation time was determined by Lee White method.[6] Red blood cell count was done by hematocytometric method using Neubaur's counting hemocytometer, thrombocyte and white blood cell count by indirect method suggested by Lucas and Jumroz (1961)[7] and ESR by Westergen method. The remaining blood was centrifuged at 6000 rpm for 10min at 4°C and the collected plasma was stored at -20°C till analyzed (Zubair Ahmad, 2011)[4]. The glucose, proteins and lactic acid content in blood plasma was assayed by Folin Wu, Lowry and Berker and Summerson method mentioned by Hawk *et al.*, (1965)[5].

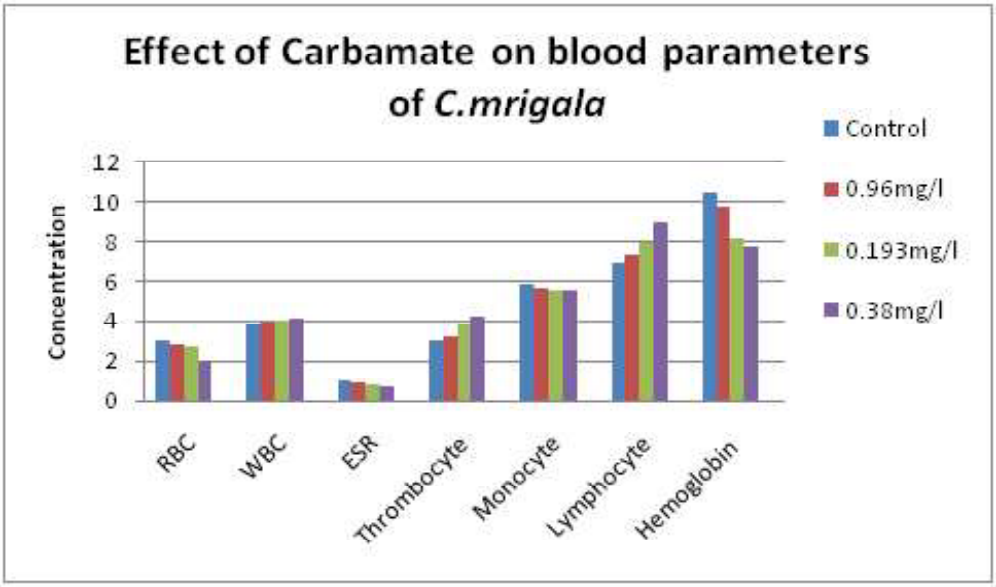
RESULTS AND DISCUSSION

LC_{50} values were calculated from the mortality rate recorded for 24, 48, 72 and 96hrs by straight line graphical interpolation method. The 95% confidence limit of LC_{50} value and slope function of the line has been calculated according to the method suggested by Litchfield and Wilcoxon (1949)[8]. The magnitude of toxic effects of pesticides also depend on length and weight, corporal surface/bodyweight ratio and breathing rate.[9,1] According to the report given by Sheela (1982)[10] carbamate is a neurotoxic agent. Oh *et al.* (1991)[11] and Murthy (1986)[1] reported that toxicity is due to detoxification and absorption. Toxicity varies with respect to species, size of fish and duration exposure (Dutta *et al.* 1995)[12].

The blood is a unique mirror in which all the internal process taking place in an organism are reflected. The observation on different hematological parameters provide a good deal in diagnosing the effects of environmental stress on an animal and in turn would give an insight in the changes induced in the circulating fluid. Investigators have also identified changes in several hematological parameters as indicators of metal exposure.[13] However, the present findings indicate that in Mrigal sub- lethal chronic exposure to carbamate altered the blood parameters.

The fish exposed to different concentration of carbamate showed increase in glucose and lactic acid and decrease in protein, hemoglobin and clotting time. Values obtained were compared with the control fish. It is documented that under stress condition, fish become hyperactive perhaps to get out of the stressful medium and would require an increased amount of oxygen to meet their energy demand, secondly fish secrete increased amount of mucus to coat the body especially gills to get relief from the irritating pollutants fig 8. This in turn reduces the gaseous exchange through the gill, thus an





increased utilization of oxygen. This condition causes glycolysis in the liver and muscles of carbamate exposed fish.

The change in the protein content in the blood plasma of Mrigal due to chronic exposure observed. The reduction in protein content was dependent on the concentration of carbamate. Among the three concentration selected the highest concentration (0.38mg/l) seem to be the most highly potent and cause significant change (-28.10%) in protein level. This could be due to the impairment of protein synthesis. Verma *et al*. (1979)[14], Singh *et al.* (1982)[9] Dutta *et al.* (2003)[15], Bucher *et al.* (1990)[3] reported similar decrease in protein content in blood of different species of fish when treated with different pollutants. Similar response was noted in other fresh water fish exposed to acute toxic level of pesticides.[16,17] The pesticide may change the function of vital organs like liver and kidney disrupting the homeostatic condition of the body. Decrease in protein content may be due to their degradation and the product which may in be fed to TCA cycle through amino transferase system to cope up with the carbamate stress. Secondly reduction in protein content level could also be attributed to histopathological damage to stomach and liver leading to its impaired functioning.

Elevation in lactic acid level was due to anaerobic pathway lead by the fish to meet the increased demand for energy under pollution stress. Similar increase in lactic acid content in *Heteropneustes fossils* was reported by Sastry and Subhadra (1982)[18]. It is noted that fish was hyperactive and this has attributed to impaired gill function. Disturbance in coagulate mechanism whether acquired or inherited may not only from alteration of coagulation but

also from abnormal fibrinolysis activity of the plasma. Depression in coagulation time between 125 and 85 sec in *Saccobranchus fossils* and between 140 and 95sec in *Labeo rohita* was observed after exposure to chloradane by Verma *et al.* (1979)[14]. The decrease in clotting time can be attributed to deletion in protein content and also due to histopathological damage caused to liver (figure 6). The LC_{50} value of Mrigal was estimated as 2.5, 2.2, 1.8 and1.5 mg/l for 24, 48, 72 and 96 hrs. The LC_{50} values recorded for the fry of tilapia was 2.8 mg/l for 24 hrs by Soman (1987)[19]. The difference in values may be related to the difference in susceptibility and tolerance related to its accumulation, biotransformation and excretion. Differences in metabolic pathway among species may result in varied patterns of biotransformation, leading to more or less toxic metabolites.[20] Blood parameters generally of fish are suitable food for evaluating the effects of chemicals.[21] In the present study, sub-lethal exposure to carbamate altered blood parameters. The fish exposed to different concentration of carbamate indicated significant reduction red blood cells (RBC's), hemoglobin as compared with the untreated control group. The decrease was 6.6% (0.096mg/l), 19% (0.193mg/l), 29% (0.38mg/l) and the decrease in hemoglobin was 7 % (0.096mg/l), 19.62% (0.193mg/l) and 28% (0.38mg/l) after 60 day exposure. In contrast to this, the white blood cell (WBC's) recorded significant increase after exposure to carbamate. WBCs counted to 31.25%, 25%,and 12.25%at 0.096mg/l,0.193mg/l and 0.38mg/l respectively. Thrombocyte count level also increased to 9.3% (0.096mg/l),25% (0.193mg/l) and 43% (0.38mg/l). A complete blood count provides detailed information about three types of blood cells, red blood cell RBC's, white blood cells and platelets. Furthermore, hematological characteristics have been widely used in the diagnosis of variety of diseases and pathologies induced by pesticides and several others.[22-25]

Red blood cells are very important for transport of oxygen and hemoglobin concentration is directly correlated with RBC count. This is due to synergistic link among these blood parameters in all vertebrates. This close correlation between erythrocyte count and hemoglobin concentration was also reported for other vertebrates including man by Harris (1972)[26]. In the present study reduction in RBC count may be due to increase rate of break down RBC or reduction rate of formation or RBCs. Similar to the present state a decrease in number of RBC, hemoglobin and hematocrit value of diazinon exposed fish was reported on Banaee *et al.* (2008)[27].

WBCs formed; either migrate to key organs such as spleen, lymph nodes, gut, or enter the blood. The increased number of leucocytes can occur abnormally as a result of a toxic chemical. Sudden increase of WBC may be due to the activation of the animal's defense mechanism and the immune system. Several chemical compounds including insecticides, generate antibodies due their interference with immune system which could be the reason for increase in WBC due to the exposure to toxic pollutants. Higher WBC count could be related to the inflammation of the stomach observed (fig. 6). Mossa (2004)[24] and Mansour and Mossa (2005)[25] in rats showed that treatment with pesticides marked by elevation in the animal's defense mechanism and immune system.

CONCLUSION

From the 96hr LC_{50} recorded, it can be concluded that carbamate is moderately toxic to Mrigal. Exposure to chronic sub- lethal concentration of carbamate resulted in significant hemato- biochemical alterations. These changes suggest that the treated fish are faced with severe metabolic stress. The results clearly indicate that usage of carbamate in the field may be a threat to fishes as well as humans. The fact is that carbamate is metabolized to toxic form in fish presumably has environmental and health implication for its use as pesticide.

REFERENCES

1. **Murthy A. S. 1986.** Toxicity of pesticides to fish. *CRC pren. Inc Boca Raton* FLP143.
2. **Dethloft G. M., Bailey H.C and Maier K.J. 2001.** Effect of dissolved copper on selected haematological, biochemical and immunological parameters of wild rainbow trout *Oncorhynochus mykiss*. *Arch,Env.Con.Toxico*. **40:**371-380.
3. **Bucher F. and Hofer R. 1990.** Effect of domestic waste water on serum enzyme activities of brown trout. *Comp. Biochem Physiol*. **97:**385-390.
4. **Zubair Ahmad. 2011.** Toxicity bioassay and effects of sublethal exposure of malathion on biochemical composition and haematological parameters of *Clarias gariepinus. Af. J. Biotech.* **10(63):**13852-59.

5. **Hawk P B., Oser B.L. and Summerson W.H. 1965.** Practical Physiological Chemistry 4th ed., Mcgraw Hill book. Comp Inc. N. Y. Toronto, London.

6. **Dacie J.A. and Lewis S.M. 1979.** Practical Hematology J and A Churchill Ltd London.

7. **Lucas A.M. and Jumroz C. 1961.** Atlas of Avian haematology. Ind. ed. Oxford IBH Pub.co., New Delhi.

8. **Litchfield T. and Wilcoxon F. 1949.** A simplified method of evaluating dose effect experiments *J. Phamac.Exp.Theory*. **96:**99-113.

9. **Singh B and Narain A. S. 1982.** Acute toxicity of Thiodan to cat fish *Heteropneustes fossils*. *Bull Environ. Contam. Toxicol.* **28:** 122 - 127.

10. **Sheela S. J. and Balakrishnan N.K. 1982.** Effect of cadmium on some aspect of carbohydrate metabolism in fresh water fish *Heteropneustes fossils. Env. Poll*. **28:**7-11.

11. **Oh H.S., Lees K., Kim Y.H., Ro H. J. K.1991.** Mechanism of selective toxicity of diazinon to kill fish (*Oryzias latipes*). *Aquat.Toxicol.Risk. Assess*. **14:**343-353.

12. **Dutta H.M.,Munshi J S D., Dutta G. R., Singh N. K., Adhikari. S and Richmonds C. R. 1995.** Age related differences in the inhibition of brain ACTH activity of *H. fossils. Comp. Biochem. Physiol*. **111a:** 331-334.

13. **Cyriac P.J., Antony.A, and Nambison P.N.K. 1989.** Haemoglobin and haematocrit value in the fish *Oreochromis mossambicus* after short term exposure to copper and mercury. *Bull. Environ. Cont. Toxicol*. **43:** 315-320.

14. **Verma S.R., Bansal S.K., Gupta A.K.and Dalela, R.C. 1979.** Pesticide induced haematological alterations in a fresh water fish *Sacchbranchus fossils. Bull Environ. Contam. Toxicol*. **22:**467 - 474.

15. **Dutta H M and Meijer H J M. 2003.** Sublethal effects of diazinon on the structure of testis of blue gill *Lepomis macrochirus*, amicroscopic analysis. *Env. Poll*. **25:**355-360.

16. **Svoboda M., Luskova V., Drastichova J. and Ziabek V. 2001.** The effect of diazinon on haematological indices of common carp *Cyprinus carpio. Acta Vet. Brno*. **70:**457-465.

17. **Rao D. S. 2010.** Carbaryl induced changes in haematological serum biochemical and immunological responses of *C. carp* with special emphasis on immunodialators. Ph.D Thesis, Andhra University India. P.235.

18. **Sastry K.V. and Subhadra. 1982.** Toxicity of certain mosquito larvicide to larvivorous fish *Macropodus cuparius. Toxicol. Lett*. **14:**45-55.

19. **Soman K. 1987.** Some observations on toxicity of insecticide Lebaycid to fresh water fish *C.fasciata*. Ph.D thesis. Mumbai University.

20. **Johnsson C.M. and Toledo M.C.F. 1993.** Acute toxicity of endosulfan to the fish *Hyphessobrycon bifasciatus* and *Brachydanio rerio. Archiv. Environ. Contam. Toxico*. **24:**151-55.

21. **Roche H .and Boge G. 1996.** Fish blood parameters as a potential tool for identification of stress caused by environmental factors and chemical intoxication. *Marine.Envi.Res*. **41:**27-43.

22. **Ali S.S. and Shakoori A.R. 1988.** Gamma BHC induced haematological and biochemical changes in blood of albino rat. *Proc. Pakistan. Cong. Zool*. **8:**61-76.

23. **Ali S.S. and Shakoori A.R. 1990.** Toxicology of aldrin in rats. *Punjab University. J. Zool.* **5:**1-56.

24. **Mossa A.H. 2004.** Genotoxicity of pesticide. Ph.D Thesis Alexandra Univ., Egypt.

25. **Mansour S.A. and Mossa A. H. 2005.** Comparative effect of some insecticides technical and formulated on male albino rat. *J. Egypt. Soc. Toxicolo*. **32:**41-54.

26. **Harris J. W. 1972.** Seasonal variation in some haematological characteristic of *Rana pipens. Comp. Biochem. Physiol.* **43:**975-89.

27. **Banaee M., Mirvaghefi A.R., Rafei G.R. and MajaziAmiri B. 2008.** Effect of sublethal diazinon conc on blood plasma biochemistry of *C. carp. Int. J. Environ. Res*. **2:**189-98.

ADDITIONAL REFERENCES

1. **Alkahem Al., Balawi H.F., Ahmad.S.,Alakel A S., Al Misned F., Soliman E M and Al Ghanim K.A. 2011.** Toxicity bioassay of lead acetate and effect of its sublethal exposure of growth, haematological parameters and reproduction of *Clarias gariepinus. Afr. J. Bio tech*. **10:**11039-11047.

2. **Beijer Kand Jornelo A. 1979.** Hand book on the toxicology of metals. Elsevier/North Holland. Biomedical press New York. pp 47-63.

3. **Cox C. 2000.** Diazinon: *Toxicology. J. Pesticide Reform*. **20:** 15 -21.

4. **Cong N.V., Phuong N Tand Bayley M. 2009.** Effect of repeated exposure of diazinon on cholinesterase activity and growth in *Channa striatus*. *Ecotoxicol. Environ. Saf*. **72:** 699-703.

5. **Gopi L. 1993.** Chronic toxic effects of fenthion-the organophosphorous insecticide to common fresh water fish *Cyprinus carpio*. Ph.D thesis Univ. of Mumbai.

6. **Kitamura S., Suzuki T., Ohta S and Fujimoto N. 2003.** Acute toxicity of endosulfan to the fish *Hyphessobrycon bifasciatus* and *Brachydanio rerio*. *Environ. Health Perspective.* **111(4):** 503-8.

7. **Kitamura S., Kadota T.,Yashida M.,JinnoN and Ohta S. 2000.** Whole body metabolism of organo phosphorous pesticide fenthion in gold fish *Carassius auratus*. *Comp.Biochem physiol. C. Toxicol. Pharmacol.* **126(3):**25-66.

8. **Lackner R. 1998.**Oxidative stress in fish by environmental pollutants. *Ecotoxicol*. pp 203-224.

9. **Miller G. G., Sweet L. I., Adam J. V., Omann G.M.,Passino D. R. and Meier. 2002.** Invitrotoxicity and interaction of environmental contaminants on thymocytes from lake trout. *Fish Shell Immunol*. **13:** 11-26.

10. **Oruce O. and Usta D. 2007.** Evaluation of oxidative stress response and neurotoxicity potential of diazinon on different tissues of *C. carpio*. *Environ. Toxicol. Pharmacol*. **23:** 48 - 55.

11. **Pandey N. B., Chanchal A. K., Singh S. B., Prasad S. and Singh M. P. 1979.** Effect of some biocide on blood and oxygen consumption of *C. punctatus* (Bloch). *Proc. Symp Environ. Biol*. 343-348.

12. **Thomas P.C. and Murthy T.L. 1976.** Studies on impact of few organic pesticides on certain fish enzymes. *Ind. J. Anm. Sci.* **46(11):** 419-29.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 247-250
An International Biannual Refereed Journal of Life Sciences

15(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Sex specific anatomical variation in the organization of Jacobson's organ (VNO) with respect to its olfactory function in common worm snake *Typhlina bramina*

Kumari Ranjana*
P.G. Department of Zoology, B.N.Mandal University, Madhepura, Bihar, India

Received : 24th December, 2019 ; Revised : 25th February, 2020

Abstract : Jacobson's organ also known as vomeronasal organ (VNO) has an important function in prey detection feeding and reproduction in the biological activities of worm snakes. In this article the anatomical and histological layout of this sensory organ in a worm snake *Typhlina bramina* has been reported. For this study, 5-7 micrometric histological sections were cut from their heads where the Jacobson's organ remain located in the mid dorsal plain of the skull having the vomar and nasal plate flooring. The micrometric measurement of the organ lodged in the vomeronasal cranial capsule also laterally flanked by baso-medial side of nasal cavity. In this species of the worm snake (*Typhlina bramina*) the Jacobson's organ has been found to possess an average linear length of 550 µm in male and 510 µm in female which is slightly smaller than other reported species. It has also a prominent connection to the nasal cavity along with a communicating tunnel with the oral cavity through 2 small apertures in the anterior palate. On close observation of the micro sections, the lachrymal duct descending from the eye (internal pocket) is also found to join the organ. Surprisingly the organs lumen since to be also lying by 2 different cellular layer likely to be sensory in function. Cellular differentiation in the sensory epithelial layer which is relatively very thick has also been noticed and discussed in the paper. The central mushroom body covered by special epithelial layer has ciliated cells. Histochemically the epithelial cells have been found to be PAS positive due to heavy deposition of glycogen.

Key words: ***Typhlina bramina*, Sensory function, Jacobson's organ/ VNO, sensory epithelium, PAS positive.**

INTRODUCTION

In comparison to general vertebrates and in contrast to the mammals, great variation is found in the organization of sense organ particularly that of snakes with reference to the blind worm snake, *Typhlina bramina*. Snake mostly use the power of smell and touch for detection of any object present in their vicinity as because they do not posses movable eye lids and rudimentary eyes remain sunken and fixed which arrest their movement. Moreover they also lack external parts of ears, middle ear and simpanic membrane excepting the ear ossicles and as a result they can only perceive the sound wave vibrations.

To supplement these lapses in the optic and auditory organ, the nature has equipped snakes with well developed organ of smell and touch in the form of accessory sensory system (ACS) known as vemero nasal organ.[1-4]

The lineage of VNOs in tetrapods and many groups of mammals including rodents has very prominent evolutionary organization but in squamates the VN

*Corresponding author :
Phone : 08789865993
E-mail : ranjanamdp@gmail.com

chemosensory system is anatomically and functionally quite distinct from the main olfactory system. Several researches have been done on the VNO of snakes[5-7] but very little has been done for the worm snakes.

MATERIALS & METHODS

6 adults (3 males & 3 females) were kept in laboratory experimental jar after their collection from local habitats of Madhepura and Saharsa. The specimens were killed with 5% chloroform and their heads were decapitated sex wise and kept separately in the smaller jars, washed with normal saline and fixed in 4% formalin for at least one week. The preserved and fixed decapitated head were later decalcified using formic acid sodium citrate buffer solution for 3-4 days as per the protocol of Smith *et al.* (1997)[8]. The tissues were dehydrated in graded series of ethanol, cleared in xyline and paraffin wax blocks were made from the decapitated head tissues. Microtomic sections were procured in transverse plain at 5-7 µm thickness.

The sections were mounted on glass slides and stained with Haematoxylin-Eosin or Alcian blue-Periodic Acid Schif (PAS). Then stained sections were scrutinized in details by the aid of a light microscope. Ocular graded graticule was used to measure specific histological parameters in the organs including the lumen's width and distance between the two capsules enclosed the organs.

RESULTS

As per the experimental work plan, the microtomic section of VNO complex have been procured from both male and female subject in order to record the sex specific variations in its structural organization. In the Typhlops bramina, the VNO consists of a paired bilateral structure located at the baso-medial side of the nasal cavity, with direct relationship with the vomer bone and palatine process of maxillary bone. In general, each organ appeared as a hollow spindle with an average length of 550µm male and 510 µm in female. Each organ begins at rostral one-third of the nasal cavity. Anteriorly, the organ is narrow but gradually widens. This increase in diameter continues to two-third of the organ while in its caudal one-third it undergoes a reverse process so that it again becomes tapered. Along with this, a marked variation in the dimension as obtained from the micrometric measurement has been found in both male and female individuals. The sex specific measurement data showing the micro level variation has been furnished in tables 1 & 2 and displayed in graphs A & B

In transversal sections the organ is observed as a dome-shaped structure which is enclosed by an osseous capsule composed of medial palatine and lateral vomer bones. The capsule is incomplete as in some regions the bone is replaced by connective or cartilage tissues. The sites and locations of these replacing tissues varies along the organ' length but mostly is seen in dorsolateral and ventromedial regions. The lumen of organ has become crescent-shaped as the result of Mushroom Body (MB) which is a bulge-like evagination protrudes from ventral or ventrolateral side into the lumen.

Along the length of the organ the lumen is lined with three types of epithelia. The dorsal or concave side of the VNO lumen is lined with a sensory epithelium whose main role is to detect chemical and elicit impulses to be conducted into the central nervous system. This is a pseudo stratified epithelium in which three types of cells can be recognized. Just beside the lumen there is a layer of supportive cells which have elongated nuclei. At the top of supportive cells a several layers of sensory cells (on average 10 layers) with rounded nuclei are observed. The receptor cells are indeed bipolar neurons whose dendrites terminate as microvilli at the luminal surface.

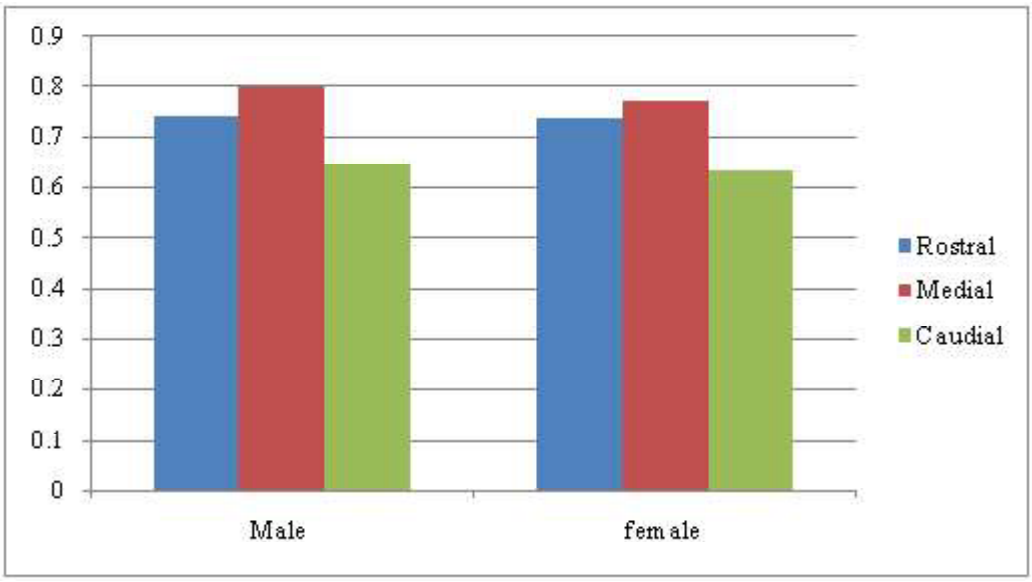
Table 1- Micrometric measurements of the nasal cavity and VNO complex dimensions in µm with VLD/ NCD ratio in male *T.bramina*

Intercapsular space of nasal cavity	VNO dimensions	VNO lumen Diameter (VLD)	Nasal Cavity Diameter (NCD)	VLD/NCD ratio
18	Rostral	162	218.2	0.742
22	Medial	215	269.0	0.799
20	Caudial	187	289.0	0.647

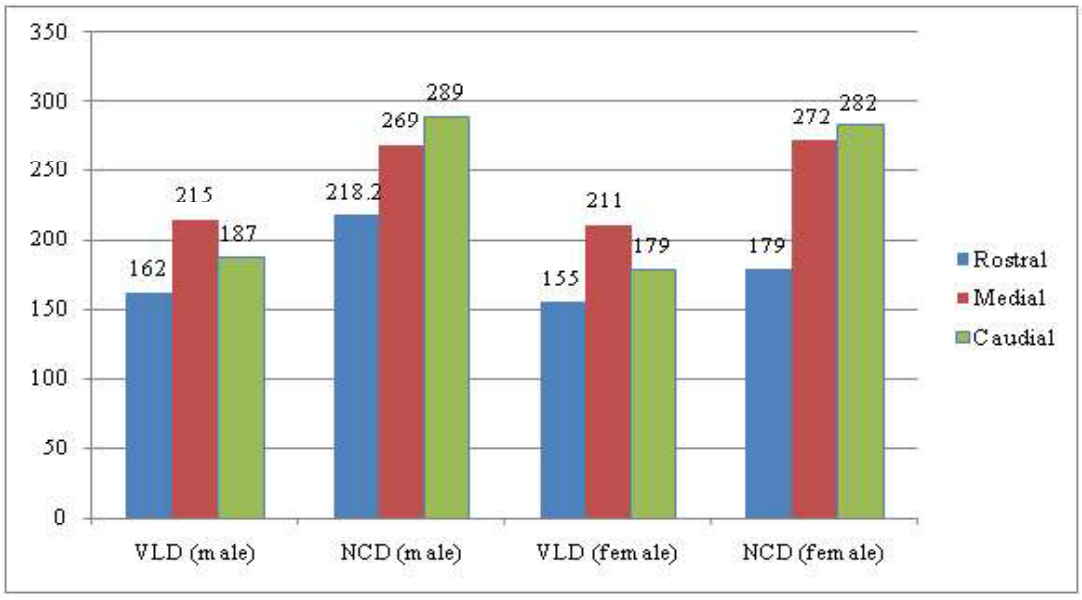
Table 2- Micrometric measurements of the nasal cavity and VNO complex dimensions in µm with VLD/ NCD ratio in female *T.bramina*

Intercapsular space of nasal cavity	VNO dimensions	VNO lumen Diameter (VLD)	Nasal Cavity Diameter (NCD)	VLD/NCD ratio
18	Rostral	155	210.0	0.738
22	Medial	211	272.0	0.775
20	Caudial	179	282.0	0.634

Graph A.- Sex specific variation in the three-dimentional VLD & NCD measurements of Jacobso's organ complex (VNO) in *T.bramina*



Graph B.- Sex specific variation in the three-dimentional VLD/NCD ratio of Jacobso's organ complex (VNO) in *T.bramina*



ACKNOWLEDGEMENT

The author is extremely thankful to her guide B.P.Yadav "Bipra" Department of Zoology, L.N.M.S. College, Birpur Supaul, Bihar and Prof. Arun Kumar, Head, Department of Zoology, B.N.M.University, Madhepura, Bihar for their support and guidance.

REFERENCE

1. **Grzimek, B., M. Hutchins, N. Schlager and D. Olendorf, 2003.** Grzimek's Animal Life Encyclopedia, Volume 7: Reptiles. 2nd Edn., Gale Group, USA., ISBN-13: 9780787657833, Pages: 593.
2. **Gove, D., 1979.** A comparative study of snake and lizard tongue-flicking, with an evolutionary hypothesis. *Zeitschrift fur Tierpsychologie,* **51:**58-76
3. **Halpern, M., 1992.** Nasal Chemical Senses in Reptiles: Structure and Function. In: Biology of the Reptilia, Gans, C. and D. Crews (Eds.). Vol. 18. University of Chicago Press, Chicago, IL., USA., pp:423-523.
4. **Halpern, M. and A. Martinez-Marcos, 2003.** Structure and function of the vomeronasal system: An update. *Prog. Neurobiol.,* **70:** 245-318.
5. **Miller, L.R. and W.H.N. Guzke, 1999.** The role of the vomeronasal organ of crotalines (Reptilia: Serpentes: Viperidae) in predator detection. *Anim. Behav.,* **58:** 53-57.
6. **Takami, S. and K. Hirosawa, 1990.** Electron microscopic observations on the vomeronasal sensory epithelium of a crotaline snake, *Trimeresurus flavoviridis. J. Morphol.,* **205:** 45-61.
7. **Young, B.A., 1993.** Evaluating hypotheses for the transfer of stimulus particles to Jacobson's organ in snakes. *Brain Behav. Evol.,* **41:** 203-209.
8. **Smith, T.D., M.I. Sigel, M.P. Mooney, A.R. Burdi, A.M. Burrows and J.S. Todhunter, 1997.** Prenatal growth of the human vomeronasal organ. *Anatomical Rec.*, **248:** 447-455.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 251-254 **16(IS).**
An International Biannual Refereed Journal of Life Sciences ***InterdiciplinaryScience***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Phytochemical studies of ethnobotanically significant plant, *Evolvulus alsinoides* Linn, from the tribal region of Deoghar district of Santal Pargnas, Jharkhand

Sanjeev Kumar Choudhary*
Department of Chemistry, T.N.B College, T.M.U Bhagalpur, Bihar, India

Received : 20th December, 2019 ; Revised : 24th February, 2020

Abstract: In the present investigation phytochemical studies of *Evolvulus alsinoides* has been done. It is an important medicinal herb which is used for the treatment of diabetes in the tribal regions of Santal Parganas. Besides diabetes, the plants are very much used by the tribal people for the treatment of Syphillis, leucorrea, Bronchitis, Asthama and liver ailments. Perusal of the data of table 1. It is clear that out of 24 amino acids only seven could be detected from the [plants under study. They were lysine, L-valine, L-argenine, Cystein, Tyrosine and DL-tryptophan. In the above list it is clear that at least 4 essential amino acids are present, they were L-argentine, lysine, L-valine and DL-tryptophan. 10 essential amino acids are necessary for proper growth and their metabolic functions in human beings among which 4 are present in this plant and rest three are non-essential amino acids. Non- essential amino acids have curative effect on a diseased person. Perhaps due to presence of these non-essential amino acids this plant is ethnobotanically important.

Key words: phytochemical studies, *Evolvulus alsinoides,* Liysine, L-valine, L-argenine, Cystein, Tyrosine & DL-tryptophan

INTRODUCTION

Santal Parganas forests are rich in indigenous plants and they are still intact due to the dependence of tribal people on herbal medicines for the treatment of various diseases. For the present investigation *Evolvulus alsinoides* Linn was selected for study. The plants were collected from Inarabaran forests of Deoghar district in the month of October-November. *Evolvulus alsinoides* is a member of convolvulaceae and has a very wide range of ecological distributions. The plant is rampantly used by the tribal people of Santal Parganas for treatment of diabetes. Besides diabetes, the plants are very much used for the treatment of syphilis, leucorrhea, bronchitis, asthama and liver ailments by the local vaidyas.

*Corresponding author :
Phone : 9852020454
E-mail : sanjeevkumarchoudhary1972@gmail.com

MATERIALAND METHODS

The plants were collected from different places of Inarabaran forest in the month of October- November. Simple chromatographic techniques were employed for detection of free and bound amino acids present in the plants following the methodology suggested by Herborne (1973)[1].

$$Rf = \frac{\text{Distance travelled by solute}}{\text{Distance travelled by solvent}}$$

The RF values were worked out on the basis of following formula:

RESULT AND DISCUSSION

Evolvulus alsinoides is a plant of the family convolvulaceae. It is a branched prostrate on the ground, 4-15" long with thinly or densely silky leaves. Leaves are

broadly elliptical -0.25 to 0.5". Flowers are sub rotate upto 2" long. One tea spoon dried flowers and leaves are given to diabetic patients in the morning for lowering blood sugar level. One tribal patient of the Inarabaran locality told us that in one week only, this herbal medicine lowered my sugar level from 390-152 pp. The data of amino acids analysis of E. alsinoides is given in table 1, perusal of the data of table 1.[2, 3] It is clear out of 24 amino acids only seven could be detected from the plants under study. They were lysine, L. argenine, Cysteion, L- valine, Tryosine and DL- tryptophan among which cystein, L- valine and DL-tryptophan could be detected in higher concentration i.e. (3+) while lysine, L- argentine, glysine and DL tryptophan were present at (+) or (2+) level on visual observation.[4] Besides this, L- argentine, cystein glycine and DL-tryptophan were present in both the forms i.e. free form

Table 1: showing data of amino acids present in the leaf extract of *E. alsinoides* collected from Inarabaran of Deoghar district.

Amino acids	Rf value	Concentration on visual observation							
		Free form				Bound form			
		T_1	T_2	T_3	Mean	T_1	T_2	T_3	Mean
Lysine	0.04	+	+	+	+	-	-	-	-
L. histidine	0.06	-	-	-	-	-	-	-	-
Aspartic acid	0.09	-	-	-	-	-	-	-	-
L-argenine	0.12	+	+	-	+	+	+	-	+
Cystine	0.16	3^+	2^+	3^+	3^+	+	-	-	-
DL- Aspargine	0.18	-	-	-	-	-	-	-	-
Proline	0.21	-	-	-	-	-	-	-	-
L –glutamine	0.23	-	-	-	-	-	-	-	-
Scrine	0.24	-	-	-	-	-	-	-	-
Hydroxyproline	0.26	-	-	-	-	-	-	-	-
DL-Threonine	0.28	-	-	-	-	-	-	-	-
Glysine	0.31	2^+	2^+	2+	2+	+	+		+
L – glutamic acid	0.33	-	-	-	-	-	-		-
Ornothine	0.34	-	-	-	-	-	-		-
Dihydroxy – Phenylamine	0.37	-	-	-	-	-	-	-	-
DL- alanine	0.45	-	-	-	-	-	-	-	-
DL- methionine	0.49	-	-	-	-	-	-	-	-
Amino butyric acid	0.56	-	-	-	-	-	-	-	-
L – Valine	0.61	2^+	3^+	3+	3+	-	-	-	-
Tyrosine	0.63	2^+	2^+	+	2^+	-	-	-	-
Phenylalanine	0.67	-	-	-	-	-	-	-	-
DL-tryptophan	0.71	2^+	3^+	3^+	3+	+	2^+	2^+	2^+
Isoleucine	0.76	-	-	-	-	-	-	-	-
L -leucine	0.80	-	-	-	-	-	-	-	-

and bound form while the rest three were restricted to free form only.[5]

From the perusal of the table 1, it is clear that in this plant out of 10 essential amino acids 4 types were present. They were argenine, lysine, valine and tryptophan. Ten essential amino acids most important for human growth and proper metabolic functions.[6] Perhaps, due to the presence of these four essential acids in this plant, the plant is useful as medicine in various diseases.

Rest three amino acids are non-essential amino acids present in this plant, these were cyctein, tyrosine and glysine. Non-essential amino acids specially cystein, glycine glutamic acid, DL methionine and ornithine have profound curative efforts on disease persons.[7] Perhaps, due to the presence of four essential and three non-essential amino acids in Evolvulus alsinoides, this plant is ethnobotanically important.[8,9] Cystein is a non-essential sulphur containing amino acid and its presence in the diet decreases the methionine percentage in the body.[10]

ACKNOWLEDGEMENT

The author is very thankful and gratitude to Professor and Head of Department of Chemistry, T.M.B College,

T.M.U Bhagalpur for their immensed co-operation & motivation to complete this research paper.

REFERENCE

1. **Herborne, J. B. 1973.** Phytochemical methods, Singapore, Topan printing co. pvt.,Ltd.
2. **Boge, G. 1960.** Amino acid composition herring and herring meal. Destruction of amino acid during processing. *Jour: Sci. Agril.* **11:**362-365.
3. **Martin, D.D Mayer, P.A and Rodwell, V. W 1981.** Hapers view of biochemistry Pp.261-273
4. **Anwar, S.Y. and Reddy , G. M. 1974.** studies on free amino acid pattern in relation to polyploidy in *O. sativa.* L *Jour. Cytol. Genel.* **14:**95-98
5. **Jha, B.N. 1988.** Chemocytogenetical studies in cucurbitaceae, Ph.D. Thesis, Bhagalpur University, Bhagalpur.
6. **Malhotra, A. 1985.** Assessment of amino acids in some ethnobtanical plants through chromatography. *Ind. Bot.Soc.* **59:**69-74
7. **Alstead, S. and Girdwood, R.H. 1994.** Text book of medical treatment (English language book society) living stone building great Britain.
8. **Patridge, S. M 1991.** Ethnobotanical and biochemical studies of *Achyranthes aspera*. *Nature*. **207:**271-275
9. **Sinha, J.P. 1988.** Ethnobotanical exploration of some cytologically significant plants. Biol. of India.
10. **Vedas and Brahman-** Rigveda, 21.3.35,4.83.9:Atharb veda, 10.8.31; Yayur veda, 2.1.35; Brahman.2.2.45.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 255-260
An International Biannual Refereed Journal of Life Sciences

17(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of copper sulphate on protein content in mature female *Hydrophilous olivaceous*

Sudhanshu Kumar *
Department of Zoology, Rajendra College, Chapra, J.P. University, Chapra, Bihar, India

Received : 20th December, 2019 ; Revised : 26th February, 2020

Abstract : Proteins are complex organic macromolecules that contain carbon, hydrogen, oxygen, nitrogen and usually sulphur and are composed of one or more chains of amino acids. These are fundamental components of all living cells and include many substances, such as enzymes, hormones and antibodies that are necessary for proper functioning of an organism. Various aspects of protein metabolism in insect development, such as the pattern of free amino acid pool, the intermediary pathway of individual amino acids and their derivatives, qualitative and quantitative changes in haemolymph and whole body protein synthesis and activity of specific enzymes have been studied by many insect's biochemists.[1-7] It is well known fact that the copper sulphate is an important hazardous water pollutant. The effect of which can be felt at all bio-chemical parameters of an aquatic life. The bio-assay experiment in the present investigation, it was realized that Copper Sulphate badly affected the protein metabolism at different levels in mature females of *H. olivaceous*. Copper sulphate act as major toxicants which affect all body actions directly or indirectly of all aquatic organisms. Almost all heavy metals are toxic at higher concentration and some are severely poisonous even at low concentration. In present investigation on mature female *Hydrophilous olivaceous*, it was observed that copper sulphate had an initial protein lowering effect and latter elevating trend on the haemolymph, fatbody and ovary of female insect. Overall initial decline of the protein content in haemolymph, fatbody and ovary were observed in insect exposed to sub-lethal concentration of copper sulphate.

Key words: Copper sulphate, protein, *Hydrophilous olivaceous*, haemolymph, fatbody, ovary.

INTRODUCTION

Proteins are complex organic compounds that contain carbon, hydrogen, oxygen, nitrogen and usually sulphur and are composed of one or more chains of amino acids. They include many substances, such as enzymes, hormones and antibodies that are necessary for proper functioning of an organism.

Aquatic environment is said to be the ultimate sink for the toxic compounds. Heavy metals from motor vehicles via urban storm, water runoff reaches to water reservoirs.

A number of works on the effect of pesticides and other chemicals indicate that although in low concentration, they may not lead to the death of aquatic organisms but can cause alterations in their metabolic processes and chemical constituents. Roan and Hopkins 1961 & O'Brien 1967 gave review of insecticides action on metabolism.

Effects of heavy metals on protein and free amino acids of certain aquatic insects on proteins and free amino acids were also studied by Chaudhary (1996)[3].

*Corresponding author :
Phone :
E-mail : sudhanshusmiles@gmail.com

Copper is a highly toxic metal posing great threat to aquatic organisms. The impact of copper and its salt can be felt at all trophic levels. The toxicity of copper and its compound to different aquatic organisms have been determined. It has been reported that copper sulphate was highly toxic and that there was tissue accumulation in the stone loach *Noemachelius barbatulus* in hard water, Solbe and Cooper, 1976. The acute lethality of copper over wide range of combinations of hardness and pH to rainbow trout has been determined,Howrath and Spraque, 1978. Thus it is expected that the copper sulphate causes alterations in quantitative and qualitative levels of proteins involved in metabolism. Keeping this in view, in present investigation mature *Hydrophilous olivaceous* and *Laccotrephes maculatus* both female and male were exposed to sub-lethal doses of Copper Sulphate and quantitative estimation of proteins were done at 24 hrs, 48 hrs, 72 hrs and 168 hrs of treatment in each case separately.

The present investigation deals with the biochemical studies of sub-lethal doses of copper sulphate on the insect *H. olivaceous* and *L. maculatus* and the main emphasis has been given on the quantitative change of carbohydrates, proteins, alkaline phosphatase and lactate dehydrogenase content in the haemolymph, fatbody, ovary and testes of the mature female and male insects.

MATERIALS AND METHODS

Quantitative Estimation of Protein

Quantitative estimation of haemolymph proteins, fatbody proteins, testes proteins of treated and control insects in mature stages were done by Lowry's method.

Principle of Lowry's Method

Proteins react with Folinciocalteaue reagent to give a coloured complex. The colour so formed is due to the reaction of Alkaline Copper with the proteins and then reduction of the Phosphomolybdate by Tryosine and Tryptophans.

Reagents

(a) Alkaline Sodium Carbonate Solution (2% of Na_2CO_3 in N / 10 NaOH).

(b) Copper Sulphate Sodium potassium tartarate (5% $CuSO_4$ in 1% Na, K tartarate) was freshly prepared by mixing stock solution.

(c) Alkaline Solution- It was prepared on day or working by mixing 50 ml of Alkaline Sodium Carbonate solution and 1 ml of Copper Sulphate Sodium Potassium tartarate solution.

(d) Folinciocalteau Reagent - The commercial reagent was diluted the equal volume of water on the day of use. This is a solution of Sodium Tongstate and Sodium Molybdate in Phosphoric Acid and HCL.

(e) Standard Protein Solution - Albumin solution 0.2 mg / ml [Bovine serum albumin Loba (Ltd.)]

Technique

(a) Technique of Haemolymph Protein: Insects head were pricked with needle and were kept inverted in centrifuge tubes, one insect in each centrifuge tube and were centrifuged at 1500 rpm to collect haemolymph. Then 1 ml of haemolymph was taken in separate centrifuge tube and added 5 ml 10 % TCA. It was centrifuged at 3000 rpm for supernatant, from supernatant test for carbohydrate was carried out.

(b) Techniques of Fatbody and ovary Protein :Insects were dissected, fatbody and ovaries or testes were taken out and crushed in 2 ml of distilled water with the help of mortar and pestle separately. The crushed material was collected in a centrifuge tube and 5 ml of 10 % TCA was added to obtain the precipitate of protein. It was centrifuged at 3000 rpm. The supernatant was kept separately for performing quantitative assay of proteins.

Precipitated protein was dissolved in 5 ml N / 10 NaOH. The protein samples obtained by the above procedure were then subjected to quantitative assay colorimetrically by employing Lowry's method.

Procedure

5 ml of alkaline solution was added to 0.1 ml of test solution. It was mixed thoroughly and .5 ml of diluted folinciocalteau reagent was added to it. The solution was kept at room temperature for 20 minutes, N / 10 NaOH were used as a blank and the absorption was read at 660 nm in colorimeter.

Calculations

The protein content was calculated as follows:

$$\frac{\text{OD of Unknown Solution}}{\text{OD of Known Solution}} \times \text{Concentration of Standard Solution} \times \text{Dilution of Unknown Solution}$$

Where,
OD = Optical Density.

RESULTS AND DISCUSSION

Observation of Protein content in maure female H. olivaceous. The total protein content in the haemolymph, of mature female *H. olivaceous* were 35 ± 7 SD mg / 100 ml and 20 ± 3 SD mg / 100 ml respectively in control insect.

Protein Content in Mature Female *H. olivaceous*

Haemolymph Protein Content

The total protein content in the haemolymph of mature female *H. olivaceous* were 35 ± 7 SD mg / 100 ml in control insect.

On treatment with copper sulphate the haemolymph protein content of mature female declined significantly in comparison to the control insects as 20 ± 4 ($P < 0.01$) SD mg / ml at 24 hrs, 15 ± 3 ($P < 0.001$) SD mg / ml at 48 hrs, 23 ± 4.5 ($P < 0.01$) SD mg / ml at 72 hrs of treatment. A slight but significant ($P > 0.05$) rise of 30 ± 6 SD mg / ml was observed at 168 hrs of treatment, (Table 1, Graph 1).

Fatbody Protein Content

The mature female *H. olivaceous* had a fatbody protein content of 200 ± 9 SD mg / gram in control insects.

On treatment with copper sulphate the fatbody protein content in mature female significantly declined up to 48 hrs of treatment. These were 150 ± 8 ($P < 0.001$) SD mg / gram at 24 hrs and 132 ± 7 ($P < 0.001$) SD mg / gram at 48 hrs of treatment. A highly significant and unusual rise of 140 ± 7.5 ($P < 0.001$) SD mg / gram in protein content of fatbody was noted at 72 hrs of treatment. At 168 hrs of treatment the fatbody protein content showed rising trend as 160 ± 8 ($P < 0.001$) SD mg / gram but still remained significantly lower than that of control insect, (Table 1, Graph 1).

Ovarian Protein Content

The ovary of mature female *H. olivaceous* had protein content of 185 ± 7 SD mg /gram of ovary in control insects.

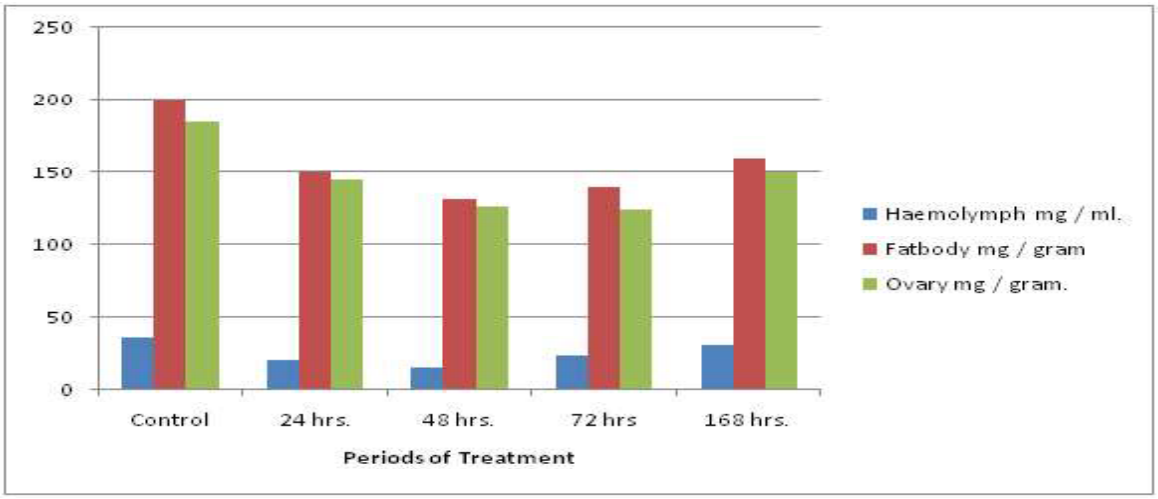
In mature female, similar decline in the ovarian protein content was noted up to 72 hrs of treatment like fatbody, as 145 ± 6 SD mg / gram at 24 hrs, 125 ± 5 SD mg / gram at 48 hrs, 123 ± 4.5 SD mg / gram at 72 hrs of treatment respectively. A significant rise of 150 ± 7 SD mg / gram was noted at the 168 hrs of treatment. All the differences were highly significant ($P < 0.001$) as compared with that of control insects, (Table 1, Graph 1).

Table 1: Protein content in haemolymph, fatbody and ovary of mature female *H. olivaceous* during control and after different periods of treatment with Copper sulphate

Periods of Treatment	Haemolymph mg/ml	Fatbody mg/gram	Ovary mg/gram
Control	35 ± 7	200 ± 9	185 ± 7
24 hrs	20 ± 4**	150 ± 8***	145 ± 6***
48 hrs	15 ± 3***	132 ± 7***	125 ± 5***
72 hrs	23 ± 4.5**	140 ± 7.5***	123 ± 4.5***
168 hrs	30 ± 6	160 ± 8***	150 ± 7***

Mean ± SD
SD = Standard Deviation
* = $P < 0.05$ (Significant)
** = $P < 0.01$ (Significant)
*** = $P < 0.001$ (Significant)

Graph – 1



On observing effect of copper sulphate on protein content of mature female *H. olivaceous* it was found that the total protein content in the haemolymph, fatbody and ovary of mature female *H. olivaceous* were 35 ± 7 SD mg/100 ml, 200 ± 9 SD mg/gram and 185 ± 7 SD mg/gram respectively in control insect.

In mature *H. olivaceous* copper sulphate had a protein lowering effect in haemolymph up to 48 hrs but in fatbody there was protein lowering effect up to 72 hrs of treatment. In ovary, the lowering tendency was maintained up to 72 hrs of treatment. A gradual rise in protein content in female haemolymph was observed from 72 hrs in all the tissues.

The overall drop in protein content in all cases female *H.olivaceous* in early stages might be due to the interference of this chemical in protein synthesizing machinery or due to rapid catabolic breakdown of protein into glucose to produce energy during chemical stress. The lesser food intake and rapid utilization of protein under stress may also be a factor leading to the loss of protein in haemolymph, fatbody, ovary.

But after 72 hrs of treatment gradual rise in protein content of haemolymph, fatbody, ovary was observed. It showed active entry of protein from ovary and fatbody into haemolymph. It suggests stability in the protein synthesis of the insects after facing stress of sub-lethal dose of Copper Sulphate.

After treatment with Copper Sulphate a sharp decline in protein content at 24 hrs was noted in all the experimental tissues in female. Then a gradual rise was noted up to 7 days of the treatment except fatbody where declining trend was maintained up to 72 hrs

The overall drop in protein content in early stages of treatment confirmed the inhibitory effects of copper sulphate in protein synthesis machinery. The rise in protein content after 72 hrs of treatment confirmed the metabolic stability of the insects due to detoxification. A similar result was observed by Islam and Roy (1983)[5] in Chrysocoris stolli exposed to various concentrations of Cadmium Chloride.

Copper Sulphate thus affected the protein synthesizing machinery of the mature female and male *H. olivaceous* and *L. maculatus*. Fatbody being the main site of protein synthesis had been affected in the initial stages of the treatment. This accordingly showed effects in the decline in haemolymph, testis and ovarian protein contents. Increase in protein content in later stages suggests some stability in protein synthesizing of insects after facing initial hazards.

The study of physiological changes due to the responses of individual insects or other aquatic animals in the presence of various types of pollutants is of immense help to aquatic toxicologists. It is obvious that the changes in an aquatic organism that occurs as a result of pollution also have real practical use. An organism makes a marvelous integration of the effect of the any type of changes in chemical and physical factors in aquatic environment. The usual approach of performing chemical analysis and physical measurements of natural water may or may not detect changes that are harmful to the aquatic organisms. In real sense aquatic organisms themselves are the best judge of that different physiological parameters are generally used as tools for the study of aquatic pollution by chemical and other means. The physiological parameters are Hematology, measurement of haematochrit, hemoglobin concentration, erythrocytes and leucocytes count being the main aspects; and blood chemistry, which ranges from the determination of blood clotting time to the estimation of hormones and tissues chemistry.

Effects of heavy metal on the blood proteins of fishes have been reported by various workers like Baskaran (1991)[8], Choudhary (1998)[9], Winteringham (1959)[10] Ehteshamuddin (1985)[11].

REFERENCES

1. **Chen, P. S. 1966.** Amino acid and protein metabolism in insect development. *Adv. Insect. Physiol.* **3:**53-132.
2. **Rao, J. P. and Tiwari, L. D. 1980.** Haemolymph proteon pattern of drones and workers of Indian honey bee *Apis indica*. 2nd International conference on apiculture in Tropical climates. Abstract. pp. 47
3. **Choudhary, N. 1996.** Effect of Heavy metals on the metabolism of aquatic insects. Ph. D. Thesis, submitted in B. R. A. Bihar University, Muzaffarpur.
4. **Islam, A. and Roy, S. 1983.** Effect of cadmium chloride on the quantitative variation of carbohydrate, protein, amino acids and cholesterol in Chrysocoris stoli wolf *Curr. Sci.* **52 :** 25.
5. **Sharma, S. and Ehteshamuddin, S. 1988.** Protein concentration in immature and adult *Spharodema*

rusticum (Belostomidae: Hemiptera). *Environ. And Ecol.* **6 (30):**765-767.

6. **James, A and Soni, V. C. 1991.** Changes in tissue proteins due to administration of mercuric chloride and two chelators in mice. *J. Comp. Physiol. Eco.* **16(1):** 32 - 37.

7. **Kumari, L. 1993.** Effefct of Chemicals on the growth of an aquatic living system; Ph. D. Thesis. Submitted in B. R. A. Bihar Univ. Muzaffarpur.

8. **Baskaran, P., Gopalakrishnasamy and Sathigahama, 1991.** Effects of metac (Organo phosphate pesticides) on the physiology and biochemistry of *Oreochromis mossambicus.* pp. 209-218. B. Gopal and V. Asthama editors. *Aquatic science in India. Indian Assoc. for Limnol. Oceanography.* India.

9. **Choudhary, N., Kumari, C., Khan, S. and Ehteshamuddin, S. 1998.** Effect of copper sulphate on protein and free amino acid concentrations of aquatic beetle *Hydrophilous olivaceous* (Hydrophilidae: Coleoptera). *Environment and Ecology.* **16:**573 - 578.

10. **Ehteshamuddin, S. 1985.** Embryonic development of *C. stolli* Wolf. (Hemptera: Pentatomidae) and role of neurosecretion on the maturation of gonads (Ovary). Ph.D. Thesis Submitted in B. R. A. Bihar University, Muzaffarpur.

11. **Winteringham, F. P. W. 1959.** Comparative Aspects of insect Biochemistry with particular reference to insecticidal action. In "Biochemistry of insects" Edited by L. Leven book published by symposium division New York.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 261-264
An International Biannual Refereed Journal of Life Sciences

18(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Studies on the hepatotoxicity due to prolonged use of colchicine in albino rat

Suman Swaraj*
Department of Zoology, B.N.Mandal University, Madhepura, Bihar, India

Received : 6th December, 2019 ; Revised :15th February, 2020

Abstract : Derived from meadow-saffron plant *Colchicum autumnale,* colchicine is an alkaloid of high potential also called mitotic poison. It is a powerful inhibitor of somatic cell multiplication in mitotically active tissues and organs like liver thereby causing significant hepatotoxicity. In the present article effect of toxic doses of colchicine injected intraperitonially in the albino rats for longer period on the liver functions has been biochemically assayed through standard LFT (liver function tests) package which includes AST(Aspertate aminotransferase test), ALT(Alanine aminotransferase), and ALP(Alkaline phosphatase). Prolonged injection of colchicine resulted in increased concentration of these liver enzymes (AST, ALT& ALP) in the blood of the treated rats. These changes have also led to histopathological abnormalities in the liver such as hepatocyte necrosis, inflammation of central hepatic vein & development of prominent cytoplasmic vacuoles.

Key words: Hepatotoxicity, LFT (AST, ALT, ALP) Colchicine, plant alkaloid, prolonged use

INTRODUCTION

Alkaloids are group of specialized nitrogenous substance having wide range of pharmacological activities. They are usually found in plant world but may be produced by bacteria, fungi, and insects. The colchicine is an important plant alkaloid derived from derived from the meadow saffron plant (*Colchicum autumnale*) and therapeutically used extensively in the treatment of gout, arthritis, Mediterranean fever[1], in the treatment of primary biliary cirrhosis & Behcet disease[2]. Colchicine interferes with the formation of microtubule by binding to microtubular synthesis complex and causing depolymerization of microtubules in diverse functions, including cell movement, vesicle transport and chromosome segregation during mitosis. It has also been reported that colchicine can be either cytotoxic or protective against cytotoxicity. Colchicine has been reported to protect against a variety of hepatotoxic stresses & improves the survival of the rats in a clinical test for alcoholic cirrrhosis.[3] As it has a narrow range of therapeutic index, the overdose of colchicine may lead to high rate of mortality.[4] Studies on the effect of mal absorption on the functions of the small intestine would be greatly facilitated if such states could be routinely induced in the laboratory. A number of efforts to achieve such states in experimental animals have met with little success[5,6] but the use of antimitotic drugs, such as aminopterin and colchicine, cause a pathological villous architecture similar to that found in many types of malabsorption.[6-8] Studies on the effect of colchicine on intestinal absorption have, however, only been carried out in conscious animals[8,9] where absorption depends not only on the activity of the mucosal cells of the intestinal epithelium but also on factors such as gastric

*Corresponding author :
Phone : 9955875200
E-mail : drsumanswaraj@gmail.com

emptying, intestinal motility, blood flow, etc. In order to avoid the influence of a number of these factors the effect of colchicine on intestinal function has been investigated using a recently developed technique in the anaesthetized rat in vivo and on sacs of isolated, everted intestine incubated in vitro.

MATERIALS & METHODS

Thirty adult albino healthy rats weighing (200–250 g) were taken for the study in the laboratory & kept in standard laboratory conditions having free access to water & food. The entire population was dived into three equal groups of 10 individuals and they were marked as -first group as control receiving only distilled water orally; second group, treated with 2 mg of colchicine per kg body weight/day and third group, treated with 3 mg of colchicine per kg body weight/day dissolved in distilled water. The treatment of colchicine concentration in rats of second & third group was done through intraperitoneal injection.

Blood samples from individuals of treated group were collected 24 hours after administration of last dose of colchicine which were centrifuged at 3000 rpm for 10 min to separate the serum.

The separated serum was subjected to biochemical analysis for following liver function tests:

1. Aspartate Aminotransferase Test (AST)

AST was performed by mixing the 5ml serum to 10ml buffered solution of L- aspartic acid and 2-ketoglutarate and then incubated for one hour at 37° C. After incubation, 1 mm of DNPH and 0.4m of NaOH was added.

2. Alanine Aminotransferase Test (ALT)

ALT was performed by mixing 5ml serum to 10 buffered solution of DL- alanine and 2- ketoglutarate, and then incubated for thirty minutes at 37° C. After incubation, 1 mm of DNPH and 0.4m of NaOH was added.

3. Alkaline Phosphatase Test (ALP)

ALP was performed by using p- nitrophenol phosphate as substrate, in alkaline buffer with fresh unhemolysed serum for 45 min at 12°C.

Table 1- Showing results of 3 enzyme biochemical assay test under LFT of albino rat exposed to different concentration of colchicine in group 2 & 3, group 1 being control.

L.F.T Group	ALT (IU/L)	AST (IU/L)	ALP (IU/L)
Group 1	10.16 ± 3.1	30.7±2.8	85.8±5.3
Group 2	22.1±2.1	34.13±1.	89.1±2.3
Group 3	45.9±12.1	51.3±8.2	144.4±9.1

RESULT & DISCUSSION

Liver Examination

Liver of control, untreated rat revealed no morphological changes. But in treated rats belonging to group 2 & 3 with 2 & 3 mg of colchicine per kg body weight/day dissolved in distilled water respectively, marked liver lobe swelling and distention have been noticed. These morphological abnormalities in the liver of rat may be collectively called as fatty liver which is related with internal biochemical changes of the blood existing in the hepatic lumen and the vein nourishing organ.

LFTs-

Significant elevation in the ALT, AST & ALP enzymes concentration have been found in members of group 2 & 3 treatment whose blood biochemical assay were done as per the standard liver function test (Table 1). The ALP concentration in group 3 having 3mg cholchicine exposure dose displayed very sharp rise compared to the control which is metabolically detrimental for related creatinine biosynthesis at kidney level.

DISCUSSION

Colchicine is a safe alkaloid if it is used for short period and under recommended dose. However, its use in the treatment of chronic diseases such as gout and familial Mediterranean fever, the higher recommended dose may be taken for prolonged period of time under clinical supervision.

The present study investigates its toxic effects on liver being metabolically very efficient organ which is also involved in the breakdown and excretion of colchicine when taken in the system. The biochemical assay of the enzymatic concentration of 3 important enzymes synthesized in the liver has been assayed by LFT blood examination of the rats.

The present study showed a statistical significant increase of liver enzymes such as ALT, AST and ALP in third group which received 3mg/kg/day of colchicine in comparison with the control group and the second group which received 2mg/kg/day. Sherlock referred that the use of liver function tests is associated with high specificity

ACKNOWLEDGEMENT

The author is extremely thankful to her guide Dr. R.B.Jha & Prof. Arun Kumar, Head, Department of Zoology, B.N Mandal University Madhepura for their valuable guidance and help in completion of this research paper.

REFERENCE

1. **Zemer D, Revach M, Pras M, 1974.** A controlled trial of colchicine in preventing attacks of familial mediterranean fever. *N Engl J Med* . **291 (18):** 932-4.
2. **Kaplan, MM, Alling, DW, Zimmerman, HJ, Wolfe, HJ, Sepersky, RA, Hirsch, GS, Elta, GH, Glick, KA, & Eagen, KA.1986.** A prospective trial of colchicine for primary biliary cirrhosis. *N Engl J Med*. **315:** 1448-1454.
3. **Maxwell MJ, Muth P and Priiy PE. 2002.** Accidental colchicine overdose. A case report and literature review. *Emerg., Med., J.* **19(3):**265-7.
4. **Lainé M, Mourissoux G and Camou F. 2012.** Early Onset Cardiogenic Shock in Acute Colchicine Overdose. *J., Clinic., Toxicol.*, **2:**2-5.
5 **Althausen, T. L. & Grodsky, G. M. 1961.** Effect of wheat proteins on intestinal absorption in rats. *Gastroenterology*. **40:**665.
6. **Williams, J. N. Jr., & Laster, L. 1963.** Investigation of the use of the albino rat for studying gliadin-induced steatorrhea. *Metabolism*, **12:** 467-471.
7. **Baserga, A., & Morsiani, M. 1958.** Non regenerative enteropathy. *Rass. giul. Med.*, **35:**208-215.
8. **Clark, P. A., & Harland W. A. 1963.** Experimental malabsorption with jejunal atrophy induced by colchicine. *Brit. J. exp. Path.*, **44:**520-523.
9. **Luketic, G. C., and Shapiro, M. 1964.** Effect of colchicine on xylose absorption in dogs. *Clin. Res.,* **12:**31.

***

MS- Young Scientist/ Senior Scientist/Life Time Achivement Award - 2020 Nomination Form

(*Jointly Sponsored by* MADHAWI SHYAM EDUCATIONAL TRUST & ICCB)

"Mandaar-Pushp", Pragatisheel Colony, Chitragupta Nagar,
Baragain,Ranchi - 834009
Jharkhand, India
(Tel. : Mob. : 94313-89253; 94317-08322)

Name of the applicant : .....

Date of birth : .....

Academic Qualifications : .....

.....

Area of Research : .....

No. of research Publications : 1) .....

(Titles of best two) .....

2) .....

.....

Other outstanding Achievements : .....

.....

Place of Working with full .....

address for communication : .....

.....

E-mail : ..... Tel./Mob. : .....

Are you a life member of the Journal or willing to accept the same : Yes / No

Signature of the applicant

Name & Address of 2 Reference : 1)..... 2) .....

....

....

....

Signature & Seal of Referees : ....

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 265-268
An International Biannual Refereed Journal of Life Sciences

19(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Diversity of Oscillatoriaceae in some chours (water reservoirs) of Supaul district

Sneha Kumari*
Department of Botany, M.L.T.College, Saharsa, B.N.M.University, Madhepura, Bihar, India

Received : 19th December, 2019 ; Revised : 25th February, 2020

Abstract : pH value and organic waste increases in summer season in different chours which favour the luxuriant growth of different species of Oscillatoriaceae during summer pH was recorded in between 8 to 9.20 which is most suitable for *Oscillatoria*, *Lyngbya* and *Phormidium*. Altogether 7 Species of *Oscillatoria*, 2 Species of *Phormidium* and 6 Species of *Lyngbya* were recorded.

Key words: Chours, organic waste, pH.

INTRODUCTION

Oscillatoriaceae in a family of Blue-green Algae growing as slender unbranched filament often tangled masses in water, Members of this family commonly exhibit oscillating movements. They reproduce only asexually by hormogonia. They grow in polluted water in which organic waste is present. Genus *Oscillatoria* is considered as pollution indicator.

River Kosi and its several tributaries pass through several villages of Supaul district. During Flood River water is deposited in low lands. These water reservoirs are locally known as Chours Water remain logged around the year in these chours, local people use these chours for fishing, bathing, washing cloths and cleaning their domestic animals. Organic waste gradually increases in these Chours which favour the luxuriant growth of Blue-green Algae perticularly members of family Oscillatoriaceae.

*Corresponding author :
Phone : 79924 20237
E-mail : sneha09mallik@gmail.com

MATERIALAND METHOD

Algal samples were collected at regular intervals in Polythene bags and brought to laboratory. Samples were identified microscopically with the help of standard monograph. Camera Lucida diagram were drawn.

RESULT

***Oscillatoria acutissima* Kufferath, 1994**

Triochomes are solitary and scattered or loosely entangled in the mucilage of other algae, gradually tapering to the apex, which is curved or bent slightly. Apical cell is acute-conical with a calyptra. Cells 1.2-2µ in diameter, 1½ to 3 times longer than wide; not constricted at the cross walls, which are not granular.

***Oscillatoria agardhii* Gomont, 1992a**

Trichomes interwoven to form a blue-green plant mass; occasionally found floating free; straight throughout their entire length, briefly tapering at the anterior end, which is usually capitate, but frequently smooth. Apical cell is truncate-conical with or without a calyptra. Cells (3.4)-

5.5-6µ in diameter and not constricted at the cross, walls, which are granular; their length from ½ to approximately equal their width (sometimes 1% times their width in length).

This species shows a great deal of variation in the morphology of the apex of the trichome. The straight interwoven filaments, the cell proportions, and the granular cross walls help to identify it.

***Oscilatoria amoena* (Kuetz.) Gomont, 1892a**

Trichomes usually forming a thin, submerged, weft-like, blue green mass, or sometimes scattered among other algae; usually straight but not rigid, slightly tapering toward the apex. Apical cell is broad capitate with a cone-shaped calyptra. Cell (2.5)-4-5/A in diameter, 2.5-4.2µ long; usually slightly constricted at the cross walls, which are granular.

***Oscillatoria amphibia* C.A. Agardh, 1827**

Trichomes are straight or curved and interwoven to form thin, blue-green plant in mass on submerged objects or in moist aerial habitats; not tapering toward the apex. Apical cell is broadly rounded, smooth, with a convex outer membrane. Cells (1.5)-2-2.8-(4)µ in diameter, 2-4 times their diameter in length (4-8µ); not constricted at the cross walls, which have a single large granule on either, side, a series of paired granules showing throughout the length of the trichome.

***Oscillatoria limosa* (Roth) C A Agardh, 1812**

Trichomes are usually forming a very dark blue-green or brownish plant mass attached to submerged objects or forming films on sandy bottoms. Although, rarely solitary or loosely entangled among filamentous algae; straight, tapering little or not at all toward the apex Apical cell rotund, the outer membrane thickened but without a definite calyptra Cells 12-18-(20)µ in diameter, 3.7-5µ, long, not constricted at the cross walls, which are usually granular. Trichomes not infrequently enclosed in a homogonous sheath.

***Oscilatoria princeps* Vaucher, 1808**

Trichomes were solitary or loosely entangled to form small floating plant masses; which are black green in colour; individual plants.

***Oscillatoria splendida* Greyille, 1824**

Trichomes are solitary and scattered, rarely aggregated in small, flake-like masses; straight or curved, tapering for a long distance to a fine hair at the apex. Apical cell is conical and capitate. Cells 2.2-2.8µ in diameter, 7.2-9µ long, not constricted at the cross walls; cell contents finely granular or homogeneous, pale blue-green.

Phormidium mucicola Naumann & Huber - Pestalozzi, 1929

Filaments short, clustered or scattered in the mucilage, Sheaths thin, Trichomes parallel or scattered; up to 50µ long; Cells quadrate 1.3-2µ in diameter, 1.8-3µ long; constricted at cross walls.

***Phormidium ambiguum* Comont 1892a**

Filaments forming a blue-green, mucilaginous layer; straight or gracefully curved, either parallel or somewhat entwined; individual sheaths usually distinct and lamellate but becoming confluent with the mucilage of the plant mass; trichomes curved or rarely straight at the apices, which are not tapering; apical, cell broadly rounded, not capitate, but with a thickened outer membrane, which may give a slightly pointed appearance; cells short, disc-like constricted at, the cross walls, 4-6µ in diameter, 1.2-2.7µ long; cell contents finally granular.

***Lyngbya aestuarii* (Mert.) Liebmann 1841**

Plants aggregated, entangled, forming extensive layers on merged substrates, or upon moist earth and stones, sometimes becoming free-floating. Filaments varying greatly in diameter, 10-261µ wide Trichomes 8-20-(28)µ in diameter, tapering a little at the apices, which vary, in shape, conical, truncate, or somewhat capitate. Cells 2.5-5µ in length, not constricted at the cross walls. Sheaths firm, becoming thickened, lamellose, and discolored with age.

***Lyngbya birgei* G.M. Smith, 1916**

Plants solitary, planktonic, scattered among water bloom Organisms; filaments straight, (18)-24-(25)µ in diameter; with many horinogona when aged; sheaths thick and firm, not larnellated, extending, far beyond the apices of, the trichomes, which are broadly rounded; trichome 20-24µ in diameter, not constricted, at the cross walls; cells very short and disciform, 3-7µ long.

***Lyngbya heironymusii* Lemmermanu, 1905**

Plants solitary, scattered among other algae or in littoral plankton. Trichomes straight and not tapering at the apices, which are broadly convex; 11-13µ in diameter. Cells 1/5-1/3 as long as wide (2.7-4µ long); usually with pseudovacuoles; not constricted at the cross walls, which are granular. Sheath rather thick, but homogeneous, not lamellose. Filaments (12)-14-19µ wide.

***Lyngbya latissima* Prescott, 1944**

Plants solitary, planktonic, entangled among other floating algae, trichomes straight, not tapering toward the

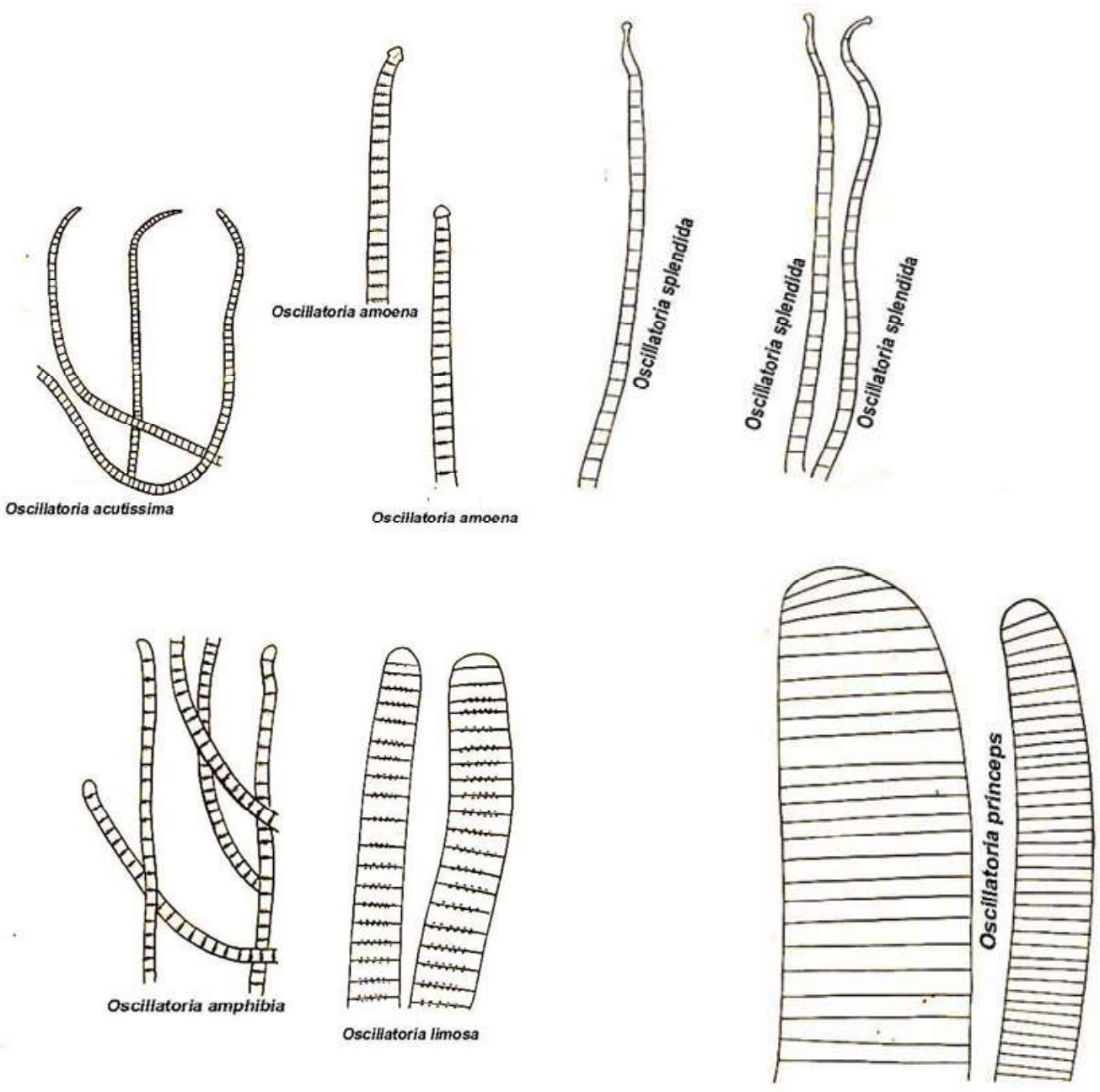
apices, cells disc-like, with contents finally and evenly granular (sometimes with coarse granules in old plants), 37-40.7µ in diameter, 3.7-7.4µ long; sheaths thick, 3.7-51µ wide, lamellated, with outer layers wrinkled and roughened in age; filaments 44-58µ in diameter.

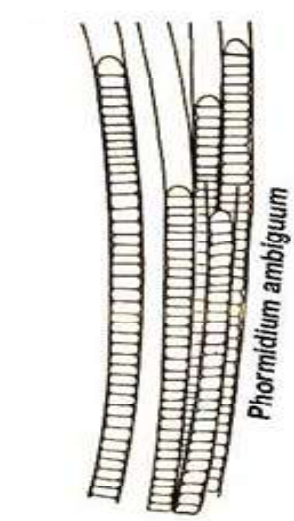
***Lyngbya limnetica* Lemmermann, 1898d**

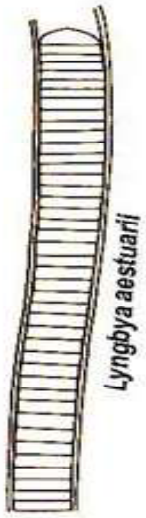
Plants straight, solitary, planktonic; trichomes 1-2-(2.5)µ in diameter, not tapering at the apices; cells 6-12µ long, not constricted at the cross walls; cell contents coarsely granular; sheaths thin and colorless; filaments 2-2.2µ wide.

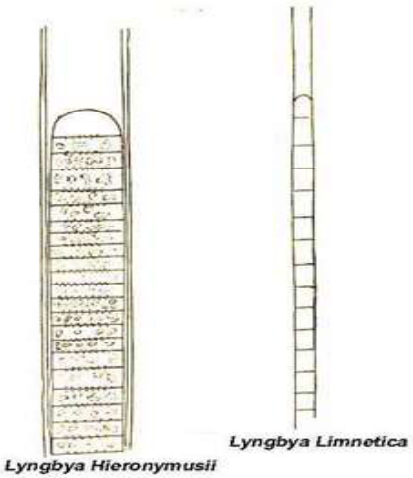
***Lyngbya major* Meneghini, 1837**

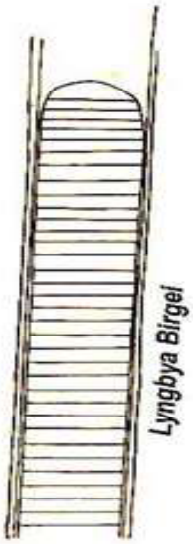
Plants solitary among other algae, or somewhat gregarious, but not forming expansions or plant masses; filaments straight; trichomes not or but very slightly tapering to the apices, which are indistinctly capitate in older plants, 11-17µ in diameter; cells 1/5-1/4 as long as wide, (2)-3.5-4µ in length, not constricted at the cross walls, which are definitely granulose; cell contents homogeneously granular; sheaths; thick (3-3.7µ), firm, and usually lamellated, becoming roughened in age; filament 22-26µ in diameter.

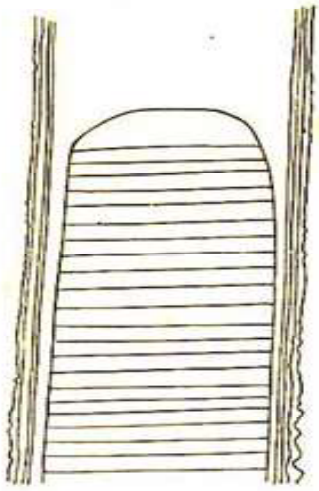


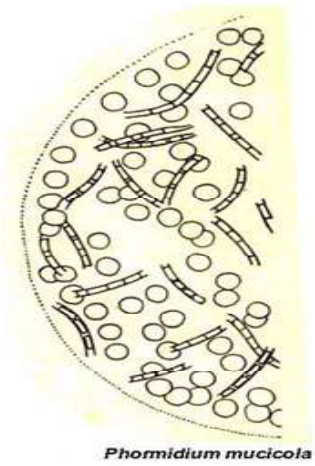


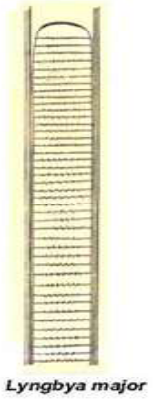


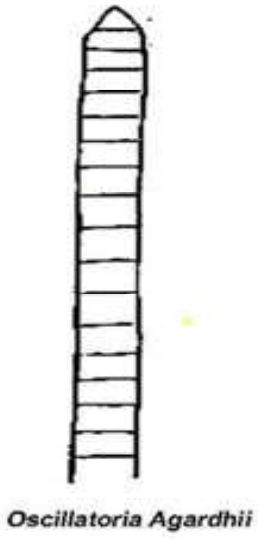


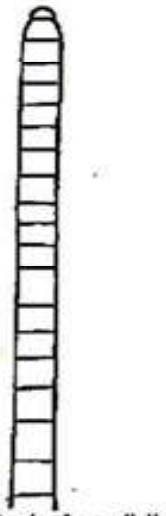












REFERENCES

1. **Reddy P M. 1983.** Changes in Pholyphosphate Bodies during Sporulation and Spore Germination in Cyanobacteria. *Biochem. Physiol. Pflanzen.* **178:**77-79.
2. **Reddy P M, Conway C M, Fisher R W. 1989.** Tunicamycin suppesses differentiation and stimulates dedifferentiation of heterocysts in the cyanobacterium Anabaena azollae Stras. *Current Microbiology.* **19:**109-114.
3. **Prasad, V. 1996.** Blue-green algae of Birganj (Nepal). Short term project. Research Division, Tribhuvan University, Nepal.
4. **Rath, J. and Adhikary S.P. 2005.** Alagal flora of Chilka Lake. Daya Publication house, Delhi.
5. **Shameel, M. 2008.** Change of divisional nomenclature in the Shameelian Classification of algae. *Int. J. Phyco.l Phycochem.* **4:** 225-232.

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 269-272
An International Biannual Refereed Journal of Life Sciences

20(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Mushroom technology as new food resource for rural development

Nayeema Khatoon*
University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

Received : 29th December, 2019 ; Revised : 24th February, 2020

Abstract : Over the past 25 years, new natural drugs have been approved for the treatment of human diseases. Natural products play a very important role in the process of discovery and development of drugs, including the treatment of chronic diseases such as cancer. For hundreds of years, medical mushrooms are used as decoctions and essences and are applied ad alternative medicine in Korea, China, Japan and Eastern Russia. Geologically, mushrooms existed on the earth even before man appeared on it as evidenced from the fossil records of the lower cretaceous period. Thus anthropologically speaking, there is every possibility that man used the mushrooms as food when still a food gathered and hunter on the chronology of Cultural Revolution. Mushrooms offer tremendous applications as they can be used as food and medicines besides their key ecological roles. They represent as one of the world's greatest untapped resources of nutrition and palatable food of the future. Mushrooms had long been used for medical and food purposes since decades. It is new increasingly recognized that correct diet, controls and modulates many functions of human body and consequently participates in the maintenance of state of good health, necessary to reduce the risk of many diseases. Mushrooms have a great nutritional value since they are quite rich in protein with an important content of essential amino acids and fiber. The dietary fiber present in *L.edodes* (Shiitake) consists of soluble and insoluble structures. In the water-soluble are found the β-glucans and proteins. In the non soluble fraction, salts are extracted only with acids or alkalis and found the polyuronide (acidic polysaccharide), hemicelluloses, β-glucan chains with heterosaccharide, lignin and chitin. They also provide a nutritionally significant content of vitamins (B1, B2, B12,C, D and E). Numerous mushrooms and their ingredients have been known to be beneficial to the skin and hair. The representative ingredients are as follows: phenolics, polyphenolics, terpenoids, selenium, polysaccharides, vitamins and volatile organic compounds. These compounds show excellent antioxidant, ant-aging, anti-wrinkle, skin whitening and moisturizing effects, which make them ideal candidates for cosmetics products. Present review is aimed to discuss the high nutritional and therapeutic potential of mushrooms and their applications as functional foods or as a source of nutraceuticals for maintenance and promotion of health and life quality.

Key words: Mushrooms, functional foods, bioactive compounds, ant-aging, antioxidant nutritional, cosmeceuticals and therapeutic agents

INTRODUCTION

A mushroom is the fleshy, spore bearing fruiting body of fungi, typically produced above ground, on soil, or on its food source, mostly in forests. It is perhaps the most well-known and documented edible forest product[1]. The word mushroom means different things for different people in different countries[2]. Therefore the term is used to describe the fleshy fruiting bodies of some Ascomycota. These gills produce microscopic spores that help the fungus spread across the ground or its occupant surface.

*Corresponding author :
Phone : 94711 63476
E-mail : nayeemaru@gmail.com

Mushrooms are considered as "god's food" by the Romans. The Greeks regarded them as providing strength for worriers in battles. Mushrooms are mysterious, cultural, traditional and legendary. Cultivation of saprophytic species such as Oyster mushrooms and shiitake has grown rapidly in recent decades.[3] Mushrooms have been widely used as foods[4,5] and very often as delicious and nutritious foods.[6] For example, The Pharaohs prized mushrooms as delicacy food[7] and valuable resources for medicines as well.[8] More than thousands of mushroom biodiversity present in nature, our ancestors, biologists, and scientists have isolated approximately 100s of mushroom species that have a unique combination of nutrition that improves our health. The commonly cultivated mushrooms are button mushrooms, oyster mushroom, paddy straw mushroom, milky mushroom, portobello mushroom, and shiitake mushrooms. The consumption of mushrooms by humans is an age-old practice. However, in recent days mushroom has become a part of every continental dish because of its good taste, flavor and nutritional value.



Mushroom cultivation is increasing day by day and becoming more popular because it is not only meets the deity requirements but also adds to the income especially of growers with insufficient land. It is playing a vital role in helping rural community make stronger their livelihoods. It is well organized income for change of agricultural wastes like hey, paddy straw etc. into precious protein and making added profits as well as reduce unemployment. It is considered to be a very fascinating hobby for the person who is unemployed or retired. Mushroom growing is one agricultural activity in which women can play a vital role without sacrificing their household responsibilities. In present scenario, mushroom cultivation faces fewer difficulties. Cultivation is also independent of weather, and can recycle agricultural by products as compound substrate which, in turn, can be used as organic mulch in growing other horticultural crops, including vegetables.[9] A small quantity of spawn planted in a suitable growing medium it is amazing that within a six weeks, it grow into a profitable crop inside a room, where no other crop can be grow. However, mushrooms have more uses in present time than any other food crops. It is carried out indoor in room, basement, shades, garage etc., which should be well ventilated. The commonly cultivated mushrooms are button mushrooms, oyster mushroom, paddy straw mushroom, milky mushroom, Portobello mushroom, and shiitake mushrooms. The consumption of mushrooms by humans is an age-old practice. However, in recent days mushroom has become a part of every continental dish because of its good taste, flavor and nutritional value.

In India, there is only three kinds of mushrooms are specially cultivated in rural areas, which are as following:-

1. Button mushroom (*Agaricus bispores*)
2. Straw mushroom (*Voluariella uoluacea*)
3. Oyster mushroom (*Pleurotus sajor-caju*)

The major focus has been given on the production of button mushrooms in India, even though the productivity, yield potential, quality, and medicinal aspects make oyster mushrooms more challenging. These mushrooms are a good quality source of non-starchy carbohydrates and medium quantity of proteins, minerals and vitamins.[10] The content of protein differs from 1.6 to 2.5%. Oyster mushrooms are rich in vitamin C, B complex, and mineral salts, which is necessary to the human being.[11]

Table.1 Mushroom species and their corresponding cultivation med

Growing Medium	Mushroom Species
Rice straw	Straw (Volvariella), Oyster (Pleurotus), Common (Agaricus)
Cotton waste from textile industry	Oyster (Pleurotus), Straw (Volvariella)
Sawdust	Shiitake (Lentinus), Oyster (Pleurotus), Lions Head or pom pom (Hericium), Ear (Auricularis), Ganoderma (Reishi), Maitake (Grifolia frondosa), Winter (Flammulina)
Oil palm waste	Straw (Volvariella)
Bean straw	Oyster (Pleurotus)
Cotton straw	Oyster (Pleurotus)
Banana leaves	Straw (Volvariella)
Sawdust-rice bran	Nameko (pholiota), Ear (Aurcularis), Shaggy Mane (Coprinus), Winter (Flammulina), Shiitake (Lentius)
Cotton seed hulls	Oyster (Pleurotus), Shiitake (Lentius)

This article is going to focus on cultivated mushrooms. The objectives of this paper are to describes the fast growth of the mushroom economy and provide

useful knowledge of the technological innovation and dissemination for rural development. The information used in this paper is more based on existing literature.

The Oyster mushroom is a member of the genus Pleurotus, commonly known as "Dhingri" in India. It has got its name "oyster" from the resemblance of its fruiting bodies to sea oyster shells that come in different colors (white, cream, grey, pink and yellow). Some countries such as China, Italy and the USA are leaders in oyster mushroom production and China alone produces 85% of all (*Pleurotus* spp.) that are grown worldwide. Oyster mushrooms are decomposer of wood and vegetable residues. At present India produces annually 10,000 tons of this Oyster mushroom. It is popularly grown in the states of Orissa, Karnataka, Kerala, Maharashtra, Andhra Pradesh, Madhya Pradesh, West Bengal and in the North-Eastern States. Oyster mushrooms are rich sources of protein (1.6-2.5%- fresh weight, 20-30% protein-dry weight), vitamin C and vitamin B complex, potassium, sodium, phosphorus, iron, and calcium. The protein quality of oyster mushrooms is considered higher than that of fruits and vegetables. Oyster mushrooms contain beta-1,3/1,6-glucan which stimulate the immune system and mevinolin which is important in lowering cholesterol level. The oyster mushroom is used fresh, dried, canned or in powder form.



Although there is no great demand in the national market, it is easy to sell the small produce by the growers. The international demand is so huge that it is difficult for a small grower to meet. There is a need to develop a chain of producers so that the production from different farmers could be collected and supplied to the international market as per export standards. Cultivation of edible mushrooms is one of the most economically viable processes for the bioconversion of lingo-cellulosic wastes. *Pleurotus* spp. can easily be grown by rural women with minimum efforts. 'Dhingri' is grown in northern plains from October to March when temperature ranges between 22-28°C. High labour-land ratio and an alarming rate of population growth may pose a threat to our food security in the very near future. The situation has led to a decrease in labour wages pushing the community to below the poverty line. Oyster mushroom cultivation is considered as an alternative source of income to uplift the living standards of poor farmers and also to add high-quality protein in their daily diets to eradicate malnutrition problems.

CONCLUSION AND DISCUSSION

Mushroom cultivation has been described as the most versatile and prolific agriculture and forestry venture all over the world. Growing evidence shows that mushroom growth has great potential in many other countries, particularly in developing countries. Approximately 14,000 described species of the 1.5 million fungi estimated in the world produce fruiting bodies that are large enough to be considered as mushroom.[12] Mushroom cultivation can provide employment in both the semi-urban and rural areas. Mushroom cultivation will improve the socio-economic condition of farmers, families and solve employment problems of both literate and illiterate, especially women. Mushrooms could be an important sector for our future agriculture and forestry. Huge quantities of vide varieties of organic waste are generated from agriculture, forestry and food processing. Bio-innovation is very important for mushroom cultivation and so is the technological dissemination. Technological development can largely increase production capabilities and even significantly reduce the costs. Mushroom cultivation is an effective bioconversion technology of transforming wastes and woods into potential valuable resources. Mushroom cultivation could also be an important part of sustainable agriculture and forestry.

REFERENCES

1. **Chamberlain, J.; Bush, R.; Hammett, A. 1998.** Non-Timber Forest Products: The other Forest products For. *Prod. J.,* **48:**10-19.

2. **Chang, S.T. ; Miles,P.G. 1992.** Mushroom biology-A new discipline. *Mycologist*. **6:**64-65.

3. **Arora, D.; Shepard, G.H. 2008.** Mushrooms and Economic Botany 1. *Econ. Bot.* **62:** 207-212.

4. **Falconer, J.; Koppel, C.R.S. 1990.** The major significance of 'minor' forest products: the local use and value of forests in the West African humid forest zone. In FAO Community Forestry Note; Food and Agriculture Organization of the United Nations Roma, Italy.

5. **Gibert, F.A., Robinson, R.F. 1957.** Food from fungi. *Econo. Bot*. **11:**126-145.

6. **Vinseti, B.; Termote, C.; Ickowitz, A.; Powell, B.; Kehlenbeck, K,; Hunter, D. 2013.** The contribution of forests and trees to sustainable diets. *Sustainability*. **5:** 4797-4824.

7. **Daba, A.S,; Kabeil, S.; Botros, W. A.; El-Saadani, M. 2008.** Production of mushroom (*Pleurotus ostreatus*) in Egypt as a source of nutritional and medicinal food. *World J. Agric. Sci*. **4:**630-634.

8. **Lakhanpal, T.N.; Rana, M. 2005.** Medicinal and nutraceutical genetic resources of mushrooms. *Plant Genet. Res*. **3:** 288-303.

9. **Marshall, E. and Nair, N.G. 2009.** Make money by growing mushroom. Rural Infrastructure and Agro-Industries Division Food and Agriculture Organization of the United Nations Rome.

10. **Croan, S.C. 2004.** Conversion of conifer waste into edible and medicinal mushrooms. *Forest Products Journal*. **54:** 68-76.

11. **Randive, S.D. 2012.** Cultivation and study of growth of oyster mushroom on different agricultural waste substrate and its nutrient analysis. *Advances in Applied Science Research*. **3:** 1938-49.

12. **Chang, S.T. 2006.** The world mushroom industry: Trends and technological development. *Int. J. Med. Mushrooms*. **8:** 297-314.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 273-276
An International Biannual Refereed Journal of Life Sciences

21(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Sewage contamination and diminished water quality of Naubatpur, Patna: A microbiological approach

Kanchan Kumar* & Shailendra Kumar
Department of Zoology, T.S. College, Hisua, Nawada, Magadh University, Bodh-Gaya, Bihar, India

Received : 18th December, 2019 ; Revised : 14th February, 2020

Abstract : Water is considered as the elixir of life. Apart from its central role in governing biological activities, it has also contributed a lot in the evolution of the human civilization. Water bodies like springs, rivers, lakes etc. have provided suitable sites for human settlements between states & countries and with advent of time, these sites later developed as political centers. Microbial analysis is one of the basic tools to evaluate the pollution status of the water bodies. The presence of coliform bacteria is considered as the bio-indicator of fecal contamination of water. The findings of the present study give an insight to the diminished water quality of the Naubatpur Block of Patna, Bihar due to sewage contamination. The increased incidence of water borne diseases in the local inhabitants should be thoroughly managed. A judicious intervention of the local authorities and the stakeholders are the need of hours to restrict unscientific and untreated sewage disposal in the water bodies of the said.

Key words: Faecal contamination,coliform bacteria, Bioindicator, Naubatpur block,Patna, Judicious, Management

INTRODUCTION

Water and aqueous medium are essential and unique among the chemical substance in the evolution and support of biomass More than 80% of our body and more the 60% of our eye is water. In past the biologists had looked upon water merely as a liquid filled up the "holes" in living cells and Tissues.[2,3] But later on they realised that this most peculiar solvent plays a major possible even the predominant role in the determination of biomacro molecular conformation and changes of conformation, and that is also control in a wide spectrum of biological processes, in particular those involving chemical mass transport through the biomembranes. Water is very important for the activities of the human civilization and it is apparent from the fact that apart from the direct human consumption and other domestic usage, water finds its applicability in a majority of industrial, agricultural and medical activities as well as in the production of various types of energy for the civilized world. Failing home septic systems can allow coliforms in the effluent to flow into the water table, aquifers, drainage ditches and nearby surface waters.[4-6] Sewage connections that are connected to storm drain pipes can also allow human sewage into surface waters. Pets, especially dogs, can contribute to fecal contamination of surface waters. Runoff from roads, parking lots, and yards can carry animal wastes to streams through storm sewers. Birds can be a significant source of fecal coliform bacteria.[7-9] Swans, geese, sea gulls, and other waterfowl can all elevate bacterial counts, especially in wetlands, lakes, ponds, and rivers. Besides, agricultural practices such as allowing livestock to graze near water

*Corresponding author :
Phone : 9334025085
E-mail : drshankarkr@gmail.com

bodies, spreading manure as fertilizer on fields during wet periods, using sewage sludge biosolids and allowing livestock watering in streams can all contribute to fecal coliform contamination. The pathogenic bacteria, associated with the sewage mixed water, are generally inherent in the coliacrogenous or coliform group of bacteria of which *B. coli* or *E.coli* is important. A faecal Coliform is a facultatively anaerobic, rod- shaped, gram- negative, non- sporulating bacterium. They generally originate in the intestines of warm- blooded animals. They are capable of growth in the presence of bile salts or similar surface agents, are oxidase negative, and produce acid and gas from lactose within 48 hours at 44 ± 0.5°C.[1] The term "thermotolerant coliform" is more correct and is gaining acceptance over "fecal coliform".[2] Coliform bacteria basically includes the genera originated in faeces e.g. *Escherichia* as well as genera not of fecal origin e.g. *Enterobacter*, *Klebsiel la*, *Citrobacter*. The assay is intended to be an indicator of fecal contamination; more specifically of *E.coli* which is an indicator microorganism for other pathogens that may be present in feces.

The pathogenic bacteria are generally inherent in the coliacrogenous or coliform group of bacteria of which *B. coli* or *E.coli* is important.

The *E. coli* bacteria inhibit the intestinal tracts of warm blooded animals and human beings. They appear in very large number in their daily faecal discharges and also in crude sewage. They are not harmful but their presence indicate the presence of other bacteria like typhoid bacillus tec. Large quantities of fecal coliform bacteria in water indicate a higher risk of pathogens being present in the water.[3] Some waterborne pathogenic diseases that may coincide with fecal coliform contamination and originate from the intestinal disorder of sick persons. Such diseases include cholera, typhoid, paratyphoid, dysentery, viral and bacterial gastroenteritis, and hepatitis and ear infections. The part played by water in the spreading of these diseases is like a mechanical agent. Thus the faces and the urine of a sick person, is not properly disposed, are conveyed by storm water or other agencies to a stream, well or other sources of water. They are then conveyed to persons who use the contaminated water. Untreated organic matter that contains fecal coliform can be harmful to the environment. Aerobic decomposition of this material can reduce dissolved oxygen levels, if discharged into rivers or waterways. This may reduce the oxygen level enough to kill fish and other aquatic life. The coliform bacteria should not exceed more than coliform colony per 100 ml. of water sample i.e. M.P.N. of *B.coli* should not exceed 1 per 100 ml. The bacterial quality should not be relaxed as to avoid the set of any serious water disease.

The objective of the present study is to determine the bacterial load of coliform in the local domestic water of Naubatpur block, Patna and suggest ways of its removal.

MATERIALS & METHODS

Microbiological examination is regarded as the main component of analysis of water which produces unique information on water quality. The main function of this examination is to detect the potentially dangers faecal pollution and the presences of organisms that causes diseases. The coli form bacteria is the most important bacteria causing the faecal pollution in drinlang and domestic water. Its determination is made as below:

Sampling of water for microbiological analysis:

The sterilized glass bottles of 300 ml capacity were used for sampling of water especially for coli form bacterial analysis. The bottle was kept stoppered until the moments it is required for filling. After collection of the sample, the bottle was immerately stoppered. It was kept in ice container and transported to the laboratory.

MPN Method for coli form bacterial analysis:

The coli form bacterial analysis in different water sample was carried QUL MPN method i.e., Most Probable Number method which is popularly known as Multiple Tube method or Sometimes it is also termed as Dilution Method.

In the MPN method, the measured volumes of samples (50 ml, 10 ml, 1 ml) are transferred to tubes containing a liquid differential medium. On incaution at specified temperature for specified period, the microorganism will either produce acid or gas or both which is considered as positive. The positive tubes are counted and the probable number of coliform organisms present in the sample.

Procedure: The sample containing bottles were shaken well and the required volume of the sample into different tubes containing the modified format Glutamate Lactose medium of double strength. 10 ml of ample was added to five tubes containing the media, 1 ml of sample to mother five tubes containing the same media and 0.1 ml of sample was transferred to five tubes containing the

same media or modified Formate Gulamate Lactose medium of Single strength. Now these tubes having their moll the carved with cotton are incubated at 37°C for 48 hours. The tube in which gas is entrapped in the inner Durham tube is considered positive. The positive tubes was then subjected confirmatory test which was performed as below.

With the help of 3 mm loop, the culture of the positive tubes was transferred to formate ricinoleat unoleats broth in tubes. They were further scubated at 37°C for 48 hours. Any amount of gas produced and entrapped A the Durhan tube contibuted a positive confirmed test.

The number of positive tubes was countered and from it the MPN Most Probable number of coliform was derived from the table 3.1 given

RESULTS AND DISCUSSION

The study of the presence of micro organism in water is highly mordant as it predicts the potentially dangerous faecal pollution and also shows the presence of micro-organisms that cause diseases. So drinking, domestic water samples need this test or analysis to ascertain its quality. Industrial water is also required to be subjected to micro biological examination to detect organisms that may contaminate or interfere in process or spoil the food products or develop unwanted growth in the water carrying structures like pipes and conduits.

The number of bacteria present in water is taken into account while assessing the quality of water for drinking domestic and industrial uses the microbiological analysis is also useful in assessing the efficacy of certain Water treatment processes like coagulation, filtration and disinfection. It also gives an early indication of pollution.

Most industries require water that contains a few numbers of microorganisms. Food processing industries need water showing a standard plate count below, fermentation industries require water with a total bacterial count below 100. The number of microorganisms should not exceed 100 per ml for water used in the manufacture of rayon.

There are several types of microorganisms which affect the quality of water in one way or other. The most important organisms are listed below:

1. Coliform bacteria
2. Proteolytic bacteria
3. Lipolytic bacteria
4. Thermophilic bacteria
5. Gelatinliquefying bacteria
6. Iron bacteria
7. Sulphete reducing bacteria
8. Sulphur reducing bacteria
9. Yeast and mould
10. Slime forming organisms.

The coliform bacteria is most important which is mainly concerned with the faecal pollution in water bodies, the presence of coliform organisms indicates faecal contamination in water. It is highly important and necessary to carry the ast for coliform bacteria in different, resources of water in Naubatpur block. Besides indicating the faecal contamination; coliform organisms are also ported to retard fermentation. Coliform organisms are the best lactose trapper and the lactic acid produced by lactose fermentation would diminish the yeast activity.

100 ml of each water samples are used for coliform analysis in MPN/100 ml of samples unit by multiple fermentation tube method (Mac Donald and David kay). As per a report in HMSO, the standards of bacteria quality of drinking water have been given in table-1.

Table - 1

Coliform, M.PN/ 100 ml	Class	Quality of Drinking Water
0	I	Excellent
1-3	II	Satisfactory
4-10	III	Suspicious
> 10	IV	Unsatisfactory

The coliform bacterial values for well water samples coincide at1.00 250 MPN/100 ml of samples while for hand pipe water samples the value is approximately 010 1.5 MPN /100 ml. The values fall between 3.50 to 3.75 MPN /100ml for pond water samples while for river water the values are approximately 3.350 MPN/100 ml.

In the present investigation, the values are maximum for pond water which may be attributed to the fact that villagers used the open defecation around the pond which contributed heavily to the faecal contamination of pond water. The situation is more or less similar for river water except for the fact that due to extreme water current in river faecal contamination is a bit less severe.

The quality of tube well water is excellent so far the faecal pollution is concerned, while water of well and hand pipe bear satisfactory quality.[10-18]

The compo native quality of water of different water resources is given in table 2.

Table-2. Comparative Values of coliform bacteria in different water sources in MPN / 100 ml of samples

Well Water Sample	Hand Pipe Water Sample	Tube Well Water Sample	Pond Water Samples	River Water Samples
1.95	0.925	0.00	3.75	3.125
AV-0.00024	0.0018		10.00052	0.000012
Quality: Satisfactory	Satisfactory	Excellent	Suspicious	Suspicious

REFERENCES

1. **Doyle,M. P. and Erickson, M. C. 2006.** Closing the door on the faecal coliform Assay"*J. Microbe*.**1:**162-163.

2. **APHA 1992.** *Standard Methods for the Examination of Water and Wastewater.* 18th edition,American Public Health Association (APHA), American Water Works Association (AWWA)and Water Pollution Control Federation (WPCF), Washington, D.C.

3. **Department of Health and Social Security. 1982.** The Bacteriological Examination of DrinkingWater Supplies. Her Majesty's Stationery Office (HMSO), London.

4. **DE Warner, HP, and Griggs, S.H; Chem. 1971.** Water pollution by biodegardabel organic substances Eng, *Prog. Symp. Ser.* **67:**310-320.

5. **Environment Canada. 1981.** Analytical Methods Manual. Water Quality Branch, Environment Canada, Ottawa.

6. **Flurez, F.1972.** Influence of O. Concentration of growth and survival of larvae and Juveniles of the Indus (L); *Rep. Inst. Freshwat. Res. Prottninghoom,* **52:** 65-73.

7. **Ford, Rs. and Fox, B.14. Wat and Wat., 1973.** Centralization of sewage treatment and sludge disposal in rural area, *Eng*. **77:** 157-159

8. **Huand, P.M. and Huang, C.P. Wat. Sewage W k s 1973.** Inorganic and organic phosphorous distribution in domestic and Municipal Sewage **120(6):** 82-83.

9. **ISO. 1984.** Water Quality C Detection and Enumeration of Faecal Streptococci. Part 1:Method by Enrichment in a Liquid Medium. International Standard ISO 7899-1, International Organization for Standardization, Geneva.

10. **ISO 1984.** Water Quality C Detection and Enumeration of Faecal Streptococci. Part 2: Method by Membrane Filtration. International Standard ISO 7899-2, International Organization for Standardization, Geneva.

11. **Jones, GL. Nature Lang 1973.** Role of Protozoa in waste purification system. **243:** 546-547 .

12. **Mara, D. and Cairncross, A. 1989.** Guidelines for the Safe Use of Wastewater and Excreta inAgriculture and Aquaculture. World Health Organization, Geneva.

13. **McGuire, JM; Alford, A.L. and Carter, M.H; U.S. 1973.** Organic pollutants identification utilizing mass spectrometry *Enviro. Protect Ag., Envir Protest Technol. Ser. E.P. A-R*, 73-234,

14. **Ongerth,H.J.spath, D.P. Grook, J. and Greanberg, A.&.E.: J. AM. Wat. Wks, Ass 1973.** Biological removal of collodial matter from waste water. **65:**495-498.

15. **P. B. Mberekpe and M. N. Eze, 2014.** "Effect of preservation on the quality of sachet water consumed by households in Nsukka zone," *International Institute for Science Technology and Education Journal*. **6(7):** 25–30.

16. **L. A. Thliza, A. U. Khan, D. B. Dangora, and A. Yahaya, 2015.** "Study of some bacterial load of some brands of sachet water sold in Ahmadu Bello University (Main Campus), Zaria, Nigeria," *International Journal of current Science*. **14:**91–97.

17. **WHO 1985.** Guidelines for Drinking-water Quality. Volume 3: Drinking-water Control in Small Community Supplies. World Health Organization, Geneva.

18. **WHO 1989.** Health Guidelines for the Use of Wastewater in Agriculture and Aquaculture. Report of a WHO Scientific Group. WHO Technical Report Series, No. 778, World Health Organization, Geneva.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 277-280
An International Biannual Refereed Journal of Life Sciences

22(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Analysis of the harmful effects of heavy metals on the body and some organs of cat fish (*Clarias batrachus*) of Kosi region with special reference to cadmium

Sanjeeta Bishwas*
Department of Zoology, B. N. Mandal University, Madhepura, Bihar, India

Received : 08th January, 2020 ; Revised : 15th February, 2020

Abstract : Discharge of heavy metals into aquatic environment from various sources, even below permissible levels, creates health hazards in aquatic organisms. The persistence and ubiquitous nature of these pollutants coupled with their tendency to accumulate in organisms ultimately produce toxic reactions in aquatic animals, especially fish. The heavy metal like cadmium, considered as one of the most toxic heavy metals.

Key words: *Clarias batrachus*, cadmium, toxicity

INTRODUCTION

Cadmium is a nonessential toxic element which enters body via a number of routes including food, water and air and by smoking or cigarettes. In blood it is mainly accumulated in the red cells and binds to a low molecular weight protein, Metallothionein plays a major role in homeostasis of cadmium More than 80% of cadmium is bound to metallothionein and found in liver and kidney There is a protein in the body called metallothionein which is formed from the amino acid cystein that transports Cd, Zn, Cu, & possibly other metals in the body. It is possible that excessive cadmium competes for transport sites with Cu & Zn and may even be preferentially transported by metallothionein and this is the reason why cadmium in excessive amounts is so toxic to the body.

Rivers and lake shores are the areas primarily affected by diluted cadmium waste from industrial facilities in big cities. The cadmium-related contamination of the aquatic habitat has greatly increased in the last decades, resulting in an increase of cadmium deposits in tissues of aquatic organisms in all food chain systems. It is important to note that cadmium is a highly toxic element for all mammals and fish. Cadmium levels have constantly been increasing, and consequently, the research on cadmium has become quite topical and urgent. Accumulation of cadmium in living organisms is a major ecological concern, especially because of its ability to accumulate very quickly. By contrast, the excretion of cadmium from living organisms is a slow process. In fish, cadmium can cause a number of structural and pathomorphological changes in various organs.

*Corresponding author :
Phone : 99316 55408
E-mail : Jonnystrongbihar@gmail.com

MATERIALS AND METHODS

The fish were procured live from non-polluted river of Koshi with the help of fishermen. They were brought to the laboratory in wide mouthed large earthen pots half filled with natural water and covered with mosquito net.

Thereafter, healthy fish of an average length (11-15 cm) and weight (25-28 g) were transferred one by one with the help of small hand net to 40 litre rectangular glass aquaria for acclimation to various laboratory condition for a fortnight.

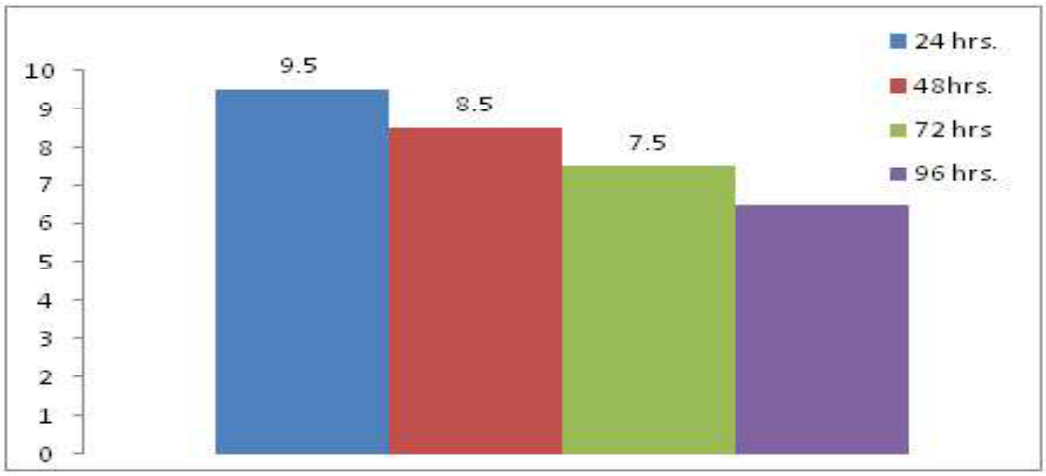
The fish were not supplied any food for the first three days. The fish were adjusted to natural photoperiod and ambient temperature. Running tab water was used in all the experiments. Running tap water was used in all the experiments and no aeration was done. They were fed with chopped goat liver every day ad libitum at 11.00 A.M sharp. The feeding was stopped 24 hr before the start of experiment and no food was supplied to them during the period of experimentation. However, the feeding schedule was strictly followed during chronic experiments to avoid starvational effects.

The fish were held in diluted water for at least one day before the start of a test, Exposure units were static and not aerated during the 96 hours test. The cadmium concentration was transferred one by one with the help of small hand net from the acclimatizing aquaria to the experimental ones. All the precautions as suggested by APHA, AWWA and WPCF (1985) were strictly followed during the bio-assay. Assays were terminated and the results discarded if and when the control mortality exceeded 10% in addition, A stock solution of cadmium chloride was prepared by simple dilution technique as described by concentrations of cadmium chloride and their percent survival at different time intervals (24, 48, 72 and 96 hours) were recorded as a measure of acute toxicity. The LC_{50} values for the aforesaid time intervals were determined by regression equation.

RESULTS AND DISCUSSION

The toxicity of cadmium chloride obtained under static system has been shown by regression equation in Table-II, Fig.-1-5. The 24hr, 48 hr, 72 hr and 96 hr of LC50 values of cadmium chloride by regression equations are 9.5 ppm, 8.5 ppm, 7.5 ppm & 6.5 ppm respectively.

Fig: - LC50 values (ppm) of cadmium chloride different time intervals obtained by regression equations



The formula of Hart *et al.* (1945) when applied to the LC_{50} values of cadmium chloride, gave the sublethal does of 2.31 ppm respectively.

Behavioural response:

The control fish shows a tendency to remain at the bottom of the aquarium with little disturbance. However, at lethal concentration of cadmium chloride the fish show prominent behaviour abnormalities viz. surfacing, rapid opercular movements, convulsions etc. Besides profuse mucus secretion with its coat deposited over the gills, gyratory swimming movements by moribund fish are the other observed behavioural responses noted during the present investigation. Loss of equilibrium before death is a symptom shown all the test fish.

Effects on organs

Total protein in liver, kidney testis and ovary:

Tables-1,reflects the results of the present study on total protein content in the cadmium chloride exposed fish *Clarias batrachus.*

Table-1: Levels of total proteins (mg/g wet tissue) in the tissue of *Clarias batrachus* chronically exposed to cadmium chloride (2.31ppm) for 30 days. Values are mean ± SE of 5 observations.

Tissue	Control	Cadmium chloride exposed
Liver	104.11±1.87	75.58 ±2.57(-27.4)
Kidney	174.926±0.1	169.972±2.532(-2.82)
Testis	81.41±1.47	70.0±1.26(-14.01)
Ovary	120.93±1.95	90.93±1.90(-24.8)

Total lipids in liver, kidney and ovary:

The results of total lipid have been found to be quite interesting is the sense that there has been depletion in its level in all the organs studied. In liver it has been found to be depleted by 3.22% is kidney by 1.96% is testes by 13.34% and in ovary 10.04% The highest account of total lipid has been estimated is kidney (173.2 ± 4.82 mg/g wet tissue) followed by liver (45.95 ± 1.89 mg/g wet tissue) ovary (21.11 ± 1.03 mg/g wet tissue) and testis (16.71 ± 0.80) mg/g wet tissue). The above decreasing order is as follows kidney> liver>ovary>testis.

As against these values to level of total lipid lowers down in the cadmium exposed fish. These are assessed to be 169.8b+-3.564; 44.47± 1.63 18.99± 0.72 and 14.48± 0.83 mg/ g wet tissue in kidney, liver, ovary & testis respectively. Thus present observation suggests that cadmium chloride either does not affect lipid molecules as severely as protein & carbohydrates or their mobilization takes place at a later stage. In treated group the decreasing orders is as follows

Kidney >Liver > Ovary > Testis.

Table-2: Levels of total lipid (mg/g wet tissue) in the tissue of *Clarias batrachus* chronically exposed to cadmium chloride (2.31ppm) for 30 days. Values are mean ± SE of 5 observations.

Tissue	Control	Cadmium chloride exposed
Liver	45.95 ±1.89	44.47 ± 1.63(-3.22)
Kidney	173.2 ± 4.82	169.8 ± 3.564(-1.96)
Testis	16.7 ± 0.80	14.48 ± 0.83(-13.34)
Ovary	21.11±1.03	18.99 ± 0.72(-10.04)

Total cholesterol, in liver, kidney, testis & ovary:

Table-3, reflects the results of the present study in total cholesterol content in the cadmium chloride exposed fish *Clarias batrachus*.

The total cholesterol content is liver, kidney, testis, and ovary of the control fish has been analysed to be 6.54± 0.73; 10.54 ±0.73; 3.43± 0.64 and 4.27± 0.42 mg/g wet tissue respectively. As evident the highest account of total cholesterol has been found is kidney followed by liver, ovary, tests. A perusal of Table-3, reveals that a considerable increase in the level of total cholesterol occurs when the fishes are subjected to cadmium stress. It elevates to 9.17± 0.65 mg/g wet tissue in liver. Which is equivalent to increase 40.21%. In kidney its level has been estimated to be 3.76± 0.63 mg/g wet tissue which account to a loss of 47.43%. The total cholesterol level in the testis and ovary of the treated fish group increases by 9.62% and 44.02% which has been measured to be 3.76± 0.63 and 6.15± 0.66 mg/g wet tissue, all the organs reflect statistically highly significant ($P<0.001$) increases is the level of total cholesterol except testis which is non-significant.

Table-3: Profiles of tissue cholesterol (mg/g wet tissue) in the tissue of *Clarias batrachus* chronically exposed to cadmium chloride (2.31ppm) for 30 days. Values are mean ± SE of 5 observations.

Tissue	Control	Cadmium chloride exposed
Liver	6.54±0.73	9.17±0.65(+40.21)
Kidney	10.54 ±0.73	15.54±0.83(+47.43)
Testis	3.431±0.64	3.76±0.63(+9.62)
Ovary	4.27±0.47	6.15±0.66(+ 44.02)

REFERENCES

1. **Ahmad, K. I., Begum, M. R., Sivaiah, S. and Rao, K. V. R. 1978.** Effect of malathion on free amino acids, total proteins, glycogen and some enzymes of pelecypod, *Lamellidens marginalis* (lamarck*). Proc. Ind. Acad. Sci. (Anim. Sci.)*. **87B(12):** 377-380.
2. **Ahsan, S.M.I. and Ahsan, S.N. 1988.** Ovarian cholesterol levels in anabas testudineus exposed to long term sub lethal concentration of aldrin. *The Indian Zoologist.* **12(3&4):** 339-343.
3. **Argese, E., Marcomini, A., Miana, P., Bettiol, C., Perin, G., 1994.** Submitochondrial particle response to linear alkylbenzene sulfonates, nonylphenol polyethoxylates and their biodegradation derivatives. *Environ. Toxicol. Chem.* **13:**737-742.
4. **Bainy, A.C.D. Saito, E, Carcalho, P. S. M. Junqueira, V. B. C. 1996.** Oxidative stress in gill, erythrocytes,

liver and kidney of Nile tilapia (*Oreochromis niloticus*) from a polluted site. *Aquat. Toxicol*. **34:**151-162.

5. **Bakthavathsalam, R., R. Ramlingam and A. Ramaswamy. 1984.** Histopathology of liver kidney and Intestine or the fish *Anabas testudineus* exposed to Furadon. *Environ and Ecol*. **2(4):**243-247.
6. **Banerjee, S. Bhattacharya, S. 1994.** Histopathology of kidney of *Channa punctatus* exposed to chronic non lethal level of elsan, mercury and ammonia. *Ecotoxicol. Environ. Saf*. **29:**265-275.
7. **Campbell, J.W. 1991.** Excretory nitrogen metabolism in Experimental and Metabolic Animal Physiology. Comparative Animal physiology, 4 edition (ed. C.L. Prosser), Wiley-Interscience, New York,pp. 277-324.
8. **Carbis, C. R. G. T. Rawlin, G. F. Mitehell, J. W. andersonand I McCauley, 1996.** The histopathology of carp, *Cyprinus carpio* L. exposed to microcystins by gage, immersion and intraperitoneal administration. *J. Fish Dis,* **19:** 199-207.
9. **Dowine, N.M. and Heath, R.W. 1970.** Basic statistical methods, Harper International Edition, New York. Evanston and London. 129-131.
10. **Draper R.P. Trimbret J.A. 1996** Urinary Creatine as a potential marker of testicular damage: Effect of Vasectomy. *Reprod. Toxicol*. **10(1):** 79-85.
11. **Dubowski, K. M. 1962.** An o-toluidine method for body fluid glucose determination. *Clein. Chem*. **8:** 215-235.
12. **Ellis EA Roberts RJ *et al*. 1978.** Fish Pathology Bailliere Tindall London. p. 318
13. **Finney DJ. 1971.** Probit Anslysis, Cambridge University press, New York. p.337.
14. **Fisher, R. A. 1950** Statistical Methods for Research Worker. IIth edition. Oliver and Boyd, London.
15. **Folch, J.; Lees, M.; Slone -Stanley, G. H. 1957.** A simple method for isolation and purification of total lipids from animal tissues. *J. Biol. Chem.* **226:** 497-507.
16. **Ghosh, R. 1990.** Toxicity of Carbamate pesticide, carboturan (technical grade) to the freshwater fish. *Heteropneustes fossilis* (Bloch) and *Clarias batrachus* (Linn). *Proc. Ind. Sci. Cong. Ass. Cochin. Abstr.*
17. **Haux C and Larsson A. 1982.** Influence of inorganic lead on the biochemical blood composition in the rainbow trout, *Salmo gairdneri*, *Ecotoxicol. Environ, Saf.* **6:**28-34.
18. **Katti, S. R. and Sathyanesan, A.G. 1983** Lead nitrate induced changes in lipid and cholesterol levels in the freshwater fish *Clarias batrachus. Toxicology Letters*. **19:**93-96.
19. **Lagwerff, J. V. and Specht A. N. 1970.** Contamination of roadside soil and vegetation with cadmium, nickel, lead and zinc. *Environ, Sci. Technol.* **4:** 583-586.
20. **Ohnishi, T., Suzuki, T., Ozawa, K. 1982.** A comparative study of plasma membrane magnesium ion ATPase activities in normal, regenerating and malignant cells. *Biochim. Biophys, Acta.* **684:**67 - 74.
21. **Oliveira Ribeiro CA, Filipak Neto F. *et al.* 2006.** Hematological findings in Neotropical fish *Hoplias malabaricus* exposed to subchronic and dietary doses of methylmercury, inorganic lead and tributyltin chloride. *Environmental Research*. **101:** 74-80.
22. **Popjak G. and Grant J.K. (eds) 1963.** The control of Lipid Metabolism. Acad. Pr. Ny.
23. **Prakash, E, Agrwal, V.P. and Nandini. 1987.** Trds life science **2(1):**51-54.
24. **Sastry K.V. and Shukla, V. 1990.** Toxic effects of cadmium on some biochemical and physiochemical parameters in the teleost fresh *Channa punctatus. Bio. Jour.* **2(2):** 325-332.
25. **Tilak, K.S. K. Veeraiah and K. Yacobu.** Studies on histopathological changes in the gill, liver and kidney of *Ctenopharyngodon idellus* (Valenciennes) exposed to technical fenvalerate and EC 20% *Pollut. Res*. **20:**387-393.
26. **Wohl M.G. and Good hart R.S. (Eds). 1968.** Modern Nutrition in Health and disease Lea and Febiger, Philadelphia.
27. **Wong, P.T.S.; Silverberg, B.A., Chan. Y.K. and Hodson, P.V. 1978.** Lead and aquatic biota. In: Biogeochemistry of lead in the environment, Part B (ed. J.C. Nriagu), Elsevier, North Holland.

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 281-282
An International Biannual Refereed Journal of Life Sciences

23(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Medicinal plant of wet land area of Supaul district

Bimala Kumari*
Department of Botany, B.N.M.University, Madhepura, Bihar, India

Received : 17th January, 2020 ; Revised : 17th February, 2020

Abstract : Wetlands are area between terrestrial and aquatic ecosystem. Fresh water wetlands are important in regulating water quality, regulating flood, control erosion, transfer sediments, filter pollutants and a source of food, fodder, fertilizer, irrigation medicine and several other flora of economic importance.

Key words: Medicinal plant, Supaul district, terrestrial and aquatic ecosystem.

INTRODUCTION

Freshwater ecosystems are the abodes of tremendous diversity of microorganisms, molluscs, arthropods, annelids and other invertebrates, amphibians, fishes, reptiles and aquatic micro- and macrophytes. Wetlands are significantly associated with many types of waterfowls and migratory birds to sponsor certain vital events of their life cycle. The wide array of living organisms in wetlands has inspired many scientists to study the micro- as well as macro-biodiversity and their various attributes, status and economic importance. Among the different types of living organisms, algae - both benthic and planktonic are probably important resources in supplying food webs in wetlands. Approximately 175,000 species of living organisms are known from freshwater sediments with unique properties of producing and processing organic carbon and also fixing and recycling nitrogen. Surface flow wetlands are valued highly for their high nutrient retention potential and their unique biodiversity. The approaches to decipher the functional aspects of wetland biodiversity in relation to nitrogen fixation, productivity, decomposition activity, pollution mitigation, microbial turnover etc. have presently gained much importance.

*Corresponding author :
Phone : 9402565435
E-mail : bimlacool@gmail.com

MATERIAL AND METHOD

A survey was conducted in five villages Mugrar, Kishanpur, Chikni, Malhad and Umarganj of Supaul District. Water remains logged around the year in these chours as river Koshi contribute. Luxuriant growth of macrophytes was observed in these chours. Medicinal Value and local name of these medicinal plants were acknowledged by the local people. Macrophytes were borrowed to laboratory and identified with the help of standard Monographs. Herbariums were prepared from identified flora. Plants with Medicinal value were identified.

RESULT

Table 1- Altogether eleven plants from studied wet lands were identified for their medicinal value.

Sl. No.	Name of Plant	Local name	Family	Medicinal use
1	*Marsilea quardifolia*	Charpatia	Marsileaceae	Paste of leaf used in snake bite
2	*Nelumbo nucifera*	Kamal	Nelumbonaceae	Decoction of flower use as cardiac and liver tonic
3	*Nymphaea pubescens*	Koka	Nymphaeaceae	Decoction of leaf in irregular menstruation.
4	*Ipomea aquatic*	Behaya	Convolvulaceae	Decoction of leaf is blood purifier.
5	*Nymphoides indica*	Panhar	Menyanthaceae	Paste of leaf used in jaundice.
6	*Polygonum Glabrum*	Jatmirch	Polygonaceae	Decoction of whole plant used in fever.
7	*Hygrophila Spinosa*	Surya Kenta	Acanthaceae	Decoctions of whole leaf treat Anemia.
8	*Hydrilla Verticillata*	Chigridal	Hydrocharitaceae	Decoction of leaves heal ulcer.
9	*Ceratophyllum demersum*	Coontail	Ceratophyllaceae	Decoctions of leaf regulate bile secretion.
10	*Eclipta Alba*	Jhucka	Compositae	Leaf extract is used as hair tonic.
11	*Bacopa Monnieri*	Brahmni	Scyophulariaceae	Decoctions of leaf with Honey strengthen nervous system.

CONCLUSION

All together eleven medicinal plant described, one belonging to pteridophite, five dicote and five to the monocot and two plants describe from the same family Nymphaeaceae.

ACKNOWLEDGEMENT

The author is thankful to her guide Dr.Shikha Choudhury, Associate Prof., Dept. of Botany, B.N. Mandal University and Sri O.P Sharma, Madhepura (Saharsa), Bihar for their technical guidance and encouragement. The author is also thankful to Prof. (Dr.) Mahendra Prasad, Retd. Professor and Head, Department of Zoology, Ranchi, University Ranchi, Jharkhand for his valuable co-operation in preparing the manuscript for publication.

REFERENCES

1. **Achariyar K.R. & C.T. Mudaliyar, 1921.** A Hand Book of Some South Indian Grasses. Madras.
2. **Ahmad Iqbal *et al.*-2006.** Modern phytomedicine (Turning Medical Plants in to Drugs) Wiley- VCH verlag G mb H & Co. K GaA weirheem.
3. **Anderson, T. 1863.** On the flora of Bihar & the mountains of Parasnath, *J. Asiat. Soc. Beng.* **327:**189-218.
4. **Anderson, C.J., and B.C. Cowell. 2004.** Mulching effects on the seasonally flooded zone of waste-central florida. USA wetlands. **24:**811-819.
5. **Anonymous, 1948-1972.** Weath of India (Raw materials), Vols. **1-9**, C.S.I.R., New Delhi.
6. **Anonymous, 1962.** Recommendation made by the members of the summer school in Botany, Kodaikanal. *Bull. Bot. Surv. India.* **4:**303-304.
7. **Arber, A. 1937.** The interpretation of the Hower, a study of some aspects of morphological thoughts *Bio.* **12:**157-184.
8. **Babu, C.R. 1977.** Herbaceous Flora of Dehara Dun, C.S.I.R. New Delhi.
9. **Brooks, R.P., D.H. Wardrop, C.H. Cole, and D.A. Cambell, 2005.** Are we purveyors of wetlands homogeneity? A model of degradation and restoration to improve wetlands mitigation performance. *Ecological engineering.* **24:**331-340.

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 283-284
An International Biannual Refereed Journal of Life Sciences

24(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of automobile pollutions on chlorophyll content of avenue trees along state highway between Saharsa to Supaul

Rupak Kumari*
Department of Botany, RJM College, Saharsa, B.N.M.University, Madhepura, Bihar, India

Received : 22nd December 2019; Revised : 21st February 2020

Abstract: Automobile exhaust contains a large number of air pollutions. Among the important constituents of pollutants are carbon monoxide, unburnt hydrocarbons, Nitrogen oxide, particulates and lead. These vehicle pollutions adversely affect Avenue trees by chlorosis, damage leaves and reduces Nutrient uptake. In the present study chlorophyll content of four Avenue trees- *Ficus religiosa, Ficus bengalensis, Delonix regia* and *Cassia fistula* was estimated and compared with that of trees far away from the road side.

Key words: Automobile, pollutant, chlorosis, Avenue trees

INTRODUCTION

Automobiles are "necessary evil" while they have made living easy and convenient, they have also made human life more complicated and vulnerable due to toxic emissions. Human beings, animals and vegetation are being adversely affected by the air pollutants emitted from vehicles.

Petrol engine automobiles emit carbon monoxide, unburnt hydrocarbons, nitrogen oxides, sulphur oxides, particulates and lead while diesel engine automobiles emit unburnt hydrocarbons, Nitrogenoxides, sulphur oxides smoke and odour.

The highest emission rate occurs during motor idling, deceleration and at slower speed. During idling and slow speeds the greatest emission is of carbon monoxide and Hydro carbons with Nitrogen.

These pollutants cause chlorosis, leaf damage leaf curling and other several adverse effects on vegetation.

*Corresponding author :
Phone : 8207620024
E-mail : rupak852201@gmail.com

MATERIAL & METHODS

The study area was selected from Saharsa to Supaul (40km). Branches of selected Avenue trees (*Ficus religiosa, Ficus bengalensis, Delonix regia* and *Cassia fistula*) were collected in summer, rainy and winter seasons.

Chlorophyll content of leaves from each plant was estimated in laboratory. 250 mg of fresh leaves were cut into small pieces and transferred in 25 ml 80% acetone. It was incubated in dark for 24 Hrs. absorbance at the extract was taken at 645 nm and 663 nm with spectrophotometer. Chlorophyll was estimated by the formula (Arnon, 1949).

$$\text{Total Chlorophyll}\left(\frac{\text{mg}}{\text{gm}}\text{Fw}\right) = \frac{20.2(A645) + 8.02(A663)}{1000X\ Fw} X\ V$$

A_{645} = Absorbance at 645 nm
A_{663} = Absorbance at 663 nm
V = Volume of Chlorophyll extract
Fw = Weight of fresh leaves

RESULT

Chlorophyll estimation of four different plants from road side between Saharsa and Supaul State Highway and chlorophyll estimation of same plants far away from road side was estimated in different seasons - summer, winter and rainy season. The result is tabulated in Table No.-01 and 02.

Table No.-1 Chlorophyll estimation at Road side Trees

SN. No.	Botanical Name	Common Name	Family	Season	Total Chlorophyll
1	*Ficus religiosa* L.	Peepal	*Moraceae*	Summer Winter Rainy	0.0179 0.0181 0.0184
2.	*Ficus bengalensis* L.	Burgad	*Moraceae*	Summer Winter Rainy	0.0226 0.0230 0.0238
3.	*Delonix regia* Raf.	Gulmohar	*Caesalpinaceae*	Summer Winter Rainy	0.0235 0.0241 0.0245
4.	*Cassia fistular* L.	Amaltash	*Ceasaltiniaceae*	Summer Winter Rainy	0.0205 0.0209 0.0214

Table No.-2 Chlorophyll estimation of Trees far away from Road side

Sl. No.	Botanical Name	Common Name	Family	Season	Total Chlorophyll
1	*Ficus religiosa* L.	Peepal	*Moraceae*	Summer Winter Rainy	0.0187 0.0188 0.0191
2.	*Ficus bengalensis* L.	Burgad	*Moraceae*	Summer Winter Rainy	0.0233 0.0236 0.0242
3.	*Delonix regia* Raf.	Gulmohar	*Caesalpinaceae*	Summer Winter Rainy	0.0240 0.0247 0.0251
4.	*Cassia fistular* L.	Amaltash	*Ceasaltiniaceae*	Summer Winter Rainy	0.0213 0.0216 0.0222

CONCLUSION

A large number of Truck, Bus, Car, Auto, Tractor and several other heavy vehicles continuously passes through State Highway between Saharsa and Supaul. The pollutant released from these vehicles adversely affect avenue trees present at the road side. The Chlorophyll content of leaves decreases and chlorosis may occur. The chlorophyll content of leaves is greatly effected in summer as the air remain hot and pollutant passes upwards while in winter season the air remain cold so the density of natural gasses of air is higher hence, pollutant passes upward slowly. In rainy season due to rain most of the heavy pollutant including Nitrogen and Sulphur dioxide dissolve in water and settle down at ground. As the pollutant are very low in country side so the chlorophyll content of these trees are higher.

REFERENCES

1. **Guderian, Robert. 1977.** Air pollution, Ecological studies vol.-22, publisher-Springer, New York.
2. **Mansfield, T.A.** Effect of air pollutants on plants. Publisher- Cambridge University Press.
3. **Troyanowsky, C. 1986.** Air pollution and plants. Publisher- VCH.
4. **Sharma, A. & A. Roychowdhury. 1996.** Slow Murder- The Deadly story of Vehicular pollution in India. Center of Science & Environment, New Delhi P-148.
5. **Perry, G. 1982.** Introduction to Environmental Toxicology, Elsevier, Netherland.

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.285-286
An International Biannual Refereed Journal of Life Sciences

25(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of atmospheric pollutants on lichens in different zones in Ranchi city

Shweta Kumari*
University Department of Zoology, Ranchi University, Ranchi, Jharkhand, India

Received : 26th December, 2019 ; Revised : 19th February, 2020

Abstract : Atmospheric pollutants concentration effects lichen. The presence of lichens on the bark of trees and on the walls at all the three sampling sites indicates that the level of concentration of gaseous pollutants is not alarming.

Key words: air pollution, lichens, cuticle, stomata

INTRODUCTION

Lichens comprise a unique group of plant that consists of two unrelated organism, a fungus and an alga, growing together in a close symbiotic association. Lichens have long been recognized as sensitive indicators of environmental conditions. The decline of lichens around city centers is attributed to air pollution.

Lichens lack a protective cuticle and stomata. They have direct dependence on atmosphere for nutrients (poikilohydrous nature), longevity, stability, and perennial condition of thallus. They have high degree of sensitivity to changes in substrate pH and pollutants which makes them ideal organisms to study the impact of air pollution. Lichens are good accumulators of many elements, particularly heavy metals and radio nuclides.

Some sensitive lichen species develop structural changes in response to air pollution including reduced photosynthesis and bleaching. Pollution can also cause the death of lichen algae, discoloration and reduced growth of the lichen fungus, or kill lichen completely. Hence the species of lichens present in a location and the concentration of pollutants measured in those lichens can tell us a lot about air quality.

Lichens are the plants that occur in most adverse conditions of climate and substrate and like little sponges take up everything that comes their way, including air pollution. They are, therefore, suitable for use as pollution monitors. Extensive work has been done in the temperate Europe and North America for assessing the usefulness of lichens as bio-indicators of atmospheric pollution. A qualitative scale has also been constructed on the basis of frequency cover of lichens for the estimation of air pollution. It has been found that pollutants consisting of SO_2, NO, NO_2, O_3, etc. have different effects on the lichen species.

Importance of lichens

Lichens are universally distributed organisms occurring in varied climatic conditions ranging from the poles to the tropics in earth. They may look like crust, spreading rapidly over the surface (crustose lichens) or leafy and loosely attached to the surface (foliose lichens)

*Corresponding author :
Phone : 82927 14318
E-mail : shwetasinhaa2018@gmail.com

and branched and shrubby, hanging from tree twigs or branches, with a single attachment (fruticose lichens). A lichen is a composite organism that arises from algae or cyanobacteria (or both) living among filaments of a fungus in a symbiotic relationship. The combined life form has properties that are very different from the properties of its component organisms. The study of lichen remains quite neglected throughout the world, through they together with mosses form dominant organisms in ecosystem covering over 10% of the earth terrestrial habitats, particularly at higher elevations. Lichens with cynobacterial blue green symbionts, contribute significantly for forest nitrogen fixation.

The sensitivity of lichens to air pollution is reflected in various ways such as decline in diversity, absence of sensitive species, and morphological, anatomical and physiological changes. The high sensitivity of lichens is related to their physiology and morphology. The stronger the dependency of the mycobiont on the photobiont, the more sensitive the lichen is to air pollution. The photobiont upon exposure to air pollution may use metabolic energy for repair of its cellular structures that would otherwise be used for maintenance of its photosynthetic activity. It, thus, leaves less metabolic energy available for the mycobiont. The alteration of the balance between the photobiont and mycobiont can lead to the breakdown of the symbiotic association. Therefore, lichen decline may result not only from the accumulation of toxic substances, but also from altered nutrient supplies that favor one symbiont over the other.

The quality of air quality is getting worse all over the world. The presence and absence of lichens and different species around different pollution sources has been used for problems connected to air pollution.

Sampling sites

Ranchi city is located at 23.23^{0}N latitude & 85.23^{0}E longitude and around 654 meters above sea level in the plateau of Ranchi. The total area covered by Ranchi urban agglomeration is 35 sq. km. The population is about 846, 454.

Sampling location:

Ashoknagar - Residential zone,
Kantatoli - Commercial and Residential zone
Tupudana - Industrial zone

The Sampling and identification of Lichens

The lichen specimens were collected with chisel and hammer. The host tree type, location of host tree, and location of the lichens thallus (on trunk, branch, twigs or leaves, soil or rock substratum) together with other ecological notes were recorded. The collected specimens were investigated for taxonomic purposes.

RESULTS

Both crustose and foliose variety of lichens was seen at the sampling sites. Table below lists the types of substratum where lichens were observed at the three sampling sites.

Table: Types of substratum where lichens were observed at the three sampling sites.

Sl. No	Type of substratum	Kantatoli	Ashoknagar	Tupudana
1.	Bark of trees	Present	Present	Present
2.	Leaves of trees	Absent	Absent	Absent
3.	Wall both plastered and unplastered	Present	Present	Present
4	Soil	Absent	Absent	Absent
5.	Rock	Absent	Present	Present

Effect of Air Pollution on Lichens

The presence of lichens on the bark of trees and on the walls at all the three sampling sites indicates that the level of concentration of gaseous pollutants is not alarming.

ACKNOWLEDGEMENT

The author is thankful to her guide, Prof. M.K. Jamuar & Prof. S.M. Shamim, Head, Department of Zoology, Ranchi University, Ranchi, Jharkhand for their holistic support and guidance in my research work.

REFERENCES

1. **Slack, N.G. 1988.** The Ecological Importance of lichen and Bryophyte. *Bib. Lichenol.* **30:** 25 - 53.
2. **Awasthi, D.D., 2000.** A hand book of Lichens Bishen Sing Mahendra Pal Singh Publication.
3. **Awasthi, D.D., 2000.** Lichenology in Indian subcontinent, a supplement to "A Hand Book of Lichenss". Published Bishen Singh Mahendra Pal Singh, India.
4. **Nash, T.H.(Ed.). 1996.** Lichen Biology. Cambridge Univ. Press. 303pp.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.287-290
An International Biannual Refereed Journal of Life Sciences

26(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

A comparative study of monthly and seasonal variations in productivity of molluscs in Hardia wetland of Saran District of North Bihar, India

Chitralekha Sinha*
P.G. Department of Zoology, J.P. University, Chapra, Bihar, India

Received : 27th December, 2019 ; Revised : 20th February, 2020

Abstract : Other faunastic survey of the wetland was also made. It was focused on macro invertebrates: arthropods, molluscs and annelids. Sharma *et al*., (1983)[1] have studied molluscan fauna, Singh *et al*., (1994)[2], Sharma and Munshi (1995)[3] have studied other fauna of wetlands also. Altogether 85 species of macro invertebrate were recorded during the present investigation belonging to three major groups-arthropods, molluscs and annelids. The wetland is rich in molluscs fauna. Its abundance is due to profuse macrophytic growth. Rich organic matter in the bottom provides good habitat for molluscs growth. The macro invertebrates abundance in the wetland is directly controlled by the seasonal appearance and disappearance of the macro vegetation of the area, nature of bottom, rainfall and flooding (Subba Rao, 1989)[4]. Molluscs were represented by the members of single class Gastropoda. Gastropoda were represented by 11 species belonging to five families. The molluscan density varied between 29 no./m^2 (August) and 160 no./m^2 (Feb.). An increasing trend in the molluscan density was observed in winter and late monsoon while decreasing trend was observed in summer and early monsoon season. The maximum production of molluscs during post monsoon season may be due to high macrophytic growth and availability of dead and decaying organic matter (Reavell, 1980)[5].

Key words: Molluscan density, monthly & seasonal variations, organic matter, Hardia wetland, increasing & decreasing trend

INTRODUCTION

This study is about molluscs of Hardia wetland. Literature related to the productivity of molluscs are scanty but notable among those are of Haniffa (1980)[6] on *Pila globosa*, Vivekananda *et al*., (1974)[7] on the energy transformation of *Pila globosa*, Raut and Bhattcharya, (1986)[8] on the growth indicates of *Bellamya bengalensis*, Singh, (1990)[9] on density, biomass and production of apple snail and *Bellamya*. Sharma & Munshi (1995)[3] on some molluscs of Kawar Lake and others.

*Corresponding author :
Phone : 8210708417
E-mail : chitralekhasinhateacher@gmail.com

The present study relates to the biomass and secondary productivity of freshwater molluscs of Hardia Wetland.

MATERIALAND METHODS

Molluscs were collected by hand and Ekman Dredge. The samples associated with macrophytes were collected using a nylon cloth net (75-80 meshes/cm^2). The collected samples were brought to laboratory, washed, counted and weighed then they were dried at 65^0C until weight constancy for estimating biomass. The secondary productivity was determined by dry weight method

(Haniffa, 1978)[10] and expressed as g. dry wt/m^2/year and g dry wt/m^2/day.

Molluscs were collected by specially designed cube of 1mx1mx1m size. The entangled molluscs were collected with the help of nylon net. The wet weight of the collected molluscs were taken. Specimen were preserved in 10% formalin. They were identified in the lab with the help of available literature of Needham and Needham (1966)[11], Ward and Whipple (1959)[12] and Subba Rao (1989)[4].

RESULTS AND DISCUSSION

Monthly samples of molluscs comprising *Bellamya bengalensis* were collected and biomass and secondary productivity were determined during 2011-13 (Table:1)

The secondary productivity of *Bellamya bengalensis* during Nov'11 to Oct'12 was recorded highest 93.2 g/m^2/yr and 0.250 g/m^2/day maximum biomass 115.80 g dry wt./m^2 during Sept' 12 and minimum biomass 22.60 g dry wt./m^2 during May' 12. During the second year of investigation November' 12 to October' 13 the secondary production was also found to be highest 77.40 g/m^2/yr and 0.212 g/m^2/day with maximum biomass 108g dry wt./m^2 during October' 13 and the minimum biomass 30.60g dry wt./m^2 during February 2013. (Table:1, Fig:1)

Table: 1: Biomass and Secondry Productivity of *Bellamya bengalensis* of Hardia wetland in (2011-13) values in g dry wt./m^2

	Biomass	
Month	2011-12	2012-13
Nov	72.2	78
Dec	76.5	62
Jan	60.4	42.3
Feb	62	30.6
Mar	55	40.2
Apr	42	38.7
May	22.6	*
Jun	*	*
Jul	30.4	66
Aug	68.4	78.8
Sep	115.8	90.6
Oct	88	108
Productivity	93.2 g/m^2/yr.	77.40 g/m^2/yr.
	0.255g /m^2/day	0.212g/m^2/day

Note: "*" Site was dry.

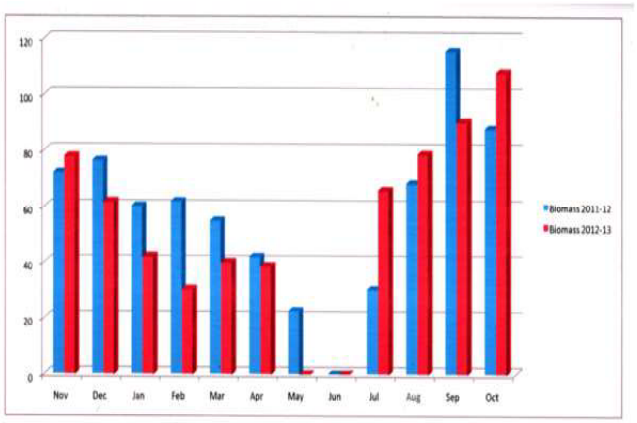


FIG.1 : Biomass and Secondary Productivity of Bellamya bengalensis of Hardia wetland in (2011-13) values in g. dry wt./m^2.

DISCUSSION

The freshwater molluscs form a high proportion of biomass and occupy a strategic position in the food chain and energy flow of the freshwater ecosystem. The maximum production and biomass of *Bellamya bengalensis* during post monsoon season may be attributed to high macrophytic growth and presence of decaying organic matter at the bottom of the wetland. Similar observations also have been made by Reavell (1980)[5] in British freshwater and Sharma and Munshi (1995)[3] in Kawar lake wetland. Minimum biomass production was observed during summer months it may be due to predation and migration of species. Similar observation has been made by Singh, (1990)[9] and Sharma and Munshi, (1995)[3]. Biomass and growth of mussel are influenced by many factors such as temperature food intake, salinity and population density. Temperature is a vital factor for the development of molluscs. *Bellamya* particularly shows distinct temperature preferences (Kulkarni *et al*., 1985)[13]. Although observation of Pidgaiko *et al*., (1972)[14] favours that temperature variation could have either positive or negative effect on secondary production depending upon geographic location and basin morphometry.

REFERENCES

1. **Sharma, U.P., Roy, S.P. and Rai, D.N. 1983.** Aquatic molluscs of Bhagalpur, *Biol.Bull. India* **5(2):** 147-155

2. **Singh, D.K., Sinha, A.K., Baruah, A. and Sharma, U.P. 1994.** Freshwater Gastropods of Kawar Lake wetland (Begusarai) North Bihar, India. *J. Freshwater Biol.* **5(1):** 41-47

3. **Sharma, U.P. and Munshi, J.S.D. 1995.** Ecology, conservation and Management of Kawar Lake. A major tropical wetland of South East Asia. 189p.

4. **Subba Rao, N.V. 1989.** Handbook of freshwater molluscs of India. *Zool. Survey of India*, Calcutta. 289p.

5. **Reavell, P.E. 1980.** A study of the clients of some British freshwater gastropods. *J. Conchol*. **30:**253-271.

6. **Haniffa, M.A. 1980.** Studied on life span, growth increments and energy budget of the freshwater Snail *Pila globosa* (Swainson) in an artificial pond. *Proc. Ind. Acad. Sci. (Animal Sci.)* **89(3):** 275-286.

7. **Vivekananda, E., Haniffa, M.A., Pandian, T.J. and Raghuraman, R. 1974.** Studies on energy transformation in the freshwater snail *Pila globosa*. Influence of feeding rate. *Freshwater Biol.* **4:**275-280.

8. **Raut, S.K. and Bhattacharya, R.K. 1986.** Growth Indices of Bandel pond snail *Bellamya bengalensis*. *Environ. and Ecol.* **4(2):** 295-296.

9. **Singh, R. 1990.** Density, biomass and secondary productivity of the prosobranch *Pila* and *Bellamya bengalensis* in a feeder tank Jamalpur. *J. Freshwater Biol.* **2(3)**: 251-256.

10. **Haniffa, M.A. 1978.** Secondary productivity and energy flow in a tropical pond. *Hydrobiol*. **59:** 839-928.

11. **Needham, J.G. and Needham, P.R. 1966.** A Guide to the study of Freshwater Biology. Holden Day Inc. San Fransisco. 108p.

12. **Ward, H.B. and Whipple, G.C. 1959.** Freshwater Biology. (ed. W.T. Edmondson) John Wiley and Sons. Inc. New York.

13. **Kulkarni, K.M., Baramatiwalla, D.O. and Holawala, D.Z. 1985.** To acclimatation temperature response to heat tolerance of the Snail *Bellamya bengalensis*. *Environ. and Ecol.* **3(2):** 254-256.

14. **Pidgaiko, M.L., Grin, V.G., Kititsina, L.A., Lenchina, L.G., Polivannaya, M.F., Sergeva, O.A. and Vinogradskaya, T.A. 1972.** Biological productivity of Kurakhov's power station cooling reservoir. In: Productivity problems in freshwaters. Led Kajak and Illowskas. Proc. of IBP, UNESCO. *Polish. Sci. Publ.*

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.291-294
An International Biannual Refereed Journal of Life Sciences

27(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Comparative skeletal anatomy of the Human, Rabbits and Horses

Shreshta Sinha*
Sarvodaya Senior Secondary School, Kota, Rajasthan, India

Received : 12th January, 2020 ; Revised : 20th February, 2020

Abstract : This paper looks upon the comparative skeletal difference between three different members of the class Mammalia. When different animals of the same class are looked upon, the bone positioning and count ratio are similar in some spheres. Human anatomy was taken as the primary reference in this study. The three mammals taken in the study are human (*Homo sapiens*), European Rabbits (*Oryctolagus cuniculus*) and Horse (*Equus caballus*). Despite being the members of the same class, evolution led to the current difference in three. Survival conditions and habitat have also played a dominant role in the bone count or number. The skeletal difference was majorly noticed in the bone number and density.

Key words: Bones, mammals, skeletal system, density

INTRODUCTION

The skeleton is the body part that provides support, shape, and protection to delicate organs and soft tissues of animals. There are many different skeletal types, the first being the exoskeleton (stable outer shell of an organism), the second one being the endoskeleton (the support structure inside the body), the hydro skeleton (flexible skeleton supported by fluid pressure) and the cytoskeleton which present in the cytoplasm of all the existent cells in the world. The skeletal system of mammals is divided into two planes namely axial plane bones and appendicular plane bones. Axial plane bones consist of all the bones in the head and trunk. It mainly consists of the braincase (cranium) and the backbone and ribs, and it serves primarily to protect the central nervous system. Appendicular plane bones consist of all the bones in upper and lower spheres mainly limbs and their girdles. The different planes have a different number of bones in each subcategory. All the 6,595 of the recognized mammalian species reflect common features including anatomy. The animals referred to in the study reflect each plane in them clearly. The mammalian skeletal system shows several advances over that of lower vertebrates.

Bones develop and grow differently in mammals. Growth of bones happens in parts of cartilage between epiphyses (The rounded end of a long bone) and diaphysis (The tubular shaft that runs between the proximal and distal ends of the bone). Mammalian skeletal growth is definitive, once the actively growing part of cartilage degenerates, growth in length stops. Just like all bony vertebrates, of course, there is a continual renewal of bone throughout life. The epiphyseal ossification has an advantage because the bones have strong articular (joint correlating) surfaces before the skeleton becomes mature. In general, the skeleton of the adult mammal has less structural cartilage than that of a reptile. Besides, there are skeletal portions derived from the gills of primitive vertebrates, termed as the visceral skeleton. Visceral elements in the mammalian skeleton include the Hyoid Apparatus supporting the tongue, the auditory ossicles of the middle ear, and jaws. The axial skeleton under the cranium part in mammals generally has remained intact during the course of evolution. The vast majority of mammals have seven cervical vertebrae (in the neck), there are few exceptions like the

*Corresponding author :
Phone : 9358868347
E-mail : sinhashreshta7@gmail.com

Sloth (Folivora). Thoracic vertebrae consist of ribs and are variable in number. The anterior ribs converge toward the ventral midline to join with the sternum, breastbone, forming a semirigid thoracic cup for the protection of the lungs and heart. Behind the thoracic region are the lumbar vertebrae, they range from 3 to 21 in number. The two main cervical vertebrae are the atlas and axis. Joints of these two bones allow complex movements of the head on the trunk Mammals have no lumbar ribs. Caudal (tail) vertebrae range from 5 to 50 (fused elements of the human coccyx). The skull of mammals differs markedly from that of reptiles because of the size and functioning of the brain. This information makes our way further into the discussion.

DISCUSSION

Comparative study

The subcategories given below have detailed information on the endoskeletonal structure of each considered species. After studying the comparative analysis of the animal's skeletal system a graph has been drafted to depict the difference clearly.

Human (*Homo sapiens*)

The skeletal system is our body's central framework. It consists of bones and connective tissue, including cartilage, tendons, and ligaments. But the main focus we're undertaking is on bones. Bones provide the structure to our body. The adult human skeleton is made up of 206 bones. These include the bones of the skull, spine (vertebrae), ribs, arms, and legs. Bones are made of connective tissue reinforced with calcium and specialized bone cells. Most bones also contain bone marrow, where blood cells are made. Bones work with muscles and joints to hold our bodies together and support freedom of movement. This is called the musculoskeletal system. The skeleton supports and shapes the body and protects delicate internal organs such as the brain, heart, and lungs. Bones contain most of our body's calcium supply. The normal ranges for bone mineral density are 0.96-1.39 g/cm^3 for women and 0.92-1.35 g/cm^3 for men. The skeleton makes up about 30-40% of an adult's body mass. The skeleton's mass is made up of a non-living bone matrix and many tiny bone cells. Roughly half of the bone matrix's mass is water, while the other half is collagen protein and solid crystals of calcium carbonate and calcium phosphate.

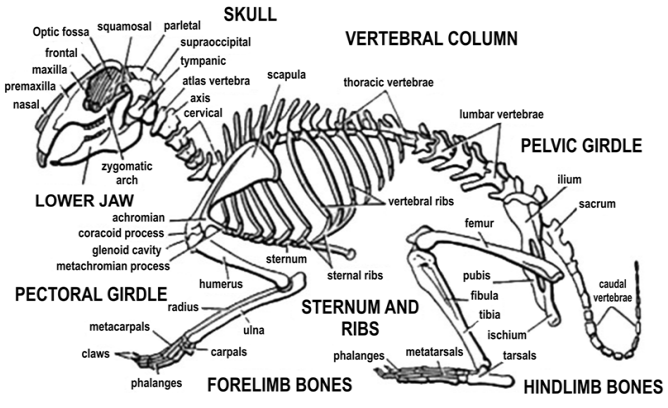
Table 1. The list of all the bones in human body with locations in their respective planes.

Human Body (206)			
Axial Skeleton (80)		Appendicular Skeleton (126)	
Skull 28	Torso 52	Upper Extremity 32 x 2 = 64	Lower Extremity 31 x 2 = 62
Paired Bones (11 x 2 = 22) 1. Nasal 2. Lacrimal 3. Inferior Nasal Concha 4. Maxiallary 5. Zygomatic 6. Temporal 7. Palatine 8. Parietal 9. Malleus 10. Incus	Paired Bones (12 x 2 = 24) 1. Rib 1 2. Rib 2 3. Rib 3 4. Rib 4 5. Rib 5 6. Rib 6 7. Rib 7 8. Rib 8 (False) 9. Rib 9 (False) 10. Rib 10 (False) 11. Rib 11 (Floating) 12. Rib 12 (Floating)	1. Scapula 2. Clavicle 3. Humerus 4. Radius 5. Ulna 6. Scaphoid 7. Lunate 8. Triquetrum 9. Pisiform 10. Hamate 11. Capitate 12. Trapezoid 13. Trapezium 14. Metacarpal 1 15. Proximal Phalange 1 16. Distal Phalange 1 17. Metacarpal 2 18. Proximal Phalange 2 19. Middle Phalange 2 20. Distal Phalange 2 21. Metacarpal 3 22. Proximal Phalange 3 23. Middle Phalange 3 24. Distal Phalange 3 25. Metacarpal 4 26. Proximal Phalange 4 27. Middle Phalange 4 28. Distal Phalange 4 29. Metacarpal 5 30. Proximal Phalange 5 31. Middle Phalange 5 32. Distal Phalange 5	1. Hip (Ilium, Ischium, Pubis) 2. Femur 3. Patella 4. Tibia 5. Fibula 6. Talus 7. Calcaneus 8. Navicular 9. Medial Cuneiform 10. Middle Cuneiform 11. Lateral Cuneiform 12. Cuboid 13. Metatarsal 1 14. Proximal Phalange 1 15. Distal Phalange 1 16. Metatarsal 2 17. Proximal Phalange 2 18. Middle Phalange 2 19. Distal Phalange 2 20. Metatarsal 3 21. Proximal Phalange 3 22. Middle Phalange 3 23. Distal Phalange 3 24. Metatarsal 4 25. Proximal Phalange 4 26. Middle Phalange 4 27. Distal Phalange 4 28. Metatarsal 5 29. Proximal Phalange 5 30. Middle Phalange 5 31. Distal Phalange 5
1. Frontal 2. Ethmoid 3. Vomer 4. Sphenoid 5. Mandible 6. Occipital	1. Hyoid 2. Sternum 3. Cervical Vertebrae 1 (atlas) 4. C2 (axis) 5. C3 6. C4 7. C5 8. C6 9. C7 10. Thoracic Vertebrae 1 11. T2 12. T3 13. T4 14. T5 15. T6 16. T7 17. T8 18. T9 19. T10 20. T11 21. T12 22. Lumbar Vertebrae 1 23. L2 24. L3 25. L4 26. L5 27. Sacrum 28. Coccyx		

Rabbit (*Oryctolagus cuniculus*)

The endoskeleton of the rabbit is chiefly formed of bone and the cartilaginous part is very little. Since there is a general tendency to increase in the size of the brain, the skull has a short posterior cranial part for lodging the brain and the long anterior facial part comprising mainly the jaws. In higher mammals, the facial part lies below the cranial part. A zygomatic arch on either side of the skull is formed by squamosal, jugal, and maxillary bones. The various bonesconstituting the cranium can be grouped into three segments, a posterior occipital segment, middle parietal segment, and anterior frontal segment. Basis phenoid is a flat, median triangular cartilage bone. Its apex joins anteriorly with the presphenoid and its broad posterior end is connected with the basioccipital by a thin plate of cartilage. It is a small, median wedge-like, triangular bone on the dorsal side in between parietals and supraoccipital. The parietal bones are a pair of thin, slightly arched bones protecting thedorsolateral brain. It is roughly a rectangular bone situated ventral to parietal on either side. The vertebral column of rabbit, like that of birds and lizards, is differentiated into the following five regions. The total number of vertebrae in the rabbit is about 45-47. The pectoral girdle consists of two separate halves placed dorsal to the anterior thoracic ribs in between the forelimbs, it supports forelimbs and protects the soft parts of the body from the ventral side. The bone of the upper arm is the humerus, which is a long bone having a proximal large rounded head for the articulation in the glenoid cavity of the scapula. The bones of the forearm are radius and ulna which are closely held together at the two ends so that they cannot move over each other. The wrist bones or carpals are 9 small bones arranged in two rows.The bone density of rabbits is around 1400 mg/cm^3.

Rabbit Skeleton



Horse (*Equus caballus*)

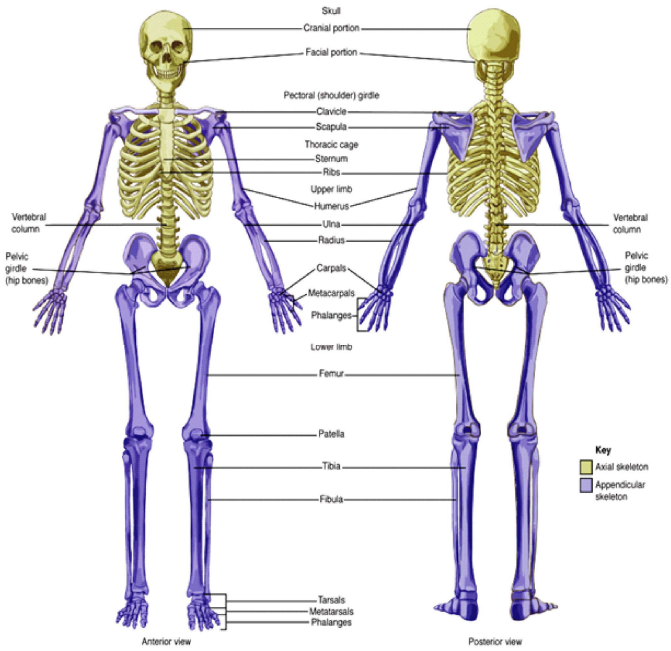
The horse's body contains 205 bones. The smooth alignment of these bones decides the horse's speed, movement, mechanics, and efficiency. The bones of the horse skeleton are held together with ligaments, tendons, and muscles. When the skeleton structure is in true aligned proportioned the joints work smoothly. One bone works about another. If the angle at which these bones are working is compromised, the joint becomes overstressed and injury to the tendons and ligaments become more likely. There are two main parts to the horses' skeleton, axial and appendicular. The axial skeleton protects the horse's vital parts and includes the skull, the ribcage, and the backbone. The appendicular skeleton supports the body and consists of the shoulders, pelvis, hind legs, and forelegs. The vertebral column usually contains 54 bones. The skull consists of 34 bones and contains four cavities (the cranial cavity, the orbital cavity, oral, and the nasal cavity). Bones of the lower limb, present in both the front and hind legs, include the Cannon bone, Splint bones, proximal sesamoid bones, long pastern (proximal or 1st phalanx), short pastern (middle or 2nd phalanx), coffin bones (distal or 3rd phalanx), and navicular bones (distal sesamoid). This is of particular significance in young horses, which at birth have a bone density of about 17%. This increases to about 74% at a year of age. Furthermore, bone is continually remodelling and adapting to periods of exercise and rest until horses reach skeletal maturity at the age of five to six years.

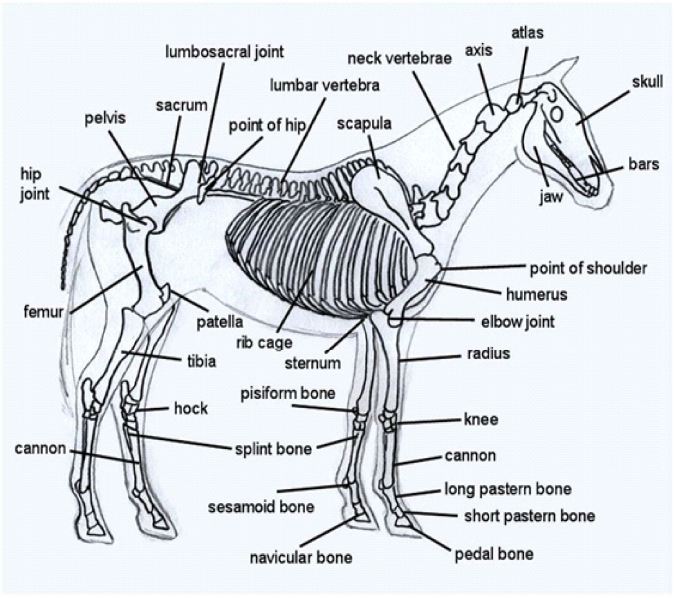
Table 2. The difference chart

	Human	Rabbit	Horse
Number of Bones	206	260	205
Axial Plane Bones	80	132	88
Appendicular Plane Bones	126	128	117

Common features in all three:-

1. Skulls contain around 29 bones that make the cranium, face, and ears of mammals.
2. The rib cage of the mammalian skeleton protects the heart, lungs and in larger mammals it supports the abdomen.
3. The pelvis connects the spine to the legs and can be used to identify the sex of a mammal.
4. Arms include the humerus, radius, and ulna bones plus the bones of the wrist and hand.
5. The leg bones are the femur, tibia, and fibula plus the tassels, metatarsals, and phalanges of the ankle and foot.





ACKNOWLEDGEMENT

The author wants to extend his deep sense of gratitude and thanks to Dr. Mrs. Supriya Sinha, Assistant Director, Dept. of Commercialization Relation, UT Southwestern Medical Centre, Dallas, Texas, U.S.A., & Prof. (Dr.) Mahendra Prasad, Retd. University Professor, Department of Zoology and former Dean, Faculty of Science, Ranchi University, Ranchi, Jharkhand for their motivation and academic support.

REFERENCES

1. **Irth, E.C., P.R. Van Weeren, D.U. Pfeiffer, J. Delahunt, and A. Barneveldt. 1999.** Effect of age, exercise and growth rate on bone mineral density (BMD) in third carpal bone and distal radius of Dutch Warmblood foals with osteochondrosis. *Equine Veterinary Journal Suppl.* **31:**74-78.
2. **Searfoss, G. 1995.** Skulls and Bones: A Guide to the Skeletal Structures and Behaviour of North American Mammals
3. **Cedo M Bagi, Edwin Berryman, Maria R Moalli, 2011.** Comparative Bone Anatomy of Commonly Used Laboratory Animals: Implications for Drug Discovery
4. **Stephen B Adams. 2019.** Components of the Musculoskeletal System of Horses
5. **https://www.britannica.com/animal/mammal/Skeleton**
6. **Aline Schafrum Macedo, Caroline Cezaretti Feitosa, Fernando Yoiti Kitamura Kawamoto, Paulo Vinicius Tertuliano Marinho, Ísis dos Santos Dal Bó, Bianca Fiuza Monteiro, Leonardo Prado, Thales Bregadioli, Gabriel Antonio Covino Diamante and Cassio Ricardo Auada Ferrigno. 2019.** Animal modelling in bone research-Should we follow the White Rabbit.
7. **Wei Yao. 2019.** Data for: Selective inhibition of progesterone in osteochondral progenitor cells, but not in mature chondrocytes, modulated subchondral bone structures.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.295-298
An International Biannual Refereed Journal of Life Sciences

28(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Profile of phytoplanktonic composition of Matashya Beez Prachhetra water body of Dighra, Muzaffarpur, Bihar, India

Kumar Abhinesh*
Department of Education, T.P Verma College, Narkatiaganj, West Champaran, Bihar, India

Received : 15th January, 2020 ; Revised : 26th February, 2020

Abstract: Although, the Matashya Beez Prachhetra water body of Dighra, Muzaffarpur is a fresh water low dimension pond used for fisheries yet its profile of planktonic composition has not been looked into so far. The planktonic composition of any water body is made up of both phytoplanktons which are the members of autotrophic group & zooplanktons as members of heterotrophic group of organisms. They construct the fundamental biotic framework upon which higher order organization of aquatic plants & animals rest. One way or the other, the planktons behave as food resource for the higher order fauna like annelids, mollusks, amphibians & fishes. They are also indicators of pollution level in the water body & hence treated as bioindicators. In the present investigation, the profile of phytoplanktons has been developed from the microscopic examination of stirred or agitated water samples in the laboratory under high resolution microscope & the numbers have been counted with Sedgewick Rafter plankton counter slide. Water samples have been randomly but punctually taken from different corners & mid points of the pond. For collection of phytoplanktons, special mesh of 50-80 µm size plankton net has been used. As many as three species Chrysophytes, three Chlorophytes (green algae), two Cyanophytes & have been isolated from these water samples as phytoplanktons.

Key words: phytoplanktonic composition, Chrysophytes, Chlorophytes (green algae), Cyanophytes, Matashya Beez Prachhetra pond, Dighra, Muzaffarpur, Bihar

INTRODUCTION

Plankton community is a heterogeneous group of floating plants (phytoplanktons) and animals (Zooplanktons) adapted to suspension in the sea and fresh coaler & usually forms directly or indirectly food for fishes & bigger animals. Fresh water planktons include representatives from the photosynthetic algae, bacillariophyceae (diatoms), myscophyceae (blue- green algae), chlorophyceae (green-algae) and occasionally other from the non-photosynthetic bacteria, and other fungi and among zooplankton, all classes of protozoa, (except sporozoa), rotatoria, entomostraca, some immature diptera, the gemmules of bryozoans and sponges and occasional aquatic termites, Gantrotridis and others.

Phytoplankton are a flora of freely floating, often minute organisms, autotrophic (self-feeding) components of the plankton community and a key part of oceans, seas and freshwater basin ecosystems that drifts with water current. The name comes from the Greek words (phyton), meaning "plant", and (planktos), meaning "wanderer" or "drifter".[1] Most phytoplankton are too small to be individually seen with the unaided eye. Various studies have been carried out since 1930s & probably it is even being continued today.[2]

*Corresponding author :
Phone : 9931691410, 9430231195
E-mail : drkumarabhinesh@gmail.com

Like land vegetation, phytoplankton uses carbon dioxide, releases oxygen, and converts minerals to a form animals can use. In fresh water, large numbers of green algae often colour lakes and ponds, and cyanobacteria may affect the taste of drinking water.Phytoplankton is a predominant type of plant found in most fish pond. The quality and quantity of phytoplankton is a good indicator of water quality. The high relative abundance of chlorophyta is a indicator of productive water. Zooplanktons form an intermediate step in grazing food chain in aquatic bioloop and an ecosystem.[3] The primary object of this investigation is to get a general idea of phytoplankton composition of Matashya Beez Prachhetra water body of Dighra.

The different species of planktons vary in their occurrence with respect to seasonal changes in water temperature and physico-chemical properties as well as the number of generations of planktons produced per year in time and space.[4] Every organism of water body whether plant or animal or whether small or large is a link in food chain or food web and thus play an important role in the flow of energy in the system.

MATERIAL & METHODS

The study was seasonally planned and samples were collected during the winter, autumn, summer & rainy season. Water samples were collected from four different corners viz. east, south, west, north of standing water body (fishery pond) of Matashya beez prachhetra from Dighra of Muzaffarpur district, Bihar India. The samples were collected from each site once a day with the help of special mesh of 50-80 µm size planktonic net has been used. For the statistical analysis of these phytoplanktons the standard methods given by different scientists were used.[5-7] Some of the parameters were analyzed in the field while for most of the parameters the samples were preserved using the suitable preservatives. The samples were collected in two liter polythene bottles during the morning hours between 8.00 am to 11.30 am.

After concentrating the samples by sedimentation method 1 ml of the sample was taken & examined with a sedgewick rafter, a counting chamber under a compound microscope with different magnification.



Fig 1 :-Phytoplankton net (mesh size:50-80 µm)

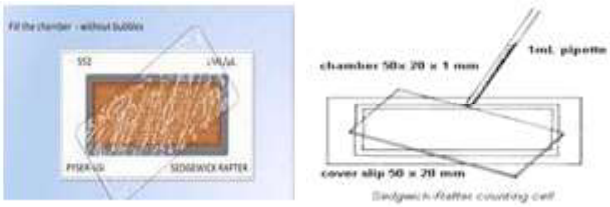


Fig2: Sedgewick Rafter plankton counter

Identification of sampled phytoplanktons:

The water sample containing the phytoplanktons were examined under high resolution stereoscopic microscope (MSZ 20) & the numbers per sample were counted in the plankton counter Sedgewick Rafter slide.

The numbers of each group of phytoplanktons were recorded in the observation Table no.1 also showing the seasonal variation.

Statistical Analysis

Analysis of relative abundance of phytoplanktons with respect to various numbers of sampling being the observation number of collected data had also been done by using the relative abundance statistical formula, $pi = n/N \times 100$, where pi is i^{th} number of phytoplanktonic species found, n = number of individuals of a species, N = total number of individual of all the species. The statistical result has also been reflected in bar diagram or histogram as shown in graph 1.

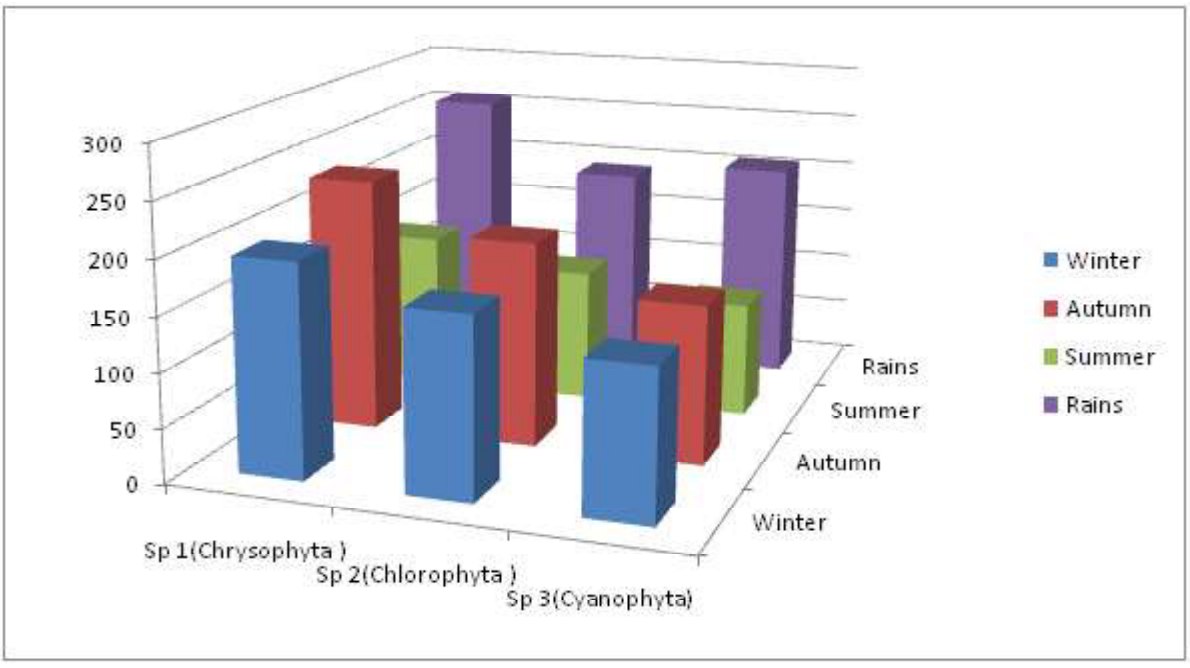
RESULTS & DISCUSSION

The phytoplankton communities of the Matashya Beez Prachhetra were represented by three groups of species namely chrysophyta, chlorophyta and cyanophyto, of which three of the taxa belonged to chrysophyta (Navicula sp., Diatoma sp, Nitzschia) 3 belonged to chlorophyta (Spirogyra, Pediastrum, Cosmarium) and 2

Table 1: Seasonal mean population & relative abundance of phytoplanktons sampled from Matashya Beez Prachhetra water body of Dighra

Seasons	Winter		Autumn		Summer		Rains	
Group & genus	Individual sample population per SNS	Relative abundance %(N/n*100)	Individual sample population per SNS	Relative abundance %(N/n*100)	Individual sample population per SNS	Relative abundance %(N/n*100)	Individual sample population per SNS	Relative abundance %(N/n*100)
(A) Chrysophyta								
(i) *Navicula sp.*	28	14.21	35	14.89	20	13.69	37	14.34
(ii) *Diatoma sp.*	69	35.02	80	34.04	48	32.87	91	35.27
(iii) *Nitzschia*	100	50.76	120	51.06	78	53.42	130	5.38
Total	197		235		146		258	
(B) Chlorophyta								
(i) *Spirogyra*	58	34.93	62	32.63	35	28	65	34.03
(ii) *Pediastrum*	32	19.27	40	21.05	22	17.6	38	19.89
(iii) *Cosmarium*	76	45.78	88	46.31	68	54.4	88	46.07
Total	166		190		125		191	
(C) Cyanophyta								
(i) *Oscillatoria sp.*	69	50	75	51.02	52	48.59	102	49.27
(ii) *Microcystic sp.*	69	50	72	48.97	55	51.40	105	50.72
Total	138		147		107		207	
Grand total	501		572		378		656	

Graph 1: Phytoplantonic composition of Matashya Beez Prachhetra water body of Dighra



belonged to the cyanophyta (*Oscillatoria* sp. & *Microcystic* sp.).

Thus from the above finding it is clear that the overall seasonal mean population density of phytoplanktons was found to be higher during the rainy season followed by autumn & winter season but minimum during the summer season. And the reason behind it may be high temperature as well as ponds physicochemical property dependent nutrient depletion and increasing zooplankton grazing that typically causes breakdown of spring blooms and maintain

low phytoplankton biomass during summer in nutrient-rich systems grazing-resistant algal species can give rise to a second bloom of phytoplanktons later in the year.8 Group wise composition revealed maximum contribution of Chrysophyta followed by Chlorophyta & then Cyanophyta.

ACKNOWLEDGMENT

The author is highly thankful to his, guide Dr. R.K Singh, retired Professor & Head, Department of Zoology, A.N. College Patna (Magadh University) & Prof. Siddha Nath Prasad Yadav 'Dean', Head University department of Zoology Magadh University Bodh Gaya for their guidance and encouragement for completing this research work.

REFERENCES

1. **Welch, P.S. 1952.** Limnology Me. Graw Hill Book co. 2nd Edition, New York.
2. **Dybas cheril Lyn. 2006.** On a collision course : oceans planktons and climate change. *Bioscience* **56:** 642-646
3. **Sugunan, V.V. 2000.** Ecology and fishery management of reservoirs in India. *J. Hydrobiologia.* **430(1-3):**121-147.
4. **Khanna D.R., Bhutiani Rakesh, Matta Gagan, Singh Vikas and Bhadauriya Gaurav 2012.** Study of planktonic diversity of river Ganga from Devprayag to Roorkee, Uttarakhand (India), *Environment Conservation Journal.* **13(1&2):**211-217
5. **APHA. 1998.** Standard Methods for Examination of Water and Wastewater, 20th Ed., American Public Health Association, Washington, DC, New York.
6. **Golterman, H.L. 1969.** Methods for Chemical Analysis of Fresh Waters. I.B.P. Handbook No. 8. Black Well Scientific Publ., Oxford. 166 pp.
7. **Trivedi, R.K. and Goel, P.K., 1984.** Chemical and biological methods for water pollution studies, Environmental Publications, Karad.
8. **Sommer U., Gliwicz Z. M., Lampert W., Duncan A.1986.** The PEG-model of seasonal succession of planktonic events in fresh waters. *Arch. Hydrobiol.* **106:**433-471.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.299-302
An International Biannual Refereed Journal of Life Sciences

29(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of growth regulators in baby corn

Sarita Kumari*
Department of Botany, B.N.M.University, Madhepura, Bihar, India

Received : 16th January, 2020 ; Revised : 25th February, 2020

Abstract : A filed experiment was conducted at the Rajendra Agriculture University Pusa (Bihar) during Rabi and Kharif Season of 2013. The soil of the experiment area was sandy and clay loam with alkaline PH: low in Organic Carbon (0.34 and 0.32%) and available N (210.4 and 233.4 kg ha-1) medium in available P (12.4 and 11.8 kg ha-1) and high in available K (438.6 and 414.1 kg ha-1) during late Rabi and Kharif 2013 season respectively. Three herbicides were selected for growth effect on baby corn with different aspect viz. leaf width leaf height total height of plants (cm) and total number of leaf each plant Herbicides 24-D atrzine and Glyphosate were used of different concentration most effective concentration was applied for further studies.

Key words: Organic carbon, Nitrogen, P. width of leaf height of leaf 2,4D.

INTRODUCTION

Despite its great importance in Indian agriculture as well as export potentiality a very little work has been done on baby corn in India. In the past, the practice has been to use any genotype of maize for the cultivation of baby com and detailed studies have not been conducted to identify and/or develop varieties for baby corn. In India, only few single cross hybrids have been found to be preferred for baby corn cultivation (HM-4, HQPM-1, PEHM-2 and Kanak etc.) All these hybrids were originally developed for grain purpose and they are considered for baby corn usage on account of some characteristic features. At present, exclusive and specific single cross baby corn hybrids are not available under public domain. Developing baby corn cultivars/hybrids specifically adaptable to Indian conditions might be one of the approaches, especially for fulfilling short and/or medium term goals. Here also emphasis needs to be given on early maturity considering the fact that many crops can be taken under the Indian conditions due to reasonably favorable weather throughout the year in most of the states. Early maturity is also an important factor for determining comparative advantage especially in relation to other vegetables in specific season.

*Corresponding author :
Phone: 8540079860, 7762943139
E-mail : Saritasaharsa1984@gmail.com

MATERIALAND METHOD

A field experiment was conducted at the Rajendra Agriculture University, Pusa (Bihar) during Rabi and Kharif season of 2013. The soil of the experiment aria sandy clay loam with alkaline pH; low in organic carbon (0.34 and 0.32%) and available N (210.4 and 233.4 kg ha-1), medium in available P (12.4 and 11.8 kg ha-1) and high in available K (438.6 and 414.6 kg ha-1) during late Rabi and Kharif 2013 season, respectively. Baby corn variety Surya (Hybrid) with field duration 65-70 days was used in the trial.

The experiment was laid out in Randomized Complete Block Design (RCBD) with three replications. The experiment consist of six treatments with three herbicides (2, 4-D, atrazine and glyphosate) each four varied concentration for along with water spray (control) and no spray (absolute control). The seed ware pre-treated with fungicide (Thriam @ g kg-1 seeds) and bio-fertilizer (Azospirillum @ 600 g ha-1 seeds) and sown at a spacing 45x25 cm. The recommended fertilizers (150:60:40NPK ha-1) were applied in the form in the Urea (N), single super phosphate (P) and muriate of potash (K). Full dose of phosphorus and half of nitrogen and potassium were applied at the time of sowing. Remaining of 50% of nitrogen and potassium were applied at 25 Days After Sowing (DAS) as top dressing. Three herbicides selected with different concentration. These herbicides 2, 4D @20-50 ppm, atrazine @ 70-100 ppm and glyphosate @ 05-20 ppm were prepared by diluting with distilled water sprayed in evening hours by hand operated sprayer at 25 and 45DAS. Two - hand weddings (20 and 40 DAS) were done to check the weeds. Need based plant protection measures were carried out timely. Detaselling was done before the emergence of tassel from the flag leaf. The cobs were harvested after 2-3 cm length of silk emergence.

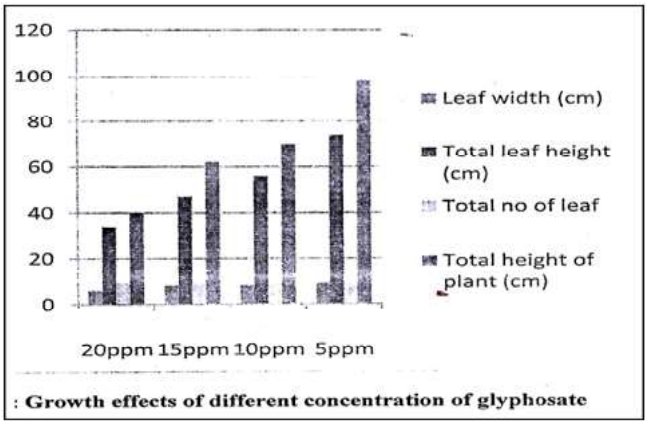
For recording various biometric observations on baby corn, from each net plot a sample consisting of five plants were selected at random and tagged. From the tagged plants, observations on plant height, number of green leaves plant-1, Leaf area Index (LAI) and Dry Matter Production (DMP) at harvest have been recorded. Plant height was measured from ground level to the tip of flag leaf. Number of green leaves plant-1 was recorded by counting the number of fully opened green leaves borne on main stem of a plant. Dried and yellow leaves are not considered for counting. The collected leaves from the sample plants were taken of the laboratory and the leaf area was measured by using leaf are meter (LICOR Model 3100). From the leaf area, LAI was calculated.

Leaf Area Index = $\frac{Leaf\ Area\ Plant - 1 \times 100}{Ground\ area\ occupied\ plant - 1}$

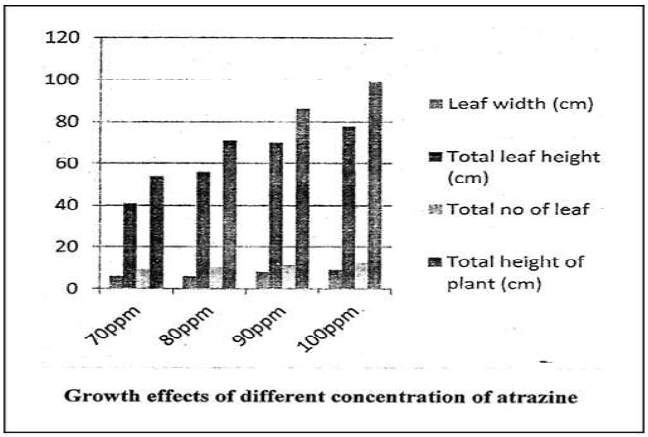
RESULTS

Three herbicides were selected for growth effect on baby corn with different aspect viz. leaf width, leaf height, total height of plant (cm) and total number of leaf each plant. Herbicides 2, 4-D, atrazine and glyphosate were used of different concentration. Most effective concentration was applied fro further studies. The results recorded in the revealed that glyphosate were found most effective at concentration 5ppm followed by 10ppm for maximum growth aspects as such as total number of leaf, width of leaf height of leat and height of each plant (cm). In case of atrazine result related that most effective concentration was at 100 ppm followed by 90 ppm for maximum growth aspects as such as total number of leaf, width of leaf, height of leaf and height of each plant (cm) and 2, 4-D most effective at 50ppm concentration followed by 40ppm for maximum growth aspects as such a total number of leaf, width of leaf, height of leaf and height of each plant (cm)

Leaf height, leaf width, Total no. of Leaves and plant height are shown in graph no. 01 and no. 02.



: Growth effects of different concentration of glyphosate



Growth effects of different concentration of atrazine

CONCLUSION

Three herbicides were selected for growth effect on baby corn with different aspect viz. leaf width, leaf height, total height of plant (cm) and total number of leaf each plant. Herbicides 2, 4-D, atrazine and glyphosate were used of different concentration. Most effective concentration was applied for further studies. The results recorded in

the revealed that glyphosate were found most effective at concentration 5ppm followed by 10ppm for maximum growth aspects as such as total number of leaf, width of leaf height of leaf and height of each plant (cm).

REFERENCES

1. **ABS. 2011.** Agricultural Commodities Australia 7121.0. Canberra.
2. **Emam, Y, Shekoofa, A. 2009.** Response of barley plants to drying soil under the influence of chlormequatchloride. *Research on Crops.* **10:**516-522.
3. **Tripathi, SC, Sayre, KD, Kaul, JN, Narang, RS. 2004.** Lodging behavior and yield potential of spring wheat (*Tritcum aestivum* L.): effects of ethephon and genotypes. **Field Crops Research. 87:**207-220.
4. **Bahram B. 2009.** Amelioration of chilling stress by paclobutrazol in watermelon seedlings. *Sci Hortic-Amsterdam.* **121:**144-148
5. **Cai KZ, Gao D, Luo SM, Zeng RS, Yang JY, Zhu XY. 2008.** Physiological and cytological mechanisms of Silicon induced resistance in rice against blast disease. *Phys Plasmas.* **131:**324-333

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 303-306
An International Biannual Refereed Journal of Life Sciences

30(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Study of intestinal parasitic infection among school going children in eastern zone of Patna district of Bihar, India

Afsana Begum*
Parasitology Research Lab, Department of Zoology, Patna University, Patna, Bihar, India

Received : 18th January, 2020 ; Revised : 26th February, 2020

Abstract : Intestinal parasitic infections caused by protozoan and helminth are one of the major health problems in children of the developing countries like India. Around 450 million children are affected due to these parasites. Therefore the present study was focused to determine the current status of intestinal parasitic infections among school going children. A total of 468 stool samples were randomly collected and examined by simple saline preparation technique. In addition stained preparation of Lugol's iodine was also used for the identification of nuclear character of the cysts. Of the total examined cases 231 (49.3%) were found to be harbor. The incidence of infections for various intestinal parasites was as follows *Entamoeba histolytica*, (13.9%) *Entamoeba coli* (4.4%) and *Giardia lamblia* (6.5%), *Ascaris lumbricoides* (22.5%), *Trichuris trichiura* (3.0%), *Hymenolepis nana* (2.1%) and *Hymenolepis diminuta* (0.3%). About 3.2 % sample contained mixed infections. The study revealed that the prevalence of intestinal parasitic infections among children in Patna district is very high and required special control measures, such as improvement in personal hygiene, sanitation, water supplies, socio-economic conditions and education.

Key words: Intestinal parasitic infection, helminth, protozoan, school children, Patna.

INTRODUCTION

Parasite is a living organism which depends on a living host for survival and derives nutrition from the host without giving any benefit to the host. Intestinal parasitic infections are devastating and prevalent throughout the world including India.[1] They affect an estimated 3.5 billion persons and cause clinical morbidity to approximately 450 million people[2], majority of the cases occur among school age children.[3,4] These parasites are highly prevalent in the population with low socio-economic status, poor hygiene and sanitation, scarcity of potable drinking water.[5,6] The most common parasites causing infections globally are *Ascaris lumbricoides* (20%), Hookworm (18%), *Trichuris trichiura* (10%), and *Entamoeba histolytica* (10%) [7]. These parasites are associated with diverse clinical manifestation among school going children such as poor growth, reduced physical activity, impaired cognitive function, hampered learning ability, intestinal obstruction and nutritional deficiency particularly iron and vitamin A as well as mental and physical growth retardation[8,9]. In India particularly in Bihar there is scant information available regarding the parasitic infections among school going children, whereas a study related to soil transmitted helminth among school children in Bihar have been published in recent past year[10] and another study reported that helminth infections to be greater than protozoan infection in eastern part of Bihar.[11]

*Corresponding author :
Phone : 8294036786, 7632911275
E-mail : afsana.begum2106@gmail.com

Therefore the objective of the study was to estimate prevalence and current status of intestinal parasitic infections in the school going children in and around the school in Patna.

MATERIALS AND METHODS

This study was conducted among 468 school children of aged 6 - 12 years between 2017- 2018 in Patna district of Bihar. Fresh fecal samples were collected in wide mouthed, sterilized container with and without 10% formalin bearing necessary code. A pre tested questionnaire based on risk factors including information such as age, sex, religion, food habits, living status, source of drinking water, existence of latrines in their homes and hand washing after defecation were also recorded.

First of all a macroscopic examination of all the collected samples was carried out for color, consistency and presence of mucus and blood. For microscopic examination, Saline wet mount and Lugol's iodine wet mount were prepared as per standard protocol[12] for the identification of eggs, cysts, trophozoites, larvae and proglottids of intestinal parasites. The data was statistically analyzed by applying Chi-Square test (χ^2 - test).

RESULTS

The present study showed that out of 468 children 231 (49.3%) students were harbored with various species of intestinal parasites (Fig. 1). Although girls show higher infection rate (56.3%; 98/174) than boys (45.2%; 133/294) and the difference was found to be significant ($p < 0.01$) (Table.1). Out of total 231 infected cases 101 (21.6%) contained only protozoan, 115 (24.5%) only helminth and 15 (3.2%) were having mixed infection (Table 2). The prominent parasites detected were *Ascaris lumbricoides* (22.5%), followed by *Entamoeba histolytica* (13.9%) *Giardia lamblia* (6.5%), *Entamoeba coli* (4.4%) *Trichuris trichiura* (3%), *Hymenolepis nana* (2.1%) and *Hymenolepis diminuta* (0.3%) (Table 3).

DISCUSSIONS

In the present study 49.3% of the school going children revealed positive result of intestinal parasites and seven species (Table.3) of parasite were detected. The findings of our result are consistent with the previous studies conducted in various parts of India, have reported a prevalence of intestinal parasites 30% to 50% [13-16] among school going children. From this study it is clear that prevalence and diversity of intestinal parasitic infections vary greatly and the maximum of both these incidence occur in the lower age group 6-12 years.[17,18] The high incidence of these pathogens in the lower age group is highly associated with the lack of awareness pertaining to hygiene and sanitation, both of which can be expected to provide maximum chances of positive exposure in contaminated surroundings. In our study helminths infection (24.5%) was greater than protozoan infection (21.6%), this result is in line with those reported in the eastern part of Bihar[11]. In our study *Entamoeba histolytica* (13.9%) was the commonest protozoa, this result is in contrast with other studies where *Giardia lamblia* was the commonest protozoan.[14,16,19] Among helminth *Ascaris lumbricoides* (22.5%) was more prevalent than other, this finding is similar to the other studies where *Ascaris lumbricoides* was the commonest helminth[13,20,21] but it is in contrast with those reported 1.9% in school going children in other part of India.[16] In our study *Hymenolepis nana* was found (2.1%), similar to other studies where *Hymenolepis nana* was found in 2-3% of the children.[14,20] It is also observed that 3.2% mixed infections were found in this study.

This result highlights the intestinal parasitic infections is higher in school children of Patna district, so in order to prevent this infection appropriate education should be given to children and their parents concerning healthy and hygienic living habit, disease transmission, safe drinking water, improvement of sanitation coupled with a nutritional program at the primary school level[22] as well as periodic administration of anti parasitic drugs. The findings of this study might be helpful for fortifying the information available so far and encourage policy makers to design effective strategies to prevent intestinal parasitic infections.

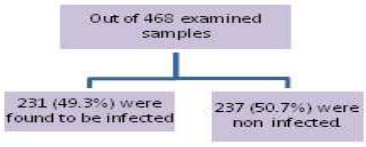
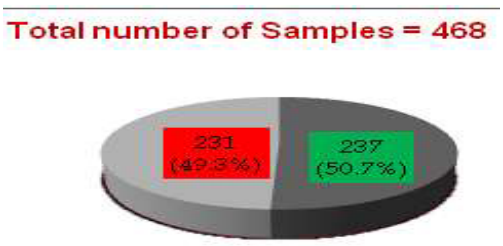


Fig. 1



Pie chart with reference to Fig. 1

Table 1. Prevalence of intestinal parasites in the population studied

Variables	Total no. of sample Studied (468)	No. of infected samples (231)	Prevalence (%) 49.3	*P value*
Sex				
Boys	294	133	45.2	
Girls	174	98	56.3	<0.01

Table 2. Prevalence of protozoan and helminths infection

Parasitic group	Number	percentage (%)
(A)Samples containing only protozoan	101	21.6
(B)Samples containing only helminths	115	24.5
(C) Mixed infections	15	3.2

Table 3. Species wise prevalence of protozoan infection out of 468 samples studied.

Parasites	No. of infected cases (231)	Prevalence (%)
Protozoans		
E. histolytica	65	13.9
E. coli	21	4.4
G. lamblia	30	6.5
Helminths		
A. lumbricoides	105	22.5
T. trichiura	14	3.0
H. nana	10	2.1
H. diminuta	1	0.3

ACKNOWLEDGEMENT

The author is extremely thankful to her guide, Prof. M. Haque and the head of the Dept. of Zoology, Patna University, Patna, Bihar for their constant encouragement & guidance.

REFERENCES

1. **Tanowitz HB, Weiss LM, Witner M. 2001.** Tapeworms, Current infectious disease reports. **3(1):**77 84.
2. **WHO. 2008.** Department of Food Safety; Zoonoses and Foodborne Disease, "First formal meeting of the Foodborne Disease Burden Epeidemiology Reference Group (FERG): implementing strategy, setting priorities and assigning the tasks" (Hebrew), http/www.who.int/iris/handle/10665/43905.
3. **World Health Organization. 1996.** *R*eport of the WHO Informal Consultation on the Use of Chemotherapy for the Control of Morbidity due to soil-transmitted Nematodes in Humans. (WHO/CTD/SIP/96.2). Schistosomiasis and Intestinal Parasites Unit, Division of Control of Tropical Diseases, WHO, Geneva, Switzerland
4. **Montresor, A., Crompton, D. W. T., Hall, A., Bundy, D. A. P. and Savoli, L. 1998.** Guidelines for the evaluation of soil-transmitted helminthiasis and schistosomiasis at community level. Division of control of Tropical Parasites unit, *World Health Organization;* Geneva, Switzerland, (WHO/CTD/SIP/98.1).
5. **Sayyari AA,Imanzadeh F, *et al.* 2005.** Prevalence of intestinal parasitic infections in the Islamic Republic of Iran. *East. Mediterranean Health J.* **11(3):**377-83
6. **Bethony J, Brooker S, Albonico M, Geiger SM, Loukas A, Diemert D, Hotez PJ. 2006.** Soil-transmitted helminths infections: *Ascariasis*, *Trichuriasis* and hookworm. *Lancet* **367:**1521-1532

7. **World Health Organization 1987.** Prevention and Control of Intestinal Parasitic Infections. Technical Reports Series WHO. 575-88.

8. **World Health Organization. 2006.** Geographical distribution and usefull facts and stats. Geneva; WHO.

9. **Girum T. 2005.** The prevalence of intestinal helminthes infections and associated risk factors among school age children in Babiletown, Easten Ethiopia. *Ethio. J. Health Dev.* **19:**140-147.

10. **Greenland K, Dixon R, Khan SA, Gunawardena K, Kihara JH, Smith JL. 2015.**The epidemiology of soil-transmitted helminths in Bihar State, India. *PLoS Negl Trop. Dis* **9:**1-14.

11. **Kumar M, Rajkumari N. 2016.** A case report on an uncommon parasitic infection of human blantidiasis. *Trop Parasitol* **6:**82-4.

12. **Paniker, C. K. J. 2007.** Textbook of medical parasitology, 6th Edn. Medical Publishers. (P) LTD, Jaypee Brothers, New Delhi.

13. **Bora D, Meena VR, *et al.* 2006.** Soil transmitted helminthes. Prevalence in school children of Pauri Garhwal district, Uttaranchal State. *J Commun Dis* **38:**112-4.

14. **Dongree AR, Deshmukh PR, *et al.* 2007.** An approach to hygiene education among rural Indian school going children. http://www.ojhas.org/issue24/2007 4-2.htm.

15. Ministry of Rural development. Government of India. Total Sanitation Campaign: Guidelines (Online) 2004 (cited on 5 Nov. 2007); Avilable from URL:http://www.ddws.nic.in/NewTSCGuidlines.doc.

16. **Aher A., Kulkarni, S. 2011.** Prevalence of intestinal parasites in school going children in rural community. *Int J Biomed Res.* **2(12):** 605-607.

17. **Mbuh JV, Ntonifor HN, and Ojong JT. 2010.** The incidence, intensity and host morbidity of human parasitic protozoan infections in gastrointestinal disorder outpatients in Buea sub division, Cameroon. *J Infect Dev Ctries.* **4(1):** 038-043

18. **Choubisa, SL, Jaroli, VJ, Choubisa, P. 2012.** Intestinal parasitic infection in Bhil tribe of Rajasthan, India. *J Parasit Dis.* **36(2):**143-148.

19. **Gujarathi UP, Goyal VR Aher AR. 1997.** Prevalence of intestinal parasitric infections in a ryral community. *Indian Med Gaz* ; **131:**277-9.

20. **Saifi MA, Wajiullah. 1997.** Intestinal parasitic infection in school children of Ujhani, Budaur. *J of parasitc Dis.* **25(1):**26-9.

21. **Mareeswaran N, Savitha AK, Gopalkrishnan S. 2018.** Prevalence of intestinal parasites among urban and rural population in Kancheepuram district of Tamil Nadu. *Int. J. Community Med Pub Health* **5(6):** 2585-2589

22. **Ross AG, Papier K, Catubig RL, Chau TN, Inobaya MT and Ng SK. 2017.** Poverty, dietary Intake, intestinal parasites, and nutritional status among school-age children in the rural Philippines. *Trop. Med. Infect. Dis.* **2:** (49) p10

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 307-310 **31(AS).**
An International Biannual Refereed Journal of Life Sciences ***Animal Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Portfolio function of fresh water crabs engineered by S-W Diversity index in the Kanwar lake, Begusarai, Bihar, India

Mousumi Dey*
Department of Zoology, B.N Mandal University, Madhepura, Bihar, India

Received : 22nd January, 2020 ; Revised : 27th February, 2020

Abstract: Portfolio function of any biotic population in an ecosystem has multidimensional control & is primarily engineered by S-W diversity index of the species. S-W diversity index is popular statistical information index as quantitative tool to quantify the species abundance, richness & bifold product value of any species that yields significant result for understanding their asset & liability value in the portfolio of biotic community. It further helps in contemplating the strategies for the management & development of sustainable environment & ecosystem. Fresh water crabs being the members of crustacean arthropods have distinct role in the wetland habitats where they successfully dwell, propagate & contribute in the portfolio of the community. Crabs sampled from the Kanwar lake of Begusarai were taxonomically identified as nine different species like-*Barytelphusa cunicularis, Barytelphusa guerini, Barytelphusa lugubris, Inglethelphusa fronto, Paratelphusa masoniona, Paratelphusa spinigera, Potamon lugubre, Sartoriana spinigera & Vanni travancorica*. Random sampling data of these nine species in terms of their per unit net sweep numerical count were recorded as species abundance (p_i)- an important asset of the wetland community portfolio accommodating these crabs. When the data were subjected to S-W diversity index tool on the basis of the species abundance, the value of H ($H= -\sum pi \times \log pi$) for all the species of crabs present in the sixty random sampling sites was found in the range of 0.8301-0.8927 during different seasons which indicate that they have profitable asset investment contribution (PF index ranging between 6.523-10.081) as edible as well as other beneficial resource in the community as per portfolio function laws.

Key words: S-W diversity, portfolio function index (PFI), freshwater crabs, profitable asset investment, edible resource, Kanwar lake

INTRODUCTION

Fresh water crabs are among the most important invertebrates inhabiting all the freshwater habitat, large lowland rivers and small water bodies.[1] There are more than 1300 species, 238 genera and eight families of freshwater crabs distributed across tropical and subtropical Asia, Africa, Central and South America, southern Europe, the Middle East and Australasia.[2,3] More than 6,700 different known species of brachyuran crabs and over 1,300 true freshwater crabs have been identified around the world. The true freshwater crabs are fully adopted for freshwater, semi-terrestrial or terrestrial appears of life. Most these crabs are found to be nocturnal scavengers or opportunistic predators, and some species have economic and ecological importance. For example, some species are rice field pests, some are a source of protein for people, and other species have medical importance serving as the second intermediate host of the parasitic human lung fluke, Paragonimus. These crabs also have a great medicinal

*Corresponding author :
Phone : 7903035949
E-mail : m5284banty@gmail.com

value and play a very important role in aquatic ecosystem.[4,5] Some species of freshwater crabs also serve as bio-indicators of ecosystem disruption; some are threatened by collection for the aquarium trade etc. Freshwater crabs inhabit in tropical and subtropical areas in many parts of the world. They can be caught in most freshwater ecosystems, from fast flowing mountain streams to slow flowing rivers, streams, stagnant ponds etc. At present crabs have become one of the important source for the establishing the fishery wealth in many countries.[6]

MATERIALAND METHODS

For the study of portfolio function of freshwater crab as an important statistical exercise, collection of the samples were done from sixty different sites of the Kanwar lake of Begusarai during morning hour between 10 am - 12 pm in four different seasons during 2016-2017. The crabs were sampled from a statistically valid number of randomly selected sites as mentioned rather than surveying the entire water body. Specimen were collected using aquatic net & were preserved in 5-10% formalin in plastic container which were then carried to the Department of Zoology, B.N Mandal university for its taxonomical identification with the help of different keys suggested by various scientists.[7,8,9] Analysis of portfolio function on the basis of random sampling data through calculation of S-W diversity index has been presented in table 1-4. The equations used for S-W diversity(H) & Portfolio function index (PFI) are as follow:-

$$H = -\Sigma pi * \log pi$$

$$PFI = V_{water} / H$$

Table 1: Portfolio function of freshwater crab of Kanwar lake during rains 2016.

Si. No.	Name	n	Per CFT volume of water in each sweep	Relative abundance %(n/N* 100)	pi(n/N)	log pi	pi*log pi	H	PF index(volume of water as host resource/diversity of crabs $H,V_{water}/H$)
1	*Sartoriana spinigera*	82	9	21.2	0.212	-0.673	-0.1426		
2	*Barytelphusa guerini*	16	9	4.1	0.041	-1.387	-0.0568		
3	*Barytelphusa lugubris*	22	9	5.6	0.056	-1.251	-0.0700		
4	*Inglethelphusa fronto*	48	9	12.4	0.124	-0.906	-0.1123		
5	*Paratelphusa masoniona*	65	9	16.8	0.168	-0.774	-0.1300	**0.8927**	**10.081**
6	*Paratelphusa spinigera*	43	9	11.1	0.111	-0.954	-0.1058		
7	*Potamon lugubre*	62	9	16	0.160	-0.795	-0.1272		
8	*Barytelphusa cunicularis*	28	9	7.2	0.072	-1.142	-0.0822		
9	*Vanni travancorica*	20	9	5.1	0.051	-1.292	-0.0658		

Table 2: Portfolio function of freshwater crab of Kanwar lake during winter 2016.

Si. No.	Name	n	Per CFT volume of water in each sweep	Relative abundance %(n/N* 100)	pi(n/N)	log pi	pi*log pi	H	PF index(volume of water as host resource/diversity of crabs $H,V_{water}/H$)
1	*Sartoriana spinigera*	65	5.8	23.7	0.237	-0.625	-0.1481		
2	*Barytelphusa guerini*	10	5.8	3.6	0.036	-1.443	-0.0519		
3	*Barytelphusa lugubris*	16	5.8	5.8	0.058	-1.236	-0.0716		
4	*Inglethelphusa fronto*	30	5.8	10.9	0.109	-0.962	-0.1048		
5	*Paratelphusa masoniona*	40	5.8	14.5	0.145	-0.838	-0.1215	**0.8301**	**6.9871**
6	*Paratelphusa spinigera*	32	5.8	11.6	0.116	-0.935	-0.1084		
7	*Potamon lugubre*	55	5.8	20	0.200	-0.698	-0.1396		
8	*Barytelphusa cunicularis*	15	5.8	5	0.054	-1.267	-0.0684		
9	*Vanni travancorica*	11	5.8	4	0.040	-1.397	-0.0158		

Table 3: Portfolio function of freshwater crab of Kanwar lake during autumn 2016.

Si. No.	Name	n	Per CFT volume of water in each sweep	Relative abundance %(n/N* 100)	pi(n/N)	log pi	pi*log pi	H	PF index(volume of water as host resource/diversity of crabs $H,V_{water}/H$)
1	*Sartoriana spinigera*	60	5.5	26.9	0.269	-0.570	-0.1533		
2	*Barytelphusa guerini*	8	5.5	3.5	0.035	-1.455	-0.0509		
3	*Barytelphusa lugubris*	11	5.5	4.9	0.049	-1.309	-0.0641		
4	*Inglethelphusa fronto*	23	5.5	10.3	0.103	-0.987	-0.1016		
5	*Paratelphusa masoniona*	32	5.5	14.3	0.143	-0.844	-0.1206	**0.8428**	**6.5258**
6	*Paratelphusa spinigera*	25	5.5	11.2	0.112	-0.950	-0.1064		
7	*Potamon lugubre*	48	5.5	21.5	0.215	-0.667	-0.1434		
8	*Barytelphusa cunicularis*	9	5.5	4	0.040	-1.397	-0.0558		
9	*Vanni travancorica*	7	5.5	3.1	0.031	-1.508	-0.0467		

Table 4: Portfolio function of freshwater crab of Kanwar lake during summer 2016.

Si. No.	Name	n	Per CFT volume of water in each sweep	Relative abundance %(n/N* 100)	pi(n/N)	log pi	pi*log pi	H	PF index(volume of water as host resource/diversity of crabs $H,V_{water}/H$)
1	*Sartoriana spinigera*	30	4.5	34.8	0.348	-0.458	-0.1593		
2	*Barytelphusa guerini*	2	4.5	2.3	0.023	-1.638	-0.0376		
3	*Barytelphusa lugubris*	4	4.5	4.6	0.046	-1.337	-0.0615		
4	*Inglethelphusa fronto*	1	4.5	1.1	0.011	-1.958	-0.0215		
5	*Paratelphusa masoniona*	14	4.5	16.2	0.162	-0.790	-0.1279	**0.7561**	**5.9515**
6	*Paratelphusa spinigera*	12	4.5	13.9	0.139	-0.856	-0.1189		
7	*Potamon lugubre*	18	4.5	20.9	0.209	-0.679	-0.1419		
8	*Barytelphusa cunicularis*	3	4.5	3.4	0.034	-1.468	-0.0499		
9	*Vanni travancorica*	2	4.5	2.3	0.023	-1.638	-0.0376		

RESULT AND DISCUSSION

From the above findings nine different species of fresh water crab has been found which includes *Sartoriana spinigera, Barytelphusa guerini, Barytelphusa lugubris, Inglethelphusa fronto, Paratelphusa masoniona, Paratelphusa spinigera, Potamon lugubre, Barytelphusa cunicularis & Vanni travancorica* maximum of which included those which were generally taken up by the people of that region. The data regarding the portfolio function of the freshwater crab were found to be 10.081, 6.9871, 6.5258 & 5.9515 that shows the presence of rich PF index level of the aquatic community in this area regardless the volume of water used for each sweep. Rise in the PF index of the aquatic community may be due to the abundance of availability of micro flora and fauna in that habitat. Therefore availability of flora and fauna in this volume of water also needs to be examined in deciding the PF index level of the habitat.

ACKNOWLEDGEMENT

Author is very much thankful to the Head of Dept of Zoology, B.N Mandal University & her guide Dr. Pankaj Kumar, Associate Prof., P.G Dept. of Zoology, T.P College Madhepura to provide the necessary facilities to carry out the research work successfully.

REFERENCES

1. **Yeo, D.C.J, P.K.L. Ng, N. Cumberlidge, C. Magalhães, S.R. Daniels & M.R. Campos. 2008.** Global diversity of crabs (Crustacea: Decapoda: Brachyura) in freshwater. *Hydrobiologia*, **595:** 275-286.

2. **Ng, P.K.L., D. Guinot & P.J.F. Davie. 2008.** Systema Brachyurorum. Part I. An annotated checklist of extant brachyuran crabs of the world. *Raffles Bull. Zool. Suppl*., **17:** 1-286.

3. **Cumberlidge, N., P.K.L. Ng, D.C.J. Yeo, C. Magalhães, M.R. Campos, F. Álvarez, T. Naruse, S.R. Daniels, L.J. Esser, F.Y.K. Attipoe, F.L. Clotilde-Ba, W. Darwall, A. Mcivor, M. Ram & B. Collen. 2009.** Freshwater crabs and the biodiversity crisis: importance, threats, status, and conservation challenges. *Biol. Conserv.,* **142:** 1665-1673.

4. **Cobb BF, Conte FS, Edwards MA. 1975.** Free Amino Acids and Osmoregulation in Penaeid Shrimp. *J Agric Food Chem* **23:** 1172-1174.

5. **Fang LS, Tang CK, Lee DL, Chen IM. 1992.** Free Amino Acid Composition in Muscle and Hemolymph of the Prawn *Penaeus monodon* in Different Salinities. *Nippon Suisan Gakkaishi*. **58:** 1095-1102.

6. **Chhapgar B. 1991.** Seashore Life of India. Nature Guides, Oxford University Press, *Bombay India*. 18-24

7. **Shafi, M. and Quddus, M. M. A. 1982.** Fisheries Resources in Bangladesh. (Bangladesher Matshaya Sampad, in Bengali), Bangla Academy, Dhaka, pp 369-396

8. **Chowdhury, S.H. and Hafizuddin, A.K.M. 1991.** Crab Fauna of Bangladesh. Part-I. Some marine crabs from the Bay of Bengal. Chittagong University Studies. Part-II, *Science.* **2(15):** 65-77

9. **Siddiqui M.Z. H. and Zafor, M. 2002.** Crabs in the Chakaria Sundarban are of Bangladesh. *The Journal of NOAMI*. **19 (2):** 61-75.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 311-312 32(PS).
An International Biannual Refereed Journal of Life Sciences **Plant Science**

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Assessment of physico-chemical parameters of soil in different area of Saharsa district

Swati Kumari*

Department of Botany, B.N. Mandal University, Madhepura, Bihar, India

Received : 28th January, 2020 ; Revised : 05th March, 2020

Abstract: Plants get nutrition from soil. Availability of inorganic salts of Phosphorus, Potash, Nitrogen etc. in soil is very important for the growth of plants. Soil pH and texture also play an important role in plant growth. In Saharsa district the pH value of soil ranged from 6.2 to 8.2 in different villages. Soil texture was loamy. Organic carbon ranged from 0.90% to 2.70% available Phosphate (P_2O_5) Traces to 6.25 K.g/hac. and available Potash (K_20) 226kg/hac to 353kg/hac.

Key words: Physico-chemical parameters, soil, Saharsa district,

INTRODUCTION

Soil is an important component of terrestrial habitat which also supports the aquatic ecosystem as well as wetlands in the biosphere.[1]

The physicochemical parameters of of soil sharply differs from other habitat media like water and air in terms of many soluble & insoluble minerals and nutrients getting deposited in layers forming the soil bed. This scientific principle makes the soil fertile or sterile & sometimes toxic due to deposition of undesirable substances.[2,3]

The present investigation is an attempt to bring on record different physicochemical parameters of soil samples collected from Saharsa district - a wetland pocket of Kosi division

MATERIAL AND METHOD :-

Soil Samples were Collected from 5 villages of Saharsa district 1) Mehshi 2) Mohanpur 3)Salkhua 4) Bangaon 5) Paharpur Samples were brought to laboratory and test was Performed for pH, Organic Carbon, texture, Potash and Phosphorus.

pH :- pH was determined as per the method described by Jackson (1958)[4]. Soil water suspension of 1.2 ratio was prepared by dissolving 10gm soil in 20ml. distilled water and stirred well for 5 minutes. It was kept undisturbed for some time and then again stirred. pH was measured on systronic pH meter after calibration with buffer of pH 4, 7 and 9.2

Potassium:- Potassium was Measured using Flame Photometer.

Phosphorus:-Phosphorus was measured using systronic spectrophotometer.

Organic Carbon :-

Total Organic Carbon was Measured as per method of Walkley and Black (1934)

01.One gram soil sample was added in a 500ml conical flask. 10ml. N Potassium dichromate was added in it.

02.20ml. Conc. Sulphuric Acid was added and left for 30 minutes.

*Corresponding author :
Phone : 8002995667, 9839312828
E-mail : swatisantan1983@gmail.com

03.200ml. distilled water was added to the mixture.

04.10ml. Orthophosphoric Acid, 0.5gm. Sodium Fluoride and one ml. diphenylamine was added to mixture.

05.Mixture was titrated against .5M ferrous ammonium sulphate till the end point appear from blue. Voilet to green.

06.A blank also run without soil sample.

Organic Carbon was calculated by the formula

Organic Carbon % = $\frac{10\ (A-B)\ x\ 0.003\ x\ 100}{A\ x\ wt.of\ soil\ in\ gram}$

A = Vol. of ferrous amm. sulphate used against mixture

B = Vol. of ferrous amm. sulphate used in blank

RESULT

The pH value of soil ranged from 6.2 to 8.2 in different villages. Soil texture was loamy. Organic Carbon ranged from 0.90% 2.70% available phosphate (P_2O_5) Traces to 6.25 k.g/hac. and available Potash (K_2O) 226 k.g/h to 353 k.g/hac.

The result is tabulated in Table No. -01

Table 1- Table showing soil pH, soil taxture Organic Carbon, Phosphorus and Potash in fields of Saharsa District

S.No.	Site	Soil pH	Soil Texture	Organic C%	Available P_2O_5	Available K_2O
1	2	3	4	5	6	7
01	Mehshi	7.5	Loamy	1.12	2.3	293.3
02	Mohanpur	6.2	Loamy	1.07	6.25	353.1
03	Salkhua	8.2	Sandy Loamy	2.70	6.15	230.7
04	Bangaon	6.2	Loamy	1.00	Trace	232.5
05	Paharpur	6.5	Loamy	0.90	6.25	226.4

CONCLUSION

Plants get their nutrition from soil inorganic form for the healthy growth of plants soil pH, Taxture, Organic Carbon, Phosphate and Potash are important factors of Soil. In different villages of saharsa district soil in loamy with a pH range of 6.2 to 8.2. Organic Carbon is also in good quantity in most of the fields. However in Bangaon available phosphate is very low. Available potash is also in a satisfied quantity in different fields.

REFERENCES

1. **Hand Book of Agriculture. 2001.** Indian Council of Agriculture research.
2. **Ce miller. LM Turk. 2002.** Fundamentals of soil science Biotech Books. 1123/74 Trinagar, Delhi, India PP-157.
3. **Parveens, Tariq M. Farmanulla JK, Hamid A. 1993.** *Journal of Agriculture* **9(5):** 467
4. **Jackson Mi. 1958.** Soil Chemical analysis.
5. **Andrews Ss. Karlen Dl. Cambardella CA. 2004.** *Soil sci*. **500** Am.jp – 68
6. **Kabata A Pendias P. 2001.** Trace element in Soils and plants (3rd) CRC *Press, Boca Raton*, F1 pp. 413

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 313-316
An International Biannual Refereed Journal of Life Sciences

33(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

On a trematode *Allocreadium hanumanthai* from the intestine of *Channa punctatus* from Tirhut division of Bihar

Amritesh Gautam* & Sneha Raj
Department of Zoology, S.N.S. College Hajipur, Vaishali, Bihar, India

Received : 2nd February, 2020 ; Revised : 7th March, 2020

Abstract: Diversity in the world is staggering. India being a tropical nation provides suitable conditions for the multiplicity of varied fauna and their parasites. Already people from all around the country have studied and contributed a lot, but a lot of work still has to be done from Bihar, in particular from Tirhut division, Bihar. High humidity and tropical conditions enough leeway for a rich helminth fauna in this region. In this report studies have been made to obtain quantitative data on the Trematode parasites of a cold blooded animal, the air breathing fish *Channa punctatus.*

Key words: Trematode, *Allocreadium hanumanthai*, *Channa punctatus*, Tirhut Division, Bihar

INTRODUCTION

The state of Bihar occupies a unique place among the states of India. It is the 3rd most populous state of the nation covering about 8.07% of the total population. The northern part of the state has rich, fertile soil which is ideally suited for agriculture. There are nine divisions in Bihar.[1-3] Tirhut is one, out of them. Tirhut lies approximately between $25^0$29' to $26^0$1' N and $85^0$4' to $85^0$39' E latitude and longitude respectively. The total area of Tirhut division is 17,668 sq. kms.[4-6] Fishes particularly air breathing fishes are an important sources of animal protein, to supplement food by human beings, from the prehistoric age till today.[7,8]

Channa punctatus inhabits the fresh water, generally muddy streams, tanks and ponds and is carnivorous. It has accessory supra-branchial cavity used for breathing atmospheric air.

*Corresponding author :
Phone : 9430432405
E-mail : amriteshsms@gmail.com

Fourteen trematodes were collected from small intestine of a single *Channa punctatus* from Sitamarhi. These on further study were found to belong to genus *Allocreadium*.[9-11]

Species belonging to the genus *Allocreadium* are generally parasite in fresh water and marine fishes. Looss in the year 1890 first of all proposed genus *Allocreadium* with type species *Allocreadium isoporum*. After Looss, 1890 a good number of species have been proposed by different workers.[12,13]

Twenty four species have been reported so far of the genus *Allocreadium* up till now. *Allocreadium hanumanthai* is one of them reported by other scientists.[14,15]

MATERIAL & METHODS

Fourteen *Channa punctatus* were collected from different places of Sitamarhi district of Tirhut division. Fishes were brought to the laboratory for trematode examination. These animals (fishes) were dissected in saline solution. The internal parts such as liver, intestine etc.

thoroughly examined. The trematodes collected were placed in saline solution. Different kinds of fixatives and preservatives were used. Trematodes were stained by Gower's stain and finally mounted in D.P.X.

Gower's stain - 10 gms of carmine powder was boiled in 100ml of 45% glacial acetic acid. The solution after cooling was filtered. The filter paper was removed carefully and dried for the preparation of the stain. The process of stain's preparation is as follows -

Acidified carmine	-	1g
Potash alum	-	10g
Distilled water	-	200ml

The above ingredients were mixed and dissolved by heating then cooled and filtered. The process of filtering clear liquid (stain), a few thymol crystals were mixed in the stock solution to avoid any subsequent fungal growth. After dehydration the trematodes were treated with methyl salicylate and benzene and then mounted in DPX.

DESCRIPTION

Body is aspinose, blunt and elongated and rounded at both the ends. The length of the body is 1.859-3.336 in measurement and breadth is 0.431-0.673 in measurement. The body is wide in the region of slightly ahead of acetabulum. The oral sucker is subterminal, small and 0.192-0.256 and 0.176-0.256 in measurement. The acetabulum measures 0.145-0.193 and 0.144-0.192 in length and breadth respectively. The prepharynx is absent. The pharynx is muscular and well developed and is horse-shoe shaped and is 0.082-0.128 and 0.116-0.119 in measurement. The caeca is simple and extends upto the posterior extremity of the body. The testes are longitudinally ovoid or round entire. The anterior testis (T_1) is 0.191-0.225 and 0.160-0.250 in measurement while the posterior testis (T_2) is 0.193-0.208 and 0.144-0.178 in measurement. The anterior and posterior testis (T_1 and T_2) are situated at a distance of 0.944-1.426 and 1.152-1.854 mm from anterior end respectively.

The cirrus sac is preacetabular lying in a curved position or somewhat transeversely and measures 0.076-0.265 in length.

The ovary is oval, Pretesticular, Preequatorial in position and situated 0.704-1.060 mm from anterior end of the body.

The ovary measures 0.128-0.160 and 0.066-0.113 in length and breadth respectively. Receptaculum seminis is not visible.

The vitelline follicles commences from below the acetabulum. The right vitellarian commencement is at 1.008 mm while the left vitellarian commencement is 1.216-1.752 mm from anterior end. The total vitellarian extension of right side is 1.280-1.920 mm while that of the left side is 1.168-1.964 mm. However, in posterior part of the fluke the vitellaria of two sides appears to converge.

DISCUSSION

The present flukes have been collected from *Channa punctatus*. These differ from *A. dollfusi, A. guptai, A. kosia, A. batrachai* and *A. calbasi* in the extension of cirrus pouch. The cirrus sac extends beyond ventral sucker in the above noted species but not so in the flukes under investigation. However, the cirrus pouch in *A. hirnai* does not extend beyond acetabulum since this species also differ from the flukes under investigation.

The flukes under discussion also differ from *A. schizothoracis, A. nicolli, A. nemachilus, A. mehrai, A. makundi, A. singhi* and *A. heteropneustusius* where the cirrus pouch extend upto the middle of ventral sucker but similar is not in the case of present flukes which are under investigation. However, it shows resemblances with *A. annendalei, A. handiai, A. thapari, A. kamlai, A. fasciatusi, A. catlai, A. hanumanthai, A. manteri* and *A. glossogobii* where the cirrus pouch extends close,to acetabulum and similar is the case in the slides under investigation.

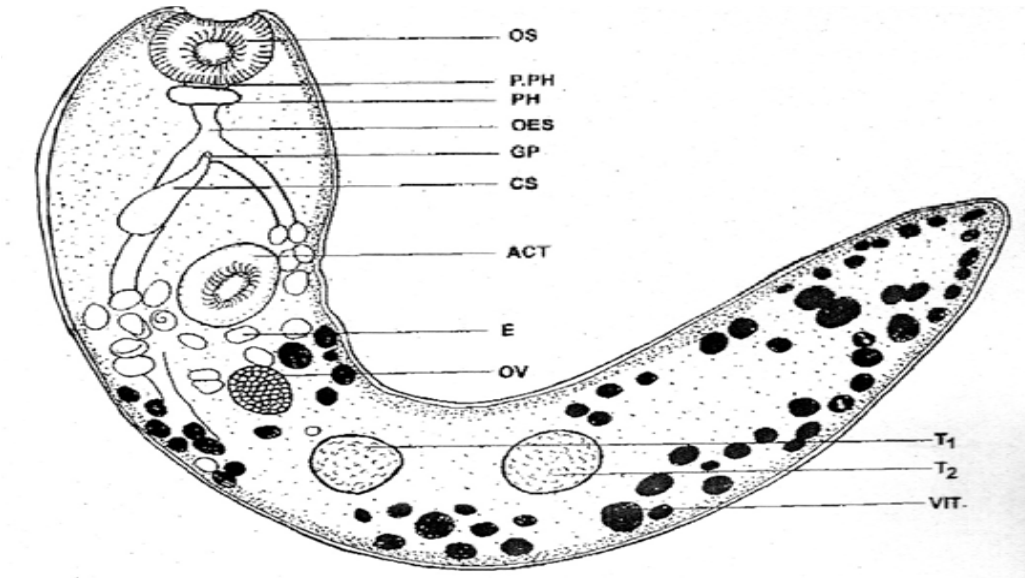
The flukes under discussion differ from *A. kamlai, A. fasciatusi, A. annadatei* and *A. monteri* in the extension of vitelline follicles where vitellaria extends from behind the anterior sucker upto the posterior end of the fluke in *A. kamlai,* vitellaria extends from the point of intestinal bifurcation to hind end of the body in *A. fasciatusi,* vitellaria extends from behind the ventral sucker to the posterior part of the body. The character also met within the flukes under discussion, but testes lying in the extreme posterior part of the body in *A. annandalei* but similar is not in the flukes under investigation, vitellaria extend from behind the ovary of acetabulum upto the posterior end of the body. This character shows similarities with the present flukes but ovary is situated just behind the acetabulum but not so in the slides under investigation.

The fluke under discussion shows similarities with *A. handiai* as far as the extension of vitellaria but differs in location of testes which are very closely situated to

each other in *A. handiai* but similar is not the case in the slides under investigation. Vitellaria extend from Pharynx to the little ahead of posterior end of the body in *A. catlai* but not so in the flukes under study the vitellaria occupy most of the area of Postacetabular part and also overlaps the intestinal caeca having a distance of 0.14 mm from posterior end in *A. hanumanthai* Sinha, 1973. Vitellaria extend from the middle region of the Pharynx upto the hind end of body and also the testes are situated in posterior part of the body in *A, manteri,* Vitellaria extend from anterior sucker upto the little ahead of anterior end of body and anterior testis (T_1) is preequatorial, posterior testis (T_2) postequatorial and ovary is pretesticular in position in *A. glossogobii.*

The fluke under discussion resembles *A. hanumanthai* Sinha, 1973 in most of its details though its occurrence from Saran district is a new record. These variations, however, are considered to be variations within the species.

Host	-	***Channa punctatus***
Location	-	Intestine
Locality	-	Sitamarhi



Allocreadium hanumanthai

	Abbriviations used		Abbriviations used
ACT -	Acetabulum	O.S -	Oral sucker
Pr.Ph -	Prepharynx	Ph_L -	Pharynx length
Ph_L -	Pharynx	T_1 -	Anterior testis
Oes. -	Oesophagus	T_2 -	Posterior testis
OV -	Ovary	CS_L -	Cirrus sac length
E -	Egg	G.P.D. -	Distance of gonopore from anterior end

REFERENCE

1. **Aliyu, M.D. and Solomon, J.R. 2012.** The intestinal parasite of *Clarias gariepinus* found at lower Usman Dam, Abuja (Nigeria). *Researcher*. **4(9):** 38-44
2. **Baugh, S.C. 1956.** Contributions to our knowledge of digenetic trematode I. *Proc.Nat.Acad.Sci* (India) 26, sect. B, Pt.**6 :** 417-430
3. **Basirullah, A. and Elahi, K.M. 1972b.** Three trematodes *(Allocreadiidae)* from the fresh water fishes of Dacca, Bangladesh, Norwegion. *Journal of Zoology.* **20:** 205-208.
4. **Banerji, M. and Chandra, K.J. 1992.** Digenetic trematodes from fresh water fishes of Mymensingh, Bangladesh Family *Allocreadiidae*. *Riv Di Parassitol* **53:**361-371
5. **B. Laxma Reddy and G. Benarjee. 2013.** Intestinal Histopathology of trematode infected fish, ***Channa striatus***. *Biolife* **1(1):** 29-31
6. **Chandra, K.J. and Yasmin, R. 2003.** Some rare and new trematodes from air breathing fresh water fishes of Bangladesh. *Indian.J.Anim.Sci.* **73:**113-118

7. **Choudhary A. Hoffnagle T.I. and Cole, R.A. 2004.** Parasites of native and non native fishes of little Colorado river grand Canyon Arizona. *The Journal of Parasitology.* **90:** 1042-1053

8. **Dayal, J. 1950.** Trematode Parasites of India fishes. Part III. Two new trematodes of the family *Allocreadiiae* from fresh water fishes in India. *Ind.J.Helm.* **2(1):**1-10

9. **Diesing, K.M. 1965.** Digenetic trematodes of fishes from North Borneo (Malaysia). *Proc.Helm.Soc.Wash.* **32(1):** 63-7

10. **Gupta, S.P. 1962.** On two new trematodes ***(Allocreadiidae)*** from the intestine of fresh water fishes of Banaras, U.P. *Proc.Helm.Soc.Wash.* **30(1):** 96-100

11. **Hamed, A.A. 1993.** Helminthes of some fresh water fish in the Sudan *Dissertation Thesis, University of Khartoum, Sudan*.

12. **Humbe Atul, Jadhav Swati & Borde Sunita. 2014.** Study of Endoparasitic infection in fresh water fishes collected from Sina Kolegoan Dam District Osmanabad (M.S.) *Science Park Research Journal* **1(40):** 01-04

13. **Jadhav, S., Anarse, S. and Borde, S.N. 2011.** Studies on the Incidence of helminth parasites in fresh water fishes from Sinakolegoan Dam, Dist. Osmanabad (MS) India. *Recent Research in Science and Technology*. **3(3):** 49-50

14. **Khalil and Thurston, J.P. 1973.** Studies on the helminth parasites of fresh water fishes in Uganda including the description of new sps. of Digeneans. *Rev.Zool.Bot.Africa* **87:** 209-248.

15. **Kumar, Sunil. 2002.** Studies on the helminth parasites of two selected commonly available vertebrates of Hajipur and its suburbs.*Ph.D. Thesis, B.R.A. Bihar University, Muzaffarpur.*

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 317-324 **34(AS).**
An International Biannual Refereed Journal of Life Sciences ***Animal Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Hematological changes in *Heteropneustes fossilis* exposed to organophosphate pesticides

Veena Kumari & Arvind Kumar*
Department of Zoology, B. N. M. V. College, B. N. M. University, Madhepura, Bihar, India

Received : 10th February, 2020 ; Revised : 20h March, 2020

Abstract: Defilement of anthropogenic xenobiotic chemicals Viz., pesticides as organophosphate–the dichlorvos alter the circulatory fluid, blood in the heterotrophic, cold blooded nektonic culturableexotic fish, Hetropneustes fossilis The fish were exposed to both lethal and sublethal concentrations of technical grade and 76% EC (Nuvan). In both the concentrations the branchial single circuit blood pumped through venous blood, right from the early point showed imbalance of homeostasis with a dire consequence of haemostasis. The alterations studied were RBC, WBC, Hb, Haematocrit, PCV, MCV and MCHC. A decrement and increment of the parameters studied were reported herewith.

Key words: *Heteropneustes fossilis*, Dichlorvos, 76% EC (Nuvan) RBC, WBC, Hb, Haematocrit, PCV, MCH and MCHC, lethal

INTRODUCTION

All pollutants are not toxicants but all toxicants are pollutants. The toxicants cause the death if it is in lethal where as in sublethal concentration; render the animal not suitable to maintain its life, the active buoyant swimmers especially the fish, heterotrophic cold blooded animals. Due to the presence even in low concentrations bring changes and in aquatic toxicology any variations in the blood serve as biomarker. Such rapid changes in the characteristics of the fish blood measured are used as indices of pollution by pesticides. The provide assessment of pollution and will be warning sign of the inhabiting organism. Such alterations either increase or decreases in the blood were studied and reported and even mentioned in the review articles.[1-4]

*Corresponding author :
Phone : 9304906785
E-mail : chiragveena010@gmail.com

Different chemicals of the four classes of pesticides organochlorines, organophosphates, carbamates and synthetic pyrethroids are reported and such changes in the fish are varied being species specific.[5-8] How ever paucity of information is available for the grass carp *Heteropneustes fossilis,* for the pesticide dichlorvos. Hence the present study is contemplated to assess the effects of the toxicant dichlorvos both in lethal and sublethal concentration using technical grade as well as 76% EC as Nuvan.[9-11]

MATERIAL & METHODS

Collection and maintenance

The freshwater fish *Heteropneustes fossilis* 3 to 5cm in length 4 to 5 gms in weight irrespective of their sex, have been chosen as the test organisms for the present investigation. Healthy and active fish were obtained from Madhepura fish hatcheries, Bihar, India in June 2015. The

fish were acclimatized to the laboratory conditions in large plastic water tanks for three weeks at a room temperature of 28±1°C.

During the period of acclimatization, the fish were fed *(ad libitum*) with groundnut oil cake and rice bran. Feeding was stopped one day prior to the acute toxicity test. All the precautions on toxicity tests to aquatic organism were followed and such acclimatized fish only were used for experimentation.[12,13]

The technical grade which was 95-98% pure which was supplied by the Syngenta India Ltd, Mumbai-400 020. The pesticide 76%EC Nuvan is locally purchased manufactured by Hikal Limited 629/630 GIDC Industrial estate Panoli Bharuch Gujarat marked by Syngenta India., 14, J. Tata Road, Mumbai. The organophosphate pesticide was introduced into water from where the pesticide entered into the fish through gill. A total of 50 fish were taken each in sublethal and lethal concentrations for both technical grade and 76% EC Nuvan. The 96hrs LC_{50} values are determined by Finneysprobit analysis as mentioned in APHA.[14,15]

Fish were euthanized by an overdose of MS-222 and then weighed and measured. Blood sample was collected by caudal severance from the disease free test fish during early hours of the day and stabilized with 50 IU sodium heparin (anticoagulant)/ml blood.

The haematological variables analyzed were red blood cells count (RBC), hemoglobin (Hb), white blood cells count (WBC), haematocrit (Ht), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH) and mean corpuscular hemoglobin concentration (MCHC).

RBC count was determined with an improved Neubauer crystalline counting chamber.[16] The blood was sucked up to 0.5mark on the RBC pipette and immediately, Hayem's solution as a diluent stain was drawn up to 101 mark and the pipette was rotated between the thumb and the forefinger to facilitate adequate mixing of the solution (dilution: 1:200). The counting chamber and the cover glass were cleaned thoroughly and cover glass was placed in position over the ruled area. The fluid from the stem of the pipette was expelled as it contains only the diluting fluid. The pipette was then held at an angle of 45^0 with the tip of the pipette at the junction of the edge of cover glass and the counting chamber. A drop of blood was placed from the tip of the pipette on the central platform near the edge of the cover slip, so that the drop was sucked up between the central platform and cover slip by the capillary force. The cells were allowed to settle for 2 to 3 min. The ruled area of the counting chamber was focused under the microscope and the numbers of RBC were counted in 80 small squares (4 squares of 16 at the four corners and one of 16 at the centre). The cells touching the upper and left hand lines were counted. The cells touching the lower and the right hand lines were omitted.

The numbers of RBC per sq mm were calculated as follows

The area of Small Square: 1/400 sq mm

The depth of the counting chamber: 1/10 mm

Therefore the volume of a small square is: 1/400 x 1/10 = 1/4000 cu mm

The dilution of blood is 1:200

Total number of RBC = n x 4000 x 200/80

n =Number of cells counted in 80 small squares

WBC count was determined.[17] The blood was drawn up to 0.5 mark of WBC pipette and immediately the diluting fluid was drawn up to the 101 mark above the bulb (the dilution fluid consists of 1.5ml of glacial acetic acid and 1 ml of aqueous gentian violet solution and made up to 100 ml with distilled water). The solution was mixed thoroughly by shaking gently and allowed to stand for 3 min. Cleaned Neubauer counting chamber and cover glass were placed over the ruled area. Excess solution was expelled and a drop of fluid was allowed to flow under the cover slip by holding the pipette at an angle of 40° and allowed to stand for 2 to 3 min. The WBC was counted in the four corner square millimetres and the number of WBC per cubic millimetre was calculated.

Hb concentration in the blood was estimated by cyanmethaemoglobin method.[18] Hb is converted into cyanmethaemoglobin by the addition of potassium ferricyanide (KCN) and the colour was read in a spectrophotometer at 540 nm against a reagent blank.

Packed cell volume was determined by micro haematocrit method.[19] The heparinised blood was filled up to the mark 100 of the haematocrit tube with the help of Pasteur pipette and centrifuged at 3000 rpm for 30min. The relative volume of the height of the RBC's packed at the bottom of the haematocrit tube was recorded as packed cell volume in terms of percentage of total blood column taken in the haematocrit tube.

Mean corpuscular volume (MCV) MCV indicates the average size of the blood cell in a given sample of blood.

MCV was calculated by the following formula and expressed as femtoliter (fL).

MCV = Haematocrit (%) x10/RBC count.

Mean corpuscular haemoglobin MCH represents the average content of the HB in each red blood cell. MCH is influenced by the HB concentration and the number of RBC. MCH was calculated by the following formula and expressed in picogram (pg).

MCH = haemoglobin (g/dL) x 10/ RBC count

Mean corpuscular haemoglobin concentration MCHC reflects the average concentration of the haemoglobin in the red blood cells in a given of the blood. MCHC was obtained by the following formula and expressed in terms of gram percent (g%).

MCHC = haemoglobin (g/dL) x 100/ haemoglobin (%)

RESULT & DISCUSSION

The blood parameters in both lethal and sublethal concentration of technical grade as well as 76% EC Nuvan are presented in the Figure-1 and represented as graph in Figure-1. Marked changes are observed in exposed fish in the commercial formulation experimented fish resulted more percentage of changes rather than the technical grade, because of the ingredients mixed imparting additive toxicity. The RBC count decreased in both lethal and sublethal concentration and more percentage of decrement in 76% EC. The WBC count increased in both lethal and sublethal more in 76% EC. The haemoglobin content decreased in both lethal and sublethal concentration more in 76% EC. The haematocrit value altered accordingly to the RBC count. Accordingly the calculated values of MCV, MCH and MCHC also showed changes.

The review article mentioned about organochlorines organophosphates, carbamates and synthetic pyrethroids alter the haematological changes as effects in fish. The work of organophosphates was reviewed and gave a thought that the pesticides do effect the biochemical changes in the blood and also parameters of the blood change.[20-22] In the opinion of the review articles on toxicity of pesticides in fish documented the work where pesticide induce alteration in blood parameters and stressed the impact of such changes.[23,24] Haematobiochemical changes induced by pyrethroid insecticides in avian, fish and mammalian species.[25] The effects of pesticides in the blood of fish serve as indices of toxicity. The review on dichlorvos toxicity in fish indirectly opined that the toxic action result in fish alteration are likely and that is mentioned. The dichlorvos toxicity on fish, a review published and commented on the toxic effect of biochemical nature of the circulating fluid.[26] The review mentioned that the haematological parameters are important in toxicological research and indicative of toxic stress. After exposure to two sublethal concentration 203 µg/L and 68.0 µg/L of chlorpyrifos for 1, 3, 7, 14 and 21 days a decreased erythrocyte count, leucocyte count, haemoglobin and haematocrit mean values in *Channa punctatus* (Bloch)

Toxication altered the haematological parameters in *Cyprinus carpio* particularly ESR count due to chlorpyrifos toxicity which is another organophosphate.[27-29] The same was opined, for the same chlorpyrifos to *Cyprinus carpio*.[30] In bony fish *Tilapia guineensis* exposed to common pesticides produced characteristic alteration in the blood .[31]The review on the toxicity and other effects of dichlorvos an organophosphates pesticides to freshwater fish, documented exposure of *Orieochromis niloticus* to dichlorvos for 96 hours duration marked changes of packed cell volume (PCV), haemoglobin (Hb), Red Blood cell (RBC), neutrophil, monocyte and lymphocytes count. The review also mentioned the work done on the fish *Cyprinus carpio* exposed to dichlorvos.[32] The other works also emphasized the haematological changes are no doubt serve indices of toxicity.[33-36]

Haematological changes of silver carp *Hypoph thalmichthys molitrix* in response to diazonin pesticide after the exposed to half of the lethal concentration was reported.[37] The results showed that leucocytes (WBC), haematocrit (Ht) haemoglobin (Hb), MCHC and lymphocyte cortisol and glucose increased where MCV and MCH were significantly decreased. They also reported to significant differences in RBC Monocyte and Eosinophils among the fish exposed to the toxicant which are derivative changes. Haematological parameters of *Cyprinus carpio* exposed to Monocrotophosan organophosphate exposed to 4.5 ppm, 6.7 ppm and 13.5 ppm was studied and found that a significant decrease in all the sublethal concentrations and maximum decrease at 12.5 ppm RBC content, ESR PCV and a significant increase in RBC and WBC count was reported.[38] The effect of Glyphosite (Roundup 41%) on the haematology of the freshwater fish *Catla catla* was studied and the RBC, WBC, haemoglobin and haematocrit value showed marked decline while MCV, MCH and MCHC showed fluctuating pattern.[39] The present work can be

correlated with this work, but both of them belong to same group of pesticides i.e., organophosphate.

Changes in haematology of the freshwater fish *Channa punctatus* (Bloch) exposed to delta methrin for 45 days was reported, where there was a significant decrease.[40]

In haemoglobin content, total erythrocyte count, PCV, MCV and MCHC. On the other hand a significant increase in total leucocyte count and MCH which coincides the present work. Toxic impact of pesticide trichlorofon on the morphological characteristics of blood cells of fish *Channa punctatus* (Bloch) and opined not only the quantitative aspects of the blood parameters are altered but also the morphological aspects of the fish where compared with control are altered. Anisocytosis deformed erythrocytes and large bulged nucleus, deformed cells are clumped and finally cell lysis, an increase in the size of MCV has been associated in response to stress.[41] Haematological changes induced by pyrethroid insecticide fenvalerate in cat fish *Clarias gariepinus* was reported a significant decrease in the haemoglobin content haematocrit value and erythrocytes whereas the leukocyte was increased.[42] The work is similar of the present report even though both of them belong to two different classes of pesticides. Haematological response of grass carp *Ctenopharyngodon idella* after exposed to endosulphan was studied.[43] The fish were exposed to two sublethal concentrations for 15, 30 and 45 days. The RBC, haemoglobin content and hematocrit values showed declined trend and MCH, MCHC also showed declined trend while MCV showed an increasing trend. She opined variation in erythrocyte count is the indices of polluted aquatic environment. The present study also showed similar results but both of them belong to different class of pesticides.

The effects of Deltamethrin on haematological indices of Indian major carp *Cirrhinus mrigala* (Hamilton) was studied by exposing the fish to lethal and sublethal concentrations and the results showed that the RBC, HB and Haematocrit values decreased in both lethal and sublethal concentration. WBC, MCV and MCH were increased consequently.[44] An increase in WBC revealed haematological toxicity coincide the present work. Haematological changes in African catfish *Clarias gariepinus* exposure to mixture of Atrazine and metolachlor in the laboratory was studied.[45] Because of mixture, the two chemicals behaved differently when compared with the present result. A significant reduction in Hb, RBC, PCV and MCV were observed. The MCHC of result showed an increase. The haematological parameters are used in monitoring health status of the fish in the wild and culture medium. Even the present study fish is a culturable one and stringent steps have to be taken not to use the pesticides in disease management of culture practices. Haematological changes in the fish *Channa punctatus* (Bloch) exposed to fenvalerate. The percentage of haemoglobin PCV, RBC showed a gradual increase while WBC showed are increase.[46] The result showed similar trend of the present study. Haematological effects of commonly used fungicide on fish *Clarias batrachus* was studied where they exposed the fish to sublethal concentration and blood was assayed for selected haematological parameters.[47] A decrease in the Hb, MCV, MCHC, RBC levels and an increase in WBC level was observed. The report in strictly different owing to chemical nature of the two toxicants i.e., Chloropyrifos and Mancozeb.

Haematological changes in the fish *Cyprinus carpio* exposed to permethrin, a synthetic pyrethroid of class I type was reported.[48] Where in a decrease in RBC, Hb, PCV was found but an increase in WBC, MCV, MCH, MCHC was observed when the fish is exposed to technical grade and 25% ECin both lethal and sublethal concentrations. Commercial formulation concentration was causing more alteration because of the ingredients present which shows similar result of the present study even though both belong to different classes. A similar report was observed in sublethal effects of dichlorvos on the freshwater fish *Cyprinus carpio* var. Communis.[49] The fish were exposed to three sublethal concentrations 50% 60% and 70% and the results showed changes of the RBC and WBC count, Hb and haematocrit and PCV. Since osmoregulatory dysfunction was the cause for increase in the rate of destructions in hematopoietic organs. The same is reiterated. Studies on freshwater fish *Channa punctatus* (Bloch) in sublethal toxicity of deltamethrin in relation to sex was reported. The fish exposed to different concentrations of deltamethrin ranging from 0.02ppm and 0.2ppm of 5 exposure periods 24, 7, 15,20 and 30 days in sub lethal concentrations and due to long duration exposure, due to change in haemostatic mechanisms the fish showing leucocytosis initially and fish might have evoked a changed and neutralized the toxicant effects of deltamethrin.

However the present study cannot be compared with this result because of a different methodology. Haematological alterations in *Cyprinus carpio* as biomarkers of cypermethrin toxicity exposed to lethal and 1/10th of 96h LC_{50} values as sublethal for 5, 10, 15 days. RBC count, Hb content, PCV showed decrement at both lethal sublethal concentrations. WBC count and MCHC exhibited increasing trend at sublethal and decreasing trend at lethal concentrations whereas MCH increased in both concentrations. The presence of the toxicant disturbs the balance in the echo systems and the fish is definitely not comfortable.

CONCLUSION

Hence it may be concluded that, even the *Heteropneustes fossilis* a grass carp cultured along with the other major carps, when pesticides contaminate the culture medium, alter the blood parameters and such alterations are more severe in EC due to the ingredients mixed. Hence stringent measures have to be taken for quality control before giving pesticide representativeness for environmental usage.

REFERENCES

1. **Sahan A, Altun T, Cevik F, Cengizler I, Nevsat E, Genc E. 2007.** Comparative study of some haematological parameters in Europeaneel (*Anguilla anguilla* L., 1758) caught from different regions of Ceyhan River (Adana,Turkey). E.U. *Journal of Fisheries and Aquatic Sciences*. **24 (1-2):** 167–171.
2. **Di Giulio RT, Hinton DE: The toxicology of fishes. Taylor & Francis Group. 2008.** Hepatic antioxidant status and hematological parameters in rainbow trout, *Oncorhynchus mykiss*, after chronic exposure to carbamazepine. *Chemico-Biological Interactions*. **183:** 98–104.
3. **Velisek J, Stara A, Svobodova. Z.2011.** The effects of pyrethroid and trazine pesticides on fish physiology. In: Pesticides in the Modern World–Pests Control and Pesticides Exposure and Toxicity Assessment, Stoytcheva, M., (Ed.). *InTech Open Access Publisher*. 377–402.
4. **Shinde BR, Borane VR, Zambare SP 2014.** Ascorbate effect on endosulfan and cypermethrin induced alterations in the RBC of the fresh water fish, *Channa orientalis* (scheinder*). Journal of Environmental Research and Development*. **8(3A):** 673-675.
5. **Ullah S, Zorriehzahra MJ: Ecotoxicology2015:** A review of pesticides induced toxicity in fish.Advances in Animal and Veterinary Sciences**. 3(1):** 40-57.
6. **Shankar Murthy K, Kiran BR, Venkateshwarlu M. 2013.** A review on toxicity of pesticides in fish. *International journal of open scientific research.* **1(1):**15-36.
7. **Chandra SekharaRao J, Neelima P, GovindaRao K2017:** A review on the toxicity and other effects of Dichlorvos, an organophosphate pesticide to the freshwater fish. *Bioscience Discovery*. **8(3):**402- 415.
8. **Das S 2013:** A review of dichlorvos toxicity in fish. Current world environment.**8 (1):** 143-149.
9. **Neelima P, Sunitha K, Gopala Rao N, Chandra S, Rao J.2015.** Haematological alterations in *Cyprinus carpio* as biomarkers of cypermethrin toxicity. *International journal of current research*. **7(8):** 18864-18870.
10. **HomairaAfreen, Toufiq-Ul Amin, Siddiqul Islam. Md, Salma Parvin, Rayhanus Salam. 2016.** In vitro phytochemical analysis and cytotoxic assay of leaves of *Solanum lycopersicum* Linn by brine shrimp bioassay. *Journal of Innovations in Pharmaceuticals and Biological Sciences*. **3 (3):**81-85.
11. **Finney DJ.** Probit analysis 3rd Ed., *Cambridge University Press, London, New York*, 333p.
12. **APHA, AWWA and WEF.1998.** Standard methods for the examination of water and waste water, 20th Edition, Clesceri, L.S. Greenberg, A.E. and Eaton, A.D. (Eds.),; American Public Health Association, American Water Work Association, Water Environment Federation, Washington DC. USA.
13. **Shaperclaus. W. 1979.** Fish Diseases, vol. 1. AkademieVerlag, Berlin.570.
14. **Donald H and Bonford RR. 1963.** Hutchinson Clinical Methods. 14th Edtion. London.; 145.

15. **Blaxhall PC, Daisley KW. 1973.** Routine haematological methods for use with fish blood. *Journal of fish biology*; **5:** 771- 781.

16. **Schalm OW, Jain NC, Carroll EJ. 1975.** Veterinary Haematology, 3rd Edition.; Lea and Febiger, Philadelphia.

17. **Saeedi FM, Roodsari HV, Zamini A, Mirrasooli E, Kazemi R. 2012.** The Effects of Diazinon on Behavior and Diazinon on Behaviour and Some Haematological Parameters of Fry Rainbow. Trout (*Oncorhynchus mykiss*). *World Journal of Fisheries and Marine Sciences.* **4(4):** 369- 375.

18. **Sampath K, James R, Velammal S. 2003.** Effect of methyl parathion on blood parameters and its recovery in a catfish, *Mystus keletius*. *Indian Journal of Fisheries.* **50(2):** 191-197.

19. **Svoboda M, Luskova V, Drastichova J, Zlabek V. 2001.** The effect of diazinon on haematological indices of common carp (*Cyprinus carpio* L.). *Acta Vetrinary Bruno*. **70:** 457-465.

20. **Satyanarayan S, Bejankiwar RS, Chaudhari PR, Kotangale JP, Satyanarayan A 2004:** Impact of some chlorinated pesticides on the haematology of the fish *Cyprinus carpio* and *Puntius ticto*. *Journal of Environmental Science.* **16(4):** 631-634.

21. **Geyer HJ, Steinberg CE, Scheunert I, Brüggemann R, Schütz W, Kettrup A, Rozman K. 1993.**A review of the relationship between acute toxicity (LC50) of gamma-hexachlorocyclohexane. (gamma-HCH, Lindane) and total lipid content of different fish species. *Toxicology*. **83 (1-3):**169-179.

22. **Ahrar khan, Latif Ahmad and Muhammad Zargham Khan. 2012.** Review article, Hemoto -Biochemical changes induced by pyrethroid insecticides in Avian, fish and mammalian species. *International Journal of agriculture & biology*. **14(5):**834-842.

23. **Deka S, Mahanta R. 2015.** Dichlorvos toxicity on fish –A review. *European Journal of Biological Research.* **5 (3):**78-85.

24. **Sunanda M, Chandra SekharaRao J, Neelima P, GovindaRao K, Simhachalam G. 2016.** Effects of Chlorpyrifos (an Organophosphate Pesticide) in Fish. *International Journal of Pharmaceutical Sciences Review and Research*; **39(1):** 299-305.

25. **Ali D, Kumar S. 2012.** Study on the effect of chlorpyrifos on acetylcholinesterase and hematological response in *Channa punctatus* (Bloch*), IIOAB Journal*. **3(5):** 12-18.

26. **Malla FA, Sharma G, Singh S. 2009.** Chlorpyrifos pesticide toxicity on erythrocyte sedimentation rate in fish, *Channa punctatus* (Bloch.) *Biology and Medicine*; **1(2):** 54-55.

27. **Ramesh M, Saravanan M 2008.** Haematological and biochemical responses in a freshwater fish *Cyprinus carpio* exposed to chlorpyrifos. *International Journal of Integrative Biology*; **3(1):** 80-83.

28. **Chindah AC, Sikoki FD, Vincent-Akpu I. 2004.** Changes in haematological characteristics of a bony fish, *Tilapia guineensis* (Bleeker, 1862), exposed to common pesticides in the Niger Delta Wetland, Nigeria, *Ghana Journal of Agricultural Science*; **37:** 59

29. **Tak AM, Bhat FA, Jan U, Shah GH 2014.** Sublethal haematological effects of dichlorvos on the freshwater fish, *Cyprinus carpio* var. communis. *International Journal of Recent Scientific Research;* **5(7):** 1334- 1337.

30. **Lakshmanan SA, Rajendran C, Sivasubramaniyan 2013.** Studies on impact of Dichlorvos on selected haematological parameters of freshwater fish, *Orechromis mossambicus* (Peters*). International Research Journal of Biological Science*s. **3(1):** 28-33.

31. **Ashade OO, Ashiru AW, Obiri CM 2011.** The comparative study of the toxic effects of 2-3 diclorovinyl dimethyl phosphate (DDVP) and chlorpyrifos on the behaviour and haematology of african- (*Clarias gariepinus). International Journal of Science and Society Yabatech*. **1(1):** 38-47.

32. **Sobhana KS, George KC, Sanil NK, Surendran KK, Paul Raj R. 2006.** Immunosuppressive action of nuvan (dichlorvos) in the banded pearl spot, *Etroplus suratensis* - evaluation of effect on haematology and humoral immune response. *Journal of the Marine Biological Association of India*; **48(1):** 29-34

33. Das BK, Mukherjee SC 2000. Sublethal effect of quinalphos on selected blood parameters of *Labeo rohita* (Ham.) fingerlings. *Asian Fisheries Science*; **13**: 225-233.

34. Hedayati A, Hassan NatajNiazie E 2015. Hematological changes of silver carp in response to Diazinon pesticide. *Journal of Environmental Health Science*

35. Vaiyanan V, Sridharan G, Raveendran S, Chairman K 2015. Impact of pesticide on haematological parameters of *Cyprinus carpio*. *World Journal of pharmacy and pharmaceutical sciences*; **4(8)**: 1424-1430*and Engineering*;**13 (52):** 1-5

36. Jerald Felix F, Saradhamani N. 2015. Impact of the herbicide glyphosate roundup (41%) on the haematology of the freshwater fish, *Catla catla* (Hamilton). *IOSR Journal of Environmental Science, and Food Technology*; **9 (4) 3**: 56-60.

37. Jayaprakash C, Shettu N. 2013. Changes in the hematology of the freshwater fish, Channa punctatus (Bloch) exposed to the toxicity of deltamethrin. *Journal of Chemical & Pharmaceutical Research*;**5(6)**:178-183.

38. Mishra A. 2017. Toxic impact of pesticides on the morphological characteristics of blood cells of fish *Channa punctatus* (Bloch*) Indian Journal of scientific Research*. **12 (2):** 068-072.

39. Sheikh Jamal AL. 2006. Haematological changes induced by a pyrethroid insecticide fenvalerate in catfish *Clarias gariepinus*. *Sci. J. Fac. Sci. Minufa Univ*.; **20**: 89 -100.

40. Renu Bala Singla. 2016. Morphometric alterations in exotic freshwater fish, *Ctenopharyngodon idella* (Cuv. and Val.) upon exposure to endosulfan *International Journal of Fisheries and Aquatic Studies*. **4(3):** 590- 593.

41. David M, Sangeetha J, Shrinivas J, Harish ER, Naik VR. 2015. Effects of deltamethrin on haematological indices of indian major carp, *Cirrhinus mrigala* (Hamilton) *International Journal of Pure and Applied Zoology*. **3(1):** 37-43.

42. George ADI, Akinrotimi OA, Nwokoma UK. 2017. Haematological changes in African Catfish (*Clarias gariepinus*) exposed to mixture of Atrazine and Metolachlor in the laboratory. *Journal of Fisheries Sciences*. **11(3):**048-054.

43. Tilak K S, Satyavardhan K. 2002. Effect of feneveralate on oxygen consumption and haematological parameters in the fish, *Channa punctatus* (Bloch). *Journal of Aquatic Biology*; **17**:81-86.

44. Jaya Shahi, Ajay Singh. 2014. Genotoxic and haematological effect of commonly used fungicide on fish *Clarias batracus*. *Journal of Biology and Earth Sciences*. **4 (2):** B137-B143.

45. GopalaRao N, Bala Krishna Naik R, SrinivasaRao G 2017: Haematological changes in the fish *Cyprinus carpio* exposed to a synthetic pyrethroid [Class I], Permethrin and its 25% EC. *Current Trends in Technology and Science*. **6 (5):** 759-763.

46. Abdul MajidTak, Bhat FA, UlfatJan, Mustafa Shah G. 2014. Sublethal haematological effects of dichlorvos on the fresh water fish, *Cyprinus carpio* var. Communis. *International Journal of Recent Scientific Research Research*. **5 (7):**1334-1337.

47. Jenkins F, SmithJ, Rajanna B, Shameem U, Umadevi K, Sandhya V, Mahadevi R. 2003. Effect of sublethal concentration of endosulfan on haematological and serum biochemical parameters in the carp *Cyprinus carpio*. *Bulletin of Environmental Contamination and Toxicology.* **70:**993-997.

48. Seth N, Saxena KK. 2003. Hematological responses in a freshwater fish, *Channa punctatus* due to fenvalerate. *Bulletin of Environmental Conta mination and Toxicology*. **71:** 1192-1199.

49. Venkata Ramudu M, Nagabhushan Reddy M, Chennaiah K, Indira P. 2008. Carbohydrate metabolism in freshwater fish *Channa punctatus* (Bloch) during sublethal toxicity of deltamethrin in relation to sex. *Bulletin of Pure & Applied Sciences Section A*. **27(1):** 7-16.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 325-330 **35(AS).**
An International Biannual Refereed Journal of Life Sciences ***Animal Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Heterogeneity and distribution patterns of ABO and RH phenotypes in the Madhepura, Bihar

Rana Pratap Ranjan*, D.N.Singh & Arun Kumar
University Department of Zoology B.N.M.University, Madhepura, Bihar, India

Received : 12th February, 2020 ; Revised : 23rd March, 2020

Abstract: Blood groups (antigens) are proteins, glycoproteins or glycolipids inherited surface markers on the red blood cell membranes, which determine the blood phenotypes of human beings. There are 36 blood group systems with over 300 antigens. Among them, ABO and Rh are of clinical significance. For a safe transfusion, a blood donor and recipient should be ABO and Rh compatible. The heterogeneity of ABO and RhD blood groups systems shows variations in different parts of the world. In Madhepura, there is limited study done on ABO and Rh D blood groups heterogeneity. The aim of this study is to determine the heterogeneity and prevalence of ABO and Rh D blood groups among population of the district. The presence of blood groups was determined by serological techniques both microtitre and tube methods. Commercial monoclonal antisera (anti-A, anti-B, anti-D and Antihuman globulin) were used. The study revealed that there is heterogeneity in the ABO and RhD blood groups.

Key words: ABO, RHD, anti-human globulin (AHG) monoclonal antibodies, heterogeneity, prevalence.

INTRODUCTION

Blood groups (antigens) are inherited surface markers on the red blood cells membrane. These antigens are proteins, glycoproteins or glycolipids on the red cell membrane surface; they determine blood phenotypes of suggesting their involvement in ammonium transport or in maintaining the symmetry of the phospholipid in the red cell membrane. Many suggestions have been made to indicate that the Rh proteins are involved in CO_2 and O_2 transportation. However, all these assumptions are un-confirmed.[1,2]

Most of the 36 defined blood groups systems exhibit the null phenotype, which is associated with lack of specific blood group antigen expression on the red cell membranes. The null phenotype is detected serologically by absence of agglutination with specific antisera.

Table 1: A Phenotype heterogeneity

Blood Type	Number	Percentage
A	97	24.5
Total	500	100

Table 2. A phenotype variation across the ABO phenotypes

Parameters	N	Percentage
Total	97	100
Variation	2.87	100

Blood groups systems consist of single or more antigens that are governed by a solitary gene locus or by

*Corresponding author :
Phone : 9955385988, 7004882230
E-mail : ranaranjan1970@gmail.com

homologous genes.[2,3] They were first discovered in 1901 by Karl Landsteiner (Nobel Prize Winner) when he saw that plasma from a few people agglutinated red cells from others but failed to agglutinate others thus he named the blood as A, B and O. Afterwards, Decastello and Sturli added a fourth group AB.[3] Forty years later, Levine and Stetson depicted the Rh blood amass framework. This was while they were investigating a Haemolytic Transfusion Reaction (HTR) in a woman who reacted to a blood transfusion from her husband.[4]

ABO and Rh blood group systems are of clinically significance in transfusion medicine because they can cause HTRs and haemolytic disease of the fetus and the newborn plus transplant rejection. By 2016, up to 346 antigens have been described of which 306 are clustered within 36 blood group systems. The remaining 38 antigens are yet to be assigned to any known blood group systems.[5]

Before 1950s, human blood groups were known as pattern of inheritance that could only be detected serologically. In the 1950s, biochemical analysis aided the revelation of the structure of carbohydrate and protein antigen, which resulted in the description of the functions of some blood group antigens. Some of these surface markers are now known to have specific functions such as membrane transporters seen in Diego (anion transporter), Kidd (urea transporter), and the Colton (water channel) receptors expressed by Duffy, complement regulatory glycoproteins seen in Cromer and Knops, enzymatic function as observed with MN antigen. Functions of the Rh proteins and the Rh-associated glycoproteins RhAG proteins are not yet fully understood, but many assumptions have been made due to their structural characteristics of membrane transporters is caused by mutations, including deletions, missense mutations, frame shift mutations, introns splice-site and promoter mutations which cause lack of antigen expression on the surface of the red cell. The null phenotypes are rare but clinically significant. Individuals with null phenotypes are at risk of immunization due to transfusion of incompatible blood or pregnancy via fetal maternal haemorrhage. Variation of antigen expression has also been reported which is caused by nucleotide duplications and hybrid gene as seen in weak or/and partial RhD.[6-9]

MATERIALS AND METHODS

A total of 500 blood samples collected in EDTA vials were randomly sampled. The whole blood samples were typed serologically by use of commercially acquired monoclonal antisera at national blood grouping laboratory at KNBTS. Both microtitre and tube grouping methods were employed. Forward typing of the samples was done in the microtitre plate by mixing 1 drop of 2% red blood cells saline suspension that was prepared in normal saline with 1 drop of the appropriate antisera (anti-A and anti-B).

Reverse grouping was done using the microtitre plate method by mixing 1 drop of plasma of the whole blood samples with 1 drop of known A,B, and O red blood cells. Rh typing was accomplished by mixing I drop of 3% red cell saline suspension with 1 drop of anti-D in the microtitre plates. All the microtitre plates and contents were loaded into the microtitre plate centrifuge and spun at 2000 rpm for 1 min. The microtitre plates' contents were locked for 1 min and examined using a magnifying glass mirror for agglutination. The Rh D negative blood samples were retyped using the tube method and confirmed for Du positive using indirect antihuman globulin test tube method.

RESULTS AND DISCUSSION

The results showed that there was a statistically significant difference $p < 0.01$ between the positive and negative blood types in both the ABO and Rh D systems. Blood group A (24.25%): positive (23.25%), negative (0.5%), weak D (0.25%) are shown in Tables 1 to 3; Blood group B (18.75%): positive (17.25%), negative (1%), weak D (0.25%) are shown in Tables 4 to 6;

Table 3. Differences between A Phenotype categories

ABO blood group system		Blood group category		Total
		A Positive	A Negative	
Positive	N	95	0	95
%		100	0	97.94
Negative	N	0	2	2
%		0	100	2.06
Total	N	95	2	97
%		100	100	100

Table 4. Blood group B variation across the ABO Blood groups.

Parameters	N	Percentage
Total	74	100
Variation	5.38	100

Table 5. Blood group B in percentages.

Blood type	Number	Percentage
B	74	18.5
Total	400	100

Blood group O (51.75%): positive (46.75%), negative (3.5%), weak D (1.5%) are shown in Tables 7 to 9. The

least was AB (5.5%) and all were positive (Tables 10 and 11). Among the RhD system, D positive was the most common (93%), followed by D negative (5%) and weak D was the least common (2%) (Tables 12).

Knowledge of blood group ABO heterogeneity is very important in planning for recruitment of voluntary blood donors in a country. It is also essential in the management of blood stocks, be it in the transfusing facilities or the national blood service activities. Typing donors and recipients blood for ABO antigens is essential in blood transfusion, genetic research, organ transplantation, settling paternity disputes, and in cases of immunization complications.[9-11]

Table 6. Frequency of blood group O in percentages.

Blood type	Number	Percentage
O	207	51.75
Total	400	100

Table 7. Blood group O variation.

Parameters	Number	Percentage
Total	207	100.0
Variation	2.98	100.0

Table 8. Blood group O categories differences.

ABO blood group system	Blood group category		Total
	O Positive	O Negative	
Positive N	3	0	193
%	100	0	93.24
Negative N	0	14	14
%	0	100	6.76
Total N	193	14	207
%	100	100	100

Table 9. Blood group AB distribution in percentages.

Blood type	Number	Percentage
AB	22	5.5
Total	400	100

Table 10. AB phenotype variation.

Blood type	Number	Percentage
Total	22	100
Variation	0.73	100

Table 11. RhD blood group variation

RHD	Number	%
RHD Positive	372	93
RHD Negative	20	5
Weak D	8	2
Total	400	100

The finding of this study analysis revealed that A phenotype heterogeneity was 97 (24.25%) in the population. Further, the analysis of the 97 samples indicated a variation of 2.87 across the ABO blood groups. There was a statistically significant difference ($p<0.001$) between the A positive and A negative blood phenotypes in the ABO system. The outcome of other studies from different countries have also shown a wide difference in the A phenotype heterogeneity; Kurd 37.2% (positive 29.99% and negative 2.75%). In Ethiopia, prevalence is 32.74%. Different patterns of A phenotype have also been observed in other parts of the world. In Delhi India, the A phenotype heterogeneity was recorded as 24.66%.[2] The Delhi outcome was almost similar to the findings of this study of 24.25%. The frequency of the A phenotype in UK was recorded as 43% of which A positive was 36.6% and A negative was 6.5%, respectively.[12,13] This is to mean that in the ABO system, individuals who are of A phenotype are either A positive or A negative.

The study outcome revealed that the variation of blood group A in the Madhepura population is 18.5% with a variation of 5.38 across the ABO blood groups. These results were further categorized into two; B positive which showed 17.5 and 1% B negative. There was a statistically significant difference $p<0.001$ between the B positive and B negative categories in the ABO blood group system.

According to studies from other countries, wide variations were also recorded in the B phenotype. In UK, blood group B was recorded as 9%. B positive was recorded as 7.6% and negative as 1.4%. In Ethiopia, record showed that B type was 20.9%.[3] Among the Kurd population, B blood group was 23.84%.[9] India recorded 32.26% of blood B type , 36.4% was recorded in Western Rajasthan, 34.15% in Bangladesh, 32.73% in Delhi and 29.95% in Bangalore. This was also shown by studies carried out in UK, Ethiopia and Kurd. Studies carried out in India showed that blood group B was the most common in that population.

The study outcome revealed that the frequency of blood group O was 51.75% (207/400) with a variation of 2.98 across the ABO blood groups. This outcome was further separated into O positive and O negative categories. The O positive was 193 (48.25%) and negative was 14 (3.5%). The outcome also indicated that there was a statistically significant difference $p<0.001$ between the O positive and negative categories in the ABO blood system. The results showed a slight variation in the blood group O frequency when compared with other studies. Studies from other countries such as UK, Kurd and Ethiopia also showed

different frequency patterns of blood group O. In UK, the frequency of phenotype O was 45%. Among the 45% O blood type, O positive was shown to be 38.25% and negative 6.75%.[9,2] In Kurd O phenotype frequency was 37.16% of which O positive was 34.03% and O negative was 3.13%.[8] Studies from India showed that O blood group frequency was 37.1%.[14] The frequency of O phenotype among the Ethiopians was 42.1%[15] and 30.42% in Delhi.[16] The study results revealed that blood group O was the most common blood phenotype in Kenya. This compares well with other studies carried out in UK, Ethiopia and Kurd. In summary, blood group O frequency was 51.75% of which 48.25% was O positive and 3.5% was negative. The variation was 2.98 across the ABO blood groups. There was a statistically significant difference $p<0.001$ between the positive and negative categories.

The study outcome indicated that AB type was 22 (5.5%) with a variation of 0.73 among the ABO blood groups. The study results grouped all the AB blood type in the AB positive category. Most studies carried out in different countries also recorded variation in the distribution of AB blood group. In UK, AB phenotype distribution was shown to be 3%, AB positive was 2.5% and negative was 0.5%. Study outcome from Kurds indicated that AB was 6.35% in the donor population. Among Kurd population, for AB distribution pattern, 6.02% was AB positive and 0.33% was negative.[15] In an Ethiopian study, AB distribution pattern was 4.3% . In the Indian population, the AB blood phenotype distribution pattern ranges from 6.3% in Western Rajasthan to 12.2% in Delhi.[15]

The study revealed that there was variation in the RhD system. RhD positive was 93%, negative was 5% and weak D was 2%. There was a statistically significant difference $p<0.001$ between the D positive and D negative. Studies from UK indicated that 85% of the population was D positive and 15% D negative.[10] In Ethiopia, 92.8% was of D positive and 7.3% was negative.[14,17]

Study from Albania indicated that the Rh D positive was 89.0%, 10.86% was negative and 0.14% was weak D. In Indian, D antigen variation was shown to range from 91.7 to 94.61%.

REFERENCE

1. **Daniels G. 2004.** Human blood groups (2nd edn). Blackwell Science Ltd a Blackwell Publishing Company. *Transfusion Medicine* **14(4):**326- 326.

2. **Daniels G, Bromilow I. 2010.** Essential Guide to Blood Groups 2 edn Willey-Blackwell Ltd. Daniels G, Finning K, Martin P, Summers J (2007). Fetal RhD genotyping. A more efficient use of anti-D immununoglobuline. *Transfusion Clin. Biol*. **14(6):**568-571

3. **Dean L. 2005.** Blood Groups and Red Cell Antigens. [Internet] Bethesda (MD). National Center for Biotechnology Information (US); 2005. The red cell Antigens, ABO and the Rh blood groups. pp. 5-7

4. **Scott ML. 2004.** ES05.01. The complexities of the Rh system. *Vox Sanguinis* **87(S1):**58-62

5. **Story Castilho L, Chen Q, Daniels G, Denomme G, Flegel WA, Gassner C, Haas M, Hyland C, Keller M, Lomas Francis C. 2016.** International society of blood transfusion working party on red cell immunogenetics and terminology: report of the Seoul and London meetings. *ISBT Sci Ser*. **11(2):**118-122.

6. **Dean L. 2005.** Blood Groups and Red Cell Antigens. [Internet] Bethesda (MD). National Center for Biotechnology Information (US); 2005. The red cell Antigens, ABO and the Rh blood groups. pp. 5-7.

7. **International Society of Blood Transfusion ISBT. 2008.** Table of blood group systems. http:// ibgrl.blood.co.uk/isbt. 25/09 /2016

8. **Jaff MS. 2010.** ABO and rhesus blood group distribution in Kurds. *J. Blood Med*. **1:**143-146.

9. **Learoyd KR, Rogan P, Haines M. 2009.** An introduction to Blood transfusion Science and Blood Bank Practice; 5th Editon. British Blood Transfusion Society (BBTS). Printed in Great Britain; Manchester.

10. **Neil DO. 2006.** Modern Human variation: Distribution of blood types. Htm online. PDF.

11. **Periyavan S, Sangeetha SK, Marimuthu P, Manjunath BK, Seema D M. 2010.** Distribution of ABO and Rhesus-D blood groups in and around Bangalore. *Asian J. Transfusion Sci.* **4(1):**41.

12. **Rajshree B, Raj JY. 2013.** Distribution of ABO Blood Group and Rh (D) Factor in Western Rajasthan. *National J. Med. Res*. **3(1):**73-75.

13. **Behra DR, Joshi DYR. 2013.** Distribution of ABO Blood Group and Rh (D) Factor in Western Rajasthan. National J. *Med. Res*. **3(1):**73-75.

14. **Agrawal A, Tiwari AK, Mehta N, Bhattacharya P, Wankhede R, Tulsiani S, Kamath S. 2014.** ABO and Rh (D) group distribution and gene frequency; the first multicentric study in India. *Asian J. Transfusion Sci.* **8(2):**121-125.

15. **Alemu G & Mama M. 2016.** Assessing ABO/Rh blood group frequency and association with asymptomatic malaria among blood donor attending Arba Minch blood bank, south Ethiopia. *Malaria Res. treatment*.

16. **Ahuja MK, Satapathy MRK, Singh V, Singh S. 2015.** Prevalence and distribution of ABO & RH-(D) antigens in Delhi-National Capital Region.

17. **Wagner FF, Eicher NI, Jørgensen JR, Lonicer CB, Flegel WA. 2002.** DNB: A partial D with anti-D present in central Europe. *Blood.* **100(6):**2253-2256

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 331-332
An International Biannual Refereed Journal of Life Sciences

36(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Efficacy of *Nicotiana tabacum* as pesticide against *Callosobruchus maculatus* (F.) in stored gram

Nawlesh Kumar*
Department of Zoology, Magadh University, Bodh Gaya, Bihar, India

Received : 12th February, 2020 ; Revised : 23rd March, 2020

Abstract: *Callosobruchus maculatus* is a serious pest of stored pulses belonging to the order of beetles, Coleoptera hence also called pulse or bean beetle. In the present article, indigenous method of pest control by adopting the products of a pesticidal plant leaf, tobacco (*Nicotiana tabacum*) has been discussed on the basis of the findings over the natality, mortality & larval growth of pest on stored grams hampered by the tobacco plant extract. A significant result has been obtained by the decreasing trend of population growth of the pest (P> 1.05) thereby encouraging the farmers & stored grain & other product managers to adopt this safe & indigenous method pest control in a low & control dose. As tobacco product is itself very toxic & harmful to the human consumers, higher doses of application should be avoided.

Key words: *Callosobruchus maculatus,* stored gram, *Nicotiana tabacum,* efficacy & pest control.

INTRODUCTION

Many insects and mites attack foodgrain (gram) during pre and post-harvest stages. In pre harvest attack, mainly the yield is affected but attack during post-harvest period lead to both quantitative and qualitative damage.

The qualitative and quantitative losses caused to stored gram in India have been discussed by number of researchers.

Callosobruchus maculatus (F) commonly known as 'pulse beetle is & important pest of grain. It belongs to Class-insecta, Order-Coleoptera, Family-Bruchidae. Larvae of the pest eat up the grain kernel and make a cavity. Adults are short lived, harmless and do not feed on stored produce at all. Even though one grub damages only one grain, yet the pest may damage 100% of the grains (gram) in the stores. The attack of the insect not only reduces the market value of the infested grains but also adversely affects their germination.

In view of present work was undertaken to examine the efficacy of product of tobacco (*Nicotiana tabacum* against *Callosobruchus maculatus* infesting stored gram.

Tobacco is one of the important cash crops of Bihar. It is most commonly used "Khaini", 'Bidi' and cigarettes.

Nicotiana tabacum is one of the common species of tobacco in India. It is main source of nicotine. Nicotine is an alkaloid fond in leaves of tobacco. Nicotine is an intoxicating chemical.

Tobacco leaves were collected from a farmer of Vaishali District. They were first of all dried in scorching sun for 3 days. Then it was put into the incubator at 60°C temperature for 6 hours in order to make it free from any

*Corresponding author :
Phone : 87090 31596
E-mail : shailesh.sk.sk@gmail.com

moisture. The dried leaves were fed into a mechanical grinder with brought out fine brown powder of tobacco. The power was further put into oven as 50°C for 8 hours to make it free from moisture.

MATERIAL & METHODS

In order to ascertain the efficacy of tobacco powder sample of 100 gm of gram were taken in containers of equal size. Eight doses of tobacco powder i.e. 0.5gm, 1gm, 1.5gm, 2.5gm, 3gm, 3.5gm and 4gm, were taken separately with 100 gm of gram in each container. Tobacco powder was mixed with grain vigorous shaking of container. Each treatment was arranged in three replications. At the same time a set of three containers with under treated gram was established as control. 5 pairs of insect i.e. *C. maculatus* taken from laboratory culture were released in each container.

RESULT & DISCUSSION

Tobacco leaves powder (TCP) showed the significant effect.3gm, 3.5gm, and 4gm, doses brought about 100% mortality on 6 DAI where 1.5gm, 2gm and 2.5gm does caused in on 10 DAI.

The egg laying was not affected too much in lower doses. In higher doses like 3.5gm, it was reduced to 75% and in 4gm, there was no oviposition at all.

The population growth was badly affected by the treatment with tobacco *Nicotiana tabacum* leaves. In 0.5gm treatment the population growth was reduced to 75% in comparison to the control. In treatment with 1gm to 4gm TP there was no population growth at all.

Effect of Tobacco Powder (TP) Treatment

Tobacco Powder/ 100 gm. of gram	Adult Mortality Percentage							
	2 DAI	4 DAI	6 DAI	8 DAI	10 DAI	12 DAI	14 DAI	16 DAI
Control					30	63	80	100
0.5 gm.	30	50	50	56	66	80	96	100
1 gm.	33	50	70	73	80	80	93	100
1.5 gm.	50	50	60	80	100			
2 gm.	50	60	60	80	100			
2.5 gm.	50	56	60	86	100			
3 gm.	70	93	100					
3.5 gm.	80	96	100					
4 gm.	80	96	100					

DAI = Days After Infestation

ACKNOWLEDGEMENT

The author is very much thankful to the head, Dept Of Zoology, Magadh University, Bodh Gaya.

REFERENCE

1. **Cotton, R.T. 1963.** Pests of stored grain and grain. Products. Burgess Pub. Co. Minneapolis. U.SA. 93.
2. **Credcand, P.F. and Wright. A.W. 1990.** Oviposition deterrants of *Callosobruchus maculatus (Coleoptera bruchidae) Physiol Entomol*, **15:**285-298.
3. **Prasad. 1984.** Investigation on population behaviour of gram beetle *Callosobruchus chinsis* Linn. Under different storage conditions. Ph.D. thesis (Unpublished) Patna University
4. **Yadav, P.N. 1994.** Investigation on theinter – competition of some common grains on wheat Ph.D. thesis (unpublished) P.U.
5. **Verma, A.P. 1998.** The alternative use of Tobacco. *The Hindustan Time*, Dec. 5, The Competitive Edge III.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 333-334 **37(PS).**
An International Biannual Refereed Journal of Life Sciences ***Plant Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Studies on some hydrophytes of Basargadh Pond, Ranchi, Jharkhand

Merina Shweta Sweety* & Radha Krishna Jha
University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

Received : 15th February,2020 ; Revised : 1st March, 2020

Abstract: A study was conducted in Basargadh pond at Ranchi for the assessment of hydrophytes from August, 2019-July, 2020. The pond supports a number of plants species which spend their whole life or part of their life cycle in water. During the study specimens are collected from the pond. A total number of 15 hydrophytes have been collected and reported from the study area. After collection the specimens are morphotaxonomically identified by the help of experts and pertinent literatures.

Key words: Morphotaxonomically, hydrophytes, dicotyledonous, monocotyledonous

INTRODUCTION

Basargadh pond is situated in Ranchi. The water of pond is used for various domestic purposes by neighboring inhabitants. The present communication deals with the study of various hydrophytes to estimate their status and condition of the pond.

MATERIAL & METHODS

During the course of study the plants were collected from the above said area. After collection the specimens were soaked in formaldehyde solution and dried with the help of herbarium press. After drying herbarium sheets were prepared and identified by experts with the help of suitable literatures. To make the accurate study the pond was visited in regular interval of time.

RESULTS

The results are based on the morphotaxonomical studies of the collected hydrophytes. A total number of 15 aquatic and semi aquatic species are reported. Among these species, 10 belong to dicot families and 5 belong to monocot families. The taxonomic data of the plants collected and studied with their name, habitat, family and flowering/fruiting times are given below.

'A' stands for aquatic plants, 'SA' stands for semi aquatic plants and 'WL' stands for wetland species. These 'SA' and 'WL' plants grow in the surrounding wet and muddy soil of the pond.

Table 1- Name, Habitat, Family and Flowering/Fruiting time of collected Dicotyledonous hydrophytes

Si.no	Species name	Habitat	Family	Flowering/Fruiting time
1	*Nymphaea nouchali L*	A	Nymphaeaceae	Aug-Nov
2	*Ipomea aquatica* Forsk	A	Convolvulaceae	Sep-Feb
3	*Ludwigia adscendens* L.	SA	Onagraceae	Throughout the year
4	*Ranunculus scleretus* L.	A	Ranunculaceae	Nov-Feb
5	*Utricularia stellaris* L.	A	Lentibulariaceae	Sept-Nov
6	*Enhydra fluctuans* Lour.	A	Asteraceae	Dec-March
7	*Nymphoides hydrophylla* Lour.	A	Menyanthaceae	July-Nov
8	*Centella asiatica* L.	SA	Apiaceae	July-Aug
9	*Alternanthera sessilis* L.	SA	Amaranthaceae	Throughout the year
10	*Oxalis corniculata*	WL	Oxalidaxeae	April-Sept.

*Corresponding author :
Phone : 9470373562
E-mail : merina_shweta08@rediffmail.com

Table 2: Name, Habitat, Family and Flowering/Fruiting time of collected Monocotyledonous hydrophytes

Sl.No	Species Name	Habitat	Family	Flowering/Fruiting time
1	*Commelina bengalensis* L.	WL	Commelinaceae	July-Nov
2	*Hydrilla verticillata* L.	A	Hydrocharitaceae	Nov-Jan
3	*Eichornia crassipes* Mart	A	Pontederiaceae	Apr-Nov
4	*Typha angustata* Borey & Chaub	WL	Typhaceae	Apr-June
5	*Cyperus rotundus* L.	WL	Cyperaceae	June-Jan

DISCUSSION

In this present study total 15 hydrophytes have been identified and reported. Among these 10 species belong to dicot families and 5 species belong to monocot families. The plants like *Nymphoides hydrophylla, Ludwigia adscendens, Nymphaea nouchali, Enhydra fluctuans* represents that the pond is exposed to domestic sewage.

ACKNOWLEDGEMENT

The authors are thankful to the Dean and HOD, Prof.Jyoti Kumar, University Department of Botany, Ranchi University, Ranchi, Jharkhand for providing his valuable suggestions and constant encouragement.

REFERENCES

1. **Kachroo, P. 1984.**"Aquatic Biology in India", Bishen Singh Mahendra Pal Singh , 23- A, Connaught Place, Dehradun, India, P.No.27-35.
2. **Bose S.C. 1948.** The Damodar Valley Project, Phoenix Press Ltd. Calcutta
3. **SinghM.P.1990,**"Hydrophytes of Ranchi. *J.Econ. Tax. Bot.* 14 No.3.
4. **Mukharjee, P. 2001.** The Floristic & Ecological study of aquatic angiosperm of Lohardaga.Ph.D.thesis, Ranchi University, Ranchi.
5. **Ghose T.K.1971.** Flora of Ranchi. Ph.D. Thesis Ranchi University, Ranchi
6. **Jha,U.N. 1965.** Hydrophytes of Ranchi. *Trop, Ecol*. 698-105
7. **Ghose,S.K,Santra,S.C.& Mukharjee,P. K.1993.** Phenological study in aquatic macrophytic plants of lower Gangatic delta,W.B.,India.*Feddes Reportorium*. **104:**93-111
8. **Biswas, K. & Calder,C. 1936.** "Handbook of common water & marsh plants of India and Burma"Bishen Singh & Mahendra Pal Singh, 23-A,Connaught Place, Dehradun, India.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 335-338 **38(AS).**
An International Biannual Refereed Journal of Life Sciences ***Animal Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Efficacy of tulsi (*Ocimum sanctum*) leaves powder as a pesticide against *Callosobruchus maculatus* (F.) in stored gram

Nawlesh Kumar*
Department of Zoology, Magadh University, Bodh Gaya, Bihar, India

Received : 12th February, 2020 ; Revised : 03rd March, 2020

Abstract: *Ocimum sanctum* is the most popular medicinal plant with religious and sacred value. It has also unidentified but speculated insecticidal property to kill the insect pests of the stored grain products. The traditional use of tulsi powder vanalogous to that of neem has yielded considerable pest control result. As such in the present study, the efficacy of tulsi leaf powder in dried condition on the stored gram pest, *Callosobruchus maculatus* has been tested experimentally in the laboratory condition on newly emerged adults propagating on fresh gram seeds. Higher dose of sprinkling of the powder has been found to disrupt the growth & reproduction of the pest upto 70-80% (50mg::10 adults-both male & female:: 8 mortality at 30°C & 40% RH). Whereas the lower quantum powder sprinkling resulted only 20-30% pest control. (10mg::10 adults-both male & female::3 mortality at 30°C & 40% RH).

Key words: *Ocimum sanctum*, dry leaf powder, pest control efficacy, *Callosobruchus maculatus*

INTRODUCTION

The tulsi plant is a shrub of great medicinal value and is quite familiar to all. This plant, which grows in all places and under a wide variety of conditions, holds a position of sanctity and importance in our religion and generally in Indian culture as well. Nearly all of our scriptures have sung its praises. The very name 'Tulsi', that which cannot be compared, the 'Incomparable one', indicates its great value. In our ancient literature it has been variously named as Gramya, Vishnuvallabha, Subhagativra, Papaghni, Mayura, Sarla, Krishna bhootpriya, Sumanjari, Vrinda, Shweta, Vaishnavi, Kaveri etc.[1,2] Actually the names characterise the various species which possess varying characteristics.[3,4] Presumably the popular name 'Tulsi' was coined in the Medieval age with a view to propagate and popularise its miraculous medicinal properties.[5,6]

Besides India, Tulsi is also found in several European, African and Asian countries. It is found in Italy, Africa, Ethiopia, Thailand, Vietnam, Brazil, Guyana, Japan, China and Combodia. Tulsi is one of the most useful plants of India.[7] It grows wild, and can also be cultivated. The plant is 3 to 4 feet tall. It grows in all types of soils, but black sticky and moist soil is particularly suitable for it. The plant bears inflorescences about 2 inches long. The leaves are of an elongated oral shape with serrations. Modern taxonomists mention about 150 species of Tulsi and put this plant under the family Lubiateae.[8,9] Its two common varieties, Holy basil and white basil known as *Ocimum sanctum* and *Ocimum album* respectively are mostly used

*Corresponding author :
Phone : 87090 31596
E-mail : shailesh.sk.sk@gmail.com

as medicines. *Ocimum sanctum* is used against fever, respiratory ailment, cold, asthma, arthritis etc. The white basil or *Ocimum album* is mostly found in South India. Major chemical constituents of Tulsi are 0.4 to 0.8% volatile oil containing chiefly eugenol (about 21%) and β–Caryophyllene (37%). Eugenol content reaches maximum in spring and minimum in autumn.

MATERIAL & METHODS

In this set of experiment Tulsi Leaves Powder (TLP) was tested as pesticide against *C. maculatus* infesting stored gram. As described earlier TLP was already kept in laboratory as stock. 100 gm. of Bengal gram (*Cicer arietinum*) was taken from laboratory stock in separate tin container of equal size. Each grain filled container was treated with a particular dose of TLP from the laboratory stock prepared for this purpose. In this experiment 8 doses were tested, i.e. 0.5gm., 1gm., 1.5gm., 2.0gm., 2.5gm., 3.0gm., 3.5gm., and 4.0gm. Each treatment was done in three replications. Simultaneously on set of three replications was kept untreated and maintained as control. Containers with TLP treated grains were tightly closed and shaken thoroughly for 10 minutes for optimum coverage of the grain surface.

The test insects were taken from laboratory culture. Five pairs (5 males and 5 females) of freshly emerged insects were released in very container including the control. The experiments were performed at normal room temperature and humidity. At regular interval adult mortality oviposition and emergence was recorded and was statistically analysed.



Figure1-Tulsi Plant, Tulsi leaves and Tulsi leaves Powder

RESULT & DISCUSSION

Treatments of Tulsi Leaves Powder (TLP) with 0.5gm., 1gm., 1.5gm., 2gm., 2.5gm., and 3gm. did not bring about 100% mortality of adults earlier than that of control. In the control the mortality rate was 63% on 12 DAI, 80% on 14 DAI and 100% on 16 DAI. In 0.5gm.and 1gm. treatments also first mortality was observed on 12 DAI and it was 80%. In 1.5gm., and 2gm. treatments also the mortality was observed not earlier than 10th day and it was 35%. In 3gm. and 3.5gm.treatments 20% mortality of adults was observed on 4 DAI and 6 DAI respectively. Here a remarkable effect was observed. In 3.5gm and 4gm. treatments 100% mortality of adults was observed on 12 DAI.

So, in TLP treatments only higher doses i.e., 3.5gm.and 4 gm. brought about 100% mortality of insect pest earlier than that in the control.

All the TLP treatments except 4gm. treatment caused not check the oviposition completely, however, it was lowered to some extent. In 0.5gm.treatment the oviposition was similar to the control (400). The percentage reduction in oviposition in 1gm., 1.5gm., 2gm., 2.5gm., and 3.5gm. treatments was 15%, 15%, 25%, 25% and 25% respectively. There was 75% reduction in egg laying in 3.5gm. treatment with respect to the control. In 4gm. treatment there was no oviposition.

The population growth observed on 45th day showed no reduction in 0.5gm. treatment. In 1gm. and 1.5gm. treatment it was reduced to 18.75% with respect to the control. In 2gm., 2.5gm., and 3.5gm.treatment the population growth was reduced to 30%. In 3.5gm. treatment the population growth was reduced to 77.5%. in 4gm there was no growth.

Effect of Tulsi Leaves Powder (TLP) Treatment

TLP/ 100 gm. of gram	Adult Mortality Percentage							
	2 DAI	4 DAI	6 DAI	8 DAI	10 DAI	12 DAI	14 DAI	16 DAI
Control					30	63	80	100
0.5 gm.						80	90	100
1 gm.						80	90	100
1.5 gm.					35	70	80	100
2 gm.					35	70	80	100
2.5 gm.					40	80	90	100
3 gm.		20	20	40	60	70	90	100
3.5 gm.		20	20	60	80	100		
4 gm.		30	50	50	90	100		

DAI = Days After Infestation

REFERENCES

1. **Chopra, R.N. 1933.** The Indigenous drugs of India - after Sharma, B.D. (1975). Tulsi, 1st Ed. Dabur Private Limited, New Delhi, P. 114.
2. **Chopra, R.N., Roy, D.N. and Ghosh, S.M. 1941.** Insecticidal and Larvicidal action of the essential oils of *Ocimum sanctum*. *J. Malaria Inst. India.,* 4:109-112.
3. **Howe, R.W. and J.E. Currie. 1964.** Some laboratory observations on the rates of development, mortality and oviposition of several species of Bruchidae breeding in stored pulses. *Bulletin of Entomological Research.* **55:** 437 – 477.
4. **Jain, D.D. 1979.** Is Tulsi cultivation a mere superstition science Reporter, CSIR, New Delhi, Feb. 1978, P 125-126.
5. **Jain Naveen and R.K. Suri. 1980.** Insecticidal, insect repellent and Piscidal Plant of Dehradun Nagarjun, **23(8):**177-181.
6. **Maheshwari, M.L., B.M. Singh, R. Gupta and Mingjien Chine. 1987.** Essential oils of sacred Basil (*O.Sanctum*) Indian Perfumer. **31(2):** 137 – 145.
7. **Oparake, A.M. 1977.** Evaluation of comparative efficacy of some plant powder of the control of *Callosobruchus maculatus* Fab. (Coleoptera :Bruchidae) on stored cowpea. [M.Sc. Thesis]. Ahmadu Bello University, Zaria, Nigeria.
8. **Pandey, G.P., Doharey, R.B., Varma, B.K. 1981.** Efficacy of some vegetable oils for protecting green gram against the attack of *Callosobruchus maculatus*. *Indian Journal of Agricultural Science,* **51:**910-912.
9. **Wardwood, G. 1904.** Published in 'Time' 29th April, 1904. Tulsi, after Sharm, B.D. (1975). Tulsi, 1st Ed. Dabur Private Ltd., New Delhi.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.339-342 **39(PS).**
An International Biannual Refereed Journal of Life Sciences ***Plant Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

A study on aquatic and semi aquatic medicinal plants of Basargadh Pond, Ranchi, Jharkhand

Merina Shweta Sweety & Radha Krishna Jha*
University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

Received : 20th February, 2020; Revised :22nd March, 2020

Abstract: The present communication is done for the assessment of aquatic and semi aquatic medicinal plants of Basargadh Pond, Ranchi. The pond exhibits growth of these plants and serves as chief and easily available source of medicines for neighboring inhabitants. During the course of study total 10 aquatic and semi aquatic plants have been reported from the study areas which are used to cure various diseases.

Key words: Inhabitants, cardiac, antioxidant, diseases

INTRODUCTION

The pond located in Basargadh area of Tupudana, Ranchi is polluted as it serves for multipurpose activities for local dwellers.[1] The pond supports rich aquatic vegetation including ethnomedicinal species which are beings used to cure various diseases by local people.[2,3] The present study is to analyze the medicinal uses of these aquatic and semi aquatic plants and create awareness among people about the importance of pond.[4-8]

MATERIALS AND METHODS

During the course of study the specimens were collected from the study area. After collection these specimens were soaked in formaldehyde solution and dried with the help of heavy herbarium press. After drying herbarium sheets were prepared and identified with the help of experts and suitable literatures. The medicinal uses of these plants were recorded with the help of local people, Vaidyas and herbal practitioners. To make the study more accurate regular pond visit was taken into account. As a result of which the habit, habitat, local name, flowering/ fruiting time of the collected specimens can be encountered easily.

*Corresponding author :
Phone : 9470373562
E-mail : merina_shweta08@rediffmail.com

RESULTS AND DISCUSSION

The result is the outcome of the survey conducted from Aug, 2019 to Sep, 2020 in which total 10 aquatic and semi aquatic medicinal plants were collected. The collected specimens were identified by experts and available literatures. The information about plants and their medicinal uses for curing various diseases were gathered from local people, herbal practitioners and vaidyas which are mentioned below.

1. *Ipomea aquatica* Forsk.

Common name-Kalmisag

Family-CONVOLVULACEAE

Parts used- Leaf

Medicinal uses- It is used against cough, diarrhoea. Leaf of the plant is used as blood purifier. It is given to nourishing mother to promote lactation.

2. *Centella asiatica*.Linn.

Common name-Bengsag

Family-APIACEAE

Medicinal uses- It is used in blood related diseases and for treating nervousness. Plant extract is used for wound healing. It is used in stomach problems, gastic disorder, ear problem, fever and amoebic dysentery. It is also used to enhance attention span and concentration.

3. *Nymphaea nouchalli.* L.

Common name-Saluk

Family-NYMPHACEAE

Parts used-Leaf

Medicinal uses-Leaves are used against dysentery and intestinal problems. It is also used in vomiting and cough.

4. *Nelumbo nucifera* Gaerth

Common name- Kamal

Family- NYMPHACEAE

Parts used- flower, seed, rhizome

Medicinal uses- It is used in cardiac problem, liver problem and fever. Seeds are used in skin diseases and rhizome is used against piles.

5. *Eichornia crassipes* Mart.

Common name-Jalkumbhi

Family-PONTEDERIACEAE

Parts used-Leaf

Medicinal uses-It is used as antioxidant.

6. *Nymphoides indicum* (L.)O.Kuntze

Common name-Kumudini

Family-MENYANTHACEAE

Parts used-Leaves

Medicinal uses-It is used against jaundice and fever.

7. *Cyperus rotundus* Linn.

Common name- Motha

Family-CYPERACEAE

Parts used-Leaves, root

Medicinal uses- It is used against vomiting, fever, dysentery, leprosy, urine contraction and stomach problem. Root paste is used for healing wounds. The root is astringent, appetizer and diuretic.

8. *Ludwigia perennis* Linn.

Common name-Arker, prousuti gas

Family-ONAGRACEAE

Parts used-Root, leaf

Medicinal uses- Leaf is used in headache and ear pain. Root is used against fever. Decoction of plant is given to lactating mother. It is also used in treatment of parsauti fever.

9. *Enhydra fluctuans* Lour

Common name-Hingcha sag

Family-ASTERACEAE

Parts used –leaf

Medicinal uses-The leaf is used against anemia. It is antioxidant and analgesic.

10. *Eclipta alba* (L.) Hassk

Common name- Bhengraj, Kesut

Family-ASTERACEAE

Parts used-Leaf, root

Medicinal uses- Plant juice is used for treatment of jaundice, diarrohea, throat problem, haepatic problem and fever. Root paste is applied to cure cattle wound.

CONCLUSION

During the research work a number of aquatic and semi aquatic ethnomedicinal plants were reported which has potential to cure various diseases.

ACKNOWLEDGEMENT

The authors are thankful to the Dean and HOD, Prof. Jyoti Kumar, University Department of Botany, Ranchi University, Ranchi for his valuable suggestions and guidance. The authors are also thankful to local vaidyas and herbal practitioners for providing valuable information.

REFERENCES

1. **Verma S.,Pandey P.K.,Mukharjee P., & Kumar S 2007.** Some water loving plants solds as pot –herbs (sag) in the village markets of Ranchi district. *Biospectra*. **2(1):**139-141**.**

2. **Bondya S.L, Khanna K.K. & Singh K.P. 2006.** Ethnomedicinal use of leafy vegetables for the tribal folk-lore of Achanakamar-Amarkantak Biosphere Reserve (Madhya Pradesh & Chattisgarh) *Ethnobotany.* **18:**145

3. **Kachroo, P. 1984**. "Aquatic Biology in India", Bishen Singh, Mahendra Pal Singh, 23-A, Connaught Place, Deheradun, India, P.No.27-35

4. **Tiwari K.C., Majumdar R., and Bhattacharya S.Folk 1979.** lore medicine from Assam and Arunachal Pradesh.(district-Tirap) *Quart.J.Crude Drug Res*. **17:**16.

5. **Mahato S.C., and Sahu H.B. 2008.** Traditional theraputical knowledge on common sags among the indigenous people of Panch Pargana in Jharkhand, India. *International journal of plant sciences*. **3(2):**617-621

6. **Gohil T.G. and thakur A.B. 2008.** Ethnobotanical uses of plants by tribals of Kaparada taluka in Valsad District of Gujarat, India. *International journal of plants sciences*. **3(2):**650-653

7. **Reddy C.S. & Raju V.S. 2005.** *J.Bombay Nat Hist. Soc.,*102-133.

8. **Bose, S.C. 1948.** The Damodar Valley Project,Phoenix Press Ltd,Calcutta.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 343-346
An International Biannual Refereed Journal of Life Sciences

40(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Study of zooplankton in Sahasmal wetland of Koshi River, in Araria district (North Bihar), India

Kumari Ranjana* & M. Rauf Anwer
P.G. Department of Zoology, Purnea College, Purnea, B. N. Mandal University, Madhepura, Bihar, India

Received : 27th February, 2020 ; Revised : 21st March, 2020

Abstract : Zooplankton are the primary consumer of the aquatic food chain. It has been extensively studied in different water bodies of our country. Present study is related to zooplankton community of Sahasmal Wetland of Koshi River in Araria District, of North Bihar. The role of Zooplankton in the aquatic food chain is very important as it occupies the place of primary consumer and at the same time as secondary producer. The productivity of zooplankton will reflect the status of the aquatic system. In the present study of Sahasmal Wetland Araria, 11 species registered in Cladocera belong to 10 generations, 7 families, and 2 orders. Along with zooplankton, there were nematodes, aquatic insects and mosquito larvae. Among the phytoplankton, 12 species were registered with Bacillariophyceae, Chlorophyceae and Dinophyceae. Overall phytoplankton and rotifers contributed to the higher composition followed by copepods, cladocerans, and protists.

Key words: Zooplankton, Sahasmal wetland, Koshi River, Araria district

INTRODUCTION

We feel that the Earth is facing serious environmental problems along with rapid depletion of natural resources. Serious concerns have been expressed among scientists for the conservation and preservation of the world's natural resources. Lack of scientific data often leads to difficulties in decision making. Wetland is one of the important natural resources. Wetlands are areas of land that are either temporarily or permanently covered by water. This means that a wetland is neither truly aquatic nor terrestrial. It is possible that wetlands may be both at the same time depending on seasonal variability. Thus, wetlands exhibit enormous diversity according to their origin, geographical location, water system and chemistry, major plants and soil or sediment characteristics. Wetlands support a wide variety of plant and animal species that adapt to water fluctuations. The strata form wetlands of significant ecological importance. According to the utility, the wetland directly and indirectly supports millions of people in providing services such as food, fiber and raw materials, storm and flood control, clean water supply, natural beauty and educational and recreational benefits.

The Koshi River system and its tributaries are the important river basins of North Bihar. The Koshi River runs along the southern slope of the Nepal Himalaya

*Corresponding author :
Phone : 9661699606
E-mail : kumariranjan.pawan@gmail.com

(25°20' to 29°0'N and 85°20' to 89°0'E long). 70.409 covers the area of two countries spread over a catchment area of KN2. Nepal (59,570 KN2) and India (10,893 KN2 North Bihar). The river originates as 3 main from the glaciers of Mount Gaisithan (8013 m), Mount Everest (8848 m) and Kanchenjunga (8579 m) viz. Koshi broke its embankment at Kusha in Nepal on August, 2008 and flooded northern Bihar. Due to the flood, 13 km wide current spread to 100 km. More than one million people lost their crops, land and homes. The district that has suffered the tragedy of this flood also includes Araria in North Bihar.

Fish are mainly aquatic vertebrates. There is a system of adaptive interactions of fish with their environments in interrelation with biological and abiotic environments. If there is a change in the biological and abiotic environments, it usually has a profound effect on the fish and its habitat.

Zooplanktons are the primary consumer of the aquatic food chain. It has been extensively studied in different water bodies of our country by different researchers.[1-4] Present study is related to zooplankton community of Sahasmal Wetland of Koshi River in Araria District, of North Bihar. The role of Zooplankton in the aquatic food chain is very important as it occupies the place of primary consumer and at the same time as secondary producer. The productivity of zooplankton will reflect the status of the aquatic system.

MATERIALS AND METHODS

The Study was carried out in Sahasmal Wetland of Koshi River in Araria District, of North Bihar, Bihar, India. It is endowed with rich aquatic biodiversity of flora and fauna. Araria district is one of the thirty-eight districts of Bihar state, India. Araria district is a part of Purnia division.

Plankton samples were collected from Sahasmal Wetland of Araria district of Bihar by filtering 50 L of surface water through 63 LM mesh size nylon plankton net and preserved in 4 million formalin at the Zoology Laboratory of Purnea College. Surface water samples had to avoid macrophytes.

Collection of Zooplankton: It was collected through a continuous zooplankton and phytoplankton separator made out of nylon bolting silk.[5] Secondary production: It was estimated following method based on displacement volume.[6]

RESULTS AND DISCUSSION

A total of 40 species of zooplankton and phytoplankton were recorded during the study period. Sarcomastigophora with 4 species has 3 generations, 3 families, and 2 orders; Euselenzoa belongs to 2 genera with 2 species; 1 order and 1 family; Dinoflagellate of 1 species has 1 generation, 1 order and 1 family; Rotifer with 19 species belonging to 9 genera, 7 families, 2 orders; Cladocera belongs to 2 species, 2 families, with 11 genera, 9 genera, 7 families and 2 orders and copepoda; And 2 orders from 15 localities respectively Different wetlands of Jharkhand respectively (Table 1). Earlier, other authors also reported 19 species from the state of Bihar.[7] With respect to these several researches also played an important role in contribution to the study of phytoplankton, zooplankton and physico chemical parameters from ponds,

Table 1. List of reported species from the wetlands of Sahasmal, Araria

Si.no	List of examined species	Si.no	List of examined species
1	*Arcella discoides*	21	*Mytilina ventralis ventralis*
2	*Centropyxis ecornis*	22	*Lecane curvicornis*
3	*Centropyxis spinosa*	23	*L. leontina*
4	*Difflugia corona*	24	*Trichocerca diurella similis*
5	*Phacus acuminatus*	25	*Asplanchna brightwelli*
6	*Euglena acus*	26	*Polyarthra vulgaris*
7	*Ceratium hirundinella*	27	*Pseudosida bidentata*
8	*Brachionus angularis*	28	*Ceriodaphnia cornuta*
9	*B. calyciflorus f. dorcas*	29	*Daphnia carinata*
10	*B. caudatus personatus*	30	*Simocephalus vetulus*
11	*B. diversicornis*	31	*Simocephalus exspinosus*
12	*B. forficula forficula*	32	*Simocephalus vetulus*
13	*B. kostei*	33	*Bosmina longirostris*
14	*B. quadridentatus quadridentatus*	34	*Moina micrura*
15	*Keratella cochlearis*	35	*Macrothrix spinosa*
16	*K. lenzi*	36	*Chydorus sphaericus*
17	*K. tropica*	37	*Alona rectangula rectangula*
18	*Anaeureopsis fissa*	38	*Heliodiaptomus contortus*
19	*Euchlanis dilatata*	39	*Mesocyclops hyalinus*
20	*Euchlanis incisa*	40	*Mesocyclops leuckarti*

lakes and rivers of Bihar.[8-13] The origin of the rotifer species was detected in the Sahasmal wetlands of Araria district.

The species diversity of rotifers was maximum among zooplankton. Chandrasekhar and Chatterjee (2003)[10] reported 9 species of cladoceran from two lakes i.e. Dimna Lake and Jubilee Park lakes of Jharkhand. In the study of Sahasmal Wetland Araria, 11 species registered in Cladocera belong to 10 generations, 7 families, and 2 orders. Along with zooplankton, there were nematodes, aquatic insects and mosquito larvae. Among the phytoplankton, 12 species were registered with Bacillariophyceae, Chlorophyceae and Dinophyceae. Overall phytoplankton and rotifers contributed to the higher composition followed by copepods, cladocerans, and protists. Freshwater bodies have shrunk due to industrialization, human populations. Existing aquatic systems are polluted by many sources, causing water scarcity as well as non-susceptibility to the existence of aquatic organisms which is one of the main problems. Further research work will have to be explored in samples employed at regular intervals and seasonal interpretation and correlate physicochemical effects to estimate the ecological significance of wetlands.

REFERENCES

1. **Jana, B.B. 1979.** Temporal plankton succession and ecology of a tropical tank of West Bengal, India. *Int. Rev. Hydrobiol.* **64:**661-671.
2. **Sinha, A.B. and Sahai, R. 1978.** Studies on the factors affecting growth and productivity of free floating macrophytes in Ramgarh Lake of Gorakhpur (India). *In: Glimpses of Ecology* (ed. J.S. Singh and B. Gopal) *Int. Sci.* Publ., Jaipur.
3. **Patil, G.P., Kedar, G.T., Yeole, S.M. 2008.** Zooplankton biodiversity study of two water bodies in Washim district, Maharashtra. *J. Aqua. Biol*. **22(1):** 13-17
4. **Sharma, B., & Sharma, S. 1999.** Freshwater Rotifers (Rotifera: Eurotatoria). *State Fauna Series: Fauna of Meghalaya.* **4(9):**11-161.
5. **Nayar, Sasi, Gupta, T.R.C. and Gowda, G. 1991.** Secondary productivity and zooplankton biomass in a tropical coastal lagoon near Manglore. South West Coast of India. *Indian J. Fish*. **46(3):** 281-288.
6. **Manges, E. and Garcia, H. 1991.** Seasonal fluctuation of Zooplankton biomass in Lake Xolottan (Managua). *Hydrobiol. Bull.* **25(2):** 157-162.
7. **Michael, R. G., & Sharma, B. K. 1988.** Indian Cladocera (Crustacea. Branchiopoda. Cladocera). *Records of Zoological Survey of Indi*a. **108(2):**111-122.
8. **Biswas, S. 1966.** Five species of Daphnidae (Crustacea: Cladocera) from Simla Hills in India with a new record of Alona costata Sars, from Kameng Division, Nefa. *Journal of Zoological Society India,* **16:**92-98.
9. **Bohra, C., & Kumar, A. 2004.** *Plankton Diversity in the Wetlands of Jharkand*. Biodiversity and Environment APH Publishing Corp. New Delhi, 91-123.
10. **Chandrasekhar, S., & Chatterjee, T. 2003.** On the collection of Cladocera from Dimna and Jublee Park Lakes, Jamshedpur, Jharkhand. *Zoos Print Journal,* **18:**1089-1090.
11. **Edmondson, W., & Hutchinson, G. 1934.** Report on Rotatoria. Article IX. Yale North India Expedition. *Conneccuit Academic Arts and Science,* **10:**153-186.
12. **Mishra, A., & Hasan, S. 2013.** Assessment of water quality of Libri River, Purnia (Bihar). *International Research Journal of Science and Engineering*, **1(2):** 63-68.
13. **Sinha, K., & Sinha, D. 1983.** Seasonal trends in physico-chemical factors and zooplankton in a freshwater pond of Munger, Bihar. *Journal of Ecobiology*. **5(4):**299-302.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 347-350
An International Biannual Refereed Journal of Life Sciences

41(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

A case study on life history of the *Aulacophora atripennis* F. (Coleoptera:Chrysomelidae) of a cucurbitaceous - beetle on the vegetable plant *Cucurbita pepo* of Ramgarh, Jharkhand, India

Rajesh Kumar Upadhyay*
Department of Zoology, Ramgarh College, Ramgarh, Jharkhand, India

Received : 28th February, 2020 ; Revised : 22nd March, 2020

Abstract : The pest *Aulacophora atripennis* F. is a serious pest of cucurbitaceous vegetable plants, it damage the stem, buds, flowers and the fruits of the plants of cucurbitaceous family. Its life cycle was completed on cucurbitaceous plants and the roots of creepers in the soil and old fallen leaves. It damage 60% of the plant leaves and sometimes the plant become leafless. On serious infestation the damage reach up to 100% and plant die. The quality and the quantity of the fruits also decline on minor infestation. The present paper deals with the life -history of *Aulacophora atripennis* F. on a new vegetable plant *Cucurbita pepo* in Ramgarh, Jharkhand, India. The life cycle was studied during the year 2018-2020 at Ramgarh. During the study it was found that the life-history of pest was very complicated generally it completed 05 to 06 overlapping generations between March to October of a year. The beetles hibernate in cold season the period was November to February. After the warm season the beetle come out from hibernation. The female beetle laid eggs after mating in the moist soil near the host plant. The newly hatched grubs remains below the soil surface and they feeding on roots of the host plants, underground stems of the creeper and old leaves of the host plants fallen on the ground. The pumpkin is the most favourable host plant for these beetles. The Pumpkin is a healthy and tastier vegetable, rich in carbohydrate, minerals and fibres. The pest damage the cultivated vegetable plants and loss not only the quality, quantity, and taste of the vegetables but also the profit of the farmers. Ramgarh is a beautiful district town, the rural area is known for agriculture and gross vegetable production, the climatic condition is moderate and suitable for vegetable production. The experimental pest is very difficult to control. The farmers are advised to control the pest by mechanical as well as by spraying carbaryl 0.1%, endosulphan 0.05%., endrin 0.04%.

Key words: pest, minerals, quality, vegetables, fibres, hibernate

INTRODUCTION

Aulacophora atripennis F. is a serious pest of cucurbitaceous vegetable plants. This beetle is common in the warm region of Asia, America, Africa, Australia and southern Europe. The life-history of this beetle needs moist and warm climate. Ramgarh is a beautiful district town located at Lat.23.38°N and Long.85.34°E., situated near to Ranchi, the capital of Jharkhand, about 40 km south-east. The temperature is moderate with more humidity. This climate is favourable for vegetable cultivation. Cabbage, Brinjal, Cauliflowers, Lady's fingers, cucurbitaceous vegetables are the main vegetable crops of Ramgarh.

Considerable works have been done by workers on the life-history of red beetle *Aulacophora foveicollis* Lucas pest on different creeper vegetable plants, but none of the

*Corresponding author :
Phone : 9471355487
E-mail : drrajeshkumarupadhyay@gmail.com

scientist studied the life cycle of *Aulacophora atripennis* F. on *Cucurbita pepo* L. pumpkin plant in Ramgarh. The present author deals with the life cycle of *Aulacophora atripennis* F. on plant *Cucurbita pepo* L., the pumpkin vegetable plant in Ramgarh. Pumpkin is a delicious vegetable available throughout the year. It is full of carbohydrate, minerals and fibres. The pest not only decline the quality, quantity and taste of the vegetable but also the financial condition of the farmers. The pest completed 05 to 06 overlapping generation during the month March to October of a year. The beetles hibernate in cold weather and it was from the month November and prolong up to February. On the warm season come the beetle become active and after mating the females lay eggs inside the soil and continue their lifecycle. After 04 to 20 days of incubation the eggs hatched in to small grubs and they started to eat the roots, the buds and the old leaves of the host plant fallen on the ground. Prevention is very difficult of this pest, farmers are advised to manually collect and destroy the beetles as well as by spray the chemical carbaryl 0.1 % or endosulfan 0.05 % or endrin 0.04% spray is also advised.

MATERIALS AND METHODS

Standard methodology was applied for the study of life history of *Aulacophora atripennis* F. Life cycle and the nature of damage of the experimental pest were studied in the crop field as well as in the laboratory. The life history was studied during the year 2018-2020. Five healthy selected plants of *Cucurbita pepo* L. (pumpkin) were completely and carefully covered by a small mesh mosquito net separately. One pair (one male and one female) of experimental pest were introduced on the plant covered by mosquito net and make observation daily. One pair (one male and one female) of the experimental pest were kept in the cage of 20 cm x 20 cm x 30 cm and provided them fresh leaves of host plant for egg laying and hatching, the observation were recorded for further study.

Methods of prevention of crop and control of pests were applied and recorded. The records of the private agencies and the scientists of Birsa Agriculture University, Ranchi, were also keeping in mind for accurate result.

OBSERVATION

Moderate temperature and moist weather is suitable for the vegetable cultivation. *Aulacophora atripennis* F. is one of the serious cucurbitaceous beetle, it damage 60% to 90% some times 100% vegetable crops. Now the present author deals the life history of *Aulacophora atripennis* F., on vegetable *Cucurbita pepo* L. the pumpkin. It is a common vegetable of Ramgarh (Jharkhand) cultivated throughout the year.

Morphology of the pest: The experimental adult beetles were oblong in shape, about 04 to 08 mm in length and 03 to 04 in breadth. The body was dark grey coloured on the dorsal side while black from ventral side. The abdominal region was covered with whitish soft hairs.

Morphology of grubs: The newly hatched grubs were creamy yellow coloured and were provided with a brown head and prothorax.

Life cycle of pest: When the climate change towards warm and on the arrival of moderate temperature the pest become active, in the month of March the adult beetles took part in reproduction. After mating the female lays eggs singly or in cluster of 09 to 10. The eggs were orange in colours and oval in shape. The single female beetle can lays 300 to 350 eggs in the moist soil. The female died after the egg laying. The grubs were hatched from eggs after the incubation of 10 days in early March 08 days in April, 04 to 06 days in May to August and 10 to 20 days in September to November (table). The newly hatched grubs were creamy in coloured having head, prothorax and abdomen. Under the surface of soil the grubs were active and eating the roots of host plant, the buds, underground stems, the old fallen leaves and the fruits laying on the surface of the soil. The grubs grown in size and become full grown stage grub after passing 04 moulting, from newly hatched grub to full grown stage grub the duration takes 10 days in early March to 08 days in April 05 to 07 days in May to August and 10 to 20 days in September to November (table). Now the full grown stage grub move towards the inside soil and get in pupation. The pupa is smooth, thick walled, oval and waterproof earthen chamber at the depth of about 01 to 03 cm in the moist soil near the host plant. After 05 days in May to August and up to 15 days in September to November (table) of pupation the adult beetles emerges out and begins to eating the leaves of the host plant and then to buds, stem and fruits of the host plant. The infestation is so high that sometimes the host plant become leafless and then dies within even two days. The adult beetle started to egg laying after the 03 to 07 days (Table) of emergence from pupa. The life cycle was completed in 20 to 70 days. 05 to 06 generation were

recorded in a year. On the return of winter in the month of November the adult beetles enter inside the moist soil up to 01 to 03 cm or among dry weeds keep themselves in hibernation and appears again in March. Some of the pest does not hibernate and they may appear in cold also.

Table 1: Tabular representation of lifecycle of pest *Aulacophora atripennis* F., on host plant *Cucurbita pepo* L

Si. No.	Name of months	No. of eggs laid	Incubation period of eggs in days	No. of days of hatching in grubs	No. of days of pupation	No. of days as adult emergence, after the female laid eggs
1	March-April	300-350	10-08	10-08	05-08	06-07
2	May-August	300-350	04-06	05-07	05-07	03-04
3	September-November	300-350	10-20	10-25	07-15	05-07
4	December-February	Hibernating	20 days up to 03 months	40-90	40-90	

SUGESSION FOR CONTROL OF PESTS:

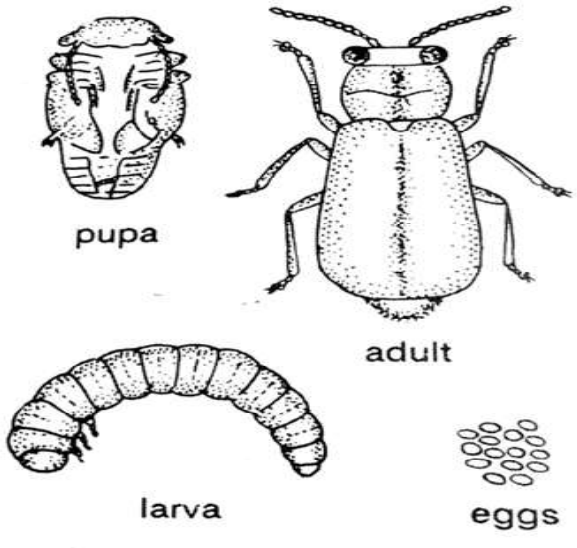
1. Early variety of cucurbitaceous vegetables should be grown.

2. Trap crop. Few scattered plants should be grown early in the field, they should be sprayed with strong insecticide, so the adult pests attracted towards the plants and die and the crop will be pest free.

3. In the morning spray the dust of mixture of ash and insecticide.

4. The field should be plough deeply after the harvesting.

5. Farmers are advised to apply pesticides as Lindane (0.1 %)@ 200-300/acre effective, Parathione (0.02%) or Malathione (0.05 %) can be dusted on host plant.



Life cycle of *Aulacophora atripennis* F. on the host plant *Cucurbita pepe* L.

DISCUSSION

The vegetable *Cucurbita pepo* L. is a delicious vegetable and main dietary component of the common people. Its production is so abundant that it controls the vegetable market in price hike. The main crops of farmers of Ramgarh are chiefly vegetable cultivation and the vegetable yielding is abundant. Ramgarh is situated on the 4 lane N.H. and good connectivity of railway hence very easy to transport make the Ramgarh good and prime market for vegetable sale.[1,2] The experimental pest *Aulacophora atripennis* F. is a serious pest of cucurbitaceous plant. The present paper deals the life cycle of the pest on *Cucurbita pepo L*, it is a common yielding vegetable crop of Ramgarh its production is abundant throughout the year. The farmers cultivated this vegetable commercially as well as for their personal consumption. The vegetable *Cucurbita pepo* is full of minerals, vitamins and fibres and less in carbohydrate, fat and proteins hence it is one of the most recommended healthy diets by the dietician for the population.[3,4]

The vegetable plant of *Cucurbita pepo* is a creeper, cultivation is very easy. The people cultivate it even at their small residence for fresh vegetable consumption of their own.

In March to April the life *cycle* completed in 03 to 04weeks, in the month of May to August the life cycle completed in 02 to 03 but when the temperature fall the life cycle delayed in the month of September to November the life cycle completed in 04 to 09 weeks. The adult and pupa hibernates during the months of December to February. All the metabolic activities directly affected by the temperature, humidity, pH etc.[5-7]

Prevention and control of the pest is a very difficult phenomenon. The population rise of pest is very fast, some times in 24 hours their attack makes the host plant leafless. The spray of carbaryl 0.1% S, endosulphan 0.05% or endrin 0.04% kill the eggs, larvae and adults on the surface.[8,9] Destroy the infected parts of plants. Cultivate the improved and resistant variety of *Abelmoschus esculentus* Moench the Lady's finger plant.

REFERENCES

1. **Upadhyay, Rajesh Kumar. 2017.** Impact of development: Threat to biodiversity of Ramgarh. *Biospectra*. **12(2):**(spl. Issue)73-74.
2. **Upadhyay, R. K. Dr. & Bakshi O.P.S.Dr. 2020.** Study of serious pest of Brinjal at Ramgarh, (Jharkhand, India): prevention and control. *Internationl Journal of Current Reasearch*. **12(02):**10340-10342.
3. **Upadhyay, R.K, Kumar,P.R, Jamuar. 2012.** Studies on the benthic fauna in the Ponds of Ramgarh, Jharkhand, India With respect to aquaculture. *Biospectra*. **7(3):**47-56.
4. **Upadhyay R.K.Dr & Bakshi O.P.S. Dr. 2019.** Studies of vegetable pests of Ramgarh (Jharkhand, India) :a case study of cabbage Pests. *International Journal of Current Reasearch*. **11(10):**7970-7971 .
5. **Upadhyay, R.K. and Verma, N.R. 2004.** Studies on the physical characteristics of Subarnarekha river of Ranchi (Jharkhand). *Proc.Zool.Soc.India.* **3(2):**67-71
6. **Upadhyay, R.K.and Verma,N.R. 2005.** Studies on common chemical characteristics of Subernarekha river of Ranchi (Jharkhand). *Proc.Zool Soc.India* **4(2):**33-41
7. **Upadhyay, Rajeshkumar. 2009.** Studies on quantitative analysis of benthic macro invertebrates of Subernarekha river at Ranchi. Jharkhand, India 230-241. Biodiversity & wild life management. Alfa publication, New Delhi.
8. **Kumar,M.and Tiwari,Shyamkumar. 2009.** Effect of celphos as fumigant on the Mortality of larva, pupa and adult of *Odioporous longicollis*– a serious Pest of Banana in Bihar**.** *Proc.Zool.Soc.India.* **8(2):**31-35
9. **Prabhakar,A.K.and Roy,S.P. 2009.** Studies on the biology and life table of *Diaphania indica* Saunders (lepidoptera:pyraustidae) on cucumber, *Cucurmi sativus*. *Proc*. *Zool.Soc.India.* **8(2):**43-50.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 351-354
An International Biannual Refereed Journal of Life Sciences

42(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of sodium fluoride on the rat liver

Satish Kumar Sah* & Arun Kumar
University Department of Zoology, B.N.M University, Madhepura, Bihar, India

Received : 02nd February, 2020; Revised : 28th February, 2020

Abstract : Sodium fluoride (NaF) is used for caries prevention in the form of fluoridated drinking water etc. Effect of fluorides on various metabolic levels in hard and soft tissues, namely respiration as well as carbohydrate, protein, enzymatic and vascular metabolism, can disturb detoxication of fluorine compounds administered orally. The study objective was to study the effect on liver of young and mature rats exposed to NaF in drinking water from conception till maturity, as well as after its withdrawal. The results observed in the liver, can help determine to what degree of oral fluoride caries prevention is safe and whether it should be implemented.

Keywords : Sodium fluoride, liver, rat

INTRODUCTION

Due to its efficacy in caries prophylaxis and straightforward application, salt (NaF) remains used for caries prevention within the sort of fluoridated beverage, fluoride tablets, fluoridated salt or milk.[1] Effect of fluorides on various metabolic levels in hard and soft tissues, namely respiration also as carbohydrate, protein, enzymatic and vascular metabolism, can disturb detoxication of fluorine compounds administered orally.[2-4]

Fluorine, considered to be one among the environmental toxins, doesn't occur free in nature but because of high affinity for the ions of calcium, sodium, magnesium and tin it forms chemical compounds with them, which are more or less soluble in water.[5,6]

Because of good solubility in water, easy absorption from the alimentary canal also as for economic reasons, salt (NaF) is that the most ordinarily used compound in collective endogenic oral caries prophylaxis. During oral exposure, it can positively affect the oral environment. However, when consumed with food via the alimentary canal it can change, counting on dose and exposure time, cell and tissue metabolism within the further stages. The organ that reacts rapidly to xenobiotics reaching the body from the surface is that the liver. It's there where detoxication processes happen and therefore the resulting pathomorphological changes are the response to the orally administered preparation.

*Corresponding author :
Phone : 9431481928
E-mail : sahsatish2018@gmail.com

MATERIAL AND METHODS

In the initial stage of the experiment, 30 female rats, 180-200 g body weight, were divided into 3 groups: one control and two experimental groups (I, II). Female rats in the experimental groups received fluorine in aqueous solutions of sodium fluoride (NaF) at a concentration of 10.6 mg NaF/dm^3 (group I) and 32.0 mg NaF/dm^3 (group II), corresponding to a dose of 1.2 mg F/kg b.w. (group I) and 3.6 mg F/kg b.w. (group II). Sodium fluoride (NaF),

was dissolved in tap water. After a two-week adaptation period, during which the rats drank an average amount of 50 ml of water, the females were covered. They received NaF with drinking water 3 days before covering, during pregnancy and lactation. On day 30 of life, young rats were separated from mothers but were still given sodium fluoride with drinking water in the concentrations as above.

Control animals received tap water ad libitum, in which the level of fluorine did not exceed 0.2 mgF/dm^3. On day 90 of life, NaF was withdrawn in groups I and II, and tap water alone was administered to all the animals. The experiment was terminated on day 120 of life.

The animals were kept in standard environmental conditions.[8] The young were subjected to the action of fluoride from conception, through the foetal period, nest period (till day 30 of life) and maturity (day 90). After 90 days of exposure, NaF was withdrawn. The experiment was terminated on day 120 of life. Rats were fed on standard granulated LSM diet.

90 young rats were included in the experiment, 5 in each study subgroup. The animals were weighed and subjected to autopsy at the following age intervals: day 4 (newborn), day 14 (young rat fed on mother's milk); day 30 (young rat, receiving mother's milk, drinking water and standard diet from day 14 to day 30 of life – on that day, the young were separated from mothers but the experimental model was continued), day 60 (mature rat), day 90 (adult rat) and day 120 (NaF was not applied for 30 days).

RESULTS AND DISCUSSION

The animals dissected on day 4 of the experiment, the liver showed the predominance of blood vessels and really numerous haemopoietic system cells that formed islets. Hepatocytes had a foamy, acidophilic cytoplasm. Cell nuclei were well stained, with distinct chromatin lumps and nucleoli. In 14-day-old rats, hepatocytes showed a definite trabecular arrangement and contained acidophilic cytoplasm with markedly stained nuclei and nucleoli. In most fields of vision, lobules were distinct. The haemapoietic cells sporadically formed islets. In 30-day-old animals, the liver structure was already mature. In 60-, 90- and 120-day rats, the structure of the organ was typical for adult animals. Slight vacuolar degeneration might be sporadically seen within the peripheral parts of the lobules.

In the animals of group 1 (NaF concentration in beverage – 10.6 mg) dissected on day 4, pictures of the liver didn't differ markedly from those seen within the control group. In some fields of vision, hepatocytes contained vacuoles. Within the livers of 14-day-old rats, in some hepatocytes the cell wall was blurred. Blood vessels were slightly dilated. In 30-day-old rats, vacuolar degeneration-type changes occurred. Necrotic lesions were sometimes seen in hepatocytes (micro focal lesion). In 60- and 90-day-old rats the liver had a traditional structure. Within the peripheral parts of rare lobules slight vacuolar degenerations were found. In 120-day-old rats during this group, i.e. 30 days after NaF withdrawal, the vacuolar degenerations persisted, particularly in hepatocytes of the peripheral lobules.

In the group II, morphological changes within the liver were more pronounced altogether the animals no matter the time of dissection. Already within the livers of 4-day-old rats an outsized number of hepatocytes with vacuoles within the cytoplasm were seen. Blood vessels were dilated and crammed with clotted acidophilic fluid. On day 14, hepatocytes showed features of injury, visible within the structure of cell and nuclear membranes. Within the vicinity of blood vessels, inflammatory infiltration of neutrophilic granulocytes was observed. Within the livers of 30-day-old rats, the changes were further intensified, and hepatocytes showed distinct vacuolar degeneration and micro necrotic foci. In 60-day-old rats, the image of intoxication within the liver was still present. Within the livers of 90-day-old rats, the changes were becoming weaker. However, vacuolar degeneration-type changes still persisted within the livers of 120-day-old rats despite NaF withdrawal.

According to research surveys conducted by the International Research Agency for Fluorination, the toxicity of fluorine compounds is usually ignored by medical doctors, dentists and paramedical staff involved in healthcare. Therefore, measures should be taken to get undoubted benefits resulting from the appliance of fluorine compounds in caries prevention and to attenuate any side-effects.[6,7,10] Assuming that the liver is involved within the metabolism of toxic compounds produced during systemic transformations and exogenous toxins going to the organism from the environment, we could expect to seek out both patho morphological and metabolic changes as reactions to NaF. Therapeutics or toxins, to which

NaF, counting on its dose, are often included, are likely to impair liver function and induce morphological changes within the liver.[15,16] Hepatotoxic action is manifested by cell respiration disorders that interfere with oxidation and reduction mechanisms, by impairment in protein, carbohydrate and lipid metabolism and by disturbances in intra- and extracellular transport.[9] In consequence, whole cell or its cytoplasmic organelles are often damaged. most often the damage is expressed as parenchymal vacuolar degeneration, necrosis of hepatocytes or disorders within the activity of metabolic enzymes.[14,15] within the livers of newborn rats in group II, slight changes were found in Browicz Kupfer cells at higher fluoride concentration. Distinct vessel dilation could suggest metabolic disturbances within the liver, thus indicating a possible toxic effect of fluoride on the organ as early as within the foetal period. The livers of 14-day old rats exposed to NaF in group I and II, receiving fluorine only with mother's milk, showed vacuolar degeneration-type changes, damage or blurring of cell or nuclear membrane and vessel dilation. The changes were more pronounced in group II, i.e. at higher fluoride concentration.

Lack of distinct morphological and enzymatic changes within the livers of 14-day-old animals of group I (lower NaF concentration) are often explained by the protective role of mother's milk .[16] Since morphological and ultrastructural changes were most pronounced within the livers of 30-day-old rats, Up to day 30 of life, the rats stayed in nests, receiving both mother's milk, standard diet and NaF-enriched water to drink, which resulted together of beverage fluorides with mother's milk fluorides. Already in group I, at the lower NaF concentration, aside from vacuolar degeneration also micronecrotic foci were observed. In group II, at the upper NaF concentration, vacuolar degeneration and various micro- necrotic foci were seen to multiply as compared to group I.

After NaF withdrawal the changes within the liver in both groups were subsiding. the quantity of glycogen increased in hepatocytes, and cell nuclei and endoplasmic reticulum were normal in appearance. Only mitochondrial polymorphism was maintained and damaged endothelial blood cells were sporadically seen.

Endoplasmic reticulum reacts rapidly to the action of toxic compounds. It's rough component undergoes vacuolization (vacuolar degeneration) and loses ribosomes, which results in a decrease in RNA. These changes are related to the impairment in protein synthesis within the cell. The graceful a part of the endoplasmic reticulum also can be subject to vacuolization or proliferation, which causes a substantial decrease within the count of glycogen granules within the affected sites.[17] This has been confirmed by our previous ultra structural findings.[13] The microscope examinations revealed dilation of channels of the rough endoplasmic reticulum in 30-day-old group II animals, i.e. those exposed to the upper concentration of fluoride ions. Similar changes within the liver found within the rough endoplasmic reticulum are described by Lavrushenko.[12] Additionally, in group II rats older than 30 days, the sinusal lumen of the liver sometimes showed collagen bundles amid micronecrosis observed within the morphological examination, which could suggest the start of liver fibrosis.

From day 30 till day 90 of NaF administration, the morphological changes showed a gradual decrease in intensity and only considerable dilation of blood vessels with endothelial swelling was observed. Taking under consideration endless exposure to NaF, the decrease might be the results of adaptive mechanisms of the organism to fluoride, which are discussed by. Hepatic hyperaemia after administration of acetate acid to rats has been observed by Luty.[10] In some systemic diseases, hyperaemia seems to be beneficial, e.g. in myocarditis. The findings of laboratory and epidemiologic studies conducted during water fluoridation period suggest that the deathrate thanks to heart infarct may have decreased thanks to fluorine compounds present in beverage.

Some literature reports as fluorine accumulates within the liver. Its blood level depends on fluoride supply, which refers to all or any fluorine forms in serum. As revealed by Chlebna- Sokó³, thanks to high homeostasis of the serum, blood fluoride levels remain constant even within the case of over dosage.[11]

The NaF-induced morphological changes in rat hepatocytes create an image almost like those observed after intoxications with other toxic compounds administered to experimental animals. It is often assumed that the liver is involved in detoxication of excessive fluorine doses. After NaF withdrawal, the changes within the liver in both groups were subsiding. The glycogen count increased, but during a considerable number of cells vacuolar degeneration persisted. Remission of most

pathomorphological changes after NaF withdrawal may suggest their transitory nature. However, at the time of exposure salt affects the event of the organism both within the prenatal and postnatal period of experimental animals.

REFERENCES

1. **Whitford GM. 1994.** Intake and metabolism of fluoride. *Adv Dental Res*.**8:** 5-14.
2. **Miller GW. 1997.** Fluoride, a toxic substance. Fluoride. **30:** 141.
3. **Burt BA. 1992.** The changing patterns of systemic fluoride intake. *J Dent Res*. **71:** 1228-37.
4. **Horowitz HS. 1995.** Commentary on and recommendations for the proper uses of fluoride. *J Public Health Dent*; **55:** 57-62.
5. **Maddrey WC, Boituott JK. 1977.** Drug-induced chronic liver disease. *Gastroenterology*. **72:** 1348-9.
6. **Martyniak R. 1979.** Ultrastruktural and biochemical changes in newborn rabbit liver after transplacental effect of phenobarbital. *Folia Morphol*. **38:** 259-75.
7. **Pohl L. 1991.** Drug induced allergic hepatitis. *Semin Liver Dis*. **10:** 305-21.
8. **De Valck V, Geerts A, Schellnick P, Wisse E. 1988.** Localization of four phosphatases in rat liver sinusoidal cells. An enzyme cytochemical study. *Histochemistry*. **89:**357-63.
9. **Koskinen-Kainulainen M, Luoma H, Tuomisto J. 1990.** The LD_{50} excretion and serum and bone levels of F after a high single F + Mg dose in rats with finding on cardiac Ca and Mg Manganese. *Trace Elem*. **9:**15-27.
10. **Philips MJ, Poucell S, Patterson J, Valencia P. 1987.** Drug and toxic effects. The Liver. *New York: Raven Press*. 159-71.
11. **Sherlock S. 1965.** Hepatic reactions to therapeutic agents. *Am Rev Pharmacol*. **5:** 429.
12. **Stenger RJ. 1970.** Organelle pathology of the liver – the endoplazma reticulum. *Gastroenterology*. **58:** 554.
13. **Dbrowska E iwsp. 2005.** The effect of natrium fluoride on selected internal rat organs – an ultrastructural study. *Acta Toxicology*. **13:** 31-6
14. **Lavrushenko LF, Trunov VI, Okunev VN. 1980.** Morphofunctional characteristics of some rat hepatocytes ultrastructures under long-period influence of sodium fluoride. *Citologia Genetica*. **14:** 10-2.
15. **Luoma H.1980.** Fluoride and magnesium, two ions in the prevention of calcium salt imbalance, including caries prevention in man and animals. *Proc Finn Dent Soc*. **76:**73-7.
16. **Whitford GM. 1996.** The metabolism and toxicity of fluoride. Vol. 16 Monographs in Oral Science.
17. **Whitford GM, Pashley DH, Reynolds KE. 1979.** Fluoride tissue distribution. Short term kinetics. *Am J Physiol*. **236:** 141-8.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 355-358
An International Biannual Refereed Journal of Life Sciences

43(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Study of dipel on development of *Diacrisia obliqua* (Walker)

Mustaque Mohammad* & S.N.Yadav
Department of Zoology, T.P.College, Madhepura, B.N.M University, Madhepura, Bihar, India

Received : 3rd February, 2020; Revised : 20th February, 2020

Abstract : *Diacrisia obliqua* Walker (Lepidoptera: Arctiidae) is a phytophagous insect causing great loss to different crops. To control these pest different concentrations of dipel were administered by Leaf Dip Method and Topical Method. It was observed that dipel reduces the percentage of pupation in larva. It prolongs the larval and pupal period. It causes a clear reduction in emergence of adult insect.

Keywords : ***Diacrisia obliquae,*** **dipel, larva, Leaf Dip Method**

INTRODUCTION

Bihar hairy caterpillar, *Diacrisia obliqua* is a notorious polyphagous pest causing damage to several crop plants. Chemical insecticides have been used by farmers to control this pest but older larva survived their toxicity. Chemical pesticides are also harmful to human and pet animals. They also cause resistance development and environmental pollution. So biopesticides are being popular day by day. Microbial agents like virus, bacteria and fungi have been widely used and tested to control several crop pests.

Bacillus thuringiensis (*B.t.*) is a gram positive bacterium, found quite effective to control lepidopteran pests. *B.t.* has been found pathogenic to more than 525 insect species. A number of toxins like endotoxin are produced by it. This toxin is a protein produced when spores are formed. It targets the insect's midgut epithelium when ingested. In the light of previous work done and literature available, following work was planned to study the effect of Dipel (a commercial preparation of *B.t.)* on growth of *D. oblique*

*Corresponding author :
Phone : 9431254871
E-mail : drmustaquemcm@gmail.com

MATERIALS AND METHODS

For the study male and female moth were collected and maintained in laboratory to ensure the regular supply of insects. Adults were maintained in glass chimneys and larvas obtained from them were kept in large petridishes. Full grown larvae were transferred to pneumatic trough having 10-15 cm thick soil on their bottom, for pupation.

Dipel is a commercial preparation of *B.t.* whose efficacy has been already evaluated to control different insects. It is a wettable powder containing $25x10^9$ viable spores per gram of final product of *B.t.* var *Kurstaki* (Serotype 3a, b strain HD-1). The concentrations of Dipel

used in this study included 0.05, 0.10, 0.50, 0.75 and 1.0%. 2% skimmed milk powder was added to Dipel to increase its adhering property. Two methods were employed to treat the insects with different bacterial preparations. Leaf Dip method, In this method, small and uniform sized leaves were treated with each concentration of Dipel and Topical Method (TM), In this method 2 hr old adults were exposed to a thin film of residue of Dipel.

RESULTS AND DISCUSSION

Any concentration of the dipel applied to larvae through food, reduced the larval survival considerably as compared the untreated adults. As regards the effect of different concentrations of this bacterial preparation on the larval survival, the percentage of pupation, varying from 25.26 to 69.42 per cent and decreasing with the increasing concentration differed from concentration to concentration as per chi-square lest.

Further, any concentration of this microbial preparation prolonged the larval stage as compared the untreated condition ($P<0.05$) and the larval stage, varying from 22.76 to 36.46 days in response to treatment by leaf dip method and showing direct proportionality to the concentration, differed with the concentrations of thuricide ($P<0.05$).[1]

Every concentration of the Dipel under leaf dip method, caused considerable reduction in emergence and marked prolongation in pupal stage ($P<0.05$). Tending to decrease with the advancing concentration, the emergence, varying from 22.12 to 71.72 per cent, differed significantly with concentrations of the thuricide ($P<0.05$).[2]

Like the emergence the pupal period, varying from 14.82 to 29.76 days among different concentrations of the Dipel also differed with different concentrations but, it tended to be directly proportional to the concentration of the microbial preparation. Further, the net mortality also varying from 52 to 94 per cent among different concentrations of the Dipel, differed from concentration to concentration ($P<0.05$) and it tended to decrease with the increasing concentrations significantly.

The life span of progeny male and female adults of untreated parents was more as compared to that earlier treated by leaf dip method with any concentration of the Dipel ($P<0.05$). The longevity of adults, varying from 3.76 to 7.92 days in male and from 3.85 to 8.24 days n female and tending to decrease with the increasing concentration, differed significantly with different concentrations ($P<0.05$).[3]

The larva of the untreated adults acquired considerably more pupation (89.43%) than that of the adults treated topically with any concentration of the Dipel ($P<0.05$). The pupation varied from 27.12 to 69.85 per cent among residue films of different concentrations and tending to decrease with the advancing concentration, the larval survival was affected by the residue film concentration significantly ($P<0.05$).[4]

In response to parent adults treatment with residue films of different concentrations of Dipel the duration of the larval stage varied from 19.64 to 38.30 days and showed a tendency of increase with the increasing concentration in comparison to the untreated adults ($P<0.05$).

The pupa of the untreated adult gained hundred per cent emergence in comparison to that of the adults treated with any concentration of the Dipel through food ($P<0.05$). In emergence, varying from 29.76 to 72.82 percent and tending to be directly proportional to the concentration differed significantly from concentration to concentration ($P<0.05$).[4]

The pupal period which varied from 14.94 to 28.70 days among residue film concentrations from 0.05 to 1.0 percent and tended to be directly proportional to the strength of the residue film was found to be effective differently by the residue films of concentrations of this microbial preparation ($P<0.05$).

The net mortality, varying from 50 to 94 percent showing direct proportionality to the concentration, differed significantly with the residue films of different concentrations of the Dipel ($P<0.05$).[5]

Every concentration of the Dipel applied as residue film to the adult reduced the life-span of both male and adults ($P<0.05$). As regards the influence of different concentrations of the Dipel as residue films on the longevity of adults, it varying from 3.68 to 8.60 days in male and 3.94 to 08.90 days in female and declining with the advancing concentration of Dipel concentration of the residue film ($P<0.05$).[6]

Table 1: Effect of dipel on development at different concentrations

Mode of treatment	Concentration (%)	Pupation (%)	Larval period (days)	Emergence (%)	Pupal period (days)
L.D.M	0.05	69.42	22.76±0.68	71.72	14.82±0.42
	0.10	58.48	26.52±0.66	65.65	15.76±0.44
	0.50	48.24	28.27±0.28	54.72	19.76±0.42
	0.75	36.37	33.32±0.37	42.65	23.86±0.56
	1.00	25.26	36.46±0.72	22.12	29.76±0.40
T.M .	0.05	69.85	19.64±0.75	72.82	14.94±0.22
	0.10	58.40	24.78±0.69	68.47	15.78±0.34
	0.50	48.32	26.85±0.42	56.32	20.87±0.46
	0.75	36.72	30.75±0.17	48.11	23.60±0.06
	1.00	27.12	36.30±0.12	29.75	28.70±0.82
	Control	89.43	16.35±0.42	100.00	12.76±0.32

Table 2: Effect of dipel on longevity in male and female at different modes of treatment

Mode of treatment	Concentration(%)	Longevity(in days)	
		Male	Female
L.D.M	0.05	7.92±0.22	8.24±0.42
	0.10	7.86±0.44	7.84±0.44
	0.50	6.06±0.42	6.06±0.28
	0.75	5.04±0.22	5.50±0.20
	1.00	3.76±0.15	3.86±0.25
T.M	0.05	8.60±0.22	8.90±0.42
	0.10	7.02±0.42	7.00±0.46
	0.50	6.06±0.45	6.87±0.37
	0.75	5.66±0.14	5.80±0.88
	1.00	3.68±0.32	3.94±0.36
	Control	11.35±0.22	13.76±0.24

Table 3: Effect of dipel on longevity in male and female at different modes of treatment

Mode of treatment	Concentration applied (%)	No. of larvae reared	No. of larvae died	No. of pupae died	Total death	%net mortality
L.D.M	0.05	60	24	12	36	52
	0.10	60	28	18	46	76
	0.50	60	34	19	53	86
	0.75	60	40	16	56	92
	1.00	60	45	12	57	94
T.M	0.05	60	19	16	35	50
	0.10	60	27	22	49	78
	0.50	60	36	17	53	86
	0.75	60	42	14	56	92
	1.00	60	43	14	57	94
	Control	60	10	NIL	10	-

CONCLUSION

The above study reveals that Dipel is a promising biopesticide. It has potential to control the pests especially lepidopteran. It is more significant in recent farming practices because of it is safe to environment.

REFERENCES

1. **Bajpai Anju. 2003.** Studies on the effects of *Bacillus thuringiensis* Ber. on the growth and development of *Leucinodes orbonalis*. *Indian society of Life Scinece*. 75-77.

2. **Krishnaiah K., Mohan N.J. and Prasad V.C. 1981.** Efficacy of *B. thuringiensis* Ber. For the control of lepidopterous pests of vegetable crops. *Entomon*, **6(2):** 87-93.

3. **Dobrivojevic K. and Injac M. 1975.** Laboratory investigation of effectiveness of the microbial insecticides Dipel T.M. and Bactospiene for the control of wooly moth (*Hypogymna morio*). *Zastita Bilja.* **26 (134):** 365-369.

4. **Chaturvedi R.K. 2003.** Studies on the effectiveness of *Bacillus thuringiensis* Ber. Against *Utetheisa pulchella* Linn. (Lepidoptera: Arctiidae). *Indian Society of Life sciences.* 68-69.

5. **Sundrababu P.C. and Subramaniam T.R. 1973.** Studies with *B.thuringiensis* Berliner on *Spodoptera litura fabricius*. *Madras Agric. J.,* **60(7):** 487-491.

6. **Sharma S.S., Kaushik H.D. and Kalra V.K. 2001.** Toxicity of *Bacillus thuringiensis* varieties *kurstaki and aizawai* against some lepidopterous pests. *Annals of Biology.* **17(1):**91-94

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 359-364
An International Biannual Refereed Journal of Life Sciences

44(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Importance of mesofauna in soil

Rima Kumari* & Arun Kumar
University Department of Zoology, B.N.M. University, Madhepura, Bihar, India

Received : 05th February, 2020; Revised : 22nd February, 2020

Abstract : Soil may be a large reservoir of varied organisms which together help in regulation of varied biogeochemical cycles. Soil biodiversity is comprised of the organisms that spend all or some of their life cycles within the soil or on its immediate surface (including surface litter and decaying logs). The soil mesofauna are a crucial a part of "> a part of terrestrial ecosystems and a connecting link between microfauna and macrofauna which together form an important part of soil decomposer community. They perform and regulate a major proportion of the organic matter transformations and nutrient fluxes in terrestrial ecosystems. The fluxes and flows are regulated to large extent by Soil mesofauna, being considered as 'Ecosystem webmasters'. The disturbance or perturbation of soils usually alters microarthropod numbers like tillage, fire, and pesticide application typically reduces populations but recovery could also be rapid and microarthropod groups respond differently.

Keywords : Acari, collembola, mesofauna, soil microarthropods

INTRODUCTION

The word Human itself has its roots within the Latin, Humus', the organic matter in soil.[1] Animal members of the soil biota are abundant and diverse. The array of species is extremely large, including representatives of all terrestrial phyla. Many groups of species are poorly understood taxonomically, and details of their explanation and biology are unknown. The easiest and most generally used system for classifying soil organisms is to group them by size (body width) into three main groups: macrobiota, mesobiota and microbiota.[2,3] Body width of fauna is also related to their microhabitats. The microfauna (protozoa, small nematodes) inhabit water films. The mesofauna inhabit air-filled pore spaces and are largely restricted to existing ones. The macrofauna, in contrast, have the power to make their own spaces, through their burrowing activities, and just like the megafauna, can have large influences on gross soil structure.[4-6] The vast range of body sizes among the soil fauna suggests that their effects on soil processes happen at a variety of spatial scales.[7,8] "Ecosystem engineers," earthworms, termites, or ants, alter the body of the soil itself, influencing rates of nutrient and energy flow. "Litter transformers," microarthropods, fragment decomposing litter and improve its availability to microbes. "Micro-food webs" include the microbial groups and their direct microfaunal predators (nematodes and protozoans). These three levels operate on different size, spatial, and time scales.

HABITAT, FUNCTION AND DISTRIBUTION OF MESOFUANA (MICROARTHROPODS)

A soil mesofauna taxon (group) also mentioned as microarthropods is an invertebrate group (aptera) found within terrestrial samples with the size ranging from 0.1-

*Corresponding author :
Phone : 9470187837
E-mail : drrimakumari@gmail.com

2mm which include organisms/orders like Acari, Collembolans, Proturans, Diplurans, Symphellids, Enchytriaeds etc. Large numbers of the microarthropod group (mainly mites and collembolans) are found in most sorts of soils. A square metre of forest floor may contain many thousands of people representing thousands of various species. Microarthropods have a big impact on the decomposition processes within the forest floor and are important reservoirs of biodiversity in forest ecosystems. Soil microarthropods are significant reservoirs of biodiversity but it's not clear exactly how diverse they'll be. Estimation of species richness may be a difficult problem for several sorts of soil organisms (fungi, bacteria, nematodes, for instance, also as micro arthropods). Unlike the macro arthropods, the mites and collembolans have little or no effect on soil structure. Their dimensions allow them to use existing spaces in soil structure thus are often also termed as, Interstitial animals. Even the massive, soft-bodied members of the mite group Prostigmata don't seem to make their own passageways. Some litter-feeding species do burrow into substrates like petioles of decaying leaves and make tunnels, but these haven't any direct effect on soil structure intrinsically. The microarthropods resemble the microfauna in this characteristic. Microarthropods also form a crucial set of linkages in food webs; many microarthropods prey on fungi and nematodes, thereby linking the microfauna and microbes with the mesofauna. Microarthropods successively are prey for macroarthropods like spiders, beetles, ants, and centipedes, thus bridging a connection to the macrofauna. Even a number of the smaller megafauna (toads, salamanders) feed on microarthropods, thus, it's essential to review soil as an ecosystem.

MESOFAUNA COMMUNITY AND THEIR ROLE IN SOIL ECOSYSTEM SURVIVAL

Generally, temperate forest floors with large accumulations of organic matter support high numbers, whereas tropical forests where the organic layer is thin contain lesser numbers of microarthropods.[10] Disturbance or perturbation of soils usually alters microarthropod numbers like tillage, fire, and pesticide application typically reduce populations but recovery could also be rapid and microarthropod groups respond differently. Soil mites usually outnumber collembolans but these become more abundant in some situations. In the springtime, forest leaf litter may develop large populations of "Snow fleas" (*Hypogastrura nivicola* and related species). Among the mites they themselves usually dominate but the fragile Prostigmata may develop large populations in cultivated soils with a surface crust of algae. Immediately following cultivation, numbers of Astigmatic mites have been seen to increase dramatically.[11] In addition, the mesofauna is mobile and migrates through different soil layers, passively transporting bacteria, fungi and their propagules within the gut or on the body surface to new microsites and substrates. Despite being minusculer compared to macrofauna species like earthworms and millipedes, which are the most bioturbators, the soil mesofauna may significantly contribute to forming the microbial habitat. Humus material (H layer) of forest ecosystems on acid soils may almost entirely contains fecal pellets of collembolans and enchytraeids. Micro- and mesofauna don't affect their food source solely by harvesting; selective grazing on certain microbial species can also change the community structure of the microflora. This alters abundance and activity of bacteria and fungi and modifies the pattern of organic matter decay.[12-14]

The soil mesofauna community in undisturbed habitats with special reference to Oribatid mites possesses large diversity of mesofauna compared to the disturbed ones.[15] While studying the results of acid rain on litter decomposition during a beech forest it had been reported that presence of mesofauna significantly reduced the facility of the acid rain to inhibit carbon mineralization.[16] Gamasid mites are good indicators of the soil quality as their high sensitivity to external impact combined with their importance for ecosystem functions make soil mesofauna extremely valuable for ecotoxicology.[17] The rainforests contain a huge kind of soil microarthropods.[18] The seasonal abundance of oribatid mites numbers are correlated positively with radiation on the day of collection.[19] The role of Orbatid mites within the decomposition of the cones of Scotch pine (*Pinus sylvestris*) is of prime importance and thus the orbatid mites are great decomposers compared to other groups of soil mesofauna.[20] The effects of summer warming on the whole population densities of soil-dwelling microartropods within the high arctic region has no significant effect of temperature elevation on orbatid mite populations while as there's negative impact on springtail numbers.[21]

The effects of manipulated soil microclimate on mesofaunal biomass and variety during a hotter, drier

summer, in contrast, experimental heating depress diversity and biomass in drier zone of the plots and variety within the moist zone but enhance biomass within the moist zone and both the biomass and thus the range are positively correlated with soil organic matter.[22] There's decline of soil mesofaunal biodiversity because of the appliance of pesticides as evident from the appliance of DDT in high-input grasslands showed a high density of microarthropods with a high fraction of thelytokous reproduction, associated with a decrease in genetic variation.[23]

The highly abundant and diverse mesofauna populations are capable of upper rates of litter fragmentation and thus the short term decreases of soil pH has no negative effect populations of collembolans.[24] While working on the impact of Collembola and Enchytraidae on soil surface roughness and properties, it had been observed that the surface roughness increased because of mesofaunal activity.[25] The role of soil microarthropods (Acari and Collembola) in organic matter decomposition and nutrient cycling during a forest ecosystem is critical and thus the tiny changes within the structure of soil microarthropod assemblages can have significant effects on the local mobilization of nutrients.[26] The influence of microarthropods on litter decomposition at three forested sites -two tropical and one temperate reveal that the microarthropod populations are considerably effective in litter decomposition which is minimal within the temperate region where the fauna tend to increase the decomposition rate only towards the highest of the year.

In contrast, the effect of fauna within the tropical regions are marked within months of the start of the start of experiment thus it become evident that the range of mesofauna is greater within the tropical regions.[27] The functional role of Collembola within the ecosystem is plant litter decomposition processes and in forming soil microstructure while as soil acidification, nitrogen supply, global global global climate change and intensive farming have negative impact on Collombolan diversity.[28] The mineral amendments on soil fauna in an acid breech forest floor show that nitrogenous amendment decrease the numbers of oribatid mites.[29] Most of the predatory mites – Gamasina or Mesostigmata are life style predators in soil and litter, on the soil surface or on plants and Mesostigmata are important predators of Nematoda, Collembola and bug larvae and should function bioindicators.[30] Soil microarthropods mainly Acari negatively answer altered soil-water availability in tall grass prairie ecosystems and are less abundant in irrigated plots and at the wetter lowland sites which confirm the importance of soil water content in affecting microarthropod densities and distributions in grasslands, and suggest complex, non-linear responses to changes in water availability.[31]

The long-term effects of compaction in arable land because of conventional soil tillage, has negative impact on collembolan number while as, the harvesting and tillage support increase of Collembola in conservation tillage. the stableness analysis of soil Oribatid mite communities from environmentally stressed habitat and relatively well preserved habitat with the attitude of consistency as a primary criteria of stability reveal that concluded that oribatid community from preserved habitat are more stable than from environmentally stressed habitat.[32] The effects of constant temperature versus diurnally fluctuating temperature and uniform versus varying moisture, on the population densities and species richness of Collembola and Mesostigmata in coniferous forest humus and birch leaf litter are well understood by the actual fact thatat fluctuating moisture and temperature regime, Collembola are most abundant, and species richness of Collembola remain higher, whereas Mesostigmata are more numerous at constant temperature.[33]

The possible host habitat specialization in two major groups of soil arthropods, the oribatid and mesostigmatid mites, under three tree species viz., *Eucalyptus pilularis*, *Eucalyptus propinqua* and *Allocasaurina torulosa* show differences between tree species are insufficient to vary species composition of mites.[34] The disturbance of vegetation and soil resulted by tropical rainforest fragmentation are the most factor affecting the range of soil mesofauna and thus the soil condition with more soil organic matter, total N and P, higher pH value and lower soil bulk density become more favorable to the soil mesofauna while because the species richness, abundance and sort of soil mesofauna in fragmented forests are above those in continuous forest, but the similarity of species composition in fragmented forest to the continual forest is minimal.[35] In case of dynamics of springtail and mite populations, there is no evidence for regulation of springtail numbers by mites or for regulation of mite numbers by macroarthropods.[36] Correlations between Collembola, total C and N are usually weak under field conditions and omnivory is probably the prevailing feeding strategy in

Collembola.[37] The relationships between Collembola, Soil chemistry and humus types in forest stands reveal that the Collembola seem to be linked closer to the body of humus than to its chemical parameters.[38] The relative abundance of Collembola and three suborders of mites (Oribatida, Mesostigmata and Prostigmata) during decomposition are greater in old litter than in fresh litter.[39] Organic matter removal and vegetation control generally cause an enormous decrease in collembolan populations; while compaction didn't significantly affect collembolan populations.[40] The groundwater level is also one of the main environmental factors influencing the composition of collembolan and oribatid mite assemblages. The responses of soil microarthropods to experimental short-term manipulates of soil moisture show that drought decrease microarthropod species richness. As the Oribatid mites and Collembola respond differently to the irrigation treatments that the latter community show species evenness and variety within the frequently irrigated plots while because the former community within the infrequently irrigated ones. The population abundance of Collembola and Acari remainlow during drought conditions and therefore the humidity is that the most vital factor determining distribution, abundance, and survival of soil Collembola in tropical forest. And high predation and low accumulation of organic matter cause low population abundance of Collembolan within the tropical habitat.

The variations within the population density of soil invertebrates are controlled by the particular soil ecological conditions. Dominant mesofauna species are morphologically and physiologically adapted for living near the soil surface. The future effects of varied regimes of repeated fertilization on fine roots, mycorrhizae, and soil mesofauna in young stands of shore pine (*Pinus contorta Dougl*. var. latifolia Engelm.) and interior spruce white spruce (Moench) Voss, Engelmann spruce Parry, and their present hybrids) show that fine root attributes and mesofauna respond differently to repeated fertilization regimes at the pine and spruce study sites. Phenanthrene affect the population dynamics of mesofauna and soil biological functioning counting on exposure duration, sort of community, or both. The impacts of invertebrate soil micro- and mesofauna (grazers and predators) on plant productivity and microbial biomass indicate that soil fauna help to manage ecosystem production, especially in nutrient-limited ecosystems.

Soil mesofauna act because the Potential Biological indicators of success in reclaimed soils for recolonization and therefore the mesofaunal densities are greater in natural soils than in reclaimed soils and community structure differ between natural and reclaimed soils.The effects of soil mesofauna and microclimate on nitrogen dynamics in leaf litter decomposition along an elevation gradient give conclusion that the rapid accumulation of N in lower elevation sites may result within the retention of mobile N in soils and therefore the effects of soil mesofauna on N dynamics could also be intimately associated with microclimate (warm and humid) and faunal diversity along the elevation gradient. The stable- isotope labeling and probing of recent photosynthates into respired CO_2, soil microbes and soil mesofauna employing a xylem and phloem stem-injection technique on Sitka spruce (Picea sitchensis) reveal that the Stem injection of huge trees with 13^oC-enriched compounds may be a successful tool to trace C-translocation belowground. In particular, the significant 13^oC enrichment of CO_2 and enchytraeids near the base of the tree and the significant 13^oC enrichment of phosphor-lipid fatty-acid (PLFAs) up to 20 m away indicate that mature Sitka skpruce (*Picea sitchensis*) have the capacity to support soil communities over large distances.

The meso-fauna foraging on seagrass pollen may serve in marine zoophilous pollination and aid within the pollination of *T. testudinum* when visiting female flowers. The diversity of acari and collembola along a pollution gradient in soils of a Pre-pyrenean forest ecosystem around a steel mill reveal that the density of acari and collembola significantly decrease with the increase in concentration of Cr, Mn, Zn, Cd and Pb. Mites appear to be more sensitive to heavy metal pollution than springtails. From the review of literature it's quite evident that the soil mesofauna are a crucial a part of terrestrial ecosystems and a connecting link between microfauna and macrofauna which together form an important a part of soil decomposer community.

CONCLUSION

Soil mesofauna are ready to use the prevailing pore space in soil, cavities or channels. They constitute important reservoirs of biodiversity and are reflectors of ecosystem metabolism. Furthermore, the soil mesofauna regulate plant productivity and microbial biomass and are key organisms to manage ecosystem production, especially in nutrient-limited ecosystems.

REFERENCES

1. **J.N. Klironomos, P. Widden and I. Deslandes. 1992.** Feeding preferences of the collembolan Folsomia candida in relation to microfungal successions on decaying litter. *Soil Biol Biochem.* **24:**685–692
2. **S. Sarker. 1990.** Studies on microarthropod community in one undisturbed habitat of Tripura (India) with special reference to oribatid mites. Revue D'Ecologie et de *Biologie du Sol.* **27:**307-329.
3. **V. Wolters. 1991.** Effects of acid rain on leaf-litter decomposition in a beech forest on calcareous soil, *Biology and Fertility of Soils.***11(2):**151-156.
4. **S. J. S. Hattar, J. R. B. Alfred and V.T Dartong. 1992.** Soil acarina and collembolan in forest and cultivated land of Khasi Hills, Meghalaya, *Record of Zoological Survey of India.* **92**:89-97.
5. **R. L. Kitching, J. M. Bergelson and L. Paul. 1993.** The biodiversity of arthropods from Australian rainforest canopies: General introduction, methods, sites, and ordinal results. *Australian Journal of Ecology.* **18:**181-191.
6. **L.C. van Nieuwenhuizen, A. J. M. Verster, I. J Horak., R.C. Krecek and J.R. Grimbeek. 1994.** The seasonal abundance of oribatid mites (Acari: Cryptostigmata) on an irrigated Kikuyu grass pasture. *Experimental and Applied Acarology.* **18(2):**73-86.
7. **N. R. Webb. 1994.** The role of Steganacarus magnus (Acari: Cryptostigmata) in the decomposition of the cones of Scots pine, *Pinus sylvestris*, *Pedobiologia,* **35:**351-359.
8. **S. J. Coulson, I. D. Hodkinson, N.R. Webb, W. Block, J.S. Bale, A. T Strathdee, M.R. Worland, and C. Wooley. 1996.** Effects of experimental temperature elevation on high arctic soil microarthropod populations, *Polar Biology.* **16(2):** 147-153.
9. **J. Harte, A. Rawa and V. Price. 1996.** Effects of manipulated soil microclimate on mesofaunal biomass and diversity, Soil Biology and Biochemistry **28 (3):**313-322.
10. **H. Siepel. 1994.** Life-history tactics of soil microarthropods, Biodiversity and Conservation, **18(4):** 263-278.
11. **R. Jandl, H. Kopeszki, and G. Glatzel. 1997.** Effect of a dense *Allium ursinum* (L.) ground cover on nutrient dynamics and mesofauna of a *Fagus sylvatica* (L.) woodland, *Plant and Soil.* **189(2):** 245-255.
12. **S. Schrader, M. Langmaack and K. Helming. 1997.** Impact of Collembola and Enchytraeidae on soil surface roughness and properities, *Biology and Fertility of Soils,* **25(4):** 396-400.
13. **L. Heneghan, and T. Bolger, 1998.** Soil microarthropod contribution to forest ecosystem processes: the importance of observational scale, *Plant and Soil.* **205(2):** 113-124.
14. **L. Heneghan, D. C. Coleman, X. Zou, D.A. Crossley and B. L. Haines. 1998.** Soil microarthropod community structure and litter decomposition dynamics: A study of tropical and temperate sites, *Applied Soil Ecology.* **9:**33-38.
15. **J. Rusek, 1998.** Biodiversity of Collembola and their functional role in the ecosystem, *Biodivers. Conserve.* **7(9):**1207-1219.
16. **S. Deleporte, and P. Tillier. 1999.** Long-term effects of mineral amendments on soil fauna and humus in an acid beech forest floor, *Forest Ecology and Management.* **118(1-3):** 245-252.
17. **H.H. Koehler. 1999.** Predatory mites (Gamasina, Mesostigmata), Agriculture, Ecosystems and Environment. **74(1-3):**395-410.
18. **H. A. O Lear and J.M. Blair. 1999.** Responses of soil microarthropods to changes in soil water availability in tallgrass prairie, *Biology and Fertility of Soils.* **29(2):**207-217.
19. **J. Chulue, and J. H. Lee. 2001.** Stability Analysis of Soil Oribatid Mite Communities (Acari:Oribatida) from Namsan and Kwan greung Deciduous Forests, Korea, *Korean J. Ecol.* **24(4):**239-243.
20. **V. Huhta, and S.M. Hanninen. 2001.** Effects of temperature and moisture fluctuations on an experimental soil microarthropod community. *Pedobiologia.* **45(3):**279-286.
21. **G.H.R. Osler and A.J. Beattie. 2001.** Contribution of oribatid and mesostigmatid soil mites in ecologically based estimates of global species richness, *Austral Ecology,* **26(1),** 2001, 70.
22. **X. Yang and L.Sha, 2001.** Species composition and diversity of soil mesofauna in the, Holy Hills

fragmentary tropical rain forest of Xishuangbanna, China, Ying Yong Sheng Ta; *Xue Bao*. **12(2):**261-265.

23. **S. H. Ferguson, and D.O Joly, 2002.** Dynamics of springtail and mite populations: the role of density dependence, predation and weather, *Ecological Entomology,* **27(5):**565-573.
24. **J. Filser, 2002.** The role of Collembola in carbon and nitrogen cycling in soil, *Pedobiologia.* **46(3-4):**234-245.
25. **N. Cassagne, C. Gers, and T. Gauguelin. 2003.** Relationships between Collembola, soil chemistry and humus types types in forest stands (France), *Biology and Fertility of Soils,* **37(6):**355-361.
26. **B.C. Reynolds, Jr. Crossley, and M.D. Hunter, 2003.** Response of soil invertebrates to forest canopy inputs along a productivity gradient, *Pedobiologia,* **47(2):** 127-139.
27. **R. J. Eaton, M. Barbercheck, M. Buford, and W. Smith, 2004.** Effects of organic matter removal, soil compaction and vegetation control on Collembolan populations, *Pedobiologia*. **48(2):**121-128.
28. **U. Irmler. 2004.** Long-term fluctuation of the soil fauna (Collembola and Oribatida) at groundwater near sites in an alder wood, *Pedobiologia,* **48(4):**349-363.
29. **MA. Tsiafouli, A.S. Kallimanis, E. Katana, G.P. Stamou and S.P. Sgardelis, 2004.** Responses of soil microarthropods to experimental short-term manipulation of soil moisture, *Applied Soil Ecology,* **29:**252-257.
30. **O. Wiwatwitaya and H. Takeda,** Seasonal changes in soil arthropod abundance in the dry evergreen forest of north- east Thailand, with special reference to collembolan communities, *Ecological Research,* **20(1):**59-70.
31 **V.I. Ubugunova, I.N. Lavrent eva, L.L. Ubugunov and T.P. Nikheleeva, 2007.** Mesofauna in soils of the ivolga depression (Western Transbaikal region), Institute of General and Experimental Biology, Russian Academy of Sciences, Buryat Republic, Russia.
32. **B. Shannon, 2008.** "Effects of Repeated Fertilization on Fine Roots, Mycorrhizae, and Soil Mesofauna in Young Lodgepole Pine and Spruce Forests in Central British Columbia." *B.C. Ministry of Forests and Range Research Branch,* Victoria.
33. **C. Jerome, J. Richard, E. Susanne, C. Michael, S 2009.** Renaud, and H. K. Paul, Interspecific relationships among soil invertebrates influence pollutant effects of phenanthrene, John Wiley and Sons.
34. **T.E Sackett, T.C. Aimée and J.S. Nathan 2010.** Linking soil food web structure to above- and below ground ecosystem processes: a meta-analysis. Dept of Ecology and Evolutionary Biology, Univ. of Tennessee, 569 Dabney Hall, University of Tennessee Knoxville, 37996-1610, USA.
35. **P. B. Jeffrey. 2011.** Exploring the World Beneath your Feet-Soil Mesofauna as Potential Biological Indicators of Success in Reclaimed Soils, Stantec Consulting, Sidney, BC Canada.
36. **W. Shaojun and R. Honghua. 2011.** Effects of soil mesofauna and microclimate on nitrogen dynamics in leaf litter decomposition along an elevation gradient. *African Journal of Biotechnology.* **10(35):** 6732-6742.
37. **C. Carolyn, W. Andrew, J. I. B. Maria, J. G. Sue 2012.** Stable-isotope labeling and probing of recent photosynthates into respired CO_2, soil microbes and soil mesofauna using a xylem and phloem stem-injection technique on Sitka spruce (*Picea sitchensis*), John Wiley & Sons.
38. **B.I van Tussenbroek, L. Veronica Monroy-Velazquez, and V. Solis-Weiss. 2012.** Meso-fauna foraging on seagrass pollen may serve in marine zoophilous pollination. Marine Ecology Progress series, Mexico.
39. **M.S Jesus, L. M. María, E. David, B, Enrique, J.Rafael , L. Esther , B. Raúl and H. A. Arturo. 2013.** Diversity of acari and collembola along a pollution Gradient in soils of a pre-pyrenean forest ecosystem" University of Navarra, Department of Chemistry and Soil Science, Pamplona (Navarra), Spain,.
40. **H.M Andre, X. Ducarme, and P. Lebrun, 2002,** Soil biodiversity: myth, reality or conning? Oikos, **96 (1):** 3-24.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.365-366
An International Biannual Refereed Journal of Life Sciences

45(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect of volatile and aromatic compounds on the growth of fungi in decay of garlic cloves

Sujata Suman*
University Department of Botany, B.N.M.University, Madhepura, Bihar, India

Received : 7th February, 2020 ; Revised : 23th February, 2020

Abstract : Garlic is one of the most important cultivated herb used as spices and medicine. During storage it is damaged by some pathogenic fungi to protect garlic from these pathogenic fungi, some volatile aromatic compounds are in use. To assess the inhibitory effect of three volatile aromatic compounds Camphor, Menthol and Thymol against pathogenic fungi during storage of garlic bulb present work was performed. It was observed that the inhibitory effect of Thymol is greatest against pathogenic fungi followed by Menthol and Camphor.

Key words: Pathogenic fungi, thymol, menthol, camphor

INTRODUCTION

Garlic is economically so important that it is variously used for medicinal purposes and as spices world over. The medicinal properties of garlic have been well documented in Ayurveda. The main chemical constituents found in the fleshy storage leaf are volatile oil which contains Allyl disulphide, Ally proylsulphide and polysulphide, Allicin and Allocetion I and II. These compounds serve as antibiotics. The action of garlic due to having these biochemicals is spermopoitic, aphrodisiac, oleative, digestive, bone healing, tonic, rejuvenator, diaphoretic, stomachic, diuretic, stimulant, expectorant, anti-bacterial and anti-septic. Due to these properties, garlic is used against flatulence, epilepsy, pulmonary tuberculosis, ulcer, skin diseases, throat trouble, cough and cold, asthma and respiratory trouble.

During storage, several pathogenic fungi attack garlic and damage bulb and leaves. The common fungi attacking garlic are *Aspergillus niger, Fusarium moniliforme, Fusarium solani, Fusarium culmorum, Aspergillus sydowi, Fusarium oxysporum, Aspergillus flavus* and *Aspergillus nidulans.* Some of them produces toxin which make garlic not useful. To protect garlic during storage, some volatile aromatic compounds are in use such as Camphor, Menthol and Thymol. In the present study, effect of these four volatile aromatic compounds against pathogenic fungi have been evaluated.

*Corresponding author :
Phone : 8140606666
E-mail : nandaniob@gmail.com, sharmaop02@gmail.com

MATERIAL & METHODS

20ml of Potato Dextrose agar medium was poured in sterilised glass petri dishes of 6cm diameter aseptically. On solidifying, the centre of the medium was inoculated with the fungal spores of *Aspergillus niger,Fusarium moniliforme, Fusarium solani, Fusarium culmorum, Aspergillus sydowi, Fusarium oxysporum, Aspergillus flavus* and *Aspergillus nidulans* isolated from the decaying cloves of garlic. Three petri dishes were inoculated from the culture of each fungus on slant of PDA with the help of sterile inoculating needle and three petri dishes were

the control. The petri dishes were kept inverted i.e. the bigger part of the pair lower and inoculated part of the pair to the upper. 20mg each of camphor, menthol and thymol was kept in the bigger part of the petri dishes and the two pairs were sealed with celluloid tape for the control none of the camphor, menthol and thymol was used for the observing growth of fungi. The inoculated petri dishes were incubated at 30 ± 1°C for 7 days.

After expiry of the incubation period, the diameter of the culture was measured in mm of scale from the outside of the petri dish. The mean diameter of the culture in three petri dishes was calculated.

RESULT

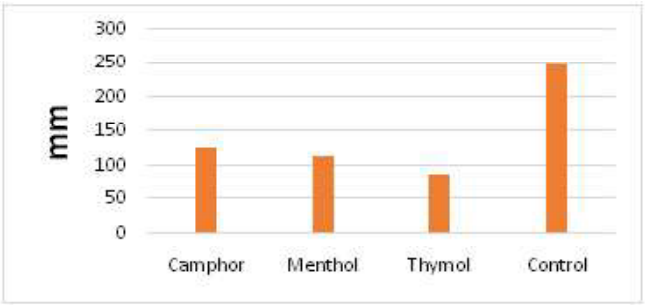
The effect of volatile aromatic compound was evaluated. It appears that growth of the fungi involved in decay of garlic bulbs is restricted by the application of volatile aromatic compounds. In this experiment three volatile aromatic compounds were used during storage, Camphor, Menthol and Thymol. It was observed that Thymol seems superior to the remaining two volatile aromatic compounds. The efficacy of these three volatile aromatic compounds in inhibiting the growth of fungi can be arranged in descending order as follows:

Thymol>Menthol> Camphor

The result is tabulated in table no. 1 and graph no. 1.

Table 1- Effect of Volatile and Aromatic compound on the growth of fungi involved in decay of garlic (diameter of culture expressed in mm)

Sl. No.	Fungi	Control	Volatile & Aromatic compound		
			Camphor	Menthol	Thymol
1.	*A. niger*	48	19	17	13
2.	*F. moniliforme*	32	16	18	14
3.	*F. solani*	27	13	15	11
4.	*F. culmorum*	25	17	13	10
5.	*A. sydowi*	33	15	13	10
6.	*F. oxysporum*	23	14	12	09
7.	*A. flavus*	41	21	16	12
8.	*A. nidulans*	20	13	11	08



Graph no. 01: Effect of Volatile and Aromatic compound on the growth of fungi involved in decay of garlic (diameter of culture based on total growth in mm)

CONCLUSION

During storage, garlic bulb and leaves are damaged due to attack of several pathogenic fungi.

To protect garlic from damage caused by pathogenic fungi, some volatile aromatic compounds are in use. The efficacy of three volatile aromatic compounds Camphor, Menthol and Thymol for the inhibition of growth of pathogenic fungi have been evaluated in the present work. It appears that growth of the fungi involved in decay of garlic bulbs is restricted by the application of volatile aromatic compounds. In this experiments three volatile aromatic compounds were used during storage, Camphor, Menthol and Thymol. It was observed that Thymol seems superior to the remaining two volatile aromatic compounds. The efficacy of these three volatile aromatic compounds in inhibiting the growth of fungi can be arranged in descending order as follows:

Thymol>Menthol> Camphor

REFERENCES

1. **Agarwal V. K. and Sinclair J.D. 1997.** Principles of seed pathology. CRC press Inc., Lewis publishers, Boca Raton, NY, London.
2. **Braida Charlene 1986.** A glorious garlic, A cookbook, Pownd, V.T. Storey Communication, 1986.
3. **Bramhavarchas 2010.** Ayurvedka Pran: Vanaushad hivigyan, YugNirmanYojna, Vistar trust, Mathura, India.
4. **Chopra, R.N., Nayar, S.L. and Chopra, I.C. 1956.** Glossary of Indian Medicinal Plants. CSIR, New Delhi.
5. **Dicklow, M.B. 2012.** UMass Extension, UMass Amherst Centre for Agriculture, Vegetable Programme, USA.
6. **Duke, J.A. 2002.** Handbook of Medicinal Herbs. *Boca Raton, CRC Press*.
7. **Kisan, Diary, 2012.** Bihar Agriculture University, Sabaur.
8. **Wheeler, H. and Luke, H.H. 1963.** Microbial toxins in Plant diseases. Ann. *Rev. Microbiol*. **17:** 223-242.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.367-370
An International Biannual Refereed Journal of Life Sciences

46(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Effect on histochemical localization of alkaline and acid phosphates in the alimentary canal of Khapra beetle, *Trogoderma granarium* (Evert) due to some environmental pollutants

Rajesh Kumar Suman[a], Surabhi Saloni[a] & Lakshman Sah[b]*

[a]P.G. Department of Zoology, Magadh University, Bodh-Gaya, Bihar, India

[b]Department of Zoology, R.K.D College Patna, Patliputra University, Patna, Bihar, India

Received : 8th February, 2020 ; Revised : 24th February, 2020

Abstract : This paper represents the distribution of enzymes alkaline and acid phosphatases, were studied in the alimentary canal of Khapra beetle. It is observed that these enzymes were localized only in the midgut region. Both foregut and hind gut showed negative reaction. In the lower part of the midgut, a very high concentration of alkaline phosphatases was observed, whereas, the upper part of midgut showed relatively lesser concentration of these enzymes. A moderate reaction was observed throughout the midgut region of acid phosphatases which was found to be localized mainly towards lumen and peripheral sides but no effect of environmental pollutants on the gut of *Trogoderma granarium* (Evert).

Key words: Khapra beetle, insect, alimentary canal, enzyme, maize

INTRODUCTION

The distribution of alkaline and acid phosphatases in the alimentary canal of various insects has been studied by Gomari (1946)[1], Frankel (1964)[2], Srivastava and Saxena (1967)[3], Asharfi *et al.* (1969)[4], Naqvi *et al.*(1969)[5], Sharan and Sinha (1975)[6], Srivastava and Sharan (1981)[7], Anon (2005)[8]. Many fold functions of both alkaline and acid phosphatases are well known. These enzymes are chiefly concerned with the energy transfer, enzymatic synthesis and hydrolysis of ester linkages. These functions are involved in the digestion and absorption of food in the alimentary canal. Considering the importance of these enzymes, an attempt has been made in the present investigation to study the localization of alkaline and acid phosphatases in the alimentary canal of *Trogoderma granarium* (Evert).

*Corresponding author :
Phone : 9931022602, 7004642268
E-mail : lakshmansah011958@gmail.com

MATERIALAND METHODS

Laboratory cultures of the insects were raised on maize (*Zea mays*) at 30 ± 2ºC temperature and 70 ± 5% R.H. only newly emerged adult insects were used in the experiment. The insects were fixed in Gomori's cold absolute acetone at 4ºC for 24 hours. Since the size of the insect is very small, hence for fixation, they were kept intact. Only elytra and legs were cut with the help of sharp blade. After fixation, materials were washed in three changes of absolute acetone and then embedded in paraffin wax (58ºC). Blocks were prepared by routine procedure.

The paraffin sections were cut on 8 microns. The Calcium cobalt method (Gomori,1956[9], after Pearse,1968[10]) and modified lead nitrate method were used for determining the sites of activities of alkaline and acid phosphatases respectively. The results were confirmed by preparing control slides.

OBSERVATIONS

Tests for alkaline and acid phosphatases gave positive results in the midgut region only. The cells of foregut and hindgut showed negative reaction for both the enzymes. An intense positive reaction was observed throughout the midgut for alkaline phosphatase which was scattered all over the cells. However, the pattern of intensity was not the same for the entire midgut region. In the upper part of the midgut, the black precipitate which is indirective of this enzyme was found relatively in slightly lesser concentration while in the lower part of the midgut, a very strong concentration of this enzyme appeared. (Fig 1,2).

A positive and moderate reaction for acid phosphatase was observed throughout the midgut region. But this was found to be concentrated mainly towards lumen and peripheral sides. (Fig 3).



Fig. 1. T.S. of upper part of midgut showing intense reaction for alkaline phosphatase

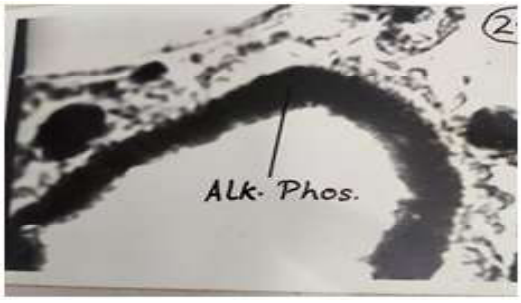


Fig. 2. T.S. of lower part of midgut showing very intense reaction for alkaline phosphatase

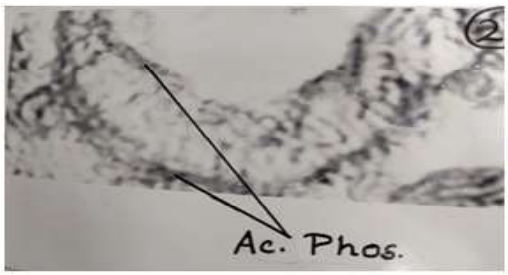


Fig. 3. T.S. of midgut showing moderate reaction for acid phosphatase

DISCUSSION

Alkaline phosphatase activity indicates an increased phosphate transfer, whereas, acid phosphatase activity indicates hydrolysis of phosphate esters.[10] According to Srivastava and Saxena (1967)[3], phosphatase is mainly responsible for dephosphorylation of hexose phosphate in the formation of glycogen. Asharfi *et al.* (1969)[4], has correlated enzyme alkaline phosphatase with the secretory activity of the digestive system of *Schistocera gregaria*. Sharan and Sinha (1975)[6], has suggested that the activity of alkaline phosphatase was confirmed to midgut region in *Gryllus testaceus*. In the present investigation, the higher concentration of alkaline phosphatase in the lower part of the midgut indicated an increased activity of secretion and absorption of metabolites in this part of the midgut. Similar results has been obtained by Srivastava and Sharan (1981)[7], for alkaline phosphatase activities in the midgut of *Tribolium castaneum*. Positive reaction of acid throughout the midgut region confirmed the hydrolysis of the ester linkages in phosphate esterase.

REFERENCES

1. **Gamori, G. 1946.** "The study of enzymes in the tissue section", *American journal of clinical pathology*, **16:**347.
2. **Frankel, H.H. 1964.** "A modified calcium cobalt method for the demonstration of alkaline phosphatase". *Amer. J. Clinical Pathology*, **42:**342.
3. **Srivastava J.P. and Saxena, S.C. 1967.** *Appl. Ent. Zool*., **2:**85-92.
4. **Ashrafi, S.H. Naqvi, S.N.H and Quadri , M.A.H. 1969.** *Ohio. J. Sci.,* **69:**183-193.
5. **Naqvi, S.N.H., Siddiqui, H.R and Yasmin, F. 1969.** *Pak. F. Zool*., **1:**27-34.

6. **Sharan, R.K. and Sinha, A. 1975.** *Indian F. Zool*., **16(1):**7-10.

7. **Srivastava, S.K. and Sharan, R.K. 1981.** F. *Ent. Res*. **5(2):**175-176.

8. **Anon. 2005.** Description, Biology and control of Khapra Beetle. Promotional website (https://www. Pest products.com/khapra beetles. html).

9. **Gomori, G. 1956.** "Histochemical methods for acid phosphatase". *J. Histochem and Cytochem*, **4:**453.

10. **Pearse, A.G. E. 1968.** In: Histochemistry, Theoretical and Applied. 3rd ED., Vol.1. Little, Brown & Co., Boston 426 pp.

ADDITIONAL REFERENCES

1. **Ackerman, G.A. 1958.** "A combined alkaline phosphatase – PA/s staining method". *Stain Technology*, **33:**269-275.

2. **French, S. and R.C. Venitte. 2005.** Mini risk assessment Khapra beetle.

3. **Szito, A. 2006.** *Trogoderma granerium*. Global invasive http://www. issg. org/ database.

4. **Wilbur, Donald, A. and Mills Robert, B. 1978.** "Stored grain insects". Fundamental of Applied Entomology 3rd Edition. Chapter -17, Macmillan Pub, New York.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.371-374
An International Biannual Refereed Journal of Life Sciences

47(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Temporal variation S-W species diversity of oriental garden lizards in Madhepura district of Bihar, India

Pranav Kumar Pratap*
P.G. Department of Zoology, B.N.Mandal University, Madhepura, Bihar, India

Received : 10th February, 2020 ; Revised : 25th February, 2020

Abstract : The status of biodiversity in general & species diversity in particular is always under anthropogenic threat throughout the world in varieties of habitats where human beings have started building their shelter. Nevertheless, the natural profile of species diversity at habitat level & biodiversity at biospheric level always displays temporal & spatial variation in time and space. The present article deals with the temporal variation of garden lizard sampled in different seasons from Madhepura during 2013. Lizards being important member of lacertilian reptiles are also exposed to different degree of endangerment and threatening pressure over their species members due to such interference. The present paper is an attempt to bring on record different species of garden lizards present in various terrestrial habitats of Madhepura district. Eight different species of garden lizards have been sampled and identified from these habitats & their cumulative species diversity index through Shannon-Weiner equation (S-W) has also been statistically computed on the basis of their individual population per unit area during the sampling period. The result reflects their survival status only.

Key words:Temporal variation, S-W species diversity, garden lizard, lacertilian reptiles.

INTRODUCTION

The word "Biodiversity" is most commonly used to for the term, species diversity and species richness.[1] Species richness and genetic diversity are the two important aspect of biodiversity that influences the overall ecosystem processes and functions.[2] Biodiversity is not distributed evenly on Earth, and is richer in the tropics.[3] These tropical forest ecosystems cover less than 10 percent of earth's surface and contain about 90 percent of the world's species. And biodiversity is significant for sustaining of the ecosystem and also for providing benefits the human society in multiple ways like food security, health, medicine, livelihood, regulating and controlling spread of diseases, disaster risk reduction, nutrient cycling, air purification, micro climate variability, etc.[4] Biodiversity is considered as economically valuable since it provides essential goods and services for the survival of human beings.[5] Reptiles are among one those species which somehow plays a very crucial role in conservation of the ecosystem. India is a country with only 2.4% of the world's land area that harbours 7-8% of all recorded species, including over 45,000 species of plants and 91,000 species of animals. According to the first global assessment 19% of the reptiles which includes crocodiles, lizards, snakes and turtles are struggling for the survival, 12% of it are critically endangered situation, 41% are endangered and 47% are vulnerable going to extinct, over

*Corresponding author :
Phone : 09404308111
E-mail : pksirmumbai@gmail.com

3800 species of lizards all across the world. Skinks which are also called lizards of the order Squamata (or scaly reptiles) are the largest and the most diverse family of lizards living today. They are also widespread, occurring throughout most of the world, wherever lizards are found in general.[6] Almost one in five of the reptiles are going to face extinction.[7] The loss of biodiversity and habitat due to rapid development is considered to be an alarming call for an urgent redressal of the issue, the Conference of the Parties of the Convention on Biological Diversity (CBD) adopted a Strategic Plan in 2002.

MATERIALAND METHODS

Study area

The study was conducted in four different season:- Winter (October to January), Spring (February to March), Summer (April to June) & Monsoon (July to September) in Madhepura district of Bihar which is situated between 25'31 and 26'20 latitude and in the middle of 86'36 to 87'07 longitudes. The maximum temperature of this district ranges from 35 to 40°C and the minimum temperature varies from 7°-9°C & average rain fall in this district is 1300 mm. Since much fluctuation is seen in temperature in this area, it function as best place for the study of variations in biotic community. To access the research based on study of temporal diversity of lizards nearby areas were selected and field trip was made during different hours depending upon the season. Since majority of lizard species are active during the day though some are active at night therefore field trip was done during the second half when the temperature was used to be little warm favourable for these species. After each encounter with these species, photo as well as video footage was taken to study their habitat, identification & species diversity.

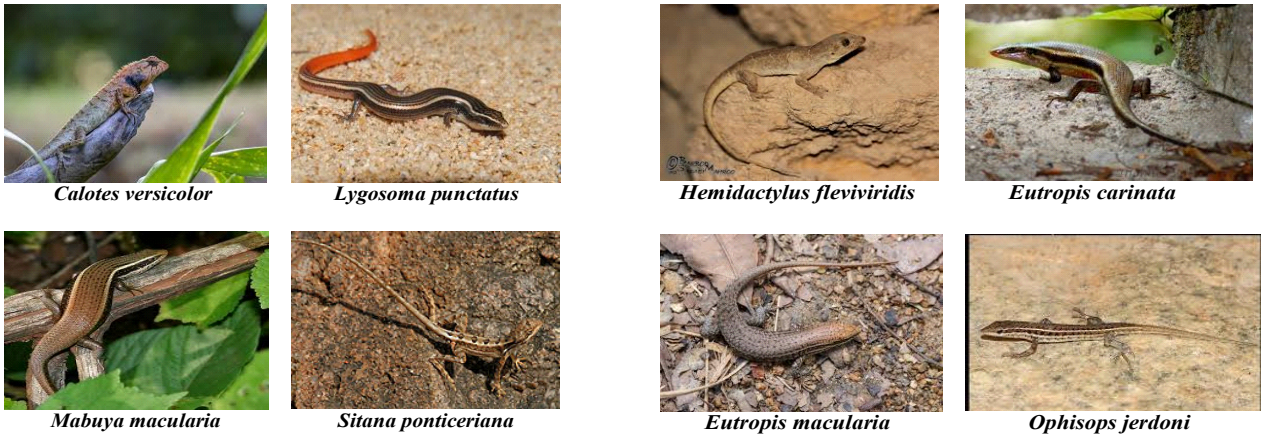


Fig. 1: Different species of lizard sampled from Madhepura

Table 1- Time specific distribution of different species of lizard

Sl no.	Name of the species	Family	Winter (Oct-Jan)	pi (n/N)	log pi	pi x log pi	H	Spring (Feb-Mar)	pi (n/N)	log pi	pi x log pi	H	Summer (Apr-Jun)	pi (n/N)	log pi	pi x log pi	H	Monsoon (July- Sept)	pi (n/N)	log pi	pi x log pi	H
1	*Calotes versicolor*	Agamidae	6	0.166	-0.779	-0.1293		10	0.175	-0.756	-0.1323		15	0.2	-0.698	-0.1396		2	0.166	-0.779	-0.1293	
2	*Lygosoma punctatus*	Scincidae	5	0.138	-0.860	-0.1186		7	0.122	-0.913	-0.1113		9	0.12	-0.920	-0.1104		3	0.25	-0.602	-0.352	
3	*Mabuya macularia*	Scincidae	4	0.111	-0.954	-0.1058		8	0.140	-0.853	-0.1194		12	0.16	-0.795	-0.1272		1	0.083	-1.080	-0.0896	
4	*Sitana ponticeriana*	Agamidae	4	0.111	-0.954	-0.1058		9	0.157	-0.804	-0.1262		11	0.146	-0.835	-0.1219		2	0.166	-0.779	-0.1293	
5	*Hemidactylus flevivirdis*	Gekkonidae	8	0.222	-0.653	-0.1449	**0.869**	7	0.122	-0.913	-0.1113	**0.8847**	8	0.106	-0.974	-0.1032	**0.8791**	1	0.083	-1.080	-0.0896	**1.0087**
6	*Eutropis carinata*	Scincidae	4	0.111	-0.954	-0.1058		7	0.122	-0.913	-0.1113		8	0.106	-0.974	-0.1032		0	00	0	0	
7	*Eutropis macularia*	Scincidae	3	0.083	-1.080	-0.0896		4	0.070	-1.154	-0.0807		5	0.066	-1.180	-0.0778		2	0.166	-0.779	-0.1293	
8	*Ophisops jerdoni*	Lacertidae	2	0.055	-1.259	-0.0692		5	0.087	-1.060	-0.0922		7	0.093	-1.031	-0.0958		1	0.083	-1.080	-0.0896	
	Total		**36**					**57**					**75**					**12**				

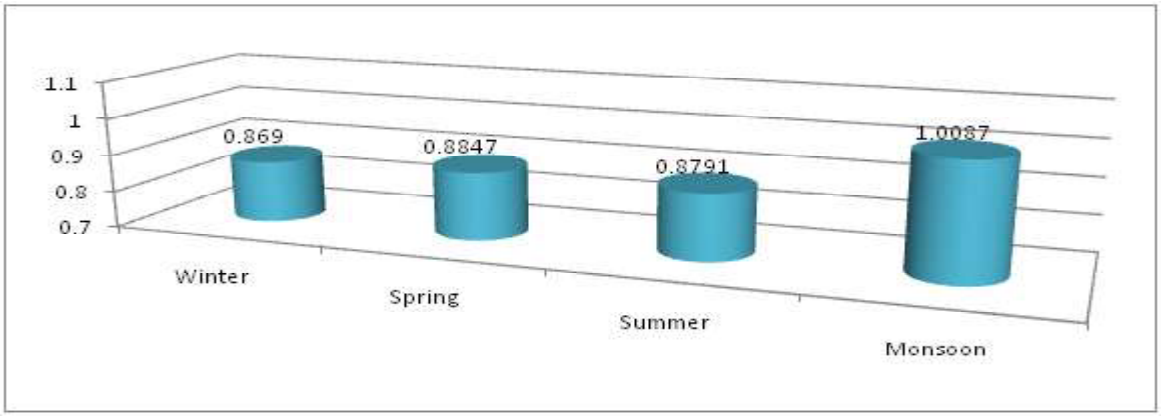


Fig. 2: Season specific temporal variation in S-W diversity of garden lizard in Madhepura District

RESULT AND DISCUSSION

On the basis of investigation done 8 various species of oriental garden lizard have been recorded in different seasons of Madhepura as denoted in table 1 & figure 1. These species includes *Calotes versicolor, Lygosoma punctatus, Mabuya macularia, Sitana ponticeriana, Hemidactylus fleviviridis, Eutropis carinata, Eutropis macularia* & *Ophisops jerdoni.* Data obtained from sampling of these species show that different seasons of a year, play an important role in breeding & population building of these species. As mentioned in table number 1 it is clear that garden lizards are seasonal breeders & their occurrence is directly related to the temperature, humidity & rainfall etc. During winter season i.e. from October to January, the S-W diversity was 0.869, in spring (February to March) it was 0.8847, in summer (April to June) 0.8791 & in monsoon (July to September) 1.0087 which has also been displayed in fig.2. Thus the above findings reflect that low temperature in water is ecologically infavourable for the survival & preparation of these reptiles while monsoon months proved to be highly congenial when the S-W diversity value climbed upto 1.0087.

ACKNOWLEDGEMENT

The author is very much grateful to Professor & Head, Department of Zoology, Dr. Arun Kumar, B.N Mandal University, Madhepura, Bihar & his supervisor Dr. R.B. Jha for giving support to complete this research paper.

REFERENCES

1. **Walker, Brian H. 1992.** "Biodiversity and Ecological Redundancy". *Conservation Biology*. **6(1):**18–23.
2. **Chapin, F. S., III, B. H. Walker, R. J. Hobbs, D. U. Hooper, J. H. Lawton, O. E. Sala, and D. Tilman. 1997.** Biotic control over the functioning of ecosystems. *Science*. **277:** 500–504.
3. **Gaston, Kevin J.; Spicer, John I. 2013.** *Biodiversity: An Introduction*. John Wiley & Sons.
4. **UN CBD COP 10 Policy brief. 2010.** The Importance of Biodiversity to human health. https://www.cbd.int/doc/health/cohab-policy-brief1-en.pdf
5. **Gamfeldt, L., Hillebrand, H., & Jonsson, P. R. 2008.** Multiple functions increase the importance of biodiversity for overall ecosystem functioning. *Ecology*. **89(5):**1223-1231.
6. **A Datta-Roy and K P Karanth. 2009.** The Out-of-India hypothesis: What do molecules suggest?, *Journal of Biosciences*. **34(5):**687-697.
7. **McCann, J., 2013.** Mail on line, Science and technology, www.dailymail.co.uk

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.375-378
An International Biannual Refereed Journal of Life Sciences

48(AS).
Animal Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

A study of physico-chemical characterisation of Kanke Dam, Ranchi

Manoranjan Kumar Mehta*
Department of Zoology, St. Columba's College, V.B.University, Hazaribag, Jharkhand, India

Received : 11th February, 2020 ; Revised : 25th February, 2020

Abstract : The back water of river Potpoto is a prime source of water for Kanke Dam, located in Kanke of Ranchi district, have been studied to project the characteristics of water with respect to different pollution sensitive parameters i.e. organic pollutants- BOD, DO, COD, Inorganic pollutants - Electrical Conductivity, Turbidity, pH, total alkalinity, Chloride, Total hardness etc. The pH (7.68), total alkalinity (180.5 mg/l), Chloride (133.12 mg/l), Total hardness(139.83 mg/l), Calcium (60.26 mg/l) and Magnesium hardness (19.38 mg/l) were found well within the prescribed standard for drinking water (BIS10500:1991) during the study period i.e. December 2017- January 2020. The study inferred that the water quality of Kanke Dam is still free from the any kind of pollutants, must be due to its higher assimilative capacity. It also indicates the higher potential for pisciculture and drinking water sources (after disinfection) for the nearby villages which may ultimately improve the economic condition of the surrounding habitation.

Key words:Assimilative capacity, Kanke Dam, Dissolved oxygen, Physico-chemical characters, Water quality

INTRODUCTION

Kanke, is well known place of the capital of Jharkhand, Ranchi. It has a dense population. This area is facing severe water crisis in this year with underground water falling deeper than ever in the state with ponds and hand pump drying up in many areas, peoples are forced to depend the water tanks provided by Ranchi Municipal Corporation. Though in general the main causes of water scarcity are climate change accompanied with high temperature, low precipitation and loss of vegetation cover but every geographical area should have some inborn causes related to its origin, structure, geographical location and setup. Water is one of the most important natural resource available to mankind.

*Corresponding author :
Phone : 9693806396
E-mail : manoranjan.mehta01@gmail.com

For sustenance of life, the need for conservation of water bodies especially the fresh water bodies is being realized everywhere in the world. Lakes, rivers and reservoirs are most important water resources and used for several purposes. Physicochemical Analysis on Kanke Dam was aimed to assess the deterioration of water quality. A number of authors have studied the physical and chemical characteristics of some Indian water bodies.[1-18] The Characteristic feature of Ranchi Jharkhand is the presence of numerous lakes, ponds and reservoir which offer immense scope for fish culture practices. In the district Ranchi itself, there are three large lakes viz.; Getalsud Reservoir, Hatia Dam and Kanke Dam. Such a vast area water is presently in a state of deletion and neglect and warrants immediate attention of the fishermen if exploitation on scientific basis. Kanke Dam is formed at Potpoto River.

The lake, extending between the parallels 23°23'50" N latitude 85°18'15"E longitude cover an area of 182.10 hectors of perennial water. The objective of the study was to monitor the water quality of the lake with respect to its suitability for drinking water and pisciculture potential.

MATERIALS & METHODS

Monitoring of the Lake carried out twice in month during December 2017- January 2020 from 8 pre-identify locations following the standard sampling procedure.[19] The samples were transported to the laboratory and analysis for the different parameters viz. Electrical conductivity, Turbidity, pH, BOD, DO, COD, total alkalinity, Chloride, Total hardness following standard procedure.[20]

RESULTS & DISCUSSION

The physicochemical parameters of the Kanke Dam have been given in the table 1. The mechanism of controlling the chemistry of surface water has been discussed by Gibbs (1970)[21]. According to him there are three basic origins for chemical load of dissolved salts in surface water, atmospheric precipitation, rock weathering and crystallization. The temperature has greater role in the dynamic of aquatic ecosystem. In the present study both the atmospheric and water temperature fluctuated from 31.0ºC to 34.6ºC and 28.3ºC to 30.0ºC respectively at different studied sites. The variation in water temperature may be due to monitoring time and fluctuation in the weather conditions of the area.[22] Temperature dictates the behavioral characteristics of organisms, solubility of gases and salts in water bodies. No other factor has so much influence as temperature.[23] The Electrical Conductivity (E.C.) ranged from 17.8 to 37.7 µmho/cm. The maximum values (37.7) were recorded in spring and minimum (17.8) in winter. The increase of E.C. is a indication of the enrichment of the ions in water bodies and may be due to natural sources or by the anthropogenic sources. Since no significant manmade contributor was notices during the study which indicated that only natural sources imparting the variation that too not significant. Similar kind of observation has been recorded by Rajyalakshmi *et al.* (1988)[24] i.e. specific conductivity varied between 8.70 to 8.90 m mhos/cm in the ponds of Chilka fringe area indicating the high salt contents. The Turbidity of the lake water did not fluctuated significantly and remained under the prescribed limit i.e. 8.78 against the prescribed highest desirable of 5 NTU and maximum permissible limit 25.0 NTU. The pH of the Kanke dam at most of the studied sites was more than 7.5 except at 2, 4 and 7 location. The pH and alkalinity goes hand to hand and are inversely proportional to water and air temperature. These are positively correlated to each other.[25] As per the water classification on the basis of pH water falls under the "A" class and is fit for drinking purpose provided all other parameters also fall under the class.[26] Rajyalakshmi *et al.* (1988)[24] recorded higher values of pH ranging from 8.3 to 8.9 in the ponds of Chilka fringe area. Wetzel (1975)[27] reported that the value of pH ranges from 8 to 9 in Indian waters. The pH of pond water was influenced by the monsoon, temperature and soil conditions. Dissolved oxygen (DO) concentration in the water bodies is a kind of indication of general health of lake. During this study the DO level was found more than 6 mg/l which showed a very good health of the lake i.e. under the "A class of water as per the Water classification. DO level in the water bodies is temperature dependent. It also depletes as the organic load increases. Similar kind of study is also reported by Rajyalakshmi *et al.* (1988)[24] which indicated DO concentration ranging from 3.40 to 6.52 mg/l in the brackish water ponds of Chilka fringe area. Thampy et al. (1988)[28] observed that DO concentration fluctuated between 1.4 to 8.2 mg/l in the saline ponds at Cochin. Devaraj (1988)[29] noted concentration of DO from 3.7 to 8.2 mg/l. In the present study, the dissolved oxygen ranged from to 6.0 to 8.0 mg/l during the first year of study while during second year it ranged from 6.1 to 8.4 mg/l. Biochemical Oxygen Demand (BOD) is the key parameter which indicates the organic load of the water body or aquatic ecosystem. BOD in the study area varied from 1.16 to 2.1 mg/l during study period which indicates that the existing ecosystem do not have any significant source of organic pollution and existing system is capable of assimilating the organic load in case any pollution occurred. Similar finding has been projected by the Paramesher *et al.* (2012)[15] at Kunigal lake in India. Chemical Oxygen Demand (COD) is a measure of oxygen equivalent to the requirement of oxidizing organic matter contents by a strong chemical oxidizing organic matter contents by a strong chemical agent. The COD test is helpful in assessing the toxic conditions and the presence of biologically resistant organic substances. The COD concentration fluctuated between 3.22 mg/l- 4.32 mg/l. The variation is not much. The COD data infer that no

chemically active substances are entering in the aquatic ecosystem which may alter the primary productivity of the system. The alkalinity of water is its capacity to neutralize acids; it ranged from the 153 mg/l to 200 mg/ l which is well within the desirable level of alkalinity for drinking water assigned by BIS (1991)[30]. Rajyalakshmi *et al.* (1988)[24], Gupta *et al.* (2008)[31] and Parameswara *et al.* (2012)[15] reported similar observation in the Kunigal Lake in Tumkur district, Karnataka of the India. The Chlorides occurs naturally in all types of water and waste water forming a major inorganic anion. In the present study the values ranged from a minimum of 115 and maximum.164 mg/l. Similar findings are projected by the Singh *et al.*, (2011)[32] for Kodaikanal Lake (Dindugal District), in India. Calcium and Magnesium were the dominant cations in the lotic aquatic ecosystem.[33] Calcium concentration was minimum of 52.5 mg/l to the maximum of 75.7 mg/l and the magnesium concentration of the present study is ranged from 13.1 mg/l to 26.8mg/l and present observation is in line with the reports of Kumar *et al.* (2006)[34].

Table 1. Physico-chemical parameter of water quality of Kanke Dam

Parameter	Unit	Site I	Site II	Site III	Site IV	Site V	Site VI	Site VII	Site VIII	Mean	SD
Atm.Temp.	°C	32.4	33.6	34.2	33.8	32.4	39.6	33.2	31	33.15	1.17
Water Temp.	°C	29	28.4	29.5	28.5	30	29.2	28.7	28.2	28.93	0.61
EC	µmhos/cm	32.3	37.7	30.8	17.8	24.5	18.8	28.2	33.2	27.91	7.055
Turbidity	NTU	6.4	7.4	6.74	8.78	7.24	8.6	6.5	8.23	7.49	7.055
pH		7.97	7.35	7.88	7.48	7.89	7.97	7.39	7.57	7.68	0.943
DO	mg/l	7.17	6	6.81	7.56	8.5	6.72	6.36	8.55	7.21	0.666
BOD	mg/l	1.58	1.16	1.74	1.78	1.43	2.1	1.71	1.28	1.59	0.302
COD	mg/l	4.32	3.67	3.32	4.12	3.34	3.86	4.1	3.22	3.74	0.42
Total Alkalinity	mg/l	165	186	155	205	187	193	153	200	180.5	20.2
Chloride	mg/l	120	115	117	145	164	134	142	128	133.12	16.74
Total Hardness	mg/l	75.0	70.4	77.5	79.3	78.0	100.0	89.0	76.0	80.65	9.42
Ca+ Hardness	mg/l	57.6	52.8	58.5	52.5	60	67.5	75.7	57.5	60.26	7.789
Mg+ Hardness	mg/l	17.4	17.2	19	26.8	18.3	24.4	13.1	18.5	19.38	4.291

CONCLUSION

The present study concluded that the characteristics of Kanke Dam is maintained with respect to the studied physico- chemical parameters and the water resources may be utilized for the propagation of the pisciculture at commercial scale which may uplift the economic status of the nearby habitation apart from the irrigation and drinking water source (after disinfection).

ACKNOWLEDGEMENT

The author extends thanks to the authorities for extending the laboratory facilities at Jharkhand State Pollution Control Board, Ranchi during study.

REFERENCES

1. **Rawat, M. S. 2002.** Diurnal variation is physico-chemical profile and fish fauna of Saudi Gad., *Geobios*. **29:**151-153.
2. **Arjariya, A. 2003.** Physic-chemical profile and plankton diversity of Ranital lake, Chhatarpur, M.P., *Nature Environment and Pollution Technology,* **2:** 327-328.
3. **Sharma, L.L. and Sarang, S. 2004.** Physico-chemical limnology and productivity of Jaisamand lake, Udaipur (Rajasthan). *Pollution Research*. **23:** 87- 92.
4. **Tiwari, A. and Chauhan, S.V.S. 2006.** Seasonal phytoplankton diversity of Kitham lake, Agra. *J. Environ. Biol.* **27:** 35-38.
5. **Tamot, S. and Sharma, P. 2006.** Physico-chemical status of reference to phosphate and nitrate concentration and their reference to phosphate and nitrate concentration and their impact on lake ecosystem. *A. J. Exp. Sci.,* **20:**151-158.
6. **Vyas, A. Mishra, D.D., Bajapai, A., Dixit, S. and Verma, N. 2006.** Environment impact of idol immersion activity lakes of Bhopal, India Asian J. Exp. Sci., **20:** 289-296.
7. **Kumari, S., Sahu, K., Mehta, A., Shukla, S. 2007.** Assessment of physico-chemical parameters of Sagar lake, India, Asian of physico-chemical parameters of Sagar lake, India, *Asian Journal of Bio Science,* **2:** 76-78.
8. **Khare, K. C. and Jadhav, M. S. 2008.** Water quality assessment of Katraj lake, Pune (Maharashtra, India): Sengupta, M. and Dalwani, R. (Editors), 2008, Proceeeding of Taal 2007: the 12th world lake conference. 292-299.
9. **Srivastava, N., Harit, G. and Srivastava, R. 2009.** A study of physico-chemical characteristics of lakes around Jaipur, India, *J. Enviro. Biol.* **30:** 889-894.

10. **Patra, A.P., Patra, J.K., Mahapatra, N.K. Das, S. and G.C. Swain. 2010.** Seasonal variation in physicochemical parameters of Chilika lake after opening of new mouth near Gabakunda, Orissa, India. *World Journal of Fish and Marine Sciences*. **2:** 109-117.

11. **Puri, P.J., Yenkie, M.K.N., Battalwar, D.G., Gandhare, N.V. and Dhanorkar, D.B. 2010.** Study and interpretation of physico-chemical characteristic of lake water quality in Nagpur city (India). *Rasayan J. Chem.,* **3:** 800-810.

12. **Koli, V.K. and Ranga, M.M. 2011.** Physicochemical status and primary productivity of Ana Sagar lake, Ajmer (Rajasthan), India. *Universal Journal of Environmental Research and Technology.* **1:** 286-292.

13. **Gupta, P. Agarwal, S. and Gupta, I. 2011.** Assessment of Physico-chemical parameters of various lakes of Jaipur, Rajasthan, India. *Indian Journal of Fundamental and Applied Life Sciences*. **1:** 246-248.

14. **Sharma, S., Yadav, R.K., Saini, Y. and Sharma, S. 2011.** Water quality status of Pushkar lake as a primary data for sustainable development. *South Asian Journal of Tourism and Heritage*. **4:** 184-192.

15. **Parameswara, T.N, Ajayan, K.V. and Lokesh, G.H. 2012.** Physcio-chemical characteristics of Kunigal lake in Tumkur Disctrict, Karantaka, India. *Int. J. Chem. Sci.* **10:** 655-663.

16. **Kulkarni, S.V. and Tapase, B.S. 2012.** Physico-chemical parameters and water quality index of Gandhisagar lake of Umrer in Nagpur district. *Indian Streams Research Journal*. **1:** 1-4.

17. **Khan, R.M., Jadhav, M.J. and Ustad, I.R. 2012.** Physicochemical analysis of Triveni lake water of Amravati District in (MS) India, *Bioscience Discovery*. **3:** 64-66.

18. **Naik, T.P., Ajayan, K.V. and Lokesh, G.H. 2012.** Physicochemical characteristics of Kunigal lake in Tumkur district, Karnataka, India. *Int. J. Chem. Sci.,* **10:** 655-663.

19. **NEERI. 1991.** Manual on water and waste water analysis. National Environmental Engineering Research Institute, Nagpur

20. **American Public Health Association. 2005.** Standard methods for the examination of water and waste water (12th edn.). Washington, DC: American Public Health Association.

21. **Gibbs, R. J. 1970.** Mechanisms Controlling World Water Chemistry. *Science*. **170:** 1088-1090.

22. **Jayaraman, P.R., Ganga D.T. and Vasudena Nayar, T. 2003.** Water quality studies on Karamana River, Thiruvananthapuram district South Kerela, India. *Pollution Research*, **22:** 89-100.

23. **Welch, P.S. 1952.** Limnological methods. New York: McGraw-Hill.

24. **Rajyalakshmi, T., Mohanty, A.N., Ravichandran, P. and Pillai, S.M. 1988.** The soil and water characteristics of confined brackish water ponds of Chilka lake fringe area. *Proc. First Indian Fisheries Forum, Mangalore.* 125-128.

25. **Ranjan, C.P., Chinmoy, C. and Raziuddin, M. 2007.** Impact of human activity on water quality of lentic water body in *Asansol. Nat. Environ. Pollut. Technol.,* **5:** 59-62.

26. **CPCB. 2008.** Guideline for water quality management, New Delhi, CPCB/MINARS/27/2007-2008: 1-35.

27. **Wetzel, R.G. 1975.** Limnology, W. B. Saunders Co., Philadelphia, USA, 74.

28. **Thampy, D.M., Jose, S. Mohan, M.V. and M.S.S. Koya. 1988.** Short-term biculture of tiger prawn, *Penaeus monodon* (Fabricius) and milkfish, *Chanos chanos* (Foskel) in a low saline ponds. *Proc. First Indian Fish. Soc. Mangalore*. 139-141.

29. **Devaraj, K.V., Mahadeova, H.S. and Fazal, A.A. 1988.** Hydrology of the Hemavathy reservoir, *Proc. First Indian Fish. Forum, Mangalore.* 323-327.

30. **BSI. 1991.** Indian standard specification for drinking water, New Delhi.

31. **Gupta, P.K. Nagdali, S.S. Tewari, P. and Singh, N. 2008.** Water chemistry of a national lake of India: Lake Nainital, Uttarakhand, Sengupta, M. And Dalwani, R. (Editors), 2008, Proceeeding Of Taal 2007: The 12th World Lake Conference. 209-216.

32. **Singh, R.P. and Balasing, G.S.B. 2011.** Limnological studies of Kodaikanal lake (Dindugal District) in special reference to phytoplankton diversity. *Ind. J. Fund. & Applied Life Sci.,* **1:** 112-118.

33. **Buckney, R T. 1980.** Chemistry of Australian waters: the basic pattern with comments on some ecological implications. In Williams, W. D. (Ed.), An Ecological Basis for Water Resources Management, Aust. Nat. Univ. Press, Canberra.

34. **Kumar DS, Sukumar NC, Jana C and Philipose MT. 2006.** Study on physico-chemical characteristics of Thunga River. *Phykos*. **32:** 27 - 39.

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp. 379-382 **49(IS).**
An International Biannual Refereed Journal of Life Sciences
Interdisciplinary Science

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Role of artificial recharge of groundwater causing changes in organic matter of soil: A case study of some villages of Palamu District, Jharkhand

Shyam Lal Singh*
University Department of Geology, Ranchi University, Ranchi, Jharkhand, India

Received : 12th February, 2020 ; Revised : 27th February, 2020

Abstract- The leaf litters, wood materials, roots and other soil organisms fall on the surface and get mixed to form organic matter after decomposition. It percolates down the land surface as solution along with percolating water during artificial recharge. The impervious rocks check percolation and the concentration of the organic matter in the soil rises upward which changes the soil composition & cause significant changes in the organic matter of soil along with increase in the water level of the area.

Key words: artificial recharge, groundwater, soil, infiltration, organic matter

INTRODUCTION

Rain water is the main source of water on the surface of the Earth. However melting ice mass also constitutes but goes as runoff and finally reaches the major water bodies. The rainfall evaporates out as fly off, part flows as surface runoff and part percolates downward enriching groundwater. The rate of percolation depends upon the porosity and permeability property of the rock. The more porous is the rock the rate of percolation is much higher through interspaces. While percolating it is not only the water but the dissolved mineral content also percolates and on saturation increases a particular element in the water. Part of it absorbed by the top soil cover and thus brings about change in the organic matter.

*Corresponding author :
Phone : 6200212325
E-mail : ssgeology11@gmail.com

This case study deals with the study about the role of artificial recharge of ground water in bringing change in organic matter of soil.

Location

The study area geographically falls within the Palamu district of Jharkhand state and is covered in the Toposheet No. F45A1 of Survey of India. The study area forms a part of a hard rock terrain and covers an area of 5246 sq kms. The location can be approached by road only and is about 27 km from Medininagar on North west side the city.

The General Geology of the Area

The Pre Cambrian crystalline rocks, Vindhyans and Gondwanas beside the top soil cover on the three main geological formations available in the district. Besides this some amount of lateritic soil is also present in part of the

area. Granitoid Gneiss as the Pre Cambrian basement, shales and sandstone of Gondwanas and in some areas the Vindhyan sandstone located far are found in the region of study area.

The northern area of the district where Koel river and adjacent Sone river is available, some amount of alluvial cover of considerable thickness are found. The thickness is variable which depends upon the topography of the area and undulating terrain. Soils are also lateritic in nature which is having high iron content, Al content, low Nitrogen, Potassium and Phosphorus. The soil is less fertile and is prone to acid reaction. Here the occurrences of ground water are confined under phreatic condition. The top soil cover allows the percolation of rainwater.

Artificial Recharge

The percolation of rainwater takes place due to natural and artificial recharge has been used because the intermittent intervals were constructed to check the surface runoff and which allows them to percolate downward recharging the ground water.

Artificial recharge requires permeable surface soil, trenches, ring ridge, ridge- ditch technique to check the surface runoff and thus provides sufficient time to percolate water to form groundwater.

The short term and long term plan is needed for this. However the low cost technology provides much higher result in artificial recharge. Moreover the slope of the area which varies between 3° to 26°C at the surface plays key role. The lesser the slope, the lower will be the runoff and thus in such area the intermittent intervals can be constructed.

Artificial recharge is directly related to porosity and permeability property of the rock of the study area, which was considered during the case study.



Role of Sustainable Resource Management (Chakriya Vikas Pranali) in Artificial recharge

The concept of Cyclic System of Development locally called as Chakriya Vikas Pranali (C.V.P.) envisaged by the noted soil scientist Padmashree Late P.R. Mishra. Initially his experiments of soil and water conservation declared successful at Sukhomajri near Chandigarh in Shiwalik hills. This stopped the siltage of Sukhna Lake of Chandigarh and got appreciation on National level.

Seeing the Forest wood cutting by the villagers for their livelihood he replicated the model in part of Palamu district by teaching villagers (were called Students) with the sharing system for equity among all the share holders. This sharing system was the key success of soil and water conservation scheme as shown below-

30% - Land owners, this includes the absentee land lords also

30% - Students working round the year on the plot

30% - Saving of the cycle which was ploughed back as capital for the next cycle

10% - Welfare fund (Kalyan Kosh) meant for emergency and to replicate such models in the nearby regions.

C.V.P. started making ridge, tie - ridges, ridge ditches technique, stone pitching, sand bag filling and other small intervals to check the surface runoff. Besides this small water storage tanks, check dams were also constructed.

This caused the enrichment of ground water to a considerable limit and made the system sustainable by using natural resources.

The core features of C.V.P. are enumerated as below-

a. Use of private fallow land by pooling them in the system.

b. Maximum utilization of natural resources particularly rainwater by using the theme- as it falls and where it falls for in situ conservation.

c. Multi layered cropping system in a single piece of land as root crops, seasonal vegetables, fruit species and forest species which yield income as various levels.

d. Mobilization of villages for regeneration of natural resources.

e. Sharing system as discussed above for bringing equity in the society.

Environmental Impact of Artificial Recharge

The effects of Artificial Recharge on the environment can be both the harmful as well as beneficial but it varies from place to place. During the process of artificial recharge some suspended solids and chemicals also leaches downwards along with the water which on precipitation clogged the top layer and reduces the percolation rate drastically. Clogging of soil matrix is a major operation problem of infiltration system for artificial recharge of ground water. Clogging can be of even greater importance in waste water lagoons or constructed wet lands where injection wells used are much more vulnerable to clogging than surface infiltration system. Remediation of clogging in boreholes and wells are much more difficult than in surface infiltration system.

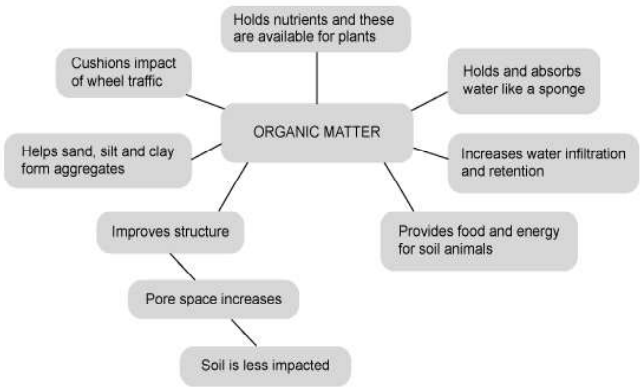
The chemicals and other constituents of soil on reaching to the groundwater along with percolating water and contaminate it badly and thus water quality becomes low. The chemical constituents and temperature of surface are different groundwater. Consequently the issue of ground water quality change. The groundwater quality presented the trend of desalination and general increase in sulphate, iron, manganese contents, organic and nitrogenous compounds after the tap of water injection. However the artificial recharge influence on the ground water quality with the radius <100m for a single well recharge and >1000m for group recharge.

On the other artificial recharge through various methods raises the water level in the area and also becomes green. Through the artificial recharge the soil enriches with organic matter present in the field.

Presence of organic matter in recharge zone

The sources of organic matters are leaf litters, woody materials, root and residue of dead bodies of animals and various soil organisms. When the leaf litters, woody materials fall on the surface of the earth and mixed to form organic matter after decomposition in the presence of water.

During the artificial recharge the organic matter present on the surface of the earth percolates along with percolating rainwater as solution downwards. When it finds impervious layer percolation stopped and the concentration of organic matter rises upward and thus the soil becomes enriched with organic water. The soil rich in organic matter also enhances infiltration of rainwater.





DISCUSSION

The rate of percolation is directly proportional to quantum and days of rainfall as well as the existing geological condition of the area. The percolating water acts as an agent in bringing concentration of organic matter, which at first stage partially dissolved organic matter and then on saturation recrystallised mixing with the top soil cover and causes changes in organic matter of soil.

CONCLUSION

The constituent unit of soil along with the chemicals present on the surface of the earth dissolves and reaches the water table. This increases the concentration and badly affects the groundwater causing lowering of water quality.

This concentration at the top soil changes the organic matter and the richness of a particular element changes the soil composition to a considerable limit. The artificial recharge process acts as catalytic agent in bringing concentration of organic matter in the soil. The soil sample analysis has also shown the positive results.

The detailed study is being carried out on pre and post monsoon seasons and the data in future will follow.

ACKNOWLEDGEMENT

The author extends thanks to his guide, Dr. P.K.Verma, Associate Professor, University Department of Geology & Dean Student's Welfare, Ranchi University, Ranchi, Jharkhand for his support and guidance.

REFERENCES

1. **Biswas, A.K. 2007. '**Water as a Human Right in the MENA Region: Challenges and Opportunities', *International Journal of Water Resources Development.* **23(2):** 209-225.
2. **Mancosu, N., Snyder, R., Kyriakakis, G. and Spano, D. 2015.** 'Water Scarcity and Future Challenges for Food Production', *Water*.**7(3):** 975-992.
3. **Muralidharan D & Athavale R.N. 1998.** The Ground Water Recharge Movement in India by Ramaswamy Sakthivadivel, pg 195.
4. **Verma P.K & Sinha S.P. 2000.** Artificial Recharge of Ground Water in hard rock terrain through Chakriya Vikas Pranali - a case study of Palamu district in Bihar. Proc of Integrated water resource management for sustainable development, 2000.
5. **Thesis, C.V. 1940.** The source of Water derived from wells: Essential factors controlling the response of an aquifer to development. *Civil Engineering*. **10(5):**277-280
6. **P. Nojd, *et al.* 2009.** "Artificial Recharge of Groundwater through Sprinkling Infiltration: Impacts on Forest Soil and Nutrients Status and Growth of Scots Pine," *Science of the Total Environment*. **407(10):** 3365-3371
7. **Dipankar Saha. 2010.** Hard rock aquifers in Jharkhand state and its development perspective. *Proc. of National Conference in ground water resource and management in hard rocks*.
8. *CGWB*.The District Administration, Geology of Mines Department, Govt. of Jharkhand, SHRMS (Chakriya Vikas) publication.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.383-386 **50(PS).**
An International Biannual Refereed Journal of Life Sciences ***Plant Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

Strategies for improving tasar (Kuchai) silk industry in the state of Jharkhand: Need for improving the working conditions of the artisans associated with reeling of cocoons and weaving

Shashi Nag[a]*, Manju Kumari[a] & Ashok Kumar Nag[b]
[a]University Department of Home Science, Ranchi University, Ranchi, Jharkhand, India
[b]University Department of Botany, D.S.P.M.University, Ranchi, Jharkhand, India

Received : 11th February, 2020 ; Revised : 26th February, 2020

Abstract : During a study on 'apparel for youth based on contemporary design on core with Tasar (Kuchai) silk of Jharkhand', information from the artisans involved in reeling the silk thread from the cocoon and the weavers of fabric for production of apparel at the capital city, Ranchi of the Jharkhand State were collected with the help of written questionnaire as well as by personal interviews. The purpose of such an investigation was to obtain the free and unbiased, but confidential opinion of the artisans with regard to their working conditions and their views regarding the steps which are required to improve better production as well as their job satisfaction. More supply of quality cocoons for reeling; improvement in the machines and continuous supply of electricity for weaving, regular disbursement of the proper wages/honorarium to the workers are the main hurdles for quality and quantity production. Such a research finding needs attention of the higher officials of the Tasar Industry in Jharkhand state for improvement of the Industry.

Key words:Tasar silk, cocoon, reelers, weavers, apparel

INTRODUCTION

Tasar (Kuchai) Silk is a wild tropical silk. The main geographical area of its occurrence is the State of Jharkhand and its adjoining and nearby other states of West Bengal, Odisha, Chhattisgarh, Andhra Pradesh, Madhya Pradesh, Uttar Pradesh and Bihar.[1] The Tasar silk worm, *Antheraea mylitta*, is a species of moth family Saturniidae, and its main host food plants: *Terminalia tomentosa, Terminalia arjuna* and *Shorea robusta* are naturally-growing tree species, found in plenty in the forests of these States.[2] Primarily because of this and to promote and boost the Tasar silk Industry in the whole of the country, just after a few years of Independence, in the year 1964, the Central Government of India established the Central Tasar Research and Training Institute, Ranchi. Its focus is to generate useful technologies through Research & Development and its effective transfer in the field, with an ultimate aim to improve the socio-economic status of the stakeholders associated with Tasar Industry.[3]

The stakeholders associated with the Tasar industry are the cocoon collectors from forest trees, the farmers who cultivate food host plants in their field and grow and rear the silk worm to obtain cocoons, the traders who purchase these cocoons to sell it to the reeling houses, the

*Corresponding author :
Phone : 09930308956
E-mail : nagshashi29@gmail.com

Management of artisans involved in reeling the silk yarn, and those associated with the weaving of textile fabric, apparel design and production keeping the contemporary fashion trends and its marketing to the consumers. Techniques of improvement in quality and quantity at all the above mentioned stages have to be continuously attempted by the researchers and managers for prosperity of the Tasar Silk Industry.[4]

MATERIALS & METHODS

During a study on 'Apparel for youth based on contemporary design on core with Tasar (Kuchai) silk of Jharkhand', information from 300 consumers and 100Artisans (Producers) of apparel were collected with the help of written questionnaires as well as through personal interviews. This paper deals with the findings concerning the attitudes, difficulties and limitations with respect to the working conditions and scope of innovations in apparel production by the artisans/craft-persons associated with garment manufacturing utilizing tropical Tasar (Kuchai silk) in and around Ranchi.

The main motivation behind the present investigation is to help popularize the apparel constructed from Tasar (Kuchai) silk which has its main home in Jharkhand so that this industry could grow and benefit all its stakeholders.

Questionnaire for Artisans associated with Tropical Tasar Silk contained questions related (after their personal introduction to feel at home and knowledge whether anybody else of the family/closed relative is also working in Tasar silk Industry) to their Occupation/Task allotted/ Duty; whether they would like to have a job change if given a better opportunity; whether some difficulties/ problems are faced during the entire process of production; what are the limitations of work allotted to them; salary/wages given per month; whether they are aware of any such schemes that the Jharkhand Government has been providing for the welfare of workers/ artisans of the Tropical Tasar Silk producers; whether any training program was given to them as per their work/ duty; whether they had attended any such training programs before joining, if yes, where and for how long; whether new machines/new technology have been introduced at their work place for better and fast production; and the last personal question was whether they wish to provide any kind suggestion or idea how to produce better and more for prosperity.

RESULTS AND DISCUSSION

The questionnaires were handed over to the artisans working at three places around and within Ranchi, namely,

1.. The Central Tasar Research & Training Institute, Ranchi located at Piska Nagri, about 18 km west of Ranchi town. (Fig.1).
2. Silk Centre, Hehal, Ranchi. (Fig.2).
3. Naya Sarai, Weaving Cluster, Ranchi. (Fig.3).



Fig.1. Separation of quality cocoon at Hehal (Ranchi)



Fig. 2 Reeling of cocoons for silk–thread



Fig.3.Weaving of Tasar silk Saree at Nayasarai

The information obtained from the Artisans were classified and analyzed. With respect to the question whether they would like to have a job change if given a better opportunity, about 90% of the workers were not eager to have a job change. Artisans working in the Weaving Cluster for the last 12 years were 16%; for the last 11years 42%, for the last 10 years 23% and for the last 7 years were 19%. This indicates that they have a liking and are interested in their profession. Usually, only one person of the family (mainly ladies) works in these Clusters of reeling and weaving; and most of them were not even matriculate!

Response of the Artisans regarding the difficulties they face during work could be classified into the following four categories:

Category 1: Yarns easily breakable, because not good quality cocoons are provided; complain of eye problem, headache.

Category 2: Workers request for help from expert masters.

Category 3: Essentials (Raw materials) are not provided on time.

Category 4: Continuous supply of electricity is needed.

With respect to the factors responsible for the limitations of work allotted to them, the Artisans response was about, good quality raw material as well as the equipment and its maintenance. It is worth mentioning here that a more or less similar finding about the reelers and weavers of the cultivated Silk (on the host food plant *Morus alba* by the silk worm *Bombyx mori*) Industry at field level in Murshidabad district of West Bengal was found by Majumdar *et al.*(2017)[5].

The salary/wages given per month to the workers in an Organization/Industry may be a looked as an index of its health and prosperity. The Artisans in the Tasar Silk Industry of Jharkhand are skilled workers; and hence it is expected that they should earn per month more than the unskilled daily wage labour would earn during a month. In the capital city of Ranchi, an unskilled daily wage lady labour is paid Rs 350/- per day. If she works for 26 working days in a month, her monthly wage would be Rs 350 x 26= Rs 9100. Compared to the expected wages of an unskilled lady daily-wage worker, it was observed that the average monthly wages of an Artisan at the Reeling & Weaving Cluster at Nayasarai (Ranchi) was Rs 5000/- only (Fig.4)

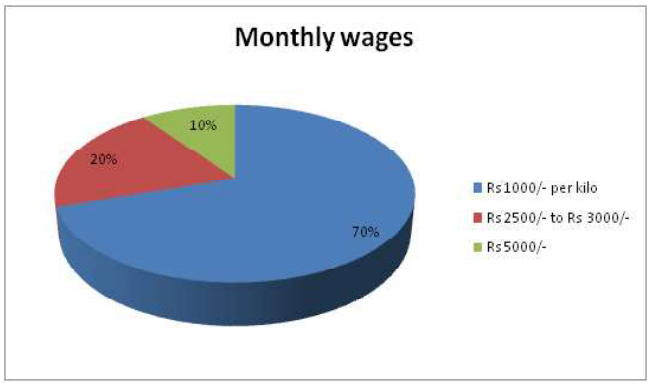


Fig.4. Salary/wages per month of Artisans

It is worth mentioning here that the reelers as well as the weavers in the silk industry in Murshidabad district of West Bengal were also found to be pitiable by the survey study of Majumdar *et al.*(2017)[5] which revealed that 90% -100% reelers and weavers were not satisfied for their wages.

With respect to the question of whether they are aware of any such schemes that the Jharkhand Government has been providing for the welfare of workers/ artisans of the Tropical Tasar Silk producers, only 58% of them answered in affirmative (Fig. 5).

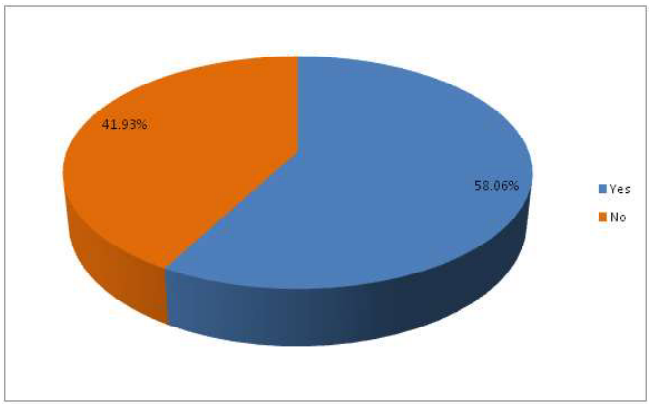


Fig.5. Awareness of the schemes provided to the workers by the Jharkhand Government

About 60% of the Artisans were given training program as per their work/duty and that training was for about a month only. There was no refresher training for these workers with new machines/new technology with expert technicians and scientists of the Research Institutions for better and fast production. In this age of fast developments in Science & Technology, it is very essential to provide training the workers off and on to

cope up with newer tools and instruments for quick and better industrial production!

The workers responded to the personal question to provide their kind suggestion or idea how to produce better and more for prosperity, and these have been summarized below:

Table 1- List of suggestions given by the workers for better production in the Tasar Silk Industry of Jharkhand

Sl. No.	Suggestions
1.	Yarn should be provided on time
2.	Advanced technology machines should be provided as manual work is time-consuming
3.	As they belong to poor families, advance payment could be of great help to them
4.	Medical facilities should be provided
5.	One time canteen should be provided
6.	"A" grade cocoon required for better production
7.	Maintenance of machines, also repairing of old and unused machines should be done
8.	Electricity from 10a.m. to 5p.m. should be provided without power cut
9.	Any new machine/technology to bring in for better production

REFERENCES

1. **Reddy, R. Manohar. 2009.** Innovative and Multidirectional Applications of Natural Fibre, Silk - A Review. *Academic Journal of Entomology.* **2(2):**71-75.

2. **Reddy, R. Manohar. 2010.** Conservation need of Tropical Tasar Silk Insect, *Antheraea mylitta* Drury (Lepidoptera: Saturniidae)- Strategies and Impact. *Journal of Entomology*. **7:**152-159.

3. **Nag, K.K. & Nag, Shweta. 2008.** Utilization of Biotechnology for Human Welfare. *In* Recent Trends in Seribiotechnology.1-8. (Eds.K.P.Chinnaswamy & A. Vijaya Bhaskar Rao) International Conference on Trends in Seribiotechnology. Department of Sericulture, Sri Krishnadevaraya University, Anantpur, Andhra Pradesh).

4. Any documentary paper of Jharcraft.

5. **Majumdar, M.; Sarkar, K & Sinha, S. 2017.** A critical analysis on major problems faced by reelers and weavers of silk industry at field level in Murshidabad District of West Bengal. *International Journal of Social Science*. **6(3):**199-210.

***

Biospectra : ISSN : 0973-7057, Vol. 15(1), March, 2020, pp.387-388 **51(PS).**
An International Biannual Refereed Journal of Life Sciences ***Plant Science***

ISSN : 0973-7057

Int. Database Index: 663 www.mjl.clarivate.com

The study of inhibiting the disease causing microbes in the leaf extract of neem plant of Mandar Block, Ranchi, Jharkhand, India

Jyoti Kumari*
P.G. Department of Botany, Maharaja College, Ara (Bhojpur), Bihar, India

Received : 12th February, 2020 ; Revised : 27th February, 2020

Abstract : All the test organisms were screened for their antibacterial activity against leaf and bark extract of Neem by agar well diffusion method. Antimicrobial activity in the leaf extract of Neem (*Azadirachta indica*) against *Salmonella typhimurium*, *Pseudomonas aeruginosa*, *Staphylococcus aureus* and inhibiting the microbes in the extract of neem leaves were used. The aim of this article mainly information on various activities like antidermatic, antiulcer, antifungal, anti-inflammatory, antimalarial, anti-bacterial, anti-allergic, insecticidal and larvicidal of neem plant and medicinal uses.

Key words: *A. indica*, pathogenic bacteria, leaf and bark extract, *S.aureus*, *S.typhimurium*, Agar disc diffusion method, *P.aeruginosa*.

INTRODUCTION

India is known as a rich repository of medicinal herbal plants. In India, Jharkhand is one of the potential natural resource states among all. There are more than 1500 medicinal herbal species are found in the state which are about 5% of the overall medicinal herbal productivity in the country. The indigenous healing practices have been culturally accepted during all phases of human culture and environmental evolution. Traditional medicine is widely used and accounts for about 40% of all health care delivered.

Mandar Block is situated in Ranchi District and lies between latitude 23°27'31" North and longitude 85°05' 32" East. It has total geographical area 238.22 Sq. Km. Total population of the Mandar Block is 128585 (as per Census 2011). Total no. of villages under this block is 69. The average sex ratio of this block is 974. The distance of Block Headquarter is 25 km. The tribal population of the Mandar Block comprises of mostly Bedia, Karmali, Birhore, Santhals, Oraon and Munda. The traditional people of this Block have absolute knowledge of herbals plants. They want to preserve this knowledge and didn't share to the outsider.

The aim of this study was to investigate the antimicrobial activity of the leaf of neem against human pathogenic bacteria, *Salmonella typhimurium* and *Pseudomonas aeruginosa*.

*Corresponding author :
Phone : 06182-248443, 9334252633
E-mail : mkpandey93342@gmail.com

MATERIALS & METHODS

Plant selection: The neem (*Azadirachta indica*) plant was selected from forestry area of Mandar Block and its leaves were collected and preserved.

Extracts of leaf and bark: The completely shaded leaves and barks were provided and allowed for successive soxhlet extraction with ethanol and methanol. The obtained liquid extracts were evaporated till dryness and stored in air tight bottle at 40°C.

Extraction of Ethanol: In a container, 50 gm of dried leaf powder were taken. Approx. 250 ml of ethanol was added into it and left for a whole day with shaking in regular interval. Further it was filtered and collected. This process was repeated three times and after that final filtrates was pooled.

Extraction of Methanol: In a container, 50 gm of dried leaf powder were taken. Approx. 250 ml of methanol was added into it and left for a whole day with shaking in

regular interval. Further it was filtered and collected. This process was repeated three times and after that final filtrates was pooled.

Microorganism: The pathogenic strains of *S. typhimurium, S. aureus* and *P. aeruginosa.*

Agar disc diffusion method: This testing developed in 1940. This method tests the effectiveness of antibiotics on a specific organism. This method is suitable for organism that grow rapidly overnight at 35-370°C. The antibiotic impregnated disc absorbs moisture from the agar and antibiotics diffuse into the agar medium. The rate of extraction of antibiotics from the disc is greater than the rate of diffusion. Antibiotic concentration reduced in logarithmic fashion as the distance from the disc increased. Now zone of inhibition of bacterial growth around each disc is measured.

Medium: The weight of 3.8 gm of agar is added in 100 ml of distilled water. Then autoclaved it for 15 minutes at 1210°C and passed in sterile petri plates to a uniform thickness of approximate mm and the agar is allowed to set at ambient temperature and used.

Inoculums: The microorganisms were inoculated in the medium of peptone and further incubated for 3 to 4 hours at 370°C. This solution was used as inoculums.

A sterile cotton swab was inserted into the bacterial suspension & then rotated and compressed against the test tube wall so as to express the excess fluid. The surface of Agar plate was inoculated with the swab. Now the swab is passed three times over the entire surface. Methanol solvent to dissolve the plant extracts and then placed upon the inoculated agar surface using sterile forceps.

Standard Disc of Tetracycline (30µg/disc) and Streptomycin (10µg/disc), 6 mm in diameter were used as positive control and the solvent used for preparing extract as negative control. The plates were incubated overnight for 18 to 24 hours at 370°C. Antimicrobial activity was calculated by using measuring High media zone scale of inhibition.

RESULT AND DISCUSSION

There are various side effects causing due to the existing synthetic drugs. Hence, drug development plant based compounds could be useful in catering this demand of drugs for minimum side effects. Leaves of neem possessed antibacterial activity and the extracts of *Azadirachta indica* when used as medicinal plant, could be useful for the growth of the bacterium. The phytoconstituents flavonoids, alkaloids, saponins and glycosides are antibiotic principles of plants.

In-vitro activities of leaves of neem in ethanol extract against pathogens

Sl. No.	1.	2.	3.
Name of organism	*Staphylococcus aureus*	*Pseudomonas aeruginosa*	*Salmonella typhimuriun*
Gentamicin 200 mg (std)	14 mm	17 mm	12 mm
Gentamicin 10 mg (std)	08 mm	15 mm	-
Ethanol	12 mm	12 mm	-

In-vitro activities of leaves of neem in methanol extract against pathogens

Sl. No.	1.	2.	3.
Name of organism	*Staphylococcus aureus*	*Pseudomonas aeruginosa*	*Salmonella typhimuriun*
Gentamicin 200 mg (std)	14 mm	17 mm	12 mm
Gentamicin 10 mg (std)	08 mm	15 mm	-
Methanol	12 mm	12 mm	-

REFERENCES

1. **Khalil Ur. Rahman, M. Sarwar, Faisal Aslam and M. Asgar. 2009.** Antibacterial activity of Various Phytoconstituents of Neem. **46(3):**456- 463
2. **Almas, K. Ansal Labi, T.R. 1995.** The natural toothbrush. *World health forum.***16:**206-210.
3. **I.J. Udeinya, A.U. Mbah, C.P. Chijioke, E.N. Shu. 2004.** Transactions of The Royal Society of Tropical Medicine and Hygiene. **98:** 435-437.
4. **Sonia Bajaj, Srinivasan B.P. 1999.** Investigation into the antibacterial activity of *Azadirachta indica*.**31:** 138-141.
5. **Kusum Harjai, Anju Bala, Ravi K. Gupta, Radhika Sharma. 2013.** Leaf extract of *Azadirachta indica*: a potential antibiofilm agent for *Pseudomonas aeruginosa*. **69:** 62-65.
6. **Okafor1, PC, Ebenso, E, Ekpe, UJ. 2010.** *Azadirachta indica* extracts as corrosion inhibitor for mild steel in acid medium. **5:**978-993.
7. **Bhola, SM, Alabbas, FM, Bhola, R, *et al.* 2014.** Neem extract as an inhibitor for biocorrosion influenced by sulfate reducing bacteria. *Eng Fail Anal.* **36:** 92-103.
8. ***Mater* Lett Valek, L, Martinez, S. 2007.** Copper corrosion inhibition by *Azadirachta indica* leaves extract in 0.5 *M sulphuric acid.* **61:** 148-151.
9. **Nahle, A, Abu-Abdoun, I, Abdel-Rahman, I, *et al.* 2010.** Neem extract as a corrosion inhibitor for carbon steel in HCl solution.UAE **9:** 460154–460154.